When Valuation Becomes a World - Complex Finance, Internal Time, and the Residue of Quantum Strangeness

Complex Finance, Internal Time, and the Residue of Quantum Strangeness

How R + iQ Evolves from a Pressure-Preserving Valuation Geometry into a Phase-Ordered, Ledger-Bearing, Backreactive Financial World—and Why That World Helps Separate Generic Observer Effects from Irreducibly Quantum Structure

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Source Note

This article extends the framework introduced in *Finance Geometry: Complex Valuation, Risk Pressure, and the Hidden Coordinate Behind Mature Finance Filters*. That earlier work began from an ordinary fact of financial practice: future economic possibilities do not enter a valuation ledger without filtration. Expected cash flows are discounted, risk-adjusted, probability-weighted, credit-filtered, liquidity-filtered, certainty-adjusted, capital-constrained, and subjected to accounting or regulatory admission rules before they appear as a scalar price or present value.

The original framework proposed preserving an orthogonal complement to that scalar result:

$$Z = R + iQ. \quad (0.1)$$

Here R is admitted value: the component that passes the declared financial filter and appears on the visible valuation axis. Q is retained pressure: the component implied by the same filter but hidden when finance reports only one scalar number. A is the declared pre-filter value amplitude, and θ is the angle induced by the mature financial filter. The defining geometry is:

$$A^2 = R^2 + Q^2. \quad (0.2)$$

$$R = A \cos \theta. \quad (0.3)$$

$$Q = A \sin \theta. \quad (0.4)$$

$$\cos \theta = R/A. \quad (0.5)$$

The earlier article was deliberately finance-first. It did not claim that markets are literal quantum systems, that Q is a second asset price, or that complex numbers replace CAPM, discounted cash flow, certainty equivalents, pricing kernels, credit models, liquidity analysis, option theory, or risk management. It proposed a coordinate extension: mature finance already filters value, and complex geometry may preserve the pressure complement that scalar valuation suppresses.

The present article begins where that static construction ended.

It asks what happens when the finance filter changes through time, when θ becomes θ(t), when

admitted value and retained pressure exchange under a moving valuation frame, when phase becomes a locally usable ordering coordinate, when financial events are committed into ledgers, and when those ledgers alter the economic field that generated them.

The article's central development is therefore not merely:

$$A \rightarrow R + iQ. \quad (0.6)$$

It is the longer runtime:

$$\text{Primary Field} \rightarrow \text{Declaration} \rightarrow \text{Projection} \rightarrow R + iQ \rightarrow \text{Phase} \rightarrow \text{Gate} \rightarrow \text{Ledger} \rightarrow \text{Backreaction} \rightarrow \text{Revision}. \quad (0.7)$$

The second source is *Handoff Full*, which develops the transition from static Finance Geometry toward dynamic phase, internal time, effective-world formation, residual leakage, backreaction, financial memory, and a comparative method for identifying what remains uniquely quantum after more general observer-bound structures are removed.

The resulting article has two linked aims.

The finance-facing aim is to distinguish changes in underlying economic amplitude from changes caused by a rotating valuation frame, and to test whether Q, phase velocity, dynamic residual, and loop memory add explanatory or predictive value beyond existing financial variables.

The physics-facing aim is more methodological. Finance provides a non-quantum system capable of reproducing complex coordinates, phase, contextual projection, order sensitivity, collapse-like commitment, trace, and observer backreaction. These features therefore cannot, by themselves, establish quantum ontology. The deeper quantum question begins only after such generic observer structures have been subtracted.

This article is not investment advice. It is a conceptual and mathematical research framework whose proposed variables require empirical testing against established financial models and null benchmarks.

Abstract

Modern finance converts large fields of economic possibility into scalar values. A future cash flow, firm, bond, project, option, collateral pool, or balance sheet does not enter the ledger raw. It passes through a declared filter: discount rate, certainty-equivalent adjustment, stochastic discount factor, credit spread, liquidity haircut, capital rule, accounting gate, or execution constraint. The visible result is usually one number.

Finance Geometry proposed preserving the complement hidden by this scalar compression:

$$Z = R + iQ. \quad (0.8)$$

R is admitted value. Q is retained pressure. A is the declared pre-filter amplitude, and θ is the angle implied by the mature financial filter:

$$Z = A \exp(i\theta). \quad (0.9)$$

$$A^2 = R^2 + Q^2. \quad (0.10)$$

This article develops the dynamic extension. When both A and θ vary:

$$Z(t) = A(t) \exp[i\theta(t)]. \quad (0.11)$$

Differentiation yields:

$$dZ/dt = [(1/A)(dA/dt) + i(d\theta/dt)]Z + \varepsilon_{dyn}. \quad (0.12)$$

Define radial economic growth and angular filter velocity by:

$$g_A = (1/A)(dA/dt). \quad (0.13)$$

$$\omega_F = d\theta/dt. \quad (0.14)$$

Then:

$$dR/dt = g_A R - \omega_F Q + Re(\varepsilon_{dyn}). \quad (0.15)$$

$$dQ/dt = g_A Q + \omega_F R + Im(\varepsilon_{dyn}). \quad (0.16)$$

This separates visible repricing into three components:

1. change in underlying economic amplitude;
2. change caused by rotation of the financial filter;
3. dynamic residual not explained by the declared world.

The angular repricing load is:

$$\Lambda_F = Q\omega_F. \quad (0.17)$$

Hence:

$$dR/dt = g_A R - \Lambda_F + Re(\varepsilon_{dyn}). \quad (0.18)$$

The framework also distinguishes three forms of temporal order. Calendar time t records external duration. Phase θ records movement through a declared valuation orientation. Ledger time k advances when a consequential event—trade, downgrade, margin call, covenant breach, impairment, default, or regulatory decision—is committed into trace.

The central philosophical proposal is that a financial representation becomes world-like when it supports not only coordinates, but also approximately closed dynamics, admissible events, commitment gates, recorded history, interventions, residual disclosure, and backreaction upon future states.

An effective financial world is therefore defined as:

$$W_P = (\mathcal{Z}_P, \mathcal{D}_P, \mathcal{G}_P, \boldsymbol{\mathcal{L}}_P, \mathcal{U}_P, \boldsymbol{\mathcal{B}}_P). \quad (0.19)$$

Here $\mathcal{Z}_P$ is the effective state space, $\mathcal{D}_P$ its internal dynamics, $\mathcal{G}_P$ its event gates, $\boldsymbol{\mathcal{L}}_P$ its ledger rules, $\mathcal{U}_P$ its admissible interventions, and $\boldsymbol{\mathcal{B}}_P$ its backreaction map.

Under constant amplitude and stable declaration, the internal phase law becomes:

$$dZ/d\theta = iZ. \quad (0.20)$$

This implies:

$$dR/d\theta = -Q. \quad (0.21)$$

$$dQ/d\theta = R. \quad (0.22)$$

$d^2R/d\theta^2 = -R$. (0.23)

$d^2Q/d\theta^2 = -Q$. (0.24)

These are classical rotational or oscillator equations. They do not derive Born probabilities, entanglement, Bell inequality violation, tensor-product nonseparability, or physical wavefunction collapse.

The classical result is not treated as a failed analogy. It relocates the difficult structure. The internal effective world may obey simple laws while its world-forming boundary remains contextual, observer-bounded, path-dependent, reflexive, and history-bearing.

Finance thereby becomes a macro control case for quantum-foundations reasoning:

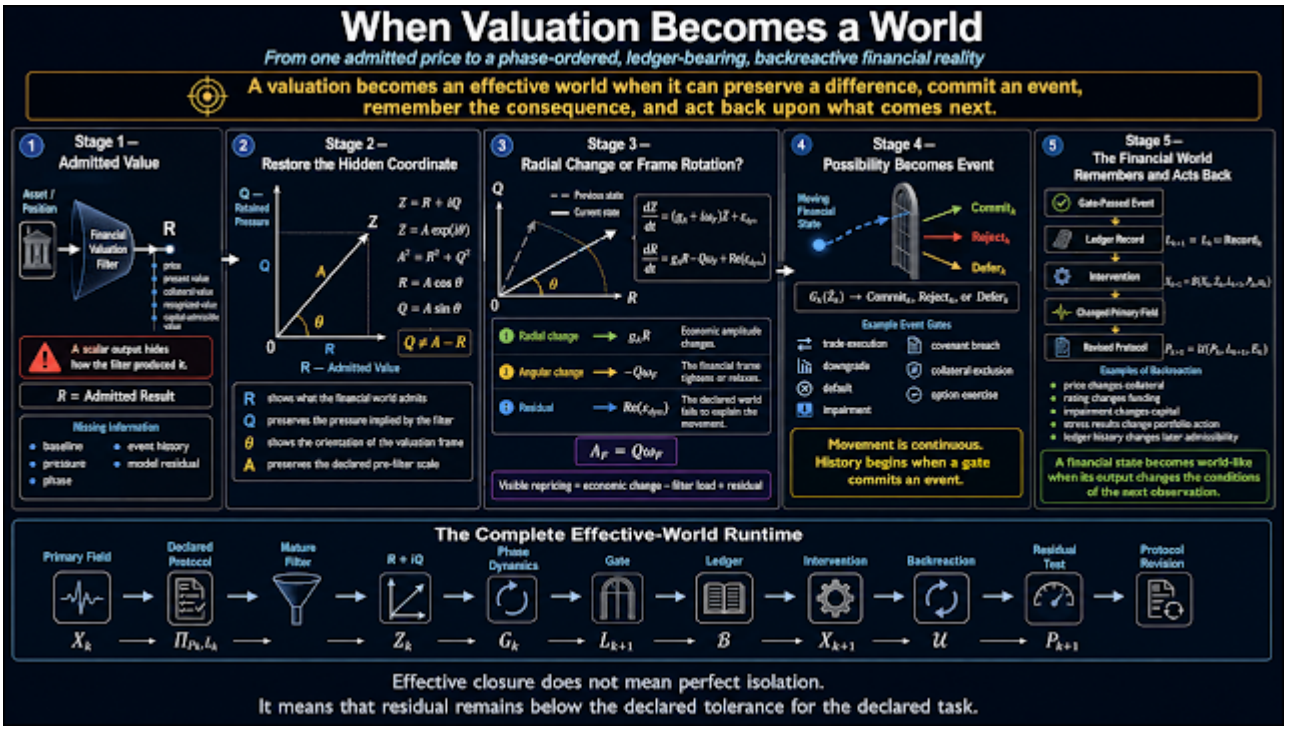
Quantum Phenomenon = Generic Observer-Bound Structure + Irreducibly Quantum Residue. (0.25)

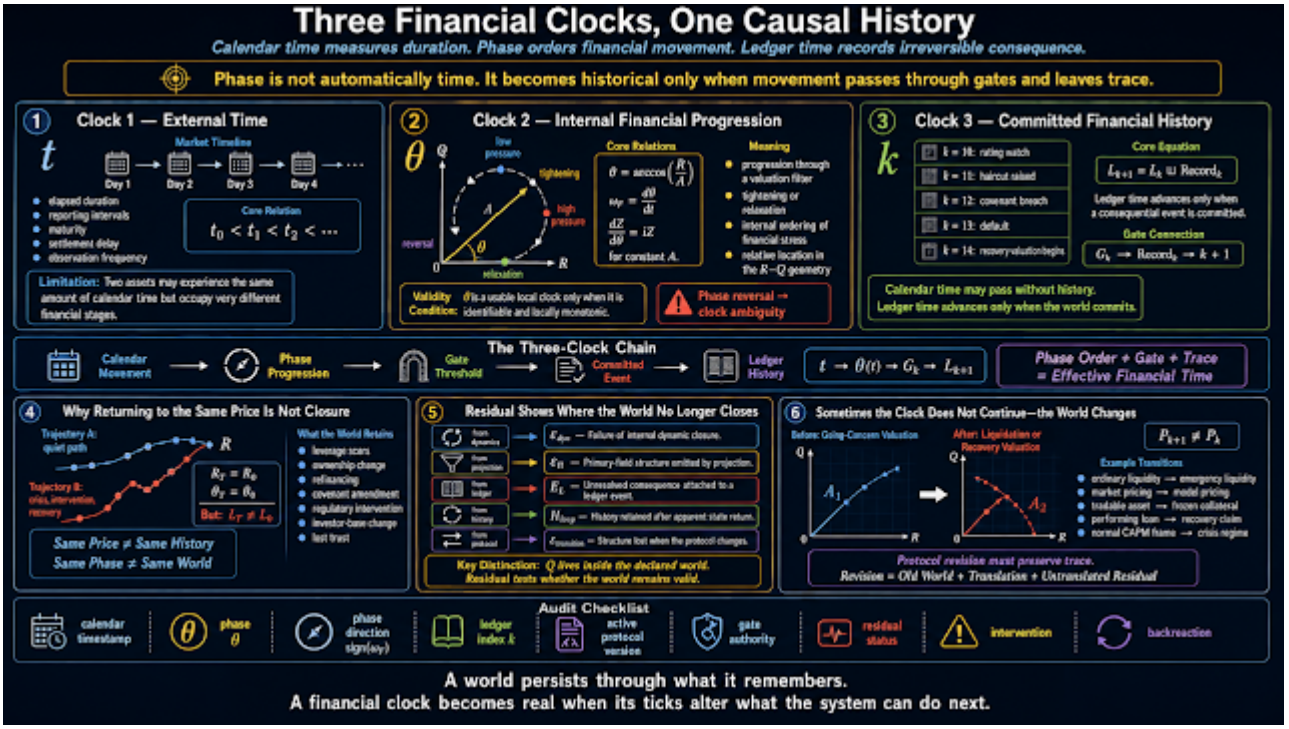
By reproducing complex coordinates, phase, contextual measurement, commitment, trace, order sensitivity, and backreaction without quantum ontology, finance helps identify which apparent mysteries are generic consequences of bounded world construction and which structures remain genuinely quantum.

The article concludes with a falsification programme. Q, Λ_F, dynamic residual, and loop residual must be tested against duration, convexity, beta, spread, volatility, liquidity measures, regime-switching models, VaR, Expected Shortfall, and conventional factor decompositions. If the new coordinates do not improve prediction, diagnosis, attribution, communication, or intervention, they remain elegant notation rather than useful finance.

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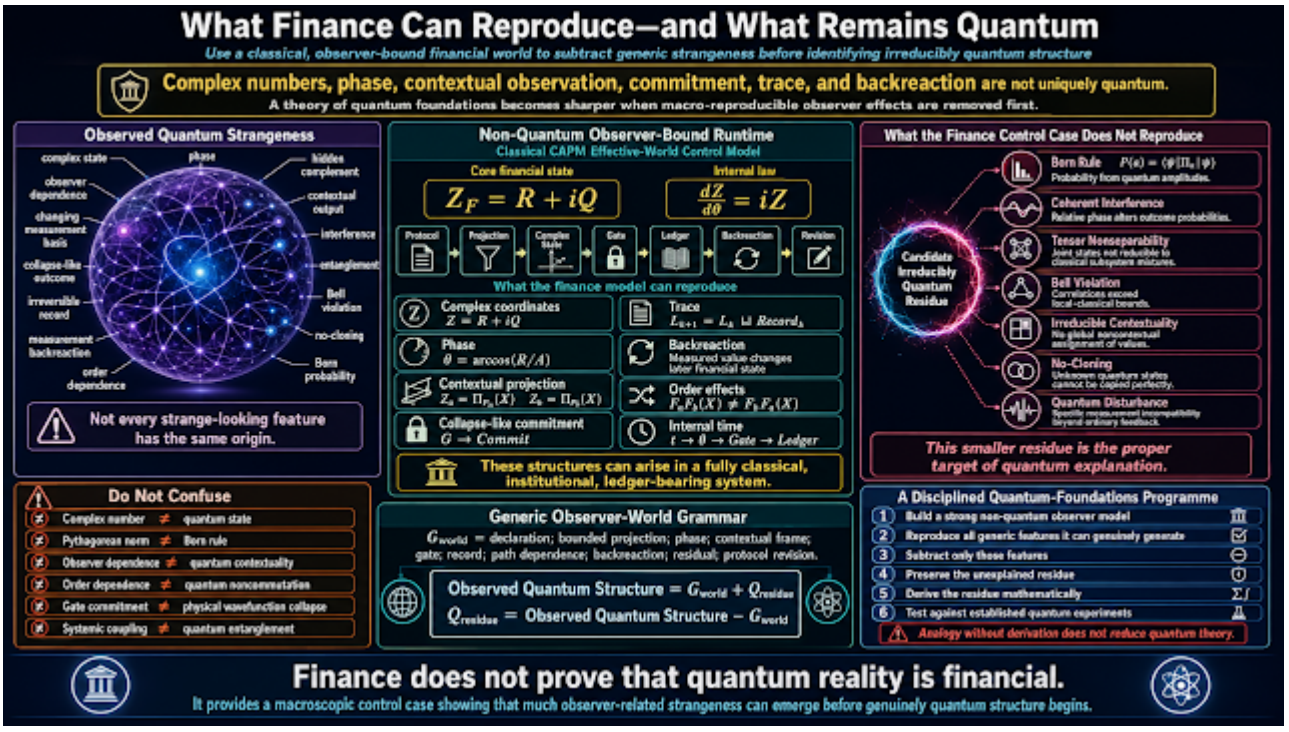
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Three Financial Clocks, One Causal History
Calendar time measures duration. Phase orders financial movement. Ledger time records irreversible consequence.
Phase is not automatically time. It becomes historical only when movement passes through gates and leaves trace.
1 Clock 1 — External Time
Market Timeline
Day 1
Day 2
Day 3
Day 4
elapsed duration
reporting intervals
maturity
settlement delay
observation frequency
Core Relation
Limitation: Two assets may experience the same amount of calendar time but occupy very different financial stages.
2 Clock 2 — Internal Financial Progression
low pressure
tightening
high pressure
reversal
relaxation
Core Relations
Meaning
progression through a valuation filter
tightening or relaxation
internal ordering of financial stress
relative location in the R–Q geometry
Validity Condition: identifiable and locally monotonic.
Phase reversal → clock ambiguity
3 Clock 3 — Committed Financial History
k = 10: rating watch
k = 11: haircut raised
k = 12: covenant breach
k = 13: default
k = 14: recovery-valuation begins
Core Equation
Ledger time advances only when a consequential event is committed.
Gate Connection
Calendar time may pass without history.
Ledger time advances only when the world commits.
The Three-Clock Chain
Calendar Movement
Phase Progression
Gate Threshold
Committed Event
Ledger History
Phase Order + Gate + Trace = Effective Financial Time
4 Why Returning to the Same Price Is Not Closure
Trajectory A: quiet path
Trajectory B: crisis, intervention, recovery
What the World Retains
leverage scars
ownership change
refinancing
covenant amendment
regulatory intervention
investor-base change
lost trust
Same Price ≠ Same History
Same Phase ≠ Same World
5 Residual Shows Where the World No Longer Closes
Key Distinction: Q lives inside the declared world. Residual tests whether the world remains valid.
6 Sometimes the Clock Does Not Continue—the World Changes
Before: Going-Concern Valuation
After: Liquidation or Recovery Valuation
Example Transitions
ordinary liquidity → emergency liquidity
market pricing → model pricing
tradable asset → frozen collateral
performing loan → recovery claim
normal CAPM frame → crisis regime
Protocol revision must preserve trace.
Revision = Old World + Translation + Untranslated Residual
Audit Checklist
calendar timestamp
phase θ
phase direction sign(ωy)
ledger index k
active protocol version
gate authority
residual status
intervention
backreaction
A world persists through what it remembers.
A financial clock becomes real when its ticks alter what the system can do next.

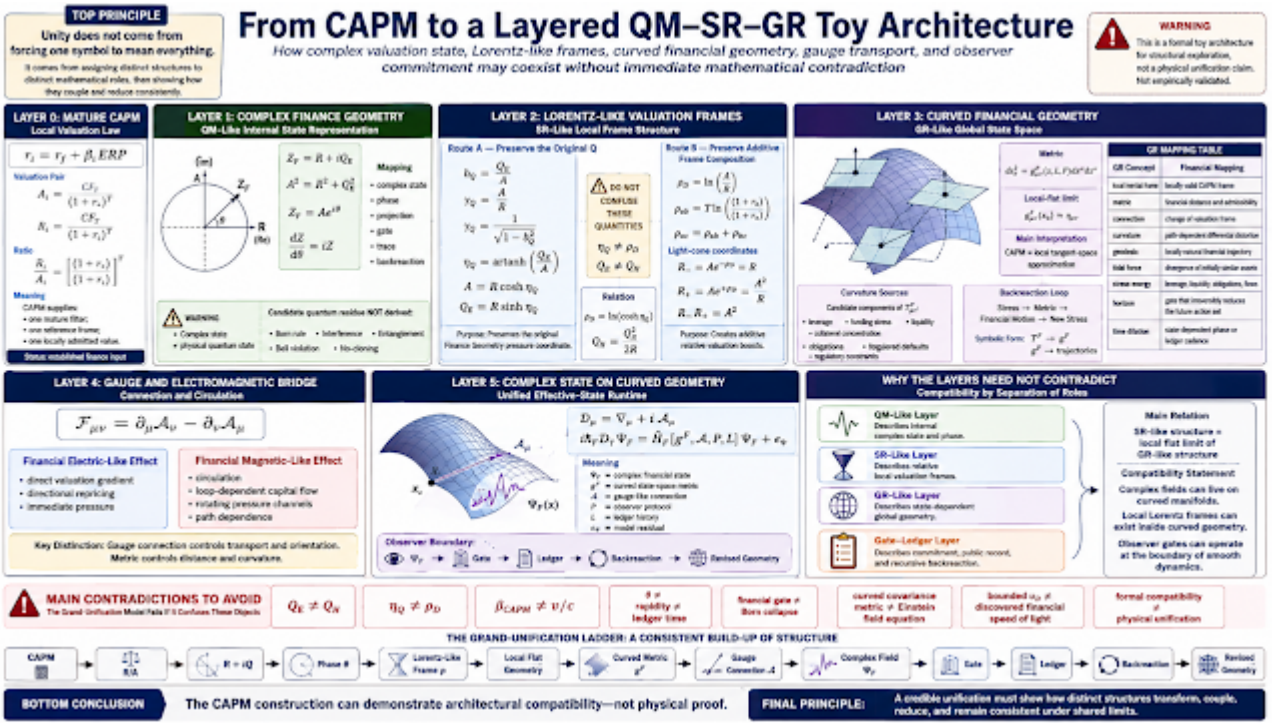
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What Finance Can Reproduce—and What Remains Quantum
Use a classical, observer-bound financial world to subtract generic strangeness before identifying irreducibly quantum structure
Complex numbers, phase, contextual observation, commitment, trace, and backreaction are not uniquely quantum.
A theory of quantum foundations becomes sharper when macro-reproducible observer effects are removed first.
Observed Quantum Strangeness
complex state
phase
hidden complement
observer dependence
contextual output
changing measurement basis
interference
collapse-like outcome
entanglement
irreversible record
Bell violation
measurement backreaction
no-cloning
order dependence
Born probability
Not every strange-looking feature has the same origin.
Non-Quantum Observer-Bound Runtime
Classical CAPM Effective-World Control Model
Core financial state
Internal law
Protocol
Projection
Complex State
Gate
Ledger
Backreaction
Revision
What the finance model can reproduce
Complex coordinates
Trace
Phase
Backreaction
Measured value changes later financial state
Contextual projection
Order effects
Collapse-like commitment
Internal time
These structures can arise in a fully classical, institutional, ledger-bearing system.
What the Finance Control Case Does Not Reproduce
Candidate Irreducibly Quantum Residue
Born Rule
Probability from quantum amplitudes.
Coherent Interference
Relative phase alters outcome probabilities.
Tensor Noneparability
Joint states not reducible to classical subsystem mixtures.
Bell Violation
Correlations exceed local-classical bounds.
Irreducible Contextuality
No global noncontextual assignment of values.
No-Cloning
Unknown quantum states cannot be copied perfectly.
Quantum Disturbance
This smaller residue is the proper target of quantum explanation.
Do Not Confuse
Complex number ≠ quantum state
Pythagorean norm ≠ Born rule
Observer dependence ≠ quantum contextuality
Order dependence ≠ quantum noncommutation
Gate commitment ≠ physical wavefunction collapse
Systemic coupling ≠ quantum entanglement
Generic Observer-World Grammar
declaration; bounded projection; phase; contextual frame; gate; record; path dependence; backreaction; residual; protocol revision.
A Disciplined Quantum-Foundations Programme
1 Build a strong non-quantum observer model
2 Reproduce all generic features it can genuinely generate
3 Subtract only those features
4 Preserve the unexplained residue
5 Derive the residue mathematically
6 Test against established quantum experiments
Analogy without derivation does not reduce quantum theory.
Finance does not prove that quantum reality is financial.
It provides a macroscopic control case showing that much observer-related strangeness can emerge before genuinely quantum structure begins.

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0. Reader's Guide: What This Article Is and Is Not

0.1 What this article extends

The earlier Finance Geometry framework was primarily static.

It began from a declared pre-filter amplitude A and a mature financial filter that produced admitted value R. It then reconstructed the orthogonal pressure coordinate Q implied by the same ratio.

The minimal chain was:

$A \to \text{Mature Filter} \to R$. (0.26)

The complex completion was:

$A \to \theta \to R + iQ$. (0.27)

The present article allows that geometry to move.

A becomes A(t).
θ becomes θ(t).
Z becomes Z(t).
Projected values are committed into ledgers.
Ledgers alter future financial states.
Large residuals may force the protocol itself to be revised.

The framework therefore expands from a coordinate map into a runtime:

$X_k \to \text{Declare}_{P_k}(X_k) \to \Pi_{P_k,L_k}(X_k) \to Z_k \to \text{Gate}_k \to L_{k+1} \to \boldsymbol{\mathcal{B}} \to X_{k+1} \to \boldsymbol{\mathcal{U}} \to P_{k+1}$. (0.28)

Here:

- X_k is the larger primary economic and financial field;

- P_k is the declared valuation protocol;
- L_k is the existing ledger;
- Π is the financial projection;
- $Z_k = R_k + iQ_k$ is the effective financial state;
- Gate_k determines whether a projected possibility becomes a committed event;
- $\boldsymbol{\mathcal{B}}$ maps ledgered financial consequences back into the primary field;
- $\boldsymbol{\mathcal{U}}$ revises the protocol when the old declaration ceases to remain adequate.

The article therefore extends Finance Geometry in five directions.

First, it distinguishes radial economic change from angular valuation change.

Second, it asks when a moving filter angle can serve as a local internal ordering coordinate.

Third, it introduces commitment gates and ledger time.

Fourth, it models backreaction: valuation outcomes may alter the economic reality being valued.

Fifth, it uses finance as a non-quantum control case for analysing quantum strangeness.

0.2 What this article does not claim

This article does not claim that financial markets are literal quantum systems.

The use of complex notation does not imply quantum ontology. Complex numbers also appear in electrical engineering, control theory, wave analysis, fluid dynamics, signal processing, stability theory, and classical oscillators.

The article does not claim that Q is a second market price.

R and Q are orthogonal coordinates under a declared geometry. Q is not normally cash, wealth, NAV, enterprise value, or a separately tradable amount.

The article does not claim:

$$Q = A - R. \quad (0.29)$$

The geometric relation is instead:

$$Q = \sqrt{(A^2 - R^2)}. \quad (0.30)$$

The scalar haircut and the orthogonal pressure coordinate are different objects.

The article does not claim that θ is a literal physical rotation angle.

θ is an angular encoding of a declared financial ratio:

$$\theta = \arccos(R/A). \quad (0.31)$$

The article does not claim that phase is automatically time.

A variable can order states without constituting historical time. Phase becomes time-like only

under additional conditions, including local monotonicity, identifiable events, commitment gates, and trace retention.

The article does not claim that observer dependence proves subjectivism.

A financial value can be protocol-dependent while remaining objective under admissibly equivalent implementations of that protocol.

The article does not claim that contextuality, order sensitivity, or measurement backreaction are uniquely quantum.

Financial systems can exhibit all three through ordinary institutional, informational, and path-dependent mechanisms.

The article does not derive:

- the Born rule;
- quantum amplitude probabilities;
- tensor-product state spaces;
- entanglement;
- Bell inequality violations;
- Kochen–Specker contextuality;
- no-cloning;
- quantum teleportation;
- physical wavefunction collapse.

Any physics-facing conclusion must preserve these limits.

0.3 Four levels of statement

The article uses four levels of claim that must not be silently mixed.

Level 1 — Mathematical identity

Examples include:

$A^2 = R^2 + Q^2$. (0.32)

$Z = A \exp(i\theta)$. (0.33)

$dZ/dt = [(1/A)(dA/dt) + i(d\theta/dt)]Z$. (0.34)

These follow from the declared representation.

Level 2 — Financial interpretation

Examples include:

- radial motion represents change in declared economic amplitude;

- angular motion represents change in the financial reading frame;
- Q can be interpreted as retained pressure or phase sensitivity;
- Qω_F can be interpreted as angular repricing load.

These interpretations depend on how A, R, and θ are constructed.

Level 3 — Empirical hypothesis

Examples include:

- Q may reveal pressure before a large scalar haircut appears;
- Qω_F may improve stress detection;
- dynamic residual may rise before a regime change;
- loop residual may capture financial memory after visible variables normalize.

These claims require data.

Level 4 — Speculative physics implication

Examples include:

- some apparent quantum strangeness may arise from generic observer-construction constraints;
- a deeper generative process may be accessible only through interfaces that it itself constructs;
- effective internal laws may be simpler than the process that generates the effective world.

These are research hypotheses, not established results.

0.4 Why use the word “world”?

The term “world” can easily become rhetorical.

A spreadsheet with two columns is not automatically a world. A coordinate transformation is not automatically a world. A model does not become a world merely because it is useful.

In this article, the term has a stricter meaning.

A reduced financial representation becomes world-like when it supports:

1. stable state distinctions;
2. usable internal coordinates;
3. approximately closed transition laws;
4. admissible events;
5. commitment gates;

6. historical trace;
7. intervention;
8. backreaction;
9. residual disclosure;
10. cross-observer reproducibility.

The proposed definition is:

An effective financial world is a protocol-bounded state space in which selected financial distinctions support approximately closed dynamics, admissible events, ledgered consequences, interventions, and backreaction upon future states. (0.35)

Its formal representation is:

W_P = ($\mathcal{Z}$_P, $\mathcal{D}$_P, $\mathcal{G}$_P, $\boldsymbol{\mathcal{L}}$_P, $\mathcal{U}$_P, $\boldsymbol{\mathcal{B}}$_P). (0.36)

A model becomes world-like when it does not merely summarize what exists. It organizes what can happen next.

0.5 The guiding question

The article's guiding question is:

When does valuation cease to be merely a number and begin to function as a causally operative world? (0.37)

The answer developed here is:

Valuation becomes world-like when a declared projection produces stable coordinates, those coordinates support lawful transitions, some transitions pass commitment gates, committed events enter a ledger, and the ledger alters future possibilities. (0.38)

The deeper physics-facing question is:

After complex phase, contextual projection, gate, trace, and backreaction have been reproduced in a non-quantum system, what remains uniquely quantum? (0.39)

Part I — From Scalar Value to Pressure-Preserving Geometry

1. Mature Finance Already Filters Reality

1.1 Financial value is never raw

A future cash flow does not possess one financially operative value independently of all conditions.

Its reported value depends on questions such as:

- At what horizon is it being evaluated?
- Relative to which baseline?
- Under which discount rate?
- Under which probability distribution?
- Under which investor or institutional constraints?
- Is the cash flow legally enforceable?
- Is it liquid?
- Can it be executed at the theoretical price?
- Is it admissible as collateral?
- Does it satisfy a regulatory capital rule?
- Does accounting recognize it now, later, or not at all?
- Does the holder have the capacity to wait?
- Is the option exercisable?
- Is the issuer expected to survive?

Modern finance has built mature tools for these questions.

CAPM alters the discount rate according to market-risk exposure.

Certainty-equivalent valuation alters the expected cash flow before discounting.

Stochastic discount factors weight outcomes according to state-dependent marginal value.

Credit spreads reduce value according to default and recovery pressure.

Liquidity haircuts distinguish theoretical value from executable value.

Option theory distinguishes latent possibility from exercisable value.

Capital rules distinguish economic value from admissible balance-sheet capacity.

Accounting standards distinguish economically imaginable value from officially recognized value.

These procedures differ, but they share one architecture:

Economic Possibility → Declared Financial Filter → Admitted Value. (1.1)

The output is usually one scalar:

- price;
- present value;
- NAV;
- fair value;

- enterprise value;
- market capitalization;
- capital value;
- collateral value;
- recognized asset value.

Scalar reporting is not a defect. It is operationally necessary. Ledgers, trades, balance sheets, limits, contracts, and decisions require committed numbers.

But commitment creates compression.

A scalar result tells us what passed the filter. It does not necessarily preserve the structural pressure implied by what failed to appear on the visible axis.

1.2 The ordinary scalar pipeline

Let A denote a declared pre-filter amplitude.

This need not mean “true intrinsic value.” It means the value amplitude before the particular filter being studied.

Let F_P denote a mature financial filter under declared protocol P.

Then:

R = F_P(A). (1.2)

The ordinary valuation pipeline is:

A → F_P → R. (1.3)

Examples include:

Base-discounted value → CAPM discount filter → CAPM value. (1.4)

Expected cash flow → Certainty-equivalent filter → Risk-adjusted cash flow. (1.5)

Default-free debt value → Credit filter → Risky debt value. (1.6)

Ideal theoretical value → Liquidity filter → Executable value. (1.7)

Unconstrained economic value → Capital gate → Admissible capital value. (1.8)

Latent project possibility → Exercise gate → Realized option value. (1.9)

The scalar R is real in the operational sense. It can influence:

- transactions;
- reporting;
- leverage;
- financing;

- tax;
- compensation;
- regulatory capital;
- collateral;
- default;
- strategic action.

But the filter has done more than produce R. It has established a relation between the larger amplitude A and the admitted value R.

Finance Geometry treats that relation as geometrically meaningful.

1.3 The missing question

Ordinary finance asks:

What value passes the filter? (1.10)

The present framework adds:

What pressure is implied by the difference between the pre-filter amplitude and the admitted result? (1.11)

The naïve answer would be:

Pressure = A − R. (1.12)

But this treats the filtered component as a scalar remainder on the same axis.

Finance Geometry proposes a different representation.

A and R are not treated as two endpoints on one line. R is treated as the visible projection of an amplitude under an angle θ.

The retained component is therefore orthogonal:

$A^2 = R^2 + Q^2$. (1.13)

$R = A \cos \theta$. (1.14)

$Q = A \sin \theta$. (1.15)

The resulting state is:

$Z = R + iQ$. (1.16)

This does not claim that Q exists as money in another dimension.

It says that a mature financial filter can be represented by two linked coordinates:

- the value it admits;
- the pressure complement implied by that admission.

1.4 Why scalar compression matters

Suppose two assets both have admitted value:

$R_1 = R_2$. (1.17)

A scalar report treats them as equal on the measured dimension.

But suppose:

$A_1 \neq A_2$. (1.18)

Then:

$Q_1 = \sqrt{(A_1^2 - R_1^2)}$. (1.19)

$Q_2 = \sqrt{(A_2^2 - R_2^2)}$. (1.20)

The assets may occupy very different positions in pressure geometry.

One may have a modest pre-filter amplitude and little retained pressure.

The other may have a much larger amplitude but face a severe filter.

The same visible value can therefore arise from different structures:

- weak amplitude with mild pressure;
- strong amplitude with severe pressure;
- moderate amplitude with moderate pressure.

Scalar equality does not imply geometric equality.

The same observation applies in reverse.

Two assets may have the same A but different R:

$A_1 = A_2$. (1.21)

$R_1 \neq R_2$. (1.22)

Then their different Q values encode the distinct pressure implied by their filters.

The purpose of the complex completion is not to deny scalar value. It is to avoid treating scalar admission as a complete description of the valuation state.

1.5 The filter must come first

The geometry is valid only if R/A arises from a meaningful mature-finance ratio.

Let:

$w = R/A$. (1.23)

If:

$0 \le w \le 1$, (1.24)

then:

$\theta = \arccos w$. (1.25)

$Q = A\sqrt{(1 - w^2)}$. (1.26)

This order is essential:

Mature Finance Ratio → θ → Q. (1.27)

The invalid order would be:

Invent Q → invent θ → search for a finance story. (1.28)

A valid Finance Geometry construction must declare:

- the system boundary;
- the baseline;
- the horizon;
- the filter;
- the meaning of A;
- the meaning of R;
- the reason R/A lies within the required domain;
- the diagnostic question being asked.

Without those declarations, θ becomes decorative and Q becomes arbitrary.

1.6 Finance Geometry as a coordinate extension

The framework does not replace the mature filter.

CAPM still calculates the discount rate.

Credit models still estimate default and recovery effects.

Liquidity models still address execution and market depth.

Option models still determine exercise-contingent value.

Accounting rules still govern recognition.

The complex extension sits above those methods:

Mature Model → Scalar Result R → Geometric Completion R + iQ. (1.29)

The value of the extension depends on whether the additional coordinate improves:

- explanation;

- attribution;
- stress analysis;
- communication;
- prediction;
- model comparison;
- intervention;
- capital allocation.

The criterion remains:

No diagnostic gain → No adoption. (1.30)

This practical discipline is central to the original Finance Geometry proposal.

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2. The Minimal Geometry of Admitted Value and Retained Pressure

2.1 Core definitions

Let $A \geq 0$ denote the pre-filter value amplitude under a declared protocol.

Let R denote the admitted value produced by a mature financial filter.

Let θ denote the implied finance angle:

$\theta = \arccos(R/A)$. (2.1)

Let Q denote retained pressure:

$Q = A \sin \theta$. (2.2)

The complex financial state is:

$Z = R + iQ$. (2.3)

Equivalently:

$Z = A \exp(i\theta)$. (2.4)

The magnitude relation is:

$|Z| = A$. (2.5)

The Pythagorean closure is:

$A^2 = R^2 + Q^2$. (2.6)

The normalized visible coordinate is:

$R/A = \cos \theta$. (2.7)

The normalized pressure coordinate is:

$Q/A = \sin \theta$. (2.8)

The pressure ratio relative to admitted value is:

$Q/R = \tan \theta$. (2.9)

These equations are exact within the declared geometry.

2.2 What A means

A is not automatically “true value.”

It is a declared pre-filter amplitude.

Its meaning depends on the construction.

In a CAPM laboratory, A may be a future cash flow discounted at a declared base rate.

In certainty-equivalent valuation, A may be expected cash flow before risk adjustment.

In credit analysis, A may be default-free value.

In liquidity analysis, A may be ideal executable value under frictionless assumptions.

In capital analysis, A may be unconstrained economic value.

In real-option analysis, A may be latent project value before the exercise gate.

The term amplitude is useful because it avoids claiming that A is a metaphysically privileged value.

It means:

The financial potential being resolved by the declared filter. (2.10)

Changing the baseline changes A.

Changing the horizon may change A.

Changing the feature map may change A.

Changing the economic scenario may change A.

Therefore:

A is protocol-relative, not arbitrary. (2.11)

2.3 What R means

R is the component admitted by the declared filter.

It is “real” in two senses.

First, it lies on the visible real axis of the representation.

Second, it often becomes operationally real through financial commitment.

Examples include:

- an executed price;
- a booked present value;
- a recognized asset;
- an approved collateral amount;
- a regulatory capital value;
- a risk-adjusted project value;
- an admitted lending capacity.

But R is not necessarily permanent.

It may be revised when:

- the economic amplitude changes;
- the financial filter changes;
- new evidence arrives;
- a gate is reversed;
- the protocol is replaced;
- a residual becomes material.

R is therefore admitted reality under a protocol, not total reality without qualification.

2.4 What Q means

Q is the orthogonal pressure coordinate implied by A and R.

It may represent different forms of retained pressure depending on the filter:

- market-risk pressure;
- credit pressure;
- liquidity pressure;
- execution pressure;
- capital pressure;
- optionality not yet admitted;
- tail pressure;
- model uncertainty;

- regulatory burden.

Q does not mean all omitted information.

It does not automatically include every unmodelled feature of the primary economic field.

That broader omission belongs to projection residual, introduced later.

Q is narrower:

Q is the pressure complement inside the declared two-coordinate geometry. (2.12)

The distinction is essential.

Internal pressure is not the same as external model error.

2.5 Why Q is not A − R

The scalar haircut is:

$H = A - R$. (2.13)

The geometric pressure is:

$Q = \sqrt{(A^2 - R^2)}$. (2.14)

These are related but not identical.

Using:

$R = A \cos \theta$, (2.15)

the haircut becomes:

$H = A(1 - \cos \theta)$. (2.16)

The pressure coordinate is:

$Q = A \sin \theta$. (2.17)

For small θ:

$\sin \theta \approx \theta$. (2.18)

$\cos \theta \approx 1 - \theta^2/2$. (2.19)

Therefore:

$Q \approx A\theta$. (2.20)

$H \approx A\theta^2/2$. (2.21)

Substituting $\theta \approx Q/A$ gives:

$H \approx Q^2/(2A)$. (2.22)

This produces an important asymmetry near the base state.

Q responds at first order in θ.

The scalar haircut responds at second order.

Therefore:

Retained pressure may grow materially before the visible scalar haircut becomes comparably large. (2.23)

This is a mathematical property of the representation.

Whether it becomes a useful early-warning signal is an empirical question.

2.6 Pressure as phase sensitivity

Under constant A:

$R = A \cos \theta$. (2.24)

Differentiating with respect to θ gives:

$dR/d\theta = -A \sin \theta$. (2.25)

Since:

$Q = A \sin \theta$, (2.26)

we obtain:

$dR/d\theta = -Q$. (2.27)

Therefore:

$Q = -dR/d\theta$. (2.28)

This gives Q a second interpretation.

Q is not only the retained pressure coordinate.

It is also the marginal sensitivity of admitted value to a tightening rotation of the declared filter.

The normalized sensitivity is:

$-(1/R)(dR/d\theta) = Q/R$. (2.29)

Therefore:

$-d \ln R/d\theta = \tan \theta$. (2.30)

This suggests a useful reading:

Q/R measures the pressure intensity of the admitted value state. (2.31)

A larger Q/R means a small additional angular movement produces a larger proportional effect on R.

Again, this is not a universal risk measure. Its meaning is only as valid as the filter that generated θ.

2.7 Boundary cases

When:

$\theta = 0$, (2.32)

then:

$R = A$. (2.33)

$Q = 0$. (2.34)

The filter admits the full declared amplitude.

When:

$\theta \to \pi/2$, (2.35)

then:

$R \to 0$. (2.36)

$Q \to A$. (2.37)

The state becomes pressure-dominant under the declared filter.

The usual financial interpretation will restrict attention to:

$0 \le \theta \le \pi/2$. (2.38)

This keeps R and Q non-negative.

Extensions involving signed Q, directional pressure, negative admitted value, liabilities, short positions, or opposing pressure channels require a richer geometry and should not be introduced casually into the minimal model.

2.8 Same R, different world

Suppose:

$R_1 = R_2 = 80$. (2.39)

Let:

$A_1 = 100$. (2.40)

Then:

$Q_1 = \sqrt{(100^2 - 80^2)} = 60$. (2.41)

But let:

$A_2 = 85$. (2.42)

Then:

$Q_2 = \sqrt{(85^2 - 80^2)} \approx 28.72.$ (2.43)

Both positions have admitted value 80.

Yet the first contains much more retained pressure relative to its pre-filter amplitude.

Their angles are:

$\theta_1 = \arccos(0.8) \approx 0.6435.$ (2.44)

$\theta_2 = \arccos(80/85) \approx 0.3446.$ (2.45)

The same scalar value can therefore occupy different locations in the effective financial geometry.

This is the first sense in which valuation begins to resemble a world rather than a number: the same visible coordinate can belong to different hidden structural states.

2.9 Same A, different admission

Suppose:

$A_1 = A_2 = 100.$ (2.46)

Let:

$R_1 = 95.$ (2.47)

$R_2 = 70.$ (2.48)

Then:

$Q_1 = \sqrt{(100^2 - 95^2)} \approx 31.22.$ (2.49)

$Q_2 = \sqrt{(100^2 - 70^2)} \approx 71.41.$ (2.50)

The second position contains much more retained pressure.

Its pressure ratio is:

$Q_2/R_2 \approx 1.0201.$ (2.51)

The first has:

$Q_1/R_1 \approx 0.3286.$ (2.52)

The visible values already differ, but the geometry adds a normalized statement about how pressure-loaded each state is.

2.10 The minimal promise

The minimal promise of Finance Geometry is not prediction.

It is better state description.

Scalar finance reports:

R. (2.53)

Finance Geometry reports:

(A, R, Q, θ). (2.54)

The richer state may reveal whether:

- a low R reflects weak underlying amplitude;
- a low R reflects severe filtering;
- a high R is pressure-light;
- a high R is supported by a large but fragile amplitude;
- pressure is accumulating faster than the visible haircut;
- a small angular movement would produce large repricing.

The dynamic article now asks the next question:

What happens when this state begins to move? (2.55)

3. CAPM as the Clean Dynamic Laboratory

3.1 Why begin with CAPM?

The purpose of using CAPM is not to declare it the final theory of asset pricing.

CAPM is used because it provides a familiar and compact mature-finance filter whose main variables are explicit:

- the risk-free rate;
- expected market return;
- market-risk premium;
- asset beta;
- valuation horizon;
- future cash flow.

This makes CAPM a useful laboratory for separating:

1. the declared pre-filter amplitude;
2. the admitted value produced by the market-risk filter;
3. the retained pressure implied by that filter;
4. changes in the underlying cash-flow amplitude;
5. changes in the valuation frame.

The basic CAPM return is:

$r_{CAPM}(t) = r_f(t) + \beta(t)[E(r_m)(t) - r_f(t)]$. (3.1)

Define the equity-risk premium:

$ERP(t) = E(r_m)(t) - r_f(t)$. (3.2)

Then:

$r_{CAPM}(t) = r_f(t) + \beta(t)ERP(t)$. (3.3)

The CAPM filter therefore changes when any of the following changes:

- r_f;
- β;
- ERP;
- the expected cash flow;
- the valuation horizon.

The original Finance Geometry article uses this structure as one of its clearest examples of a mature filter generating an implied finance angle rather than requiring an angle to be invented independently.

3.2 Declaring the base amplitude

Let CF_T denote a future cash flow expected at horizon T.

Choose a declared base rate r_*.

The base-discounted amplitude is:

$A(t) = CF_T(t)/(1 + r_*(t))^T$. (3.4)

In the cleanest laboratory, r_* and CF_T are held fixed during one episode:

$A = CF_T/(1 + r_*)^T$. (3.5)

This does not assert that the base rate is objectively correct.

It declares the reference from which the market-risk filter will be measured.

Possible choices for r_* include:

- a risk-free rate;
- a contractual funding rate;
- a policy benchmark;
- a strategic hurdle-rate baseline;
- a default-free comparison rate;
- a fixed episode reference.

Different choices produce different amplitudes.

Therefore, every application must report:

- what r_* means;
- why it was selected;
- whether it changes through time;
- whether comparisons across assets use the same base;
- whether changing r_* would materially alter the result.

The amplitude is not independent of declaration.

The discipline is:

No declared baseline → No interpretable A. (3.6)

3.3 CAPM-admitted value

The ordinary CAPM-discounted value is:

$$R(t) = CF_T(t)/[1 + r_{CAPM}(t)]^T. \quad (3.7)$$

If CF_T and r_* are fixed while the CAPM return varies, the ratio between R and A is:

$$R(t)/A = [(1 + r_*)/(1 + r_{CAPM}(t))]^T. \quad (3.8)$$

Therefore:

$$\cos \theta(t) = [(1 + r_*)/(1 + r_{CAPM}(t))]^T. \quad (3.9)$$

The implied finance angle is:

$$\theta(t) = \arccos\{[(1 + r_*)/(1 + r_{CAPM}(t))]^T\}. \quad (3.10)$$

The retained CAPM pressure is:

$$Q(t) = A\sqrt{[1 - \cos^2\theta(t)]}. \quad (3.11)$$

Equivalently:

$$Q(t) = \sqrt{[A^2 - R^2(t)]}. \quad (3.12)$$

The complex CAPM state is:

$$Z_{CAPM}(t) = R(t) + iQ(t). \quad (3.13)$$

or:

$$Z_{CAPM}(t) = A \exp[i\theta(t)]. \quad (3.14)$$

The ordinary CAPM valuation reports R.

The complex completion reports:

$Z_{CAPM}(t) = [\text{admitted CAPM value}] + i[\text{retained market-risk pressure}]$. (3.15)

3.4 What causes the CAPM angle to move?

Because:

$r_{CAPM} = r_f + \beta ERP$, (3.16)

the finance angle may change through several channels.

Risk-free-rate channel

$\partial r_{CAPM}/\partial r_f = 1 - \beta$, (3.17)

if ERP is defined as $E(r_m) - r_f$ and $E(r_m)$ is held fixed.

Alternatively, under a formulation where ERP is treated as an independently estimated variable:

$\partial r_{CAPM}/\partial r_f = 1$. (3.18)

This shows why the protocol must state how ERP is estimated.

Beta channel

$\partial r_{CAPM}/\partial \beta = ERP$. (3.19)

A rise in beta increases the required return when $ERP > 0$.

Equity-risk-premium channel

$\partial r_{CAPM}/\partial ERP = \beta$. (3.20)

A broad increase in market-risk compensation rotates the valuation frame even if the expected cash flow is unchanged.

Horizon channel

The angle depends on T through compounding:

$\cos \theta = [(1 + r_*)/(1 + r_{CAPM})]^T$. (3.21)

For the same rate difference, a longer horizon usually produces a larger angular separation.

Cash-flow channel

If CF_T changes while both discount rates remain fixed, both A and R change proportionally.

This is primarily radial movement.

The importance of these channels is that not every decline in R means the asset's underlying economic amplitude deteriorated.

R may fall because the market-risk frame rotated.

3.5 Sensitivity of the CAPM angle

Let:

$$c(t) = \cos \theta(t) = [(1 + r_*)/(1 + r_{CAPM}(t))]^T. \quad (3.22)$$

Taking logarithms:

$$\ln c = T[\ln(1 + r_*) - \ln(1 + r_{CAPM})]. \quad (3.23)$$

Differentiating with respect to calendar time:

$$(1/c)(dc/dt) = T[(1/(1 + r_*))(dr_*/dt) - (1/(1 + r_{CAPM}))(dr_{CAPM}/dt)]. \quad (3.24)$$

Since:

$$dc/dt = -\sin \theta(d\theta/dt), \quad (3.25)$$

we obtain:

$$-\tan \theta(d\theta/dt) = T[(1/(1 + r_*))(dr_*/dt) - (1/(1 + r_{CAPM}))(dr_{CAPM}/dt)]. \quad (3.26)$$

Therefore:

$$d\theta/dt = [T/\tan \theta][(1/(1 + r_{CAPM}))(dr_{CAPM}/dt) - (1/(1 + r_*))(dr_*/dt)]. \quad (3.27)$$

Under a fixed base rate:

$$dr_*/dt = 0, \quad (3.28)$$

so:

$$d\theta/dt = T/(\tan \theta)(1 + r_{CAPM}). \quad (3.29)$$

Define the filter angular velocity:

$$\omega_F = d\theta/dt. \quad (3.30)$$

Then:

$$\omega_F = T/(\tan \theta)(1 + r_{CAPM}). \quad (3.31)$$

This equation requires care near $\theta = 0$, because $\tan \theta$ approaches zero. The apparent singularity does not necessarily represent infinite economic instability. It arises because θ is a nonlinear transformation of a discount ratio near the fully admitted state.

A practical implementation may therefore use:

- finite differences;
- smoothed angular increments;
- direct differentiation of R and Q;
- a minimum-angle threshold;
- a locally regularized phase coordinate.

The point is not that θ must always be differentiated analytically.

The point is that a changing CAPM rate induces a measurable movement of the finance frame.

3.6 The sign of phase motion

Suppose r_* is fixed.

If:

dr_CAPM/dt > 0, (3.32)

then, under the ordinary positive-rate domain:

dθ/dt > 0. (3.33)

The filter rotates toward greater retained pressure.

Admitted value falls:

dR/dt < 0, (3.34)

if A remains fixed.

If:

dr_CAPM/dt < 0, (3.35)

then:

dθ/dt < 0. (3.36)

The market-risk filter relaxes.

Admitted value rises, again assuming constant A.

Thus, during a clean CAPM episode:

r_CAPM ↑ → θ ↑ → R ↓ → Q ↑. (3.37)

r_CAPM ↓ → θ ↓ → R ↑ → Q ↓. (3.38)

This is not a new CAPM prediction.

It is a geometric re-expression of the discounting relation.

Its potential value lies in linking the visible change in R to a pressure coordinate, phase velocity, and later a residual structure.

3.7 A numerical illustration

Assume:

CF_T = 121. (3.39)

T = 2. (3.40)

r_* = 0.05. (3.41)

Then:

$A = 121/(1.05)^2 \approx 109.7506.$ (3.42)

Suppose initially:

$r_f = 0.03.$ (3.43)

$\beta = 1.0.$ (3.44)

$ERP = 0.04.$ (3.45)

Then:

$r_{CAPM} = 0.07.$ (3.46)

The admitted value is:

$R = 121/(1.07)^2 \approx 105.6861.$ (3.47)

The filter ratio is:

$R/A \approx 0.9630.$ (3.48)

The angle is:

$\theta \approx \arccos(0.9630) \approx 0.2728.$ (3.49)

The retained pressure is:

$Q = \sqrt{(A^2 - R^2)} \approx 29.5987.$ (3.50)

Now suppose ERP rises from 0.04 to 0.07 while CF_T, r_*, r_f, and β remain fixed.

Then:

$r_{CAPM,new} = 0.10.$ (3.51)

The new admitted value is:

$R_{new} = 121/(1.10)^2 = 100.$ (3.52)

The new ratio is:

$R_{new}/A \approx 0.9112.$ (3.53)

The new angle is:

$\theta_{new} \approx \arccos(0.9112) \approx 0.4240.$ (3.54)

The new pressure coordinate is:

$Q_{new} = \sqrt{(A^2 - R_{new}^2)} \approx 45.2242.$ (3.55)

The visible value fell by approximately:

$\Delta R \approx -5.6861.$ (3.56)

But the declared base amplitude did not change:

$\Delta A = 0$. (3.57)

The movement is therefore angular rather than radial.

The geometric account is:

ERP increase → CAPM-rate increase → frame rotation → R decrease + Q increase. (3.58)

A scalar valuation reports the decline from 105.6861 to 100.

The complex geometry also records that retained pressure increased from approximately 29.5987 to 45.2242.

3.8 The same decline caused by amplitude deterioration

Now consider a different event.

Keep:

$r_{CAPM} = 0.07$. (3.59)

$r_* = 0.05$. (3.60)

$T = 2$. (3.61)

But reduce the expected cash flow from:

$CF_T = 121$ (3.62)

to:

$CF_{T,new} = 114.49$. (3.63)

This is a 5.38% decline in the future cash-flow amplitude.

The new base amplitude is:

$A_{new} = 114.49/(1.05)^2 \approx 103.8458$. (3.64)

The new admitted CAPM value is:

$R_{new} = 114.49/(1.07)^2 \approx 100$. (3.65)

Again, the admitted value becomes 100.

But now:

$R_{new}/A_{new} \approx 0.9630$. (3.66)

The finance angle remains approximately:

$\theta_{new} \approx 0.2728$. (3.67)

The retained pressure becomes:

$Q_{new} \approx 28.0127$. (3.68)

The same visible value of 100 has arisen through a different process.

Scenario A — Frame tightening

A remains approximately 109.7506.

θ rises from approximately 0.2728 to 0.4240.

Q rises from approximately 29.5987 to 45.2242.

Scenario B — Amplitude deterioration

A falls from approximately 109.7506 to 103.8458.

θ remains approximately 0.2728.

Q falls proportionally with the amplitude.

Scalar finance sees the same final R.

Dynamic Finance Geometry distinguishes:

- pressure-driven repricing;
- fundamental-amplitude deterioration.

This distinction will become central in Part II.

3.9 CAPM as a declared world, not an ontological truth

The CAPM laboratory must not be overread.

A CAPM-implied θ is valid only within the declared CAPM protocol.

Different pricing models may produce different admitted values:

$R_{CAPM} \neq R_{SDF} \neq R_{market} \neq R_{credit\text{-}adjusted}$. (3.69)

Therefore:

$\theta_{CAPM} \neq \theta_{SDF} \neq \theta_{market} \neq \theta_{credit}$. (3.70)

This is not automatically a contradiction.

Each angle answers a different declared question.

The correct comparison is not:

Which θ is the one true angle? (3.71)

It is:

Which protocol produced each θ, and what invariant relation survives across admissibly comparable protocols? (3.72)

This is the beginning of the world-forming interpretation.

Each mature filter constructs a specific effective financial world.

The worlds may overlap.

They may disagree.

They may become mutually translatable.

But their coordinates should not be silently mixed.

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4. Beyond CAPM: Other Mature Finance Bridges

4.1 Why CAPM is only the opening laboratory

CAPM has three advantages:

- conceptual familiarity;
- a clear discount-rate filter;
- a simple ratio between base-discounted and risk-adjusted value.

But the architecture is more general.

A valid Finance Geometry bridge requires:

1. a declared pre-filter amplitude A;
2. an admitted value R generated by a mature financial process;
3. a meaningful ratio R/A;
4. an interpretable pressure complement Q;
5. a diagnostic question that the geometry may improve.

The same pattern can be applied to several established finance domains. The original Finance Geometry article explicitly develops such bridges rather than treating CAPM as the only possible source of θ.

4.2 Certainty-equivalent valuation

Let E(CF_T) denote expected future cash flow.

Let CE_T denote its certainty equivalent.

Assume:

$0 \le CE_T \le E(CF_T)$. (4.1)

Using the same base discount rate r_*:

$A_CE = E(CF_T)/(1 + r_*)^T$. (4.2)

$R_CE = CE_T/(1 + r_*)^T$. (4.3)

Then:

$\cos \theta_{CE} = R_{CE}/A_{CE}$. (4.4)

Because the same discount factor appears in numerator and denominator:

$\cos \theta_{CE} = CE_T/E(CF_T)$. (4.5)

The retained certainty-equivalent pressure is:

$Q_{CE} = A_{CE}\sqrt{[1 - (CE_T/E(CF_T))^2]}$. (4.6)

The complex state is:

$Z_{CE} = R_{CE} + iQ_{CE}$. (4.7)

Here:

- A_{CE} is expected-value amplitude;
- R_{CE} is certainty-admitted value;
- Q_{CE} is retained risky-cash-flow pressure;
- θ_{CE} is the certainty-equivalent filter angle.

This construction differs from CAPM.

CAPM filters primarily through the discount rate.

Certainty-equivalent valuation filters through the cash-flow numerator.

Yet both produce a mature ratio that can generate θ.

4.3 Certainty-equivalent dynamics

Suppose $E(CF_T)$ remains stable while the certainty equivalent declines.

Then:

$CE_T \downarrow \rightarrow \cos \theta_{CE} \downarrow \rightarrow \theta_{CE} \uparrow \rightarrow Q_{CE} \uparrow$. (4.8)

The visible value falls because confidence in realizing the expected cash flow has weakened.

This is angular movement in the certainty-equivalent world.

Suppose instead that both expected cash flow and certainty equivalent decline proportionally:

$CE_T/E(CF_T) = \text{constant}$. (4.9)

Then:

$\theta_{CE} = \text{constant}$. (4.10)

The movement is radial.

Again, the geometry distinguishes:

- deterioration in the underlying expectation;
- deterioration in the degree of certainty applied to that expectation.

4.4 Risky debt relative to default-free debt

Let V_0 denote the value of an otherwise comparable default-free claim.

Let V_{risk} denote the value of the risky debt.

Assume:

$0 \le V_{risk} \le V_0$. (4.11)

Define:

$A_{credit} = V_0$. (4.12)

$R_{credit} = V_{risk}$. (4.13)

Then:

$\cos \theta_{credit} = V_{risk}/V_0$. (4.14)

The retained credit-pressure coordinate is:

$Q_{credit} = V_0\sqrt{[1 - (V_{risk}/V_0)^2]}$. (4.15)

The complex credit state is:

$Z_{credit} = V_{risk} + iQ_{credit}$. (4.16)

Under a simplified yield-based comparison:

$V_0 = CF_T/(1 + r_0)^T$. (4.17)

$V_{risk} = CF_T/(1 + r_0 + s_{credit})^T$. (4.18)

Therefore:

$\cos \theta_{credit} = [(1 + r_0)/(1 + r_0 + s_{credit})]^T$. (4.19)

As the credit spread rises:

$s_{credit} \uparrow \rightarrow \theta_{credit} \uparrow \rightarrow R_{credit} \downarrow \rightarrow Q_{credit} \uparrow$. (4.20)

The geometry does not replace expected-loss or structural-credit models.

It provides a pressure-preserving coordinate for the result of the declared credit filter.

4.5 Default event versus credit-phase movement

Credit analysis reveals an important distinction between smooth phase movement and gate events.

Before default, credit pressure may rise continuously:

θ_credit(t) ↑. (4.21)

Q_credit(t) ↑. (4.22)

R_credit(t) ↓. (4.23)

But default itself is not merely a further infinitesimal angle.

It is a gate event:

G_default(state_t) = Commit(Default). (4.24)

Once committed, the ledger changes:

$L_{k+1} = L_k \sqcup \text{DefaultRecord}_k$. (4.25)

That record may alter:

- acceleration rights;
- collateral control;
- recovery procedures;
- legal ranking;
- future funding;
- cross-default provisions;
- accounting recognition;
- regulatory treatment.

The pre-default credit world and post-default recovery world may use different protocols.

Thus:

Credit-pressure motion within $P_k \neq$ Default-triggered transition to P_{k+1}. (4.26)

This distinction will later separate internal dynamics from world revision.

4.6 Liquidity-adjusted value

Let V_ideal denote value under ideal execution assumptions.

Let V_exec denote realistically executable value.

Assume:

0 ≤ V_exec ≤ V_ideal. (4.27)

Define:

A_liq = V_ideal. (4.28)

R_liq = V_exec. (4.29)

Then:

$\cos \theta_{liq} = V_{exec}/V_{ideal}$. (4.30)

The retained liquidity-pressure coordinate is:

$Q_{liq} = V_{ideal}\sqrt{[1 - (V_{exec}/V_{ideal})^2]}$. (4.31)

The complex liquidity state is:

$Z_{liq} = V_{exec} + iQ_{liq}$. (4.32)

This geometry may be useful when the theoretical price remains stable but actual execution capacity deteriorates.

For example:

$V_{ideal} \approx$ constant. (4.33)

$V_{exec} \downarrow$. (4.34)

Then:

$\theta_{liq} \uparrow$. (4.35)

$Q_{liq} \uparrow$. (4.36)

The asset has not necessarily lost the same amount of long-run economic capacity.

Its immediate admissibility into executable value has weakened.

This is a clear example of frame rotation rather than radial deterioration.

4.7 Liquidity gates

Liquidity also contains discrete events.

Examples include:

- a market reopening or closing;
- collateral becoming ineligible;
- a redemption gate being activated;
- a trading halt;
- a margin threshold being crossed;
- a central bank facility becoming available;
- settlement failure;
- suspension of convertibility.

These are not merely changes in θ_{liq}.

They alter the admissible action set.

If U_k denotes admissible interventions, a liquidity gate may produce:

$U_{k+1} \neq U_k$. (4.37)

The effective world changes because what agents are allowed or able to do has changed.

4.8 Capital-admissible value

A financial institution may recognize an economic value that cannot be used fully for regulatory, collateral, or internal-capital purposes.

Let V_econ denote unconstrained economic value.

Let V_cap denote value admitted under a capital rule.

Assume:

$0 \leq V_cap \leq V_econ$. (4.38)

Define:

$A_cap = V_econ$. (4.39)

$R_cap = V_cap$. (4.40)

Then:

$\cos \theta_cap = V_cap/V_econ$. (4.41)

The retained capital-pressure coordinate is:

$Q_cap = V_econ\sqrt{[1 - (V_cap/V_econ)^2]}$. (4.42)

The complex capital state is:

$Z_cap = V_cap + iQ_cap$. (4.43)

This construction can represent pressure created by:

- risk weights;
- concentration limits;
- leverage ratios;
- haircuts;
- eligibility rules;
- internal risk appetite;
- stress buffers;
- legal-entity restrictions.

The same economic asset may have different capital-admitted values in different institutions because their protocols differ.

That does not make the values arbitrary.

Each value is objective relative to its declared rule set.

4.9 Accounting recognition

Accounting provides an especially clear example of admission into a public ledger.

Let V_{econ} denote a declared economic-value estimate.

Let V_{rec} denote the amount recognized under an accounting protocol.

Define:

$A_{acc} = V_{econ}$. (4.44)

$R_{acc} = V_{rec}$. (4.45)

Then:

$\cos \theta_{acc} = V_{rec}/V_{econ}$. (4.46)

The pressure coordinate is:

$Q_{acc} = V_{econ}\sqrt{[1 - (V_{rec}/V_{econ})^2]}$. (4.47)

The accounting state is:

$Z_{acc} = V_{rec} + iQ_{acc}$. (4.48)

The interpretation requires caution.

Accounting recognition is not simply a discount for “risk.” It reflects:

- recognition criteria;
- measurement reliability;
- contractual rights;
- control;
- probability thresholds;
- classification;
- timing;
- prudence;
- legal and institutional standards.

The pressure coordinate may therefore represent non-admitted economic possibility rather than loss.

A future benefit can be economically plausible but not yet recognizable.

Q_{acc} may contain:

- timing pressure;
- evidence pressure;
- control uncertainty;
- measurement uncertainty;
- legal uncertainty.

Again, Q is filter-specific.

4.10 Real options and exercise gates

Let A_option denote latent project or strategic optionality.

Let R_ex denote value admitted by the current exercise rule.

Define:

$\cos \theta_{option} = R_{ex}/A_{option}$. (4.49)

Then:

$Q_{option} = A_{option}\sqrt{[1 - (R_{ex}/A_{option})^2]}$. (4.50)

Before exercise, much of the option's possibility may remain in Q.

When exercise conditions are met, a gate converts part of the latent state into committed investment:

$G_{exercise}(Z_{option}) = \text{Commit}(\text{Project})$. (4.51)

The gate may depend on:

- market price;
- volatility;
- financing availability;
- strategic fit;
- regulatory approval;
- managerial authority;
- operational capacity.

After exercise, the effective world changes.

The option world becomes a project-execution world.

The ledger now contains sunk costs, contractual commitments, and path-dependent obligations.

Therefore:

Option Exercise ≠ Continuous Rotation Alone. (4.52)

It is a world-forming event.

4.11 Pricing kernels and stochastic discount factors

Let M_T denote a stochastic discount factor.

Ordinary asset value is:

R_SDF = E(M_T CF_T). (4.53)

A possible reference amplitude might be:

A_ref = E(CF_T)/(1 + r_*)^{T}. (4.54)

Then:

cos θ_SDF = R_SDF/A_ref, (4.55)

provided the ratio lies within the declared geometric domain.

This construction requires more caution than CAPM.

The stochastic discount factor already combines:

- state probabilities;
- marginal utility;
- covariance;
- intertemporal substitution;
- risk pricing.

A single Q_SDF may therefore compress several pressure channels.

Its interpretation should not be called merely “risk.”

It is more accurately:

retained state-price pressure under the declared reference amplitude. (4.56)

This illustrates a general principle:

The richer the mature filter, the more carefully Q must be interpreted. (4.57)

4.12 Market-implied versus model-implied geometry

Suppose a model produces:

R_model. (4.58)

The market produces:

R_market. (4.59)

Using the same declared amplitude A:

$Q_{model} = \sqrt{(A^2 - R_{model}^2)}$. (4.60)

$Q_{market} = \sqrt{(A^2 - R_{market}^2)}$. (4.61)

Then:

$\theta_{model} = \arccos(R_{model}/A)$. (4.62)

$\theta_{market} = \arccos(R_{market}/A)$. (4.63)

The difference:

$\Delta\theta_{model\text{-}market} = \theta_{market} - \theta_{model}$ (4.64)

may represent disagreement between:

- the declared analytical filter;
- the market's effective aggregate filter.

But this interpretation is valid only if:

- both values refer to the same asset and horizon;
- both use the same amplitude A;
- units are comparable;
- the market price is executable;
- embedded options and liquidity effects are controlled;
- no hidden protocol shift has occurred.

Otherwise, the angle difference mixes distinct worlds.

The correct discipline is:

Never subtract angles whose declarations are not aligned. (4.65)

4.13 One asset, many financial worlds

The same underlying primary asset may generate several effective states:

$Z_{CAPM} = R_{CAPM} + iQ_{CAPM}$. (4.66)

$Z_{credit} = R_{credit} + iQ_{credit}$. (4.67)

$Z_{liq} = R_{liq} + iQ_{liq}$. (4.68)

$Z_{cap} = R_{cap} + iQ_{cap}$. (4.69)

$Z_{acc} = R_{acc} + iQ_{acc}$. (4.70)

These are not necessarily competing estimates of one identical object.

They may answer different questions:

- What is the market-risk-adjusted value?
- What is the default-adjusted value?
- What is executable today?
- What is admissible for capital purposes?
- What can be recognized in the official ledger?

The asset is therefore not represented by one universal scalar.

It participates in a family of declared worlds.

The general form is:

$Z_F = \Pi_F(X \mid P_F)$. (4.71)

Where F identifies the mature filter.

This is one reason the word “world” becomes useful.

Different protocols disclose different operational realities from the same larger field.

4.14 Why the filters must not be collapsed too early

It may be tempting to define one total angle:

$\theta_{total} = \theta_{market} + \theta_{credit} + \theta_{liquidity} + \theta_{cap} + \theta_{acc}$. (4.72)

This is generally unjustified.

The filters may:

- overlap;
- use different baselines;
- act in different orders;
- change one another’s inputs;
- operate at different horizons;
- include correlated pressure;
- contain nonlinear gates.

For example, credit spread may already contain liquidity compensation.

Capital rules may respond to credit ratings.

Accounting impairment may use market prices already affected by liquidity pressure.

Adding the angles may double-count the same underlying structure.

The correct progression is:

1. define each world separately;
2. test each coordinate;
3. identify overlap;
4. residualize shared components;
5. study coupling;
6. only then construct a multi-pressure geometry.

The single-Q model is therefore not a claim that finance has only one hidden pressure dimension.

It is the minimum laboratory in which the logic can be tested cleanly.

Part II — The Moving Financial Frame

5. Does the Asset Rotate, or Does the Measurement Basis Rotate?

5.1 The elementary picture

The expression:

$Z = A \exp(i\theta)$ (5.1)

naturally invites an active-rotation image.

One imagines the financial state vector rotating through the R–Q plane:

$[A, 0] \rightarrow [R, Q]$. (5.2)

Under this picture:

- the asset begins on the real axis;
- the filter rotates it;
- admitted value becomes the horizontal component;
- retained pressure becomes the vertical component.

This is mathematically adequate for the static geometry.

But it may produce the wrong financial intuition.

It can sound as though the asset itself has physically moved into an imaginary dimension.

A stronger interpretation is passive.

5.2 Passive rotation of the valuation basis

Let the underlying amplitude vector be fixed during a declared episode:

$v_A = [A, 0]^T$. (5.3)

Let the measurement basis rotate by θ.

The observed coordinates are:

$[R, Q]^T = Rot(\theta)[A, 0]^T$. (5.4)

Using:

$Rot(\theta) = [[\cos \theta, -\sin \theta], [\sin \theta, \cos \theta]]$, (5.5)

we obtain:

$R = A \cos \theta$. (5.6)

$Q = A \sin \theta$. (5.7)

The economic interpretation becomes:

The declared amplitude need not move; the financial basis through which it is read may rotate. (5.8)

This basis can be determined by:

- market-risk requirements;
- credit standards;
- liquidity conditions;
- capital rules;
- accounting recognition;
- strategic exercise thresholds.

The asset appears differently because the financial world has changed its measurement orientation.

This passive-rotation interpretation is a central conceptual development in the handoff material.

5.3 Repricing as changing visibility

Under the passive picture, a fall in R does not automatically mean that the asset's underlying economic amplitude has fallen.

It may mean that less of the amplitude is visible under the current financial basis.

The change can be written:

$\Delta R = \Delta R_{amplitude} + \Delta R_{frame} + \Delta R_{residual}$. (5.9)

Later, the differential form will become:

$$dR/dt = g_A R - \omega_F Q + Re(\varepsilon_{dyn}). \quad (5.10)$$

This creates a disciplined attribution framework.

Amplitude deterioration

The productive or expected economic capacity declines.

Frame tightening

The market, lender, regulator, or accounting system admits less of the same amplitude.

Residual

The declared two-coordinate world fails to explain the movement.

The same visible repricing can therefore come from different causal geometries.

5.4 The measurement setting is part of the object's effective state

In ordinary intuition, one may think the asset exists first and valuation merely reports it.

In practice, the effective financial object depends partly on the setting.

A bond under one collateral regime is not operationally identical to the same bond under another.

A cash flow under one capital rule is not operationally identical to the same cash flow under another.

An asset before recognition is not institutionally identical to the same asset after recognition.

A project before exercise is not identical to the project after contractual commitment.

Therefore:

Effective Financial Object = Primary Asset + Declared Measurement-and-Admission Protocol. (5.11)

This does not mean the protocol creates the underlying cash flow from nothing.

It means that financial reality is constituted jointly by:

- the primary economic structure;
- the rules under which it becomes tradable, fundable, recognizable, admissible, or enforceable.

The measurement setting is therefore not external decoration.

It participates in the effective object.

5.5 Active and passive descriptions are mathematically equivalent but conceptually different

One may write the geometry actively:

$v_Z = \mathrm{Rot}(\theta) v_A$. (5.12)

Or passively:

coordinates in frame F_θ = coordinates of fixed v_A under rotated basis. (5.13)

The numerical outputs are the same.

But the interpretations differ.

Active interpretation

The asset state rotates.

Passive interpretation

The basis through which the asset is admitted rotates.

For finance, the passive interpretation is often more natural because:

- risk premia change;
- liquidity conditions change;
- regulation changes;
- accounting treatment changes;
- market confidence changes;
- collateral rules change.

The asset may also change, but that is a separate radial component.

This separation prepares the dynamic theory.

5.6 Observer dependence without arbitrary subjectivism

A rotated financial frame is observer- or protocol-dependent.

But protocol dependence does not imply that every observer may choose any result.

Once the protocol is declared, the calculation is constrained.

For a given A and R:

$\theta = \arccos(R/A)$. (5.14)

$Q = \sqrt{(A^2 - R^2)}$. (5.15)

Two analysts using equivalent data and equivalent protocols should recover equivalent coordinates within stated error bounds.

Therefore:

Constructed Coordinate + Declared Protocol + Reproducibility → Protocol-Relative Objectivity. (5.16)

The effective state is not observer-free.

But neither is it arbitrary.

This distinction will later become important when the framework is compared with quantum measurement.

5.7 Multiple observers and multiple frames

Let observer a use protocol P_a.

Let observer b use protocol P_b.

They obtain:

Z_a = Π_{P_a}(X). (5.17)

Z_b = Π_{P_b}(X). (5.18)

In general:

Z_a ≠ Z_b. (5.19)

The disagreement may arise because they use different:

- boundaries;
- baselines;
- horizons;
- cash-flow forecasts;
- risk models;
- liquidity assumptions;
- legal permissions;
- capital rules.

The correct response is not immediately to average the values.

First ask whether a frame map exists:

M_ab: Z_a → Z_b. (5.20)

A meaningful frame map should explain which changes in declaration transform one effective state into the other.

Objectivity then concerns invariant relations under admissible frame transformations.

For example, both observers may disagree on R while agreeing that:

- credit pressure increased;
- liquidity pressure dominates market-risk pressure;
- a gate threshold is approaching;
- dynamic residual is rising;
- the current model is becoming unstable.

Such relations may survive even when scalar values differ.

5.8 The first world-forming lesson

The passive-rotation interpretation yields the article's first broader lesson:

A world is partly defined by the axes along which it can disclose structure. (5.21)

The primary economic field may contain many possibilities.

A mature finance protocol selects:

- which distinction matters;
- which baseline is relevant;
- which value becomes visible;
- which pressure remains orthogonal;
- which event may pass the gate;
- which result enters the ledger.

Finance therefore does not merely observe a pre-given financial world.

Through its filters and ledgers, it helps constitute the effective world in which financial action becomes possible.

6. Radial Change and Angular Change

6.1 The general dynamic state

The static Finance Geometry state is:

$Z = A \exp(i\theta)$. (6.1)

The dynamic extension allows both the declared amplitude and the financial frame to move:

$Z(t) = A(t) \exp[i\theta(t)]$. (6.2)

This equation contains two distinct kinds of change.

The first is radial:

A(t) changes. (6.3)

The second is angular:

θ(t) changes. (6.4)

These changes should not be interpreted as interchangeable.

A radial change modifies the magnitude of the declared economic possibility.

An angular change modifies the proportion of that possibility admitted under the current financial filter.

The decomposition is therefore:

Financial-State Change = Amplitude Change + Frame Change. (6.5)

This distinction is one of the central developments supplied by the handoff framework.

6.2 Differentiating the complex financial state

Differentiate:

$Z(t) = A(t) \exp[i\theta(t)]$. (6.6)

Using the product and chain rules:

$dZ/dt = (dA/dt)\exp(i\theta) + iA(d\theta/dt)\exp(i\theta)$. (6.7)

Since:

$Z = A \exp(i\theta)$, (6.8)

we obtain:

$dZ/dt = [(1/A)(dA/dt) + i(d\theta/dt)]Z$. (6.9)

Define the radial growth rate:

$g_A = (1/A)(dA/dt)$. (6.10)

Define the financial angular velocity:

$\omega_F = d\theta/dt$. (6.11)

Then:

$dZ/dt = (g_A + i\omega_F)Z$. (6.12)

In an empirical system, the declared two-coordinate model will rarely explain every observed movement exactly. Introduce a dynamic residual:

$\varepsilon_{dyn} = dZ/dt - (g_A + i\omega_F)Z$. (6.13)

The full equation becomes:

$$dZ/dt = (g_A + i\omega_F)Z + \varepsilon_{dyn}. \quad (6.14)$$

This is the minimum dynamic kernel of the article.

6.3 Separating the real and pressure coordinates

Write:

$$Z = R + iQ. \quad (6.15)$$

Substitute into:

$$dZ/dt = (g_A + i\omega_F)(R + iQ) + \varepsilon_{dyn}. \quad (6.16)$$

Expanding:

$$(g_A + i\omega_F)(R + iQ) = g_A R - \omega_F Q + i(g_A Q + \omega_F R). \quad (6.17)$$

Therefore:

$$dR/dt = g_A R - \omega_F Q + \mathrm{Re}(\varepsilon_{dyn}). \quad (6.18)$$

$$dQ/dt = g_A Q + \omega_F R + \mathrm{Im}(\varepsilon_{dyn}). \quad (6.19)$$

These two equations describe the coupled motion of admitted value and retained pressure.

The visible coordinate R changes through:

1. radial growth or contraction, $g_A R$;
2. angular conversion, $-\omega_F Q$;
3. unexplained residual, $\mathrm{Re}(\varepsilon_{dyn})$.

The pressure coordinate Q changes through:

1. radial scaling, $g_A Q$;
2. angular conversion, $+\omega_F R$;
3. unexplained residual, $\mathrm{Im}(\varepsilon_{dyn})$.

The pair forms a structured exchange.

When the frame tightens:

$$\omega_F > 0, \quad (6.20)$$

the angular term tends to move value from R toward Q.

When the frame relaxes:

$$\omega_F < 0, \quad (6.21)$$

the angular term tends to move value from Q toward R.

6.4 The visible-value decomposition

The most direct finance equation is:

$dR/dt = g_A R - \omega_F Q + Re(\varepsilon_{dyn})$. (6.22)

Define the angular repricing load:

$\Lambda_F = Q\omega_F$. (6.23)

Then:

$dR/dt = g_A R - \Lambda_F + Re(\varepsilon_{dyn})$. (6.24)

This equation decomposes visible repricing into:

Fundamental-Amplitude Contribution = $g_A R$. (6.25)

Filter-Pressure Contribution = $-\Lambda_F$. (6.26)

Unexplained Contribution = $Re(\varepsilon_{dyn})$. (6.27)

The decomposition does not claim that the first term contains every “fundamental” effect or that the second contains every “technical” or “market” effect.

Its meaning is protocol-specific.

The first term is whatever the declared amplitude A captures.

The second term is whatever the declared angle θ captures.

The residual contains what the two-coordinate construction fails to explain.

6.5 Normalized growth decomposition

Divide equation (6.22) by R, assuming $R > 0$:

$(1/R)(dR/dt) = g_A - (Q/R)\omega_F + Re(\varepsilon_{dyn})/R$. (6.28)

Since:

$Q/R = \tan\theta$, (6.29)

we obtain:

$d \ln R/dt = g_A - \tan\theta \cdot \omega_F + Re(\varepsilon_{dyn})/R$. (6.30)

This gives a compact proportional-growth equation:

Visible-Value Growth = Amplitude Growth − Pressure Ratio × Frame Velocity + Residual. (6.31)

A pressure-loaded state is more sensitive to angular movement because $\tan\theta$ is larger.

A rapidly moving frame is more consequential when the existing state already contains substantial pressure.

This means:

High Q alone does not imply immediate repricing. (6.32)

High ω_F alone does not imply large repricing. (6.33)

High Q × high ω_F is the stronger angular-stress configuration. (6.34)

6.6 Radial motion

A radial change satisfies:

$dA/dt \neq 0$. (6.35)

while the angle may remain approximately constant:

$d\theta/dt \approx 0$. (6.36)

Then:

$dR/dt \approx g_A R$. (6.37)

$dQ/dt \approx g_A Q$. (6.38)

Both coordinates expand or contract proportionally.

The ratio remains constant:

$R/A = \cos \theta = \text{constant}$. (6.39)

$Q/A = \sin \theta = \text{constant}$. (6.40)

$Q/R = \tan \theta = \text{constant}$. (6.41)

Examples of primarily radial change include:

- revised expected cash flows;
- changes in productive capacity;
- new reserves or resource discoveries;
- technological improvement;
- loss of customers;
- physical asset damage;
- deterioration in expected recovery amounts;
- changes in project scale;
- revised long-run economic demand.

Under a purely radial decline, the asset becomes smaller within approximately the same financial orientation.

6.7 Angular motion

An angular change satisfies:

$$d\theta/dt \neq 0. \quad (6.42)$$

while the amplitude may remain approximately fixed:

$$dA/dt \approx 0. \quad (6.43)$$

Then:

$$dR/dt \approx -\omega_F Q. \quad (6.44)$$

$$dQ/dt \approx \omega_F R. \quad (6.45)$$

The magnitude remains approximately constant:

$$d(A^2)/dt \approx 0. \quad (6.46)$$

But admitted value and retained pressure exchange.

Examples of primarily angular change include:

- changing equity-risk premium;
- changing credit spread;
- changing liquidity haircut;
- changing collateral eligibility;
- changing regulatory capital treatment;
- changing accounting recognition probability;
- changing confidence in execution;
- changing certainty-equivalent adjustment;
- changing exercise threshold.

Under a purely angular tightening, the asset does not necessarily become economically smaller.

Less of its declared amplitude becomes financially admitted.

6.8 Mixed motion

Real financial systems usually exhibit both radial and angular change.

Expected cash flow may deteriorate while credit spreads widen.

Economic capacity may improve while liquidity worsens.

A firm may grow while regulatory admissibility falls.

A bond may retain expected repayment but lose collateral eligibility.

The dynamic equation handles the mixed case:

$$\frac{dR}{dt} = g_A R - \omega_F Q + \mathrm{Re}(\varepsilon_{dyn}). \quad (6.47)$$

Suppose:

$$g_A > 0, \quad (6.48)$$

but:

$$\omega_F Q > g_A R. \quad (6.49)$$

Then:

$$\frac{dR}{dt} < 0, \quad (6.50)$$

ignoring residual.

The underlying amplitude is growing, but the financial frame is tightening faster than the growth can compensate.

The opposite can also occur.

Suppose:

$$g_A < 0, \quad (6.51)$$

but:

$$\omega_F < 0. \quad (6.52)$$

A sufficiently rapid frame relaxation may cause:

$$\frac{dR}{dt} > 0. \quad (6.53)$$

even while the underlying amplitude declines.

This is a familiar financial phenomenon expressed in geometric form:

A weakening asset can rise in price if discount pressure relaxes sufficiently. (6.54)

A strengthening asset can fall in price if valuation pressure tightens sufficiently. (6.55)

6.9 A two-scenario attribution example

Assume initially:

$$A_0 = 100. \quad (6.56)$$

$$\theta_0 = 0.30. \quad (6.57)$$

Then:

$$R_0 = 100 \cos(0.30) \approx 95.5336. \quad (6.58)$$

$$Q_0 = 100 \sin(0.30) \approx 29.5520. \quad (6.59)$$

Scenario A — Pure amplitude decline

Let:

$A_1 = 95$. (6.60)

$\theta_1 = 0.30$. (6.61)

Then:

$R_1 \approx 90.7569$. (6.62)

$Q_1 \approx 28.0744$. (6.63)

Both R and Q decline proportionally.

The pressure ratio remains:

$Q_1/R_1 \approx \tan(0.30)$. (6.64)

Scenario B — Pure frame tightening

Let:

$A_1 = 100$. (6.65)

$\theta_1 = 0.425$. (6.66)

Then:

$R_1 \approx 91.1000$. (6.67)

$Q_1 \approx 41.2410$. (6.68)

The visible value is similar to Scenario A.

But the pressure coordinate is much larger.

The same approximate decline in R therefore corresponds to two different financial states:

Scenario A: smaller amplitude, similar orientation. (6.69)

Scenario B: same amplitude, harsher orientation. (6.70)

This is the dynamic form of the article's core diagnostic promise.

6.10 The meaning of residual

The residual term must not become a container into which every unexplained movement is placed without discipline.

A nonzero ε_{dyn} may arise because:

- A was mismeasured;
- θ was misestimated;
- more than one pressure channel is active;

- the baseline changed;
- the horizon changed;
- the filter changed discontinuously;
- the market experienced a gate event;
- the system crossed into a new regime;
- the two-dimensional state is insufficient;
- noise or sampling error is material.

Therefore:

Large ε_dyn does not prove hidden pressure. (6.71)

Large ε_dyn may mean the declared world is wrong. (6.72)

The residual must be allowed to falsify the model.

This principle follows the original Finance Geometry requirement that the complex extension justify itself through diagnostic gain rather than elegance alone.

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7. Q as Phase Sensitivity

7.1 The derivative identity

Under constant amplitude:

$R = A \cos \theta$. (7.1)

Differentiating with respect to θ:

$dR/d\theta = -A \sin \theta$. (7.2)

Since:

$Q = A \sin \theta$, (7.3)

we obtain:

$dR/d\theta = -Q$. (7.4)

Therefore:

$Q = -dR/d\theta$. (7.5)

This identity gives Q an operational interpretation.

Q measures how rapidly admitted value changes under a marginal movement of the financial angle.

A larger Q means that a given angular tightening produces a larger absolute reduction in R.

7.2 Normalized phase sensitivity

Divide by R:

$-(1/R)(dR/d\theta) = Q/R.$ (7.6)

Since:

$Q/R = \tan \theta,$ (7.7)

we obtain:

$-d \ln R/d\theta = \tan \theta.$ (7.8)

Define the pressure ratio:

$\Pi_Q = Q/R.$ (7.9)

Then:

$\Pi_Q = \tan \theta.$ (7.10)

The quantity Π_Q measures retained pressure per unit of admitted value.

It is not a probability.

It is not leverage.

It is not volatility.

It is a normalized geometric sensitivity under the declared filter.

7.3 Angular repricing load

The phase sensitivity Q becomes dynamically consequential when the frame moves.

Define:

$\omega_F = d\theta/dt.$ (7.11)

Then:

$\Lambda_F = Q\omega_F.$ (7.12)

The angular contribution to visible repricing is:

$dR/dt|_{angular} = -\Lambda_F.$ (7.13)

The sign of Λ_F matters.

If:

$Q > 0$ (7.14)

and:

$\omega_F > 0,$ (7.15)

then:

$\Lambda_F > 0$, (7.16)

and the angular contribution reduces admitted value.

If:

$\omega_F < 0$, (7.17)

then:

$\Lambda_F < 0$, (7.18)

and the angular contribution raises admitted value.

Therefore, Λ_F is directional.

It is not merely a pressure stock.

It is pressure multiplied by phase movement.

7.4 Pressure stock versus pressure conversion

Q can be large while θ remains stable.

Then:

$\omega_F \approx 0$. (7.19)

and:

$\Lambda_F \approx 0$. (7.20)

The state is pressure-loaded but not currently converting that pressure into visible repricing.

This may describe:

- a stable high-risk environment;
- a persistently illiquid asset;
- a long-standing capital restriction;
- a known but unchanged credit burden.

Conversely, θ may move quickly while Q remains small.

Then the frame is changing, but the state is not yet highly pressure-loaded.

The strongest immediate angular effect occurs when both are large.

This motivates a three-part distinction:

Pressure Stock = Q. (7.21)

Pressure Intensity = Q/R. (7.22)

Pressure Conversion Rate = Qω_F. (7.23)

7.5 A possible early-warning hierarchy

Near θ = 0:

Q ≈ Aθ. (7.24)

A − R ≈ Aθ²/2. (7.25)

Therefore:

Q grows linearly in small θ. (7.26)

A − R grows quadratically in small θ. (7.27)

This suggests a possible warning hierarchy:

Phase movement appears first. (7.28)

Q responds at first order. (7.29)

Visible haircut responds at second order. (7.30)

Qω_F may identify active conversion of pressure into repricing. (7.31)

This is a hypothesis for empirical testing.

It may fail if:

- θ is too noisy;
- A is unstable;
- the chosen filter is not economically meaningful;
- existing variables already capture the same information;
- estimated derivatives amplify measurement error.

The article therefore does not claim that Q is automatically an early-warning indicator.

It proposes a reason why it might become one under suitable protocols.

7.6 Stress configurations

The two variables Q and ω_F define four basic configurations.

Q	ω_F	Interpretation
low	low	low pressure, stable frame
high	low	pressure-loaded but stable
low	high	rapidly changing frame, low stored pressure
high	high	pressure-loaded state under rapid tightening

The final configuration is the most likely to produce large angular repricing:

$|\Lambda_F| = |Q\omega_F|$. (7.32)

A fifth dimension is residual.

A state with high $|\varepsilon_{dyn}|$ may be more dangerous than any of the four configurations because the declared geometry itself is failing.

Thus a practical dashboard should not show Q alone.

It should show at least:

$Q, Q/R, \omega_F, Q\omega_F, \varepsilon_{dyn}$. (7.33)

7.7 Acceleration of the filter

The angular velocity may itself change.

Define angular acceleration:

$\alpha_F = d\omega_F/dt = d^2\theta/dt^2$. (7.34)

Differentiate:

$\Lambda_F = Q\omega_F$. (7.35)

Then:

$d\Lambda_F/dt = (dQ/dt)\omega_F + Q\alpha_F$. (7.36)

Ignoring residual and using:

$dQ/dt = g_A Q + \omega_F R$, (7.37)

we obtain:

$d\Lambda_F/dt = g_A Q\omega_F + \omega_F^2 R + Q\alpha_F$. (7.38)

Therefore:

$d\Lambda_F/dt = g_A\Lambda_F + \omega_F^2 R + Q\alpha_F$. (7.39)

This decomposes the growth of angular repricing load into:

- amplitude scaling;
- rotational persistence;
- acceleration of the financial frame.

A rapidly accelerating frame may be especially important during:

- funding crises;
- rating cascades;
- volatility shocks;

- policy surprises;
- collateral reclassification;
- sudden accounting impairment;
- market closures.

However, second derivatives are empirically unstable.

Any implementation should use smoothing, robust estimation, and explicit uncertainty intervals.

7.8 Pressure release

If the frame relaxes:

$\omega_F < 0$, (7.40)

then the angular contribution to R is positive:

$-Q\omega_F > 0$. (7.41)

This may describe:

- falling risk premia;
- spread compression;
- restored liquidity;
- regulatory relief;
- improved collateral eligibility;
- reduced uncertainty;
- successful refinancing;
- improved execution confidence.

Pressure release does not necessarily mean the amplitude improved.

A price recovery may be entirely angular.

This is why the decomposition can help prevent mistaken narratives.

A rising price is not always evidence of stronger fundamentals.

A falling price is not always evidence of weaker fundamentals.

7.9 The sign convention

The minimal model uses:

$Q \geq 0$. (7.42)

and:

$0 \le \theta \le \pi/2$. (7.43)

Direction is carried by ω_F.

This is convenient for valuation filters where Q represents non-negative retained pressure.

A more advanced model may use signed pressure coordinates:

$Q_{signed} \in \mathbb{R}$. (7.44)

Such an extension could distinguish:

- upward versus downward pressure;
- long versus short positioning;
- positive versus negative optionality;
- opposing residual channels.

But signed Q should not be introduced into the basic framework without a mature financial derivation.

Otherwise the sign becomes another invented metaphor.

---

8. Four Modes of Financial Change

8.1 Why one category of “price movement” is insufficient

Financial commentary often compresses several distinct phenomena into one statement:

The asset price moved. (8.1)

But a price change may arise from:

- economic-amplitude change;
- filter-frame change;
- commitment or gate event;
- revision of the financial protocol itself.

These modes have different meanings and different intervention implications.

The article therefore distinguishes four regimes:

1. amplitude-dominant;
2. frame-dominant;
3. gate-dominant;

4. declaration-dominant.

8.2 Amplitude-dominant regime

The amplitude term dominates when:

$$|g_A R| \gg |\omega_F Q|. \quad (8.2)$$

and residual is controlled:

$$|\mathrm{Re}(\varepsilon_{dyn})| \ll |g_A R|. \quad (8.3)$$

Then:

$$dR/dt \approx g_A R. \quad (8.4)$$

The visible change is primarily associated with the declared underlying amplitude.

Examples include:

- major earnings revision;
- permanent loss of productive assets;
- new capacity becoming operational;
- revised recovery estimate;
- structural customer loss;
- successful product launch;
- resource discovery;
- major legal liability affecting expected cash flow.

The appropriate response focuses on:

- operating performance;
- cash-flow assumptions;
- strategic position;
- economic capacity;
- scenario revision.

8.3 Frame-dominant regime

The frame term dominates when:

$$|\omega_F Q| \gg |g_A R|. \quad (8.5)$$

and:

$|\mathrm{Re}(\varepsilon_{dyn})| \ll |\omega_F Q|$. (8.6)

Then:

$dR/dt \approx -\omega_F Q$. (8.7)

The visible change is primarily associated with the financial filter.

Examples include:

- rapid rise in risk premium;
- sudden credit-spread widening;
- liquidity drying up;
- collateral haircut increase;
- capital-rule tightening;
- accounting recognition becoming less likely;
- strategic hurdle rate increasing.

The appropriate response may focus on:

- funding;
- hedging;
- liquidity;
- capital structure;
- timing;
- collateral;
- communication;
- regime risk.

The underlying asset may remain economically viable while becoming financially difficult to hold, fund, recognize, or execute.

8.4 Gate-dominant regime

A gate-dominant event is discontinuous.

The system does not merely move smoothly through a coordinate space.

A threshold is crossed and a possibility becomes a committed event.

Let:

$G_k(\text{state}) \in \{0,1\}$. (8.8)

Where:

$G_k = 0$ means not admitted. (8.9)

$G_k = 1$ means committed. (8.10)

Examples include:

- trade execution;
- margin call;
- covenant breach;
- default;
- downgrade;
- impairment;
- insolvency declaration;
- regulatory intervention;
- collateral ineligibility;
- option exercise.

A gate event updates the ledger:

$L_{k+1} = L_k \sqcup \text{Record}_k$. (8.11)

The event may alter future dynamics even if the underlying variables later return to earlier levels.

Therefore:

Gate Event ≠ Large Continuous Price Change. (8.12)

A small numerical threshold crossing may be more consequential than a large continuous movement.

8.5 Declaration-dominant regime

A declaration-dominant change occurs when the old protocol no longer adequately defines the world.

Then:

$P_{k+1} \neq P_k$. (8.13)

The change may involve:

- a new baseline;
- a new horizon;
- a new risk model;
- a new collateral rule;

- a new accounting category;
- a new legal regime;
- a new intervention set;
- a new definition of liquidity;
- a new market structure.

Examples include:

- switching from going-concern to liquidation valuation;
- changing from ordinary liquidity assumptions to crisis execution;
- replacing historical volatility with stressed volatility;
- moving from market-value accounting to impairment testing;
- changing collateral eligibility after a rating event;
- imposing capital controls;
- entering bankruptcy protection.

The state does not simply move to a new coordinate under the same world.

The coordinate system changes.

8.6 Regime-classification table

Regime	Dominant condition	Main question
Amplitude		g_A R
Frame		ω_F Q
Gate	threshold commitment	What event became irreversible in the ledger?
Declaration	$P_{k+1} \neq P_k$	Why is the old world no longer adequate?

The residual layer cuts across all four.

If:

|ε_dyn| is large, (8.14)

the regime classification may itself be unreliable.

8.7 Transition between regimes

A system may move from one regime to another.

A common crisis sequence is:

Frame Tightening → Gate Event → Declaration Revision. (8.15)

For example:

Credit spreads widen. (8.16)

Collateral value falls. (8.17)

A margin threshold is crossed. (8.18)

Forced sale occurs. (8.19)

The market becomes illiquid. (8.20)

The old valuation assumptions are abandoned. (8.21)

The sequence begins as angular movement.

It then becomes a gate event.

Finally, it becomes a new declared world.

This layered view is more informative than treating the entire episode as one continuous increase in risk.

8.8 Misdiagnosis risk

The four-mode framework also identifies common analytical errors.

Error 1 — Treating frame movement as fundamental deterioration

This may produce unnecessary liquidation or underinvestment.

Error 2 — Treating fundamental deterioration as temporary frame pressure

This may delay necessary restructuring.

Error 3 — Treating a gate event as reversible noise

This may ignore legal or contractual path dependence.

Error 4 — Treating declaration failure as an ordinary parameter change

This may preserve a model that no longer describes the system.

Error 5 — Treating large residual as a new hidden factor without validation

This may turn falsification into storytelling.

Part III — From Phase to Internal Time

9. Three Clocks in Finance

9.1 Calendar time

Calendar time is the external chronology:

$t_0 < t_1 < t_2 < \dots$ (9.1)

It measures:

- days;
- months;
- quarters;
- maturities;
- settlement periods;
- reporting cycles;
- contractual deadlines.

Calendar time is necessary, but it does not always reflect the system's internal rate of change.

A market may remain quiet for months and transform within hours.

9.2 Phase time

Phase time is generated by the financial angle:

$\theta = \theta(t)$. (9.2)

The phase coordinate records how far the declared financial frame has moved.

A slow calendar interval may contain large phase movement.

A long calendar interval may contain almost none.

Therefore:

Equal Calendar Duration ≠ Equal Financial Phase Distance. (9.3)

The phase distance over an episode is:

$\Delta\theta = \theta(t_2) - \theta(t_1)$. (9.4)

A crisis may traverse:

$|\Delta\theta_{crisis}| \gg |\Delta\theta_{quiet}|$ (9.5)

even when:

$\Delta t_{crisis} \ll \Delta t_{quiet}$. (9.6)

Phase therefore measures internal financial progression rather than external duration.

9.3 Ledger time

Ledger time advances through committed events.

Let k index ledger entries:

$k = 0,1,2,\ldots$ (9.7)

The ledger update is:

$L_{k+1} = L_k \sqcup \text{Record}_k$. (9.8)

Examples of ledger ticks include:

- execution of a trade;
- issue of a margin call;
- official downgrade;
- covenant breach;
- default declaration;
- impairment recognition;
- regulatory order;
- collateral seizure;
- bankruptcy filing.

Ledger time does not advance simply because the clock moves.

It advances when the system commits a consequential record.

9.4 The three-clock relation

The complete temporal chain is:

$t \to \theta(t) \to \text{Gate}_k \to L_{k+1}$. (9.9)

Calendar time provides duration.

Phase time provides internal ordering.

Ledger time provides historical commitment.

The three clocks answer different questions.

Clock	Question
Calendar t	How much external time passed?
Phase θ	How far did the financial orientation move?

Clock	Question
Ledger k	How many consequential commitments occurred?

9.5 Why the distinction matters

Two episodes may have the same calendar length but different phase distances.

Two episodes may have the same phase distance but different numbers of gate events.

Two episodes may return to the same phase while retaining different ledgers.

Therefore:

Same t does not imply same θ. (9.10)

Same θ does not imply same L. (9.11)

Same L-entry count does not imply same economic consequence. (9.12)

The financial world has several forms of time because it has several forms of change.

---

10. When Phase Becomes a Local Internal Clock

10.1 Monotonicity

A variable can serve as a local internal clock when it orders states without ambiguity.

Suppose, over a declared episode:

$d\theta/dt > 0$. (10.1)

Then θ increases monotonically.

Alternatively:

$d\theta/dt < 0$. (10.2)

Then θ decreases monotonically.

In either case, the map between calendar time and phase is locally invertible:

$t = t(\theta)$. (10.3)

A system variable Y can then be written as:

$Y(t) = Y[t(\theta)] = Y(\theta)$. (10.4)

Its dynamics may be expressed as:

$dY/d\theta = (dY/dt)/(d\theta/dt)$. (10.5)

This allows the system to be studied in phase order rather than calendar order.

10.2 The ideal internal phase law

Under constant amplitude:

$Z(\theta) = A \exp(i\theta)$. (10.6)

Then:

$dZ/d\theta = iZ$. (10.7)

This law is independent of how irregularly θ moves in calendar time.

The calendar-time dynamics are:

$dZ/dt = i(d\theta/dt)Z$. (10.8)

But inside the phase description:

$dZ/d\theta = iZ$. (10.9)

The phase-indexed law is simpler than the calendar-time law.

This is the first sense in which a secondary effective world may possess a regular internal clock.

The parent financial process may be irregular in t.

The internal phase world may be regular in θ.

10.3 Phase order is not yet historical time

Monotonic phase provides order:

$\theta_1 < \theta_2 < \theta_3$. (10.10)

But order alone does not create history.

A reversible oscillator has phase order but may leave no irreversible record.

To become historical time, phase must interact with:

- gates;
- records;
- consequences;
- memory.

The stronger condition is:

Phase Order + Commitment Gate + Trace Retention → Effective Financial Time. (10.11)

This distinction parallels the wider ledger-based account of time developed in the project documents, where ordered disclosure becomes time-like only when projection is committed into trace.

10.4 Internal time as event-bearing phase

Let τ_F denote effective financial time.

A simple event-bearing construction is:

$$d\tau_F = \chi_G(\theta,L)\,|d\theta|. \quad (10.12)$$

Here χ_G indicates whether the phase movement passes through a relevant event or gate structure.

A more general weighting may be:

$$d\tau_F = w(\theta,Q,G,L)|d\theta|. \quad (10.13)$$

Where w increases when:

- phase pressure is high;
- a threshold is near;
- a gate is crossed;
- a trace is written;
- the event changes future admissibility.

This means not every phase increment contributes equally to financial history.

A quiet rotation may produce little ledger time.

A small movement across a covenant threshold may produce a major historical tick.

10.5 Phase-indexed prediction

Suppose a variable Y responds more consistently to phase than to calendar time.

Then:

$$dY/d\theta = F(Y,\theta). \quad (10.14)$$

may be more stable than:

$$dY/dt = G(Y,t). \quad (10.15)$$

Possible examples include:

- collateral usage as credit phase tightens;
- redemption behaviour as liquidity phase moves;
- option exercise probability as strategic phase approaches a threshold;
- capital usage as admissibility phase changes;
- downgrade hazard as credit pressure rotates.

The empirical question is:

Does phase indexing reduce residual or improve event alignment? (10.16)

If not, phase is merely a reparameterization.

10.6 A crisis-clock interpretation

Suppose two credit episodes each move from:

$\theta = 0.20$ (10.17)

to:

$\theta = 0.60$. (10.18)

Episode A takes six months.

Episode B takes two days.

In calendar time, they are radically different.

In phase distance:

$\Delta\theta_A = \Delta\theta_B = 0.40$. (10.19)

The internal path may therefore be comparable even though the external speed differs.

The angular velocity distinguishes them:

$\omega_{F,A} \approx 0.40/(\text{six months})$. (10.20)

$\omega_{F,B} \approx 0.40/(\text{two days})$. (10.21)

Thus phase distance measures progression.

Angular velocity measures urgency.

Ledger events measure historical consequence.

---

11. Phase Reversal, Ambiguity, and Clock Failure

11.1 Monotonicity is local

Financial phase will not usually move monotonically forever.

Risk premia tighten and relax.

Liquidity disappears and returns.

Credit spreads widen and compress.

Therefore, phase should be used as a local clock within declared episodes.

The episode boundary may be defined by:

- sign change in ω_F;
- gate event;
- structural break;
- model residual threshold;
- protocol revision;
- external intervention.

11.2 Phase reversal

A phase reversal occurs when:

ω_F changes sign. (11.1)

For example:

$\omega_F > 0$ before t^*. (11.2)

$\omega_F < 0$ after t^*. (11.3)

At t^*, the local phase clock changes direction.

This does not mean calendar time reverses.

It means the financial orientation has begun to unwind.

The episode should therefore be segmented:

Episode 1: tightening phase. (11.4)

Episode 2: relaxation phase. (11.5)

Each segment may have its own local phase ordering.

11.3 Oscillating phase

If θ oscillates:

$\theta(t) = \bar{\theta} + a \sin(\Omega t)$, (11.6)

then θ alone cannot serve as a global one-to-one clock.

The same phase value occurs repeatedly.

Additional information is required:

- direction of motion;
- cycle count;
- ledger state;

- branch label.

A richer state may be:

State = (θ, sign ω_F, n_cycle, L). (11.7)

The phase value alone is insufficient.

This demonstrates why historical trace matters.

Without the ledger, repeated phase positions may appear identical.

With the ledger, they belong to different histories.

11.4 Rapid reversals

Rapid phase reversals may indicate:

- unstable expectations;
- noisy estimation;
- competing filters;
- market manipulation;
- insufficient smoothing;
- an invalid one-dimensional coordinate;
- regime ambiguity.

A useful diagnostic is the reversal rate:

ν_rev = Number of sign changes in ω_F / Observation Window. (11.8)

High ν_rev may signal that θ is not functioning as a reliable internal clock.

It may also indicate a critical regime in which no stable direction has formed.

11.5 Phase jumps

Some financial events create discontinuous changes:

$\Delta\theta = \theta(t^{+}) - \theta(t^{-})$. (11.9)

Examples include:

- surprise default;
- emergency regulation;
- trading halt;
- sovereign intervention;

- sudden collateral rejection;
- accounting reclassification;
- major fraud disclosure.

A phase jump may be represented as:

$d\theta/dt = \text{regular component} + J\delta(t - t^*)$. (11.10)

Where J is the jump size.

The jump is not merely fast continuous rotation.

It may be evidence of a gate event or declaration change.

11.6 Competing phase coordinates

A financial system may have multiple angles:

$\theta_{market}(t)$. (11.11)

$\theta_{credit}(t)$. (11.12)

$\theta_{liquidity}(t)$. (11.13)

$\theta_{capital}(t)$. (11.14)

These may move in different directions.

For example:

$\theta_{market} \downarrow$ (11.15)

while:

$\theta_{liquidity} \uparrow$. (11.16)

A single global phase may then be misleading.

The system may require a phase vector:

$\vec{\theta} = (\theta_{market}, \theta_{credit}, \theta_{liquidity}, \theta_{capital}, \ldots)$. (11.17)

The one-dimensional clock interpretation fails when no dominant monotonic projection exists.

This is not a flaw in reality.

It is a limit of the chosen coordinate.

11.7 Dynamic residual as clock-failure detector

Under variable amplitude, the ideal phase equation is:

$dZ/d\theta = [(d \ln A/d\theta) + i]Z$. (11.18)

Define:

$\varepsilon_\theta = dZ/d\theta - [(d \ln A/d\theta) + i]Z.$ (11.19)

If:

$|\varepsilon_\theta|$ is small, (11.20)

the phase description approximately closes.

If:

$|\varepsilon_\theta|$ is large, (11.21)

possible explanations include:

- phase is not the correct internal coordinate;
- A is misdeclared;
- multiple pressures are interacting;
- a gate has changed the state discontinuously;
- the protocol has changed;
- the world model has failed.

Thus:

Clock Failure may appear as Closure Residual. (11.22)

11.8 When to abandon the phase-clock interpretation

The phase-clock model should be suspended when:

1. θ is not identifiable;
2. θ is dominated by noise;
3. phase reverses too rapidly;
4. the map $t \leftrightarrow \theta$ is not locally invertible;
5. residual exceeds a declared threshold;
6. several pressure angles conflict;
7. a declaration change occurs;
8. the model provides no empirical gain.

The discipline is:

A variable should not be called time merely because it is plotted on a horizontal axis. (11.23)

Phase becomes time-like only when it supports stable order, event structure, trace, and predictive closure.

Part IV — The Effective Financial World

12. A Reduced Model Is Not Automatically a World

12.1 The danger of inflationary language

The word “world” can be used too easily.

A valuation model is not automatically a world.

A chart is not automatically a world.

A two-dimensional coordinate system is not automatically a world.

A statistical summary is not automatically a world.

If every useful representation is called a world, the term loses analytical value.

The present framework therefore requires more than dimensional reduction.

A reduced financial representation becomes world-like only when it supports a sufficiently complete operational structure.

The minimum distinction is:

Reduced Model = Compressed Description. (12.1)

Effective World = Compressed Description + Operational Closure + Event Structure + Historical Consequence. (12.2)

The additional terms matter.

A reduced model may describe what is currently visible.

An effective world also determines:

- what states can occur;
- what transitions are admissible;
- what events can be committed;
- what records are retained;
- what interventions are possible;
- how past events constrain the future.

12.2 The declared financial world

Let P denote the declared financial protocol:

P = (B, q, φ, h, F, G, T, U). (12.3)

Where:

- B = system boundary;
- q = baseline or reference state;
- φ = feature map;
- h = valuation or observation horizon;
- F = mature finance filter;
- G = admission or commitment gate;
- T = trace and residual rule;
- U = admissible intervention family.

The primary financial field is:

$X \in \mathcal{X}$. (12.4)

The declared protocol maps X into an effective state:

$Z = \Pi_{P,L}(X)$. (12.5)

The effective state space is:

$\mathcal{Z}_P = \{Z \mid Z = \Pi_{P,L}(X), X \in \mathcal{X}\}$. (12.6)

The protocol does not create the entire primary field.

It declares which distinctions in that field become financially readable and actionable.

12.3 Minimum conditions for worldhood

A protocol-bounded financial representation becomes world-like when it satisfies at least ten conditions.

Condition 1 — State distinction

The framework must distinguish relevant states.

For example:

$Z_1 \neq Z_2$ (12.7)

must correspond to a financially meaningful difference.

A coordinate system that produces numerical variation without operational interpretation is not enough.

Condition 2 — Internal coordinates

The state must be expressible in coordinates that support analysis and action.

In the minimal model:

$Z = R + iQ = A \exp(i\theta)$. (12.8)

The coordinates must have declared meanings:

- A = pre-filter amplitude;
- R = admitted value;
- Q = retained pressure;
- θ = finance-filter angle.

Condition 3 — Transition law

The system must support at least approximate internal dynamics.

For example:

$dZ/dt = (g_A + i\omega_F)Z + \varepsilon_{dyn}$. (12.9)

or, under constant amplitude:

$dZ/d\theta = iZ$. (12.10)

A list of unrelated states is not yet a world.

A world requires some rule connecting one state to another.

Condition 4 — Event grammar

The representation must distinguish continuous change from consequential events.

Examples include:

- trade;
- downgrade;
- default;
- impairment;
- margin call;
- covenant breach;
- exercise;
- regulatory intervention.

An event is not simply a large numerical movement.

It is a transition recognized by the protocol as consequential.

Condition 5 — Admission gates

Not every possible state becomes an official event.

A gate determines whether a possibility passes into commitment:

G(state) → Commit or Residual. (12.11)

The gate may be:

- contractual;
- regulatory;
- accounting;
- legal;
- algorithmic;
- market-based;
- institutional.

Condition 6 — Trace retention

Committed events must leave records.

The minimum ledger update is:

$L_{k+1} = L_k \sqcup \text{Record}_k$. (12.12)

A stronger record includes:

$L_{k+1} = L_k \sqcup (\text{Record}_k, \text{Evidence}_k, \text{GateMetadata}_k, \text{Residual}_k)$. (12.13)

Without trace, the system may oscillate but does not accumulate history.

Condition 7 — Intervention

Agents must be able to act within the effective state space.

Let:

$u \in U$. (12.14)

Examples include:

- buy;
- sell;
- hedge;
- refinance;
- post collateral;
- exercise;

- restructure;
- impose a limit;
- change capital allocation.

A purely passive representation is less world-like than one that organizes admissible action.

Condition 8 — Backreaction

Actions and committed financial states must influence future primary states.

The backreaction map is:

$X_{k+1} = \mathcal{B}(X_k, Z_k, L_{k+1}, P_k, u_k)$. (12.15)

If valuation has no consequence for anything beyond the page, it remains description.

A world acts back.

Condition 9 — Residual disclosure

The model must preserve what it fails to explain.

Let:

$E_k = \text{Residual}(X_k, Z_k, P_k)$. (12.16)

A mature effective world does not redefine every failure as confirmation.

It allows residual to show that:

- the state space is incomplete;
- the protocol is unstable;
- the filter is wrong;
- the world no longer closes.

Condition 10 — Cross-observer reproducibility

Equivalent implementations should recover stable relations.

For observers a and b using admissibly equivalent protocols:

$P_a \simeq P_b$. (12.17)

A declared invariant should satisfy:

$\mathcal{I}(W_{P_a}) \approx \mathcal{I}(W_{P_b})$. (12.18)

This prevents “constructed world” from becoming arbitrary subjectivism.

12.4 Definition of the effective financial world

The effective financial world is represented as:

$W_P = (\mathcal{Z}_P, \mathcal{D}_P, \mathcal{G}_P, \boldsymbol{\mathcal{L}}_P, \mathcal{U}_P, \boldsymbol{\mathcal{B}}_P)$. (12.19)

Where:

- $\mathcal{Z}_P$ = effective state space;
- $\mathcal{D}_P$ = internal dynamics;
- $\mathcal{G}_P$ = admission and event gates;
- $\boldsymbol{\mathcal{L}}_P$ = ledger and trace rules;
- $\mathcal{U}_P$ = admissible interventions;
- $\boldsymbol{\mathcal{B}}_P$ = backreaction map.

This gives the formal definition:

An effective financial world is a protocol-bounded state space in which selected financial distinctions support approximately closed dynamics, admissible events, ledgered consequences, interventions, and backreaction upon future states. (12.20)

The phrase “approximately closed” is essential.

No financial world is perfectly isolated.

It is world-like only within declared limits.

12.5 Constructed does not mean arbitrary

Financial worlds are constructed through:

- definitions;
- contracts;
- measurement rules;
- pricing models;
- accounting standards;
- market conventions;
- regulatory systems;
- institutional authority.

But construction does not imply arbitrariness.

A trade price is constructed through a market mechanism, yet it can be publicly verified.

A credit rating is constructed through a rating protocol, yet the agency must apply declared

criteria.

An accounting value is constructed through recognition and measurement rules, yet it can be audited.

A collateral value is constructed through eligibility and haircut rules, yet those rules can be tested and enforced.

Therefore:

Constructed Reality ≠ Fiction. (12.21)

A constructed financial world becomes objective when:

- its declaration is explicit;
- its operations are reproducible;
- its records are auditable;
- its invariants survive admissible reframing;
- its residual is disclosed.

12.6 Effective does not mean unreal

The term “effective” means reduced relative to a larger field.

It does not mean imaginary or inconsequential.

A credit rating is an effective object.

It is not the total economic reality of the borrower.

Yet it can alter:

- interest rates;
- collateral requirements;
- investor eligibility;
- capital charges;
- refinancing options.

A market price is an effective object.

It is not the total ontology of the asset.

Yet it can alter:

- wealth;
- leverage;
- borrowing capacity;

- compensation;
- strategic decisions;
- insolvency.

A financial world is real because its records constrain action.

Its reality lies partly in consequence.

12.7 The world-formation test

A practical test is:

Does the representation merely summarize the primary field, or does it organize future admissibility? (12.22)

If it only summarizes, it remains a model.

If it determines:

- who may act;
- what may be recognized;
- what may be pledged;
- what must be paid;
- what becomes enforceable;
- what future states remain available;

then it is functioning as an effective world.

---

13. The Classical Internal Law

13.1 The clean phase episode

Consider an episode in which:

- A is approximately constant;
- P is stable;
- θ varies smoothly;
- no gate event changes the declaration;
- residual remains small.

Then:

$Z(\theta) = A \exp(i\theta)$. (13.1)

Differentiating with respect to θ:

$dZ/d\theta = iA \exp(i\theta)$. (13.2)

Therefore:

$dZ/d\theta = iZ$. (13.3)

This is the minimal internal law of the phase-ordered financial world.

13.2 Exchange between admitted value and retained pressure

Write:

$Z = R + iQ$. (13.4)

Then:

$dZ/d\theta = dR/d\theta + i(dQ/d\theta)$. (13.5)

Also:

$iZ = iR - Q$. (13.6)

Equating real and imaginary parts gives:

$dR/d\theta = -Q$. (13.7)

$dQ/d\theta = R$. (13.8)

These equations describe continuous exchange between visible value and retained pressure.

As θ increases:

- R tends to decrease;
- Q tends to increase.

As θ decreases:

- R tends to increase;
- Q tends to decrease.

The amplitude remains constant:

$R^2 + Q^2 = A^2$. (13.9)

13.3 Second-order form

Differentiate equation (13.7):

$d^2R/d\theta^2 = -dQ/d\theta$. (13.10)

Using equation (13.8):

$d^2R/d\theta^2 = -R.$ (13.11)

Similarly:

$d^2Q/d\theta^2 = dR/d\theta.$ (13.12)

Therefore:

$d^2Q/d\theta^2 = -Q.$ (13.13)

The internal coordinates obey harmonic-oscillator equations:

$d^2R/d\theta^2 + R = 0.$ (13.14)

$d^2Q/d\theta^2 + Q = 0.$ (13.15)

The solutions are:

$R(\theta) = A \cos \theta.$ (13.16)

$Q(\theta) = A \sin \theta.$ (13.17)

13.4 Classical quadrature

The R and Q components are separated by one-quarter cycle.

When R is maximal, Q is zero.

When Q is maximal, R is zero.

This is classical quadrature.

The same mathematics appears in:

- AC phasors;
- ideal LC circuits;
- classical oscillators;
- two-dimensional rotations;
- wave quadratures;
- signal processing.

Therefore:

Complex Finance Geometry ≠ Quantum Mechanics. (13.18)

The appearance of i does not establish quantum behaviour.

The existence of phase does not establish quantum behaviour.

The exchange between orthogonal coordinates does not establish quantum behaviour.

13.5 Why the classical result matters

The classical result could initially appear disappointing.

If the purpose were to prove that finance is quantum, the derivation would fail.

But the deeper framework does not require finance to become quantum.

The classical result reveals something more useful:

A secondary world may possess simple internal laws even when the process that constructs the world is irregular, contextual, and history-bearing. (13.19)

The primary economic field may contain:

- changing expectations;
- institutional conflict;
- discontinuous regulation;
- strategic behaviour;
- incomplete information;
- multiple observers;
- liquidity shocks.

Yet after declared projection into a suitable phase coordinate, the local internal law may become:

$dZ/d\theta = iZ$. (13.20)

Complex construction can generate simple internal dynamics.

This is the central reversal of the article.

13.6 What the internal law explains

The law:

$dZ/d\theta = iZ$ (13.21)

explains:

- phase rotation;
- orthogonal value-pressure exchange;
- conservation of A during a clean episode;
- marginal sensitivity $dR/d\theta = -Q$;
- periodic return in the idealized state plane;
- the existence of a simple internal phase coordinate.

It may help organize:

- stress trajectories;
- valuation-frame movement;
- pressure release;
- phase-indexed comparison;
- deviations from ideal closure.

13.7 What the internal law does not explain

The equation does not produce:

- probability amplitudes;
- the Born rule;
- superposition of physical states;
- tensor-product composition;
- entanglement;
- Bell inequality violation;
- quantum contextuality;
- no-cloning;
- quantum tunnelling;
- physical measurement collapse.

It is a classical effective law.

The article must preserve this negative result explicitly.

The original Finance Geometry framework likewise cautions that its complex plane is a representational extension of mature finance rather than a claim that markets literally instantiate quantum mechanics.

13.8 Variable amplitude

If A varies with θ:

$$Z(\theta) = A(\theta)\exp(i\theta). \quad (13.22)$$

Then:

$$dZ/d\theta = [(d \ln A/d\theta) + i]Z. \quad (13.23)$$

The radial generator in phase coordinates is:

$g_{A,\theta} = d \ln A/d\theta$. (13.24)

The internal law becomes:

$dZ/d\theta = (g_{A,\theta} + i)Z$. (13.25)

Separating coordinates:

$dR/d\theta = g_{A,\theta}R - Q$. (13.26)

$dQ/d\theta = g_{A,\theta}Q + R$. (13.27)

The clean rotational law is recovered when:

$g_{A,\theta} = 0$. (13.28)

13.9 Dynamic residual in phase coordinates

Define:

$\varepsilon_\theta = dZ/d\theta - [(d \ln A/d\theta) + i]Z$. (13.29)

Then:

$dZ/d\theta = [(d \ln A/d\theta) + i]Z + \varepsilon_\theta$. (13.30)

A small residual means the declared phase world approximately closes.

A large residual may indicate:

- omitted pressure dimensions;
- incorrect amplitude;
- protocol drift;
- gate discontinuity;
- phase ambiguity;
- model breakdown.

The residual therefore measures the distance between:

Observed Movement (13.31)

and:

Movement Predicted by the Declared Effective World. (13.32)

---

14. Internal Dynamics and World-Forming Dynamics

14.1 The decisive distinction

The article’s central structural distinction is:

Internal Effective Dynamics ≠ World-Forming Dynamics. (14.1)

Inside the effective financial world, the system may obey:

$dZ/d\theta = iZ$. (14.2)

At the world-forming boundary, the system must still perform:

- declaration;
- projection;
- gate selection;
- record creation;
- residual management;
- backreaction;
- protocol revision.

The internal law describes how the state moves after the world has been declared.

It does not explain how the world was selected, constituted, or maintained.

14.2 The internal layer

The internal layer contains:

- A;
- R;
- Q;
- θ;
- g_A;
- ω_F;
- ε_dyn;
- admissible transitions.

A compact internal state is:

$S_{int} = (A,R,Q,\theta)$. (14.3)

The internal dynamics are:

$dS_{int}/dt = \mathcal{D}_P(S_{int},u) + \varepsilon$. (14.4)

The observer inside this layer may treat the dynamics as the relevant law.

It need not have direct access to every primary-field variable that generated the coordinates.

14.3 The world-forming layer

The boundary layer contains:

- protocol P;
- primary state X;
- ledger L;
- projection Π;
- gate G;
- trace rule T;
- backreaction $\mathcal{B}$;
- revision rule $\mathcal{U}$.

The world-forming process is:

$Z_k = \Pi_{P_k,L_k}(X_k)$. (14.5)

$\mathrm{Commit}_k = G_{P_k}(Z_k,X_k,L_k)$. (14.6)

$L_{k+1} = T_{P_k}(L_k,\mathrm{Commit}_k,\mathrm{Residual}_k)$. (14.7)

$X_{k+1} = \mathcal{B}_{P_k}(X_k,Z_k,L_{k+1},u_k)$. (14.8)

$P_{k+1} = \mathcal{U}(P_k,L_{k+1},\mathrm{Residual}_k)$. (14.9)

This layer is generally more complex than the internal phase law.

14.4 Why the internal world can look simpler than the parent field

A projection may discard dimensions irrelevant to a declared task.

Suppose the primary field contains:

$X = (x_1,x_2,\ldots,x_n)$. (14.10)

The projection retains only:

$Z = \Pi_P(X) = R + iQ$. (14.11)

Many distinct primary states may map to similar effective states:

$\Pi_P(X_a) \approx \Pi_P(X_b)$. (14.12)

This compression can produce a low-dimensional law even when the primary field is high-dimensional.

Therefore:

High-Dimensional Parent Process → Low-Dimensional Effective Law. (14.13)

This is common in science and engineering.

Thermodynamics compresses molecular detail.

Circuit models compress electromagnetic fields.

Macroeconomic variables compress many transactions.

Financial valuation compresses many scenarios.

The new point is that the compressed state can become causally operative through gate, ledger, and backreaction.

14.5 The boundary is not merely outside the world

The boundary partly belongs to the world because it determines what the world can observe and do.

A collateral rule affects:

- what assets exist as usable collateral;
- which actions are admissible;
- which funding paths remain open.

An accounting rule affects:

- what assets exist in the official ledger;
- which profits can be reported;
- which capital constraints apply.

A market microstructure affects:

- which prices can print;
- which orders can execute;
- which liquidity is visible.

Therefore:

Boundary Conditions Participate in Effective Reality. (14.14)

The world is not only the states inside the boundary.

It is also the grammar that admits and transforms them.

14.6 A two-layer architecture

The framework can be summarized as:

Layer I — Internal Financial Physics. (14.15)

Layer II — World-Formation Governance. (14.16)

The first asks:

How does Z move under a stable declaration? (14.17)

The second asks:

Who declares P, which gate commits events, what is recorded, what remains residual, and when must P change? (14.18)

The first layer may be smooth.

The second may be discontinuous.

The first may be classical.

The second may be contextual and reflexive.

14.7 Why this matters for the quantum comparison

Many apparent mysteries arise not from the internal law but from the relation between:

- observer;
- measurement frame;
- admissible instruments;
- record;
- inaccessible primary state.

Finance shows that an entirely non-quantum system can possess:

- observer-dependent effective states;
- changing measurement frames;
- commitment events;
- irreversible records;
- adaptive future measurements;
- backreaction.

Therefore, these structures should not automatically be classified as uniquely quantum.

The genuinely quantum residue must be identified after the world-forming grammar is separated from the internal law.

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15. Financial Gates and the Birth of Events

15.1 Continuous possibility versus committed event

Before commitment, a financial system may contain many potential outcomes.

A trade may be quoted but not executed.

A loan may be stressed but not in default.

An impairment may be discussed but not recognized.

An option may be valuable but not exercised.

A covenant may be near its threshold but not breached.

The pre-gate state is:

$\hat{Z}_k$ = Potential Financial State. (15.1)

A gate determines whether the potential state becomes a committed event.

15.2 Gate function

Define:

$G_k(\hat{Z}_k, X_k, L_k, P_k) \in \{\text{Commit, Reject, Defer}\}$. (15.2)

A simpler binary form is:

$G_k(\hat{Z}_k) = 1$ if committed. (15.3)

$G_k(\hat{Z}_k) = 0$ if not committed. (15.4)

But real financial gates often require a third state:

Defer = insufficient evidence or incomplete authority. (15.5)

The output may therefore be:

$G_k(\hat{Z}_k) \rightarrow (\text{Decision}_k, \text{Residual}_k)$. (15.6)

15.3 Examples of gates

Market-execution gate

An order becomes a trade only when matching and execution conditions are satisfied.

Credit gate

A deterioration becomes default only when contractual or legal conditions are met.

Rating gate

A modelled decline becomes an official downgrade only when governance and evidence standards are satisfied.

Accounting gate

An economic possibility becomes a recognized asset, liability, gain, loss, or impairment only when the applicable criteria are met.

Capital gate

An economic value becomes admissible capital or collateral only when eligibility rules are satisfied.

Option gate

Latent value becomes committed investment only when exercise authority and conditions are satisfied.

Regulatory gate

A concern becomes an enforceable restriction only after the relevant authority acts.

15.4 Event as world-changing commitment

A committed event differs from an estimate because it changes future admissibility.

For example, once default is declared:

- acceleration clauses may activate;
- collateral rights may change;
- cross-default provisions may trigger;
- funding access may close;
- legal processes may begin.

The event is therefore not merely informational.

It is performative.

A useful definition is:

Financial Event = Gate-Passed Projection That Alters the Future Action Space. (15.7)

15.5 Gate metadata

A mature gate should record:

- who had authority;
- which evidence was used;
- which threshold was applied;
- when the decision occurred;

- which protocol version governed;
- which residual remained unresolved.

Define:

$M_G,k = (\text{Authority}_k,\text{Evidence}_k,\text{Threshold}_k,\text{Time}_k,\text{ProtocolVersion}_k)$. (15.8)

The ledger entry becomes:

$\text{Record}_k = (\text{Decision}_k,M_G,k,\text{Residual}_k)$. (15.9)

This makes the event auditable.

15.6 Premature gate closure

A gate may close too early.

Examples include:

- recognizing uncertain value without adequate evidence;
- downgrading on transient noise;
- triggering forced liquidation from a distorted price;
- rejecting collateral through an outdated classification;
- exercising an option before strategic conditions are ready.

Premature closure can convert manageable pressure into irreversible loss.

Represent premature closure risk as:

$\text{Risk_premature} = P(G = \text{Commit} \mid \text{State not yet admissible})$. (15.10)

15.7 Delayed gate closure

A gate may also close too late.

Examples include:

- delayed impairment;
- delayed default recognition;
- failure to enforce a covenant;
- refusal to update a risk classification;
- delayed liquidity intervention.

Delayed closure can hide residual until it becomes nonlinear.

Represent delayed closure risk as:

Risk_delay = P(G ≠ Commit | State already materially admissible). (15.11)

A well-designed gate balances:

- false commitment;
- false non-commitment;
- cost of delay;
- cost of irreversibility.

15.8 Gate and Q

A high Q may indicate that a gate is approaching, but Q does not determine the gate by itself.

The gate may depend on:

- R;
- Q;
- θ;
- ω_F;
- external evidence;
- legal terms;
- authority;
- accumulated ledger state.

A general gate condition is:

$G_k = 1$ if $H(R,Q,\theta,\omega_F,L,Evidence,P) \geq \kappa_G$. (15.12)

Where κ_G is the declared threshold.

This prevents the geometry from replacing institutional rules.

It informs the gate; it does not automatically govern it.

---

16. Ledger Time and Financial History

16.1 From event to trace

Once an event passes the gate, it enters the ledger.

The minimal update is:

$L_{k+1} = L_k \sqcup Record_k$. (16.1)

The operator ⊔ indicates that the new record is joined to the existing trace.

The ledger is not merely a database.

It is the structure through which past commitments constrain future financial states.

16.2 Strong ledger structure

A mature ledger entry should preserve:

$\text{Record}_k = (\text{Outcome}_k, \text{Evidence}_k, \text{GateMetadata}_k, \text{Authority}_k, \text{Residual}_k)$. (16.2)

Then:

$L_{k+1} = L_k \sqcup \text{Record}_k$. (16.3)

This is stronger than recording only the final number.

A weak ledger records:

$\text{Price} = 80$. (16.4)

A stronger ledger records:

- price;
- timestamp;
- venue;
- volume;
- execution conditions;
- model version;
- evidence;
- remaining uncertainty.

The stronger record supports later audit and revision.

16.3 Ledger time

Let k denote the ordered sequence of committed records.

Ledger time is:

$\tau_L = \text{order}(L_0, L_1, L_2, \ldots)$. (16.5)

A ledger tick occurs when:

$L_{k+1} \neq L_k$. (16.6)

Calendar time can pass without a ledger tick.

Many ledger ticks can occur within a short calendar interval.

Therefore:

Ledger Time ≠ Calendar Time. (16.7)

16.4 Why price history is not merely a time series

A price time series lists numerical observations.

A ledgered price history also records the events through which those observations became consequential.

For example:

- whether the price was executable;
- whether volume was meaningful;
- whether trading was halted;
- whether the price triggered margin;
- whether it changed collateral;
- whether it caused forced sales.

The same numerical path can produce different histories under different gate and ledger rules.

Therefore:

Numerical Path + Event Grammar + Ledger Consequence = Financial History. (16.8)

16.5 Trace changes future projection

The next projection depends on the existing ledger:

$Z_{k+1} = \Pi_{P_{k+1}, L_{k+1}}(X_{k+1})$. (16.9)

A borrower with a past default is not projected through the same credit frame as an otherwise similar borrower without that record.

A firm with an impairment history may face different investor expectations.

A market with a recent liquidity freeze may be valued differently even after spreads normalize.

The trace becomes part of the observer.

16.6 Irreversibility

A financial event is irreversible in the ledger sense when it changes future admissibility even if some physical or numerical state is later restored.

Suppose:

$R_T = R_0$. (16.10)

But:

$L_T \neq L_0$. (16.11)

Then the visible value returned, but the world did not.

The difference may include:

- ownership changes;
- defaults;
- realized losses;
- legal action;
- regulatory intervention;
- reputational damage;
- lost trust.

This is semantic and institutional irreversibility.

16.7 Reversal versus erasure

A later event may reverse the practical effect of an earlier event.

For example:

- a downgrade may be upgraded;
- an impairment may be partly reversed;
- a margin call may be satisfied;
- a restriction may be lifted.

But reversal does not erase the earlier trace.

The ledger becomes:

$L_{new} = L_{old} \sqcup ReversalRecord$. (16.12)

It does not become:

$L_{new} = L_{before_event}$. (16.13)

History is extended, not deleted.

16.8 The ledger as a source of identity

Institutions preserve identity partly through ledger continuity.

A bank changes employees, systems, and portfolios, yet remains the same legal entity because:

- obligations persist;
- records persist;
- licenses persist;
- liabilities persist;
- authority persists.

An effective financial observer is therefore not defined only by its current model.

It is defined by:

Observer Identity = Projection Rules + Ledger Continuity + Admissible Revision. (16.14)

This anticipates the later discussion of declaration revision and observerhood.

---

17. Backreaction: When Valuation Changes What It Values

17.1 Passive valuation is often an illusion

A textbook valuation may appear to observe an asset without changing it.

In many real financial systems, this is false.

The valuation result enters:

- prices;
- collateral systems;
- margin systems;
- lending decisions;
- capital allocation;
- accounting;
- regulation;
- compensation;
- market narratives.

Once admitted, the result changes the system that will be valued next.

The projection is therefore reflexive.

17.2 General backreaction map

Let:

$Z_k = \Pi_{P_k,L_k}(X_k)$. (17.1)

After gate and ledger update:

$L_{k+1} = T(L_k,Z_k,G_k,\text{Residual}_k)$. (17.2)

The primary field evolves as:

$X_{k+1} = \mathcal{B}(X_k,Z_k,L_{k+1},u_k)$. (17.3)

The next effective state is:

$Z_{k+1} = \Pi_{P_{k+1},L_{k+1}}(X_{k+1})$. (17.4)

The full loop is:

$X_k \to Z_k \to L_{k+1} \to X_{k+1} \to Z_{k+1}$. (17.5)

17.3 Price–collateral–leverage loop

A common loop is:

Price ↓ → Collateral Value ↓ → Borrowing Capacity ↓ → Forced Sale ↑ → Price ↓. (17.6)

This can be represented schematically as:

$C_k = h_c(R_k)$. (17.7)

$\text{Leverage Capacity}_k = h_l(C_k)$. (17.8)

$\text{Forced Sales}_k = h_s(\text{Leverage Capacity}_k,L_k)$. (17.9)

$R_{k+1} = h_p(\text{Forced Sales}_k,X_k)$. (17.10)

The measured price affects collateral.

Collateral affects leverage.

Leverage affects action.

Action affects the next price.

The observable becomes causal.

17.4 Rating–funding–default loop

Another loop is:

Downgrade → Funding Cost ↑ → Liquidity ↓ → Default Risk ↑ → Further Downgrade. (17.11)

Let rating state be:

$\rho_{\text{rating},k}$. (17.12)

Funding cost may satisfy:

$\text{FundingCost}_k = f_c(\rho_{\text{rating}}, k, \text{MarketState}_k)$. (17.13)

Liquidity then evolves:

$\text{Liquidity}_{k+1} = f_l(\text{Liquidity}_k, \text{FundingCost}_k, \text{CashFlow}_k)$. (17.14)

Default pressure changes:

$Q_{\text{credit},k+1} = f_q(Q_{\text{credit}}, k, \text{Liquidity}_{k+1})$. (17.15)

The next rating gate uses the new state.

The rating is not merely descriptive.

It participates in the dynamics.

17.5 Stress-test–capital–asset-sale loop

A stress test may produce:

$\text{StressLoss}_k = F_{\text{stress}}(X_k, P_k)$. (17.16)

Required capital becomes:

$\text{CapitalReq}_k = f_{\text{cap}}(\text{StressLoss}_k)$. (17.17)

If capital is insufficient:

$\text{AssetSales}_k = f_{\text{sale}}(\text{CapitalGap}_k)$. (17.18)

Sales affect market prices:

$R_{k+1} = f_{\text{market}}(\text{AssetSales}_k)$. (17.19)

The next stress test uses the new prices.

Thus:

Stress Result → Capital Restriction → Asset Sale → Market Price → Next Stress Result. (17.20)

The measurement changes the measured system.

17.6 Accounting backreaction

An impairment can reduce:

- reported equity;
- covenant headroom;
- distributable reserves;
- regulatory capital;
- market confidence;

- borrowing capacity.

The accounting entry is therefore not a passive mirror.

It becomes part of the economic mechanism.

Let:

I_k = ImpairmentRecord$_k$. (17.21)

Then:

Equity$_{k+1}$ = Equity$_k$ − I_k. (17.22)

FundingCapacity$_{k+1}$ = f(Equity$_{k+1}$,Covenants$_k$). (17.23)

Future cash flows may then change because financing has changed.

17.7 Model-induced behaviour

Models can also create backreaction before an official ledger event.

If many institutions use similar risk models:

- the same volatility rise may trigger similar deleveraging;
- the same rating threshold may trigger similar sales;
- the same VaR increase may reduce similar positions.

A model can therefore become an interaction channel.

Let:

u_i,k = Policy_i(ModelOutput_i,k). (17.24)

Aggregate action is:

U_k = Σ_i u_i,k. (17.25)

The market state changes:

X_{k+1} = $\mathcal{B}$(X_k,U_k). (17.26)

If many observers share the same frame, the frame may generate the reality it predicts.

17.8 Positive and negative backreaction

Backreaction can amplify pressure.

Positive feedback:

Pressure ↑ → Restriction ↑ → Forced Action ↑ → Pressure ↑. (17.27)

Backreaction can also stabilize the system.

Negative feedback:

Pressure ↑ → Capital Injection ↑ → Liquidity ↑ → Pressure ↓. (17.28)

The sign of backreaction may be represented by a local gain:

$G_B = \partial X_{k+1}/\partial Z_k$. (17.29)

A large positive reinforcing gain may indicate instability.

A negative restoring gain may indicate stabilization.

17.9 Backreaction and angular dynamics

The phase angle itself may be affected by previous projection.

Suppose:

$\theta_{k+1} = \Theta(X_{k+1}, L_{k+1}, P_{k+1})$. (17.30)

Since X_{k+1} depends on Z_k:

$\theta_{k+1} = \Theta[\boldsymbol{\mathcal{B}}(X_k, Z_k, L_{k+1}, u_k), L_{k+1}, P_{k+1}]$. (17.31)

The financial frame is therefore endogenous.

A price decline can increase perceived risk.

The increased risk rotates the filter further.

This can produce angular acceleration:

$d^2\theta/dt^2 > 0$. (17.32)

during self-reinforcing stress.

17.10 The world-maintenance loop

The complete effective-world loop is:

Primary Field → Projection → Effective State → Gate → Ledger → Action → Backreaction → New Primary Field. (17.33)

In symbols:

$X_k \to \Pi_\{P_k, L_k\} \to Z_k \to G_k \to L_{k+1} \to u_k \to \boldsymbol{\mathcal{B}} \to X_{k+1}$. (17.34)

If residual becomes too large:

$P_k \to \boldsymbol{\mathcal{U}} \to P_{k+1}$. (17.35)

The financial world is therefore not constructed once.

It is repeatedly reconstructed through its own traces.

That recursive reconstruction is what allows a valuation system to become a durable operational

reality rather than a temporary calculation.

Part V — Residual, Memory, and World Failure

18. Q Is Not the Residual

18.1 Three quantities that must not be confused

The language of hidden pressure can easily become imprecise.

Four different objects may be present:

1. retained pressure Q;
2. scalar haircut A − R;
3. model residual ε;
4. primary-field structure omitted by projection.

These objects are related only under declared assumptions.

They are not interchangeable.

The minimal Finance Geometry identity is:

$A^2 = R^2 + Q^2$. (18.1)

Therefore:

$Q = \sqrt{(A^2 - R^2)}$. (18.2)

The scalar haircut is:

$H = A - R$. (18.3)

The dynamic residual is:

$\varepsilon_{dyn} = dZ/dt - (g_A + i\omega_F)Z$. (18.4)

The projection residual is:

$\varepsilon_{\Pi} = X - \tilde{X}(Z \mid P)$. (18.5)

Each quantity answers a different question.

Quantity	Question
Q	What pressure complement is implied inside the declared geometry?
A − R	How much smaller is the scalar admitted value than the amplitude?
ε_{dyn}	How much observed movement violates the declared dynamic law?
ε_{Π}	What primary-world structure is omitted from the effective state?

A mature application must keep these categories separate.

18.2 Q belongs inside the declared world

Q is not evidence that the declared financial world has failed.

It is one of that world's coordinates.

The state is:

$Z = R + iQ.$ (18.6)

Both R and Q lie inside the declared Finance Geometry.

R is admitted value.

Q is retained pressure under the same filter.

A large Q can therefore exist in a model that closes well.

For example:

$|\varepsilon_{dyn}| \approx 0$ (18.7)

may hold even when:

$Q/R \gg 1.$ (18.8)

This would describe a strongly pressure-loaded state whose internal dynamics are nevertheless well represented by the declared geometry.

Therefore:

High Pressure ≠ Model Failure. (18.9)

18.3 Residual lies against the world's adequacy

Residual asks a different question:

Does the declared state space adequately explain the observed system? (18.10)

A residual becomes large when:

- the dynamics fail;
- omitted variables become material;
- a gate creates a discontinuity;
- the baseline is no longer appropriate;
- several pressure channels cannot be represented by one Q;
- the protocol has changed;
- the reconstruction from Z to the relevant part of X becomes poor.

Therefore:

Residual is pressure against the adequacy of the world model. (18.11)

This distinction can be summarized as:

Q = pressure within the declared world. (18.12)

ε = pressure upon the declaration of that world. (18.13)

The difference is conceptually fundamental.

18.4 Why the distinction matters operationally

Suppose Q rises while ε_dyn remains low.

The correct interpretation may be:

The world remains valid, but pressure is accumulating within it. (18.14)

Suppose Q remains moderate while ε_dyn rises sharply.

The correct interpretation may be:

The declared pressure coordinate is no longer capturing the system. (18.15)

In the first case, one may respond within the existing model.

In the second case, one may need to revise the model itself.

This is the difference between:

Intervention inside P (18.16)

and:

Revision of P. (18.17)

18.5 Uncertainty is not automatically residual

Uncertainty may already be represented inside the declared filter.

A certainty-equivalent adjustment may incorporate uncertainty.

A credit model may incorporate default probabilities.

A liquidity haircut may incorporate execution uncertainty.

In those cases, part of uncertainty contributes to θ and Q.

But uncertainty can also remain outside the declared model.

For example:

- unknown fraud;
- hidden collateral chains;

- unobserved legal restrictions;
- omitted strategic behaviour;
- unknown correlation breakdown.

That uncertainty belongs more naturally to projection residual.

Therefore:

Modelled Uncertainty → may contribute to Q. (18.18)

Unmodelled Uncertainty → may contribute to ε_Π. (18.19)

The phrase “uncertainty pressure” should not be used without specifying which category is meant.

18.6 Risk is not automatically Q

Risk is a broad financial category.

Depending on the protocol, Q may encode:

- market-risk pressure;
- credit pressure;
- liquidity pressure;
- capital pressure;
- tail pressure;
- execution pressure.

But Q is not “total risk.”

A CAPM-derived Q does not automatically include:

- credit default;
- liquidity;
- operational risk;
- legal risk;
- accounting risk;
- model risk.

The correct notation is filter-labelled:

Q_CAPM. (18.20)

Q_credit. (18.21)

Q_liquidity. (18.22)

Q_capital. (18.23)

A bare Q should be used only when the originating filter is already clear.

18.7 Residual honesty

A theory becomes unfalsifiable if every discrepancy is reclassified as a hidden form of Q.

Suppose observed movement violates:

dZ/dt = (g_A + iω_F)Z. (18.24)

The theory must permit the conclusion:

The declared two-coordinate world is inadequate. (18.25)

It must not automatically say:

A deeper invisible Q explains the discrepancy. (18.26)

The latter move merely postpones failure by inventing new dimensions after every observation.

The correct discipline is:

Residual must retain the power to reject the world model. (18.27)

This principle is consistent with the original Finance Geometry requirement that the complex extension earn adoption through measurable diagnostic gain rather than through elegance or metaphor.

---

19. Dynamic Closure Residual

19.1 Calendar-time residual

The declared dynamic law is:

dZ/dt = (g_A + iω_F)Z. (19.1)

The observed derivative is:

(dZ/dt)_obs. (19.2)

Define the dynamic residual:

ε_dyn = (dZ/dt)_obs − (g_A + iω_F)Z. (19.3)

Write:

ε_dyn = ε_R + iε_Q. (19.4)

Then:

ε_R = dR/dt − (g_A R − ω_F Q). (19.5)

ε_Q = dQ/dt − (g_A Q + ω_F R). (19.6)

The residual magnitude is:

$|\varepsilon_{dyn}| = \sqrt{(\varepsilon_R^2 + \varepsilon_Q^2)}$. (19.7)

A normalized version is:

$r_{dyn} = |\varepsilon_{dyn}|/(|dZ/dt| + \delta)$. (19.8)

Here $\delta > 0$ prevents division by a near-zero observed derivative.

19.2 Phase-time residual

When θ is used as the local internal coordinate:

$dZ/d\theta = [(d \ln A/d\theta) + i]Z$. (19.9)

Define:

$\varepsilon_\theta = (dZ/d\theta)_{obs} - [(d \ln A/d\theta) + i]Z$. (19.10)

For constant amplitude:

$d \ln A/d\theta = 0$. (19.11)

Then:

$\varepsilon_\theta = dZ/d\theta - iZ$. (19.12)

Separating coordinates:

$\varepsilon_{R,\theta} = dR/d\theta + Q$. (19.13)

$\varepsilon_{Q,\theta} = dQ/d\theta - R$. (19.14)

The ideal phase world closes when:

$\varepsilon_\theta = 0$. (19.15)

In data:

$|\varepsilon_\theta| \leq \kappa_\varepsilon$ (19.16)

may define an acceptable closure region, where κ_ε is a declared tolerance.

19.3 Interpretation of small residual

A small residual suggests that:

- A and θ have been measured consistently;
- the two-coordinate state is adequate for the episode;
- the declared filter remains stable;
- phase is functioning as a useful internal coordinate;

- omitted couplings are not currently dominant.

This does not prove that the model is ontologically complete.

It means only:

The declared effective world closes sufficiently for the stated task and horizon. (19.17)

19.4 Interpretation of large residual

A large residual can arise from several causes.

Cause 1 — Amplitude error

The assumed A may not reflect the changing economic field.

Examples include:

- stale cash-flow forecast;
- incorrect base rate;
- wrong recovery expectation;
- omitted economic shock.

Cause 2 — Angle error

The filter ratio may be mismeasured.

Examples include:

- noisy market price;
- unstable beta;
- illiquid spread;
- inconsistent horizon.

Cause 3 — Multiple pressure dimensions

One Q may be attempting to represent market, credit, and liquidity pressure simultaneously.

Cause 4 — Gate discontinuity

A default, downgrade, impairment, or regulatory intervention may cause a jump that the smooth law cannot represent.

Cause 5 — Protocol drift

The market may have changed how it values the asset even though the analyst continues using the old model.

Cause 6 — Backreaction

The projection may have changed the primary state before the next observation.

Cause 7 — Regime transition

The system may no longer belong to the same effective world.

19.5 Residual decomposition

A useful conceptual decomposition is:

$$\varepsilon_{dyn} = \varepsilon_{measure} + \varepsilon_{model} + \varepsilon_{gate} + \varepsilon_{protocol} + \varepsilon_{backreaction}. \quad (19.18)$$

Where:

- $\varepsilon_{measure}$ = sampling and estimation error;
- ε_{model} = omitted internal dynamics;
- ε_{gate} = discontinuous event component;
- $\varepsilon_{protocol}$ = declaration mismatch;
- $\varepsilon_{backreaction}$ = endogenous feedback not represented in the local law.

This decomposition will rarely be uniquely identifiable from one data stream.

Its purpose is diagnostic.

A large residual should trigger questions before it triggers a new metaphor.

19.6 Residual threshold as a world-validity gate

Define a world-validity test:

$$\mathrm{ValidWorld}_P(t) \Leftrightarrow r_{dyn}(t) \le \kappa_{dyn} \wedge r_{\Pi}(t) \le \kappa_{\Pi}. \quad (19.19)$$

Where:

- r_{dyn} measures dynamic closure error;
- r_{Π} measures projection or reconstruction error;
- κ_{dyn} and κ_{Π} are declared thresholds.

If the test passes:

Continue operating within P. (19.20)

If it fails temporarily:

Enter caution or diagnostic mode. (19.21)

If it fails persistently:

Consider declaration revision. (19.22)

This turns residual into a governance signal.

19.7 Residual trend

The level of residual matters, but so does its trend.

Define:

$v_\varepsilon = d|\varepsilon_{dyn}|/dt.$ (19.23)

If:

$|\varepsilon_{dyn}|$ is moderate (19.24)

but:

$v_\varepsilon \gg 0,$ (19.25)

the world may be approaching failure.

A rising residual may therefore precede:

- regime change;
- gate event;
- model breakdown;
- protocol revision.

This is another empirical hypothesis for testing.

19.8 Residual persistence

A one-off residual spike may reflect noise.

Persistent residual is more concerning.

Define an exponentially weighted residual:

$\bar{E}_{dyn}(t) = (1 - \lambda)\Sigma_{j=0}^{\infty} \lambda^j |\varepsilon_{dyn}(t - j)|.$ (19.26)

Where:

$0 < \lambda < 1.$ (19.27)

Persistent closure failure is indicated when:

$\bar{E}_{dyn} > \kappa_{persist}.$ (19.28)

This may justify moving from:

parameter recalibration (19.29)

to:

world-model revision. (19.30)

19.9 Residual as epistemic protection

The residual channel protects the framework from overconfidence.

Without residual, the model says:

Everything belongs to R or Q. (19.31)

With residual, the model says:

Some structure is not captured by the declared financial world. (19.32)

This is intellectually stronger.

A mature theory must know not only what it represents, but where its representation stops.

---

20. Projection Residual

20.1 Projection is necessarily selective

The primary field X may contain far more information than the effective state Z.

Let:

$X \in \mathcal{X}$. (20.1)

The projection is:

$Z = \Pi_P(X)$. (20.2)

Because Π_P is a compression:

$\dim(\mathcal{Z}_P) \ll \dim(\mathcal{X})$ (20.3)

in most practical applications.

Many primary states may map to the same effective state:

$\Pi_P(X_a) = \Pi_P(X_b)$ (20.4)

even when:

$X_a \neq X_b$. (20.5)

The effective world therefore cannot reconstruct the entire primary field.

20.2 Reconstruction map

Define a protocol-relative reconstruction:

$\tilde{X} = \mathscr{R}_P(Z,L)$. (20.6)

This is not necessarily an inverse.

It is the best reconstruction available under the declared model and ledger.

The projection residual is:

$\varepsilon_{\Pi} = X - \tilde{X}$. (20.7)

Because X may be high-dimensional, ε_{Π} may be represented through selected omitted features:

$\varepsilon_{\Pi} = [e_1, e_2, \ldots, e_m]^T$. (20.8)

A weighted magnitude is:

$\|\varepsilon_{\Pi}\|_W^2 = \varepsilon_{\Pi}^T W \varepsilon_{\Pi}$. (20.9)

Where W declares which omitted structures are operationally important.

20.3 Harmless omission versus dangerous omission

Not every omitted variable matters.

A reduced world is useful precisely because it ignores irrelevant detail.

Projection omission becomes dangerous when omitted structure materially affects:

- future transitions;
- gate outcomes;
- intervention performance;
- backreaction;
- world-validity conclusions.

Define operational relevance:

$Rel_P(e_j)$ = Expected Change in Decision Quality if e_j is Included. (20.10)

An omitted feature is material when:

$Rel_P(e_j) \geq \kappa_{rel}$. (20.11)

Thus:

Omitted Information ≠ Material Residual Automatically. (20.12)

Materiality depends on the declared task.

20.4 Examples of projection residual

Off-balance-sheet exposure

The effective valuation may omit guarantees, derivatives, or contingent obligations.

Hidden collateral networks

Two institutions may appear independent while relying on the same collateral chain.

Legal restrictions

An asset may appear economically transferable but be legally constrained.

Strategic behaviour

A borrower may alter behaviour after observing the lender's model.

Concentration risk

Individual positions may appear acceptable while the aggregate portfolio is fragile.

Settlement structure

A price may appear valid while settlement failure risk remains outside the model.

Political or regulatory intervention

The effective world may omit the possibility of sudden rule change.

20.5 Dark structure in the primary field

Projection residual can be interpreted as "dark" structure only in a restricted sense.

It is dark relative to the declared financial observer.

It is not metaphysically invisible.

A legal adviser, regulator, trader, auditor, or counterparty may observe parts of ε_Π that the valuation model does not.

Therefore:

Hidden to One Protocol ≠ Hidden to All Observers. (20.13)

This is why cross-observer comparison can reveal projection weakness.

20.6 Residual triangulation

Let several observers use different protocols:

$Z_a = \Pi_{P_a}(X)$. (20.14)

$Z_b = \Pi_{P_b}(X)$. (20.15)

$Z_c = \Pi_{P_c}(X)$. (20.16)

Their disagreements may expose omitted structure.

For example:

- valuation model;
- credit model;
- liquidity desk;
- legal review;
- accounting analysis.

A triangulation residual can be defined schematically as:

ε_tri = Dispersion{M_a→*(Z_a), M_b→*(Z_b), M_c→*(Z_c)}. (20.17)

Where each M maps the local state into a common comparison frame.

High ε_tri may indicate:

- protocol misalignment;
- genuinely omitted structure;
- inconsistent baselines;
- emerging regime change.

20.7 Residual budget

A bounded observer cannot eliminate all residual.

The practical task is to govern it.

Define a residual budget:

B_ε = Maximum Residual Exposure Acceptable Under P. (20.18)

The world remains admissible when:

‖ε_Π‖_W ≤ B_ε. (20.19)

The budget may depend on:

- decision stakes;
- reversibility;
- horizon;
- uncertainty;
- available capital;
- intervention capacity.

A low-risk exploratory model may tolerate more residual.

A capital or solvency decision should tolerate less.

20.8 Residual attachment

Projection residual should be attached to the state or ledger record that depends upon it.

A strong state report is:

$\text{StateReport}_k = (Z_k, P_k, \text{Confidence}_k, \varepsilon_\Pi,k)$. (20.20)

A weak report gives only:

R_k. (20.21)

The strong report preserves:

- what was admitted;
- how it was generated;
- how uncertain the reconstruction remains;
- what omitted structures may matter.

This is one way Finance Geometry can improve governance even before it improves prediction.

---

21. Ledger Residual

21.1 Commitment does not eliminate uncertainty

A financial event may be committed even while some questions remain unresolved.

A trade can execute while settlement risk remains.

A loan can be classified as defaulted while recovery remains uncertain.

An impairment can be recognized while the final loss remains unknown.

A regulator can intervene while legal challenges continue.

Therefore, a committed record should not imply:

$\text{Residual} = 0$. (21.1)

Instead:

$\text{Commitment} + \text{Residual Attachment} = \text{Honest Closure}$. (21.2)

21.2 Ledger residual definition

Let Record_k be the committed outcome.

Define the attached ledger residual:

$E_{L,k}$ = Unresolved Consequence Remaining After $Record_k$. (21.3)

The ledger update is:

$L_{k+1} = L_k \sqcup (Record_k, E_{L,k})$. (21.4)

The residual may include:

- disputed amount;
- uncertain recovery;
- missing evidence;
- unresolved authority;
- legal appeal;
- settlement exposure;
- model disagreement;
- unquantified tail risk.

21.3 Why the residual must remain attached

If residual is separated from the record, later users may treat the outcome as more certain than it was.

For example:

Recognized Value = 80. (21.5)

may later be read as a complete fact.

But the original decision may have been:

Recognized Value = 80 ± 15 under disputed assumptions. (21.6)

If the uncertainty is not attached, the ledger becomes progressively overconfident.

The historical record loses the conditions under which the value was admitted.

21.4 Residual decay and resolution

Ledger residual may decline when:

- evidence arrives;
- settlement completes;
- litigation ends;
- recovery is realized;

- model disagreement narrows;
- authority is confirmed.

Represent residual evolution as:

$E_{L,k+1} = \mathcal{R}_E(E_{L,k}, NewEvidence_k, Action_k)$. (21.7)

Residual resolution should produce a new record:

$L_{k+2} = L_{k+1} \sqcup ResolutionRecord_{k+1}$. (21.8)

The original uncertainty should not be retrospectively erased.

21.5 Residual accumulation

Unresolved residuals may accumulate.

Define total ledger residual:

$E_{L,total}(k) = \Sigma_{j \le k} w_j E_{L,j}$. (21.9)

Where w_j reflects:

- materiality;
- persistence;
- correlation;
- legal priority;
- ageing.

A high residual stock can indicate:

- hidden operational debt;
- model debt;
- legal debt;
- accounting ambiguity;
- unresolved counterparty exposure.

Residual accumulation may make the world increasingly fragile even when current R remains stable.

21.6 Residual ageing

Define age of unresolved residual j:

$a_j(k) = k - j$. (21.10)

A residual-age burden is:

$$B_{age}(k) = \Sigma_{j \le k} a_j(k) w_j |E_{L,j}|. \quad (21.11)$$

Old unresolved items may be more dangerous because:

- evidence decays;
- responsibility becomes unclear;
- assumptions become stale;
- future records build upon them.

A mature ledger should therefore report not only residual magnitude, but residual age.

21.7 Residual and trust

A system that preserves residual may initially appear less certain.

But it may become more trustworthy.

A system that suppresses residual appears clean until failure.

Trust should therefore be associated with:

- declared uncertainty;
- preserved evidence;
- visible revision history;
- non-erasure of prior errors.

The principle is:

$$\text{Residual Honesty} > \text{Cosmetic Certainty}. \quad (21.12)$$

---

22. Loop Residual and Financial Memory

22.1 The illusion of return

Suppose a market variable returns to its initial value:

$$R_T = R_0. \quad (22.1)$$

Or the finance angle returns:

$$\theta_T = \theta_0. \quad (22.2)$$

The state appears to have completed a cycle.

But the wider world may not have returned.

The ledger may satisfy:

$$L_T \neq L_0. \quad (22.3)$$

The primary field may satisfy:

$$X_T \neq X_0. \quad (22.4)$$

The apparent phase cycle therefore contains historical residue.

22.2 Loop residual

Let $\mathcal{S}$ extract a declared world-state summary from the ledger and primary field.

Define:

$$H_{loop} = \mathcal{S}(X_T, L_T, P_T) - \mathcal{S}(X_0, L_0, P_0). \quad (22.5)$$

If:

$$H_{loop} = 0, \quad (22.6)$$

the loop closes under the declared summary.

If:

$$H_{loop} \neq 0, \quad (22.7)$$

the financial world has retained memory.

The phase may close while history remains open.

22.3 Sources of financial memory

Leverage memory

A cycle may leave more debt even after prices recover.

Ownership memory

Forced sales may transfer assets permanently.

Collateral memory

Assets may remain ineligible after the shock passes.

Regulatory memory

Emergency restrictions may become permanent rules.

Legal memory

Defaults, disputes, or bankruptcies remain recorded.

Trust memory

Market participants may demand larger future premia.

Liquidity memory

Market depth may recover slowly after a freeze.

Organizational memory

Risk limits and management behaviour may change.

22.4 Price recovery without world recovery

Suppose:

$R_T = R_0$. (22.8)

But during the cycle:

- an institution failed;
- collateral was seized;
- ownership changed;
- regulation tightened;
- leverage was reduced;
- investors withdrew.

Then:

Same Price ≠ Same World. (22.9)

The current scalar value does not reveal the historical transformation.

This is one reason a time series alone cannot capture financial history.

22.5 Phase closure versus ledger closure

Phase closure is:

$\theta_T = \theta_0 \bmod 2\pi$. (22.10)

Ledger closure would require:

$L_T \simeq L_0$ (22.11)

under a declared equivalence relation.

In most consequential financial cycles:

Phase Closure does not imply Ledger Closure. (22.12)

The difference is financial hysteresis.

22.6 Hysteresis

Hysteresis means that the current state depends on the path taken to reach it.

Represent the state as:

$Z_t = Z(X_t,L_t,P_t)$. (22.13)

Even if:

$X_a = X_b$, (22.14)

different ledgers may produce:

$Z(X_a,L_a,P) \neq Z(X_b,L_b,P)$. (22.15)

The same current fundamentals can be valued differently because the histories differ.

Examples include:

- recent default history;
- prior liquidity freeze;
- previous fraud;
- regulatory intervention;
- broken covenant;
- failed refinancing.

The ledger curves the next projection.

22.7 A simple loop-memory index

Define a normalized loop-memory index:

$M_{loop} = \|H_{loop}\|/[PathLength + \delta]$. (22.16)

Where PathLength measures total movement during the episode.

A high M_{loop} means that substantial historical change remains after an apparently closed path.

A low M_{loop} means that the world approximately returned under the declared summary.

This is only a general template.

Different applications require different definitions of δ and norm.

22.8 Geometric interpretation

In simple Euclidean rotation, a closed loop returns to its starting state.

In a history-bearing financial world, transport around a loop may alter the object.

Schematically:

$$\text{Transport}_{\Gamma}(Z_0, L_0) \to (Z_T, L_T). \quad (22.17)$$

Even if:

$$Z_T = Z_0, \quad (22.18)$$

it may remain true that:

$$L_T \neq L_0. \quad (22.19)$$

The difference resembles holonomy in the broad geometric sense: a closed path can produce a non-trivial transformation.

This is an analogy, not a claim that financial systems instantiate a particular physical gauge theory.

22.9 Memory as future pressure

Loop residual may influence future θ.

For example:

$$\theta_{future} = \Theta(X_{future}, L_T, P_{future}). \quad (22.20)$$

A prior crisis may increase future risk premium even after present conditions normalize.

Therefore:

$$H_{loop} \to \text{Future Filter Rotation}. \quad (22.21)$$

History becomes pressure.

This is how ledger time enters future geometry.

Part VI — Beyond One Pressure Axis

23. Multi-Pressure Finance Geometry

23.1 Why one Q is insufficient

The minimal model uses:

$$Z = R + iQ. \quad (23.1)$$

This is useful when one mature filter dominates the question.

But financial systems often contain several pressure channels:

- market;
- credit;

- liquidity;
- capital;
- tail;
- execution;
- legal;
- model;
- optionality.

A more complete pressure state is:

$\vec{Q} = (Q_{market}, Q_{credit}, Q_{liquidity}, Q_{capital}, Q_{tail}, Q_{model}, \ldots)$. (23.2)

The effective state becomes:

$W = (R, \vec{Q})$. (23.3)

23.2 Generalized magnitude

A naïve extension would be:

$A^2 = R^2 + \Sigma_j Q_j^2$. (23.4)

But this assumes that pressure channels are orthogonal and equally scaled.

In practice, they may overlap and interact.

Introduce a positive semidefinite pressure metric G:

$A^2 = R^2 + \vec{Q}^{\top} G \vec{Q}$. (23.5)

The matrix G determines:

- scale;
- correlation;
- overlap;
- interaction strength;
- effective distance.

If:

$G = I$, (23.6)

the channels are orthonormal under the declared representation.

If G contains off-diagonal terms, pressures are coupled.

23.3 Interpretation of off-diagonal terms

Suppose:

$\vec{Q} = (Q_{credit}, Q_{liquidity})$. (23.7)

Then:

$\vec{Q}^{T} G \vec{Q} = g_{cc} Q_{credit}^2 + 2 g_{cl} Q_{credit} Q_{liquidity} + g_{ll} Q_{liquidity}^2$. (23.8)

If:

$g_{cl} > 0$, (23.9)

credit and liquidity pressure reinforce one another.

If:

$g_{cl} < 0$, (23.10)

one channel partially offsets the other under the declared metric.

The sign should not be assigned narratively.

It must be estimated or derived from a mature joint model.

23.4 Double-counting danger

Credit spreads may already contain liquidity compensation.

Market volatility may already reflect credit concern.

Capital charges may depend on ratings.

Accounting impairment may depend on market prices.

Therefore, a raw pressure vector can double-count the same underlying structure.

The correct sequence is:

Raw Channels → Common-Factor Removal → Residualized Pressure Coordinates → Metric Estimation. (23.11)

Let raw pressure be:

$\vec{Q}_{raw}$. (23.12)

Let C denote common components.

Then residualized pressure is:

$\vec{Q}_{res} = \vec{Q}_{raw} - \mathrm{Proj}_C(\vec{Q}_{raw})$. (23.13)

Only after residualization should the channels be combined geometrically.

The original Finance Geometry material likewise warns that multiple Q channels require careful

orthogonalization or residualization rather than simple addition.

23.5 Dynamic multi-pressure state

A generalized state may be written:

$Y = [R, \vec{Q}^{T}]^{T}$. (23.14)

Its dynamics are:

$dY/dt = \mathfrak{A}_P(t)Y + \varepsilon_Y$. (23.15)

The generator $\mathfrak{A}_P$ may contain:

- radial growth;
- rotations between R and each pressure channel;
- coupling among pressure channels;
- damping;
- gate effects.

A schematic form is:

$dR/dt = g_A R - \vec{\omega}^{T}\vec{Q} + \varepsilon_R$. (23.16)

$d\vec{Q}/dt = g_A\vec{Q} + \vec{\omega}R + C_Q\vec{Q} + \varepsilon_Q$. (23.17)

Where:

- $\vec{\omega}$ contains channel-specific angular velocities;
- C_Q captures pressure coupling.

23.6 Multi-pressure repricing load

Define:

$\Lambda_{total} = \vec{\omega}^{T}\vec{Q}$. (23.18)

Then:

$dR/dt = g_A R - \Lambda_{total} + \varepsilon_R$. (23.19)

Channel attribution is:

$\Lambda_j = \omega_j Q_j$. (23.20)

So:

$\Lambda_{total} = \Sigma_j \Lambda_j$ (23.21)

only when the declared coordinate system makes such addition valid.

With coupling, interaction terms may also be required.

23.7 Dominant-pressure regime

Define normalized channel contribution:

$$s_j = |\Lambda_j| / \Sigma_l |\Lambda_l|. \quad (23.22)$$

A dominant-pressure regime exists when:

$$\max_j s_j \geq \kappa_{dom}. \quad (23.23)$$

For example:

$$s_{liquidity} \approx 0.75 \quad (23.24)$$

would indicate that most angular repricing load is associated with the liquidity channel.

This may improve intervention selection.

Liquidity pressure suggests a different response from fundamental credit deterioration.

23.8 Pressure rotation among channels

Pressure may move from one hidden channel to another without immediately changing R.

For example:

credit concern may become liquidity concern;

liquidity concern may become capital pressure;

capital pressure may become forced-sale pressure.

This can be represented as:

$$d\vec{Q}/dt = C_Q\vec{Q}. \quad (23.25)$$

If C_Q is approximately skew-symmetric, pressure rotates among channels while preserving total pressure magnitude.

If C_Q has positive symmetric components, total pressure may amplify.

If it has negative components, pressure may dissipate.

This is a richer extension beyond the one-angle model.

23.9 Why the article begins with one Q

The one-Q model remains necessary because it provides:

- clear derivation;

- interpretable geometry;
- simple falsification;
- identifiable phase;
- manageable data requirements.

A high-dimensional pressure model introduced too early would obscure the central discovery:

Repricing contains distinct radial and angular components, and the effective world may possess its own phase order, gate structure, ledger, and residual. (23.26)

The multi-pressure model is an extension, not the foundation.

---

24. Multiple Filters and Path Dependence

24.1 Sequential filters

Financial values often pass through several filters.

Let:

F_a = credit filter. (24.1)

F_b = liquidity filter. (24.2)

A sequential valuation may be:

R_ab = F_b[F_a(A)]. (24.3)

The reverse order is:

R_ba = F_a[F_b(A)]. (24.4)

If:

R_ab = R_ba, (24.5)

the filters commute under the declared conditions.

If:

R_ab ≠ R_ba, (24.6)

order matters.

24.2 Why financial filters may not commute

The first filter may alter the input to the second.

For example:

Credit Downgrade → Collateral Ineligibility → Liquidity Haircut. (24.7)

The reverse sequence may be:

Liquidity Shock → Forced Sale → Credit Deterioration. (24.8)

The resulting values may differ because:

- populations change;
- thresholds activate;
- assets are sold;
- legal rights change;
- funding disappears;
- market depth changes.

Thus:

F_creditF_liquidity ≠ F_liquidityF_credit. (24.9)

This is ordinary path dependence in a history-bearing macro system.

24.3 The filter commutator

Define the effective filter commutator:

[F_a,F_b]X = F_a[F_b(X)] − F_b[F_a(X)]. (24.10)

If:

[F_a,F_b]X = 0, (24.11)

the filters commute for state X.

If:

[F_a,F_b]X ≠ 0, (24.12)

the sequence matters.

A normalized order-sensitivity measure is:

κ_ab(X) = ‖[F_a,F_b]X‖/[‖F_aF_bX‖ + ‖F_bF_aX‖ + δ]. (24.13)

This can be empirically estimated for declared sequential stress scenarios.

24.4 Gate-induced noncommutation

Noncommutation may arise because the first filter triggers a gate.

Suppose credit deterioration crosses a collateral threshold.

Then:

G_collateral[F_credit(X)] = 1. (24.14)

The liquidity filter now operates on a different admissible asset set.

If liquidity pressure occurs first but does not trigger the credit gate, the sequence produces a different world.

Therefore:

Order Dependence May Arise from Gate-Changed State Space. (24.15)

24.5 Ledger-induced noncommutation

The first event also changes the ledger.

Let:

L_a = L₀ ⊔ Record_a. (24.16)

The second filter becomes:

F_b(X | L_a). (24.17)

In the reverse order:

L_b = L₀ ⊔ Record_b. (24.18)

Then:

F_a(X | L_b). (24.19)

Because:

L_a ≠ L_b, (24.20)

the operations need not commute.

The order effect is produced by memory.

24.6 Example: downgrade and liquidity shock

Path A

1. rating downgrade;
2. collateral ineligibility;
3. forced selling;
4. liquidity collapse;
5. further credit deterioration.

Path B

1. temporary liquidity shock;
2. central liquidity support;
3. no downgrade;
4. collateral remains eligible;
5. recovery.

The same initial credit and liquidity variables may produce different outcomes because the event sequence differs.

The world branches at the gate.

24.7 Order sensitivity is not uniquely quantum

Quantum mechanics contains noncommuting observables with specifically quantum mathematical structure.

But financial systems can also exhibit noncommuting operations through:

- memory;
- changing constraints;
- gates;
- irreversible records;
- endogenous action.

Therefore:

Observed Order Effect ≠ Proof of Quantum Ontology. (24.21)

The quantum question is not merely whether order matters.

It is whether the order dependence has a structure that cannot be reproduced by a classical contextual, memory-bearing process.

This distinction is central to the later quantum subtraction method.

24.8 Path dependence and phase

With multiple filters, phase may become path-dependent.

The final angle may satisfy:

$\theta_{final}^{a\to b} \neq \theta_{final}^{b\to a}$. (24.22)

A single scalar θ may therefore be insufficient.

The state may require:

- a filter sequence;

- a path label;
- a ledger;
- a multi-pressure coordinate.

A richer state is:

State = $(R, \vec{Q}, \text{Path}, L, P)$. (24.23)

The effective world has curvature when sequential transformations cannot be reduced to one path-independent scalar.

---

25. Declaration Change and Inter-World Transition

25.1 Motion within one world

Under a stable protocol P:

$Z_k \to Z_{k+1}$. (25.1)

The coordinates change, but their meanings remain stable.

For example:

- A remains base-discounted value;
- θ remains the CAPM angle;
- Q remains CAPM pressure;
- the same gate rules apply;
- the same ledger schema is used.

This is motion inside one effective world.

25.2 Change of world

A declaration change is:

$P_k \to P_{k+1}$. (25.2)

The meanings of the coordinates may change.

Examples include:

- going-concern value becomes liquidation value;
- ordinary liquidity becomes stressed executable value;
- market price becomes model price after trading suspension;
- investment asset becomes non-performing exposure;

- recognized value becomes impaired value;
- eligible collateral becomes ineligible collateral.

The old and new Z may not be directly comparable.

25.3 Protocol structure

Recall:

$P_k = (B_k, q_k, \varphi_k, h_k, F_k, G_k, T_k, U_k)$. (25.3)

A declaration change may modify any component.

Boundary change

$B_{k+1} \neq B_k$. (25.4)

Example: analysis moves from one legal entity to a consolidated group.

Baseline change

$q_{k+1} \neq q_k$. (25.5)

Example: ordinary market conditions are replaced by crisis conditions.

Feature-map change

$\varphi_{k+1} \neq \varphi_k$. (25.6)

Example: liquidity variables become central after a freeze.

Horizon change

$h_{k+1} \neq h_k$. (25.7)

Example: long-run value is replaced by immediate survival analysis.

Filter change

$F_{k+1} \neq F_k$. (25.8)

Example: CAPM valuation is replaced by liquidation recovery.

Gate change

$G_{k+1} \neq G_k$. (25.9)

Example: stricter collateral thresholds are imposed.

Trace-rule change

$T_{k+1} \neq T_k$. (25.10)

Example: new reporting requirements attach more evidence.

Intervention change

$U_{k+1} \neq U_k$. (25.11)

Example: withdrawals, trading, or dividends become restricted.

25.4 Crisis as declaration failure

A crisis often begins as severe movement inside a world.

But it becomes a declaration crisis when the old model no longer answers the relevant questions.

For example:

- ordinary market price is no longer executable;
- historical correlation no longer holds;
- financing is unavailable at any ordinary spread;
- legal restrictions override economic value;
- long-term solvency becomes less relevant than next-day liquidity.

At that point:

Parameter Update is insufficient. (25.12)

World Declaration must change. (25.13)

25.5 Revision operator

Define:

$P_{k+1} = \mathcal{U}(P_k, L_{k+1}, E_k)$. (25.14)

Where E_k includes:

- dynamic residual;
- projection residual;
- ledger residual;
- external evidence;
- gate failures.

The revision operator may:

- recalibrate;
- add a pressure channel;

- change the baseline;
- shorten the horizon;
- modify the gate;
- restrict interventions;
- replace the filter entirely.

25.6 Parameter revision versus structural revision

A parameter revision changes values inside the same model.

Example:

$\beta_{k+1} \neq \beta_k$. (25.15)

A structural revision changes the form of the model.

Example:

F_CAPM → F_liquidation. (25.16)

The distinction is:

Parameter Revision: Same World, New Coordinates. (25.17)

Structural Revision: New World, New Coordinate Grammar. (25.18)

A system that confuses the two may keep recalibrating a model whose domain has disappeared.

25.7 Admissible revision

Not every revision is legitimate.

A pathological observer may revise the protocol to avoid admitting failure.

For example:

- changing the baseline after losses;
- redefining the gate after a breach;
- deleting residual;
- changing the horizon to hide deterioration;
- excluding inconvenient evidence.

An admissible revision should satisfy at least:

Trace Preservation. (25.19)

Residual Honesty. (25.20)

Frame Robustness. (25.21)

Budget Boundedness. (25.22)

Non-Degeneracy. (25.23)

The broader project's declaration framework likewise treats mature observerhood as constrained self-revision rather than unrestricted rule changing.

25.8 Trace-preserving revision

A revised world must preserve the old world's historical record.

The correct relation is:

L_{k+1},new = Translate(L_{k+1},old → new) ⊔ RevisionRecord. (25.24)

It is not:

L_{k+1},new = EmptyLedger. (25.25)

The revision record should state:

- what changed;
- why it changed;
- which residual forced revision;
- how old coordinates map into new ones;
- what cannot be translated.

25.9 Cross-world translation

Let:

M_{P→P′}: $\mathcal{Z}$_P → $\mathcal{Z}$_{P′} (25.26)

be a partial translation map.

The map may preserve some invariants while losing others.

For example:

- cash-flow amount may translate;
- market liquidity assumptions may not;
- legal priority may remain;
- CAPM angle may not map into liquidation angle.

Cross-world comparison should therefore report:

- preserved variables;

- transformed variables;
- discarded variables;
- new residual.

25.10 Inter-world discontinuity

A declaration change may create a discontinuity:

$Z_{P'}(t^*) \neq M_{P \to P'}[Z_P(t^{*-})]$. (25.27)

The mismatch is a transition residual:

$\varepsilon_{transition} = Z_{P'}(t^*) - M_{P \to P'}[Z_P(t^{*-})]$. (25.28)

A large transition residual means the old world did not contain enough structure to reconstruct the new one.

This is common in crises.

The variables that matter after collapse may have been invisible before collapse.

25.11 World death and world birth

An effective financial world “dies” operationally when:

- its coordinates no longer predict relevant transitions;
- its gates no longer correspond to actual events;
- its intervention set is obsolete;
- residual persistently exceeds tolerance;
- cross-observer invariance collapses.

A new world is born when a revised protocol establishes:

- new readable states;
- new gates;
- new intervention rules;
- new ledger consequences;
- sufficient closure.

Thus:

World Death = Persistent Failure of Declared Operational Closure. (25.29)

World Birth = Establishment of a New Protocol-Bounded Closure. (25.30)

25.12 The full world cycle

The article's runtime can now be written:

X_k
→ Declare_{P_k}
→ Z_k
→ Internal Dynamics
→ $Gate_k$
→ $Record_k$ + $Residual_k$
→ L_{k+1}
→ Backreaction
→ X_{k+1}
→ World-Validity Test
→ Continue P_k or Revise to P_{k+1}. (25.31)

In compact form:

$X_k \to \Pi_\{P_k,L_k\} \to Z_k \to G_k \to L_{k+1} \to \boldsymbol{\mathcal{B}} \to X_{k+1} \to \boldsymbol{\mathcal{U}} \to P_{k+1}$. (25.32)

This completes the finance-facing theory of effective-world formation.

The next task is comparative.

A non-quantum financial world has now reproduced:

- complex coordinates;
- hidden pressure;
- moving measurement frames;
- internal phase;
- event gates;
- irreversible trace;
- observer backreaction;
- order sensitivity;
- world revision.

The physics-facing question can therefore be stated more sharply:

After these generic world-forming structures are removed, what remains uniquely quantum?

Part VII — Finance as a Control Case for Quantum Foundations

26. What a Non-Quantum Financial World Can Reproduce

26.1 Why a control case is needed

Discussions of quantum strangeness often begin by collecting features that appear unusual:

- complex amplitudes;
- phase;
- observer dependence;
- contextual measurement;
- state disturbance;
- order sensitivity;
- irreversible records;
- collapse language.

But the presence of one or more of these features does not establish that a system is quantum.

Many non-quantum systems also contain:

- complex coordinates;
- rotating frames;
- context-dependent outputs;
- adaptive measurement;
- history-dependent state transitions;
- irreversible institutional commitment.

Finance Geometry provides a useful control case because its underlying domain is plainly macroscopic and institutionally constructed. No claim is required that a bond, company, balance sheet, or market price is secretly a microscopic quantum state.

The procedure is therefore:

1. construct the richest non-quantum observer-bound system available;
2. determine which supposedly “quantum-like” features it reproduces;
3. subtract those generic features from the quantum problem;
4. identify the residue that still requires specifically quantum structure.

In compact form:

Quantum Analysis by Resemblance → weak. (26.1)

Quantum Analysis by Controlled Subtraction → stronger. (26.2)

26.2 Complex coordinates

The financial state is:

$Z = R + iQ$. (26.3)

It also has polar form:

$Z = A \exp(i\theta)$. (26.4)

This provides:

- magnitude;
- phase;
- orthogonal components;
- rotation;
- phase sensitivity.

Yet the system remains classical.

Therefore:

Complex Number Representation ≠ Quantum Ontology. (26.5)

The symbol i may encode quadrature, delayed response, pressure complement, or phase relation without representing a quantum probability amplitude.

This lesson is already familiar from AC circuit theory, signal processing, control systems, and classical wave mechanics.

Finance adds a new example: complex notation can preserve the pressure coordinate hidden by scalar valuation.

26.3 Hidden orthogonal complements

Ordinary scalar valuation reveals R.

Finance Geometry preserves Q:

$A^2 = R^2 + Q^2$. (26.6)

The observer sees the admitted coordinate but may not ordinarily report the orthogonal complement.

This resembles a broader observer problem:

Visible Structure + Hidden Complement = Declared State. (26.7)

But hidden does not mean quantum.

Q is hidden because the reporting convention suppresses it, not because nature forbids simultaneous classical specification.

Given A and R, Q is calculable:

$Q = \sqrt{(A^2 - R^2)}$. (26.8)

Therefore:

Unreported Complement ≠ Quantum Uncertainty. (26.9)

A hidden coordinate may arise from compression, institutional convention, or limited measurement design.

26.4 Rotating measurement frames

The passive-rotation interpretation gives:

$[R,Q]^T = \mathrm{Rot}(\theta)[A,0]^T$. (26.10)

The effective state changes because the financial basis changes.

Examples include:

- market-risk frame;
- credit frame;
- liquidity frame;
- accounting frame;
- capital-admissibility frame.

The output depends on the measurement setting.

Yet this remains non-quantum.

Therefore:

Measurement-Frame Dependence ≠ Quantum Contextuality Automatically. (26.11)

A classical system may produce different projections under different frames while preserving an underlying jointly specifiable state.

The specifically quantum question is whether the contextuality can be represented by such an underlying global assignment.

26.5 Context-dependent outputs

Let two protocols observe the same primary field:

$Z_a = \Pi_{P_a}(X)$. (26.12)

$Z_b = \Pi_{P_b}(X)$. (26.13)

In general:

$Z_a \neq Z_b$. (26.14)

The same asset may have different:

- market value;
- collateral value;

- accounting value;
- regulatory value;
- liquidation value;
- strategic value.

These differences do not imply that one observer is irrational.

They arise because the protocols ask different operational questions.

Finance therefore reproduces:

Same Primary Field + Different Valid Contexts → Different Effective Outcomes. (26.15)

Context dependence alone is not uniquely quantum.

26.6 Phase ordering

A monotonic financial angle can locally order states:

$\theta_1 < \theta_2 < \theta_3$. (26.16)

An internal process may be written:

$dY/d\theta = F(Y,\theta)$. (26.17)

The phase coordinate can become a local clock.

Again, this does not require quantum mechanics.

Classical oscillators, rotating machinery, business cycles, and control loops can all use phase as an internal order parameter.

Therefore:

Phase-Based Internal Time ≠ Quantum Time Automatically. (26.18)

26.7 Collapse-like commitment

Before a gate, the financial system may contain unresolved possibilities.

After a gate:

$G(\hat{Z}) \rightarrow$ Commit. (26.19)

Examples include:

- quote becoming trade;
- deterioration becoming official default;
- potential loss becoming recognized impairment;
- option becoming exercised project;

- suspected breach becoming legal breach.

The state is converted from possibility into official consequence.

This resembles collapse language.

But the mechanism is institutional and classical.

Therefore:

Possibility → Gate → Committed Outcome (26.20)

is a general event grammar, not uniquely quantum collapse.

26.8 State-changing observation

A published price may change collateral.

A credit rating may change funding cost.

A stress test may cause asset sales.

An accounting entry may change capital.

Thus:

Observation → Record → Backreaction → New State. (26.21)

Measurement is performative.

But performative measurement is common in social, legal, biological, and control systems.

Therefore:

Measurement Backreaction ≠ Quantum Measurement Automatically. (26.22)

The important question is the mathematical form and unavoidable lower bound of the disturbance, not merely its existence.

26.9 Adaptive instruments

A financial observer may choose its next measurement based on prior trace.

Let:

I_k = Instrument selected at episode k. (26.23)

Then:

I_{k+1} = Policy(L_{k+1}). (26.24)

For example:

- a risk team increases sampling after volatility rises;
- a regulator performs a deeper review after a breach;

- a lender requests more collateral after a downgrade;
- an auditor expands testing after finding an anomaly.

The measurement process is adaptive and self-referential.

Again, no quantum ontology is required.

26.10 Order-sensitive operations

Sequential filters may fail to commute:

$F_aF_b(X) \neq F_bF_a(X)$. (26.25)

The difference may arise from:

- gate activation;
- changed collateral;
- changed legal state;
- changed admissible population;
- changed ledger;
- changed behaviour.

Thus:

Noncommuting Effective Operations ≠ Quantum Operator Noncommutation Automatically. (26.26)

A classical process with memory and endogenous state change can be order-sensitive.

26.11 Irreversible trace

A committed event enters the ledger:

$L_{k+1} = L_k \sqcup \text{Record}_k$. (26.27)

Even if the visible state later returns:

$R_T = R_0$, (26.28)

the ledger may not:

$L_T \neq L_0$. (26.29)

This creates historical irreversibility.

But again, irreversible trace does not require quantum mechanics.

Legal decisions, transactions, defaults, and institutional records are classical but historically irreversible.

26.12 Redundant objectivity

A financial record may be copied across:

- exchanges;
- clearinghouses;
- banks;
- regulators;
- auditors;
- custodians;
- counterparties.

When many independent systems preserve the same event, the event becomes publicly objective.

Let record r be encoded in fragments:

$E_1(r), E_2(r), \ldots, E_n(r)$. (26.30)

If multiple observers can recover r from independent fragments:

$\text{Recover}(E_j) = r$, (26.31)

then practical objectivity emerges through redundancy.

This resembles a general objectivity mechanism in which many observers agree because a stable record has been redundantly broadcast.

But redundant agreement also exists in entirely classical ledgers.

Therefore:

Redundant Consensus ≠ Quantum Darwinism Uniquely. (26.32)

Quantum Darwinism proposes a specifically quantum mechanism for redundant environmental encoding. The general structural role of redundancy is broader.

26.13 Simple internal laws from complex construction

The internal phase law is:

$dZ/d\theta = iZ$. (26.33)

The world-forming runtime is much richer:

$X_k \rightarrow \text{Declare} \rightarrow \text{Project} \rightarrow \text{Gate} \rightarrow \text{Ledger} \rightarrow \text{Backreact} \rightarrow \text{Revise}$. (26.34)

The internal observer may encounter a simple lawful world even though its construction depends on:

- many hidden variables;

- multiple institutions;
- changing protocols;
- historical records;
- nonlinear feedback.

This establishes a general possibility:

Complex Parent Process → Simple Effective Internal Law. (26.35)

That possibility is relevant to physics.

The simplicity of an observed physical law does not by itself reveal the simplicity of the process through which the observable world was constituted.

26.14 Summary of the macro control case

Finance can reproduce the following without quantum ontology:

Feature	Non-quantum financial realization
complex state	$R + iQ$
phase	finance-filter angle θ
hidden complement	retained pressure Q
frame dependence	different mature valuation protocols
phase time	monotonic θ as local order
collapse-like event	gate-passed commitment
trace	transaction, accounting, legal, or credit ledger
backreaction	price, collateral, leverage, and funding loops
order effect	sequential noncommuting filters
objectivity	redundant auditable records
irreversibility	ledger changes future admissibility
adaptive measurement	instruments selected from prior trace
simple internal law	$dZ/d\theta = iZ$

The conclusion is not that quantum mechanics has been explained.

It is narrower:

These features are not sufficient to identify quantum behaviour. (26.36)

---

27. The Quantum Subtraction Method

27.1 Why resemblance is too permissive

An analogy-based approach may say:

- finance uses complex numbers;
- quantum mechanics uses complex numbers;

- therefore finance is quantum-like.

Or:

- a trade converts possibility into outcome;
- quantum measurement converts possibility into outcome;
- therefore trade execution is collapse.

Such mappings may be inspirational, but they do not establish structural equivalence.

Many domains share broad patterns:

- selection;
- commitment;
- memory;
- feedback;
- phase;
- uncertainty.

The scientific task is not to count similarities.

It is to identify which mathematical and operational structures survive controlled comparison.

27.2 The subtraction architecture

Let QP denote a quantum phenomenon.

Let GOBS denote generic observer-bound structure.

Let QR denote candidate quantum residue.

Then:

QP = GOBS + QR. (27.1)

Therefore:

QR = QP − GOBS. (27.2)

This is not literal arithmetic.

It is a comparative research programme.

The task is to construct the strongest possible non-quantum model containing:

- bounded observers;
- contextual protocols;
- adaptive instruments;

- complex coordinates;
- commitment gates;
- trace;
- residual;
- backreaction;
- noncommuting effective operations.

Any quantum feature reproduced by that control model belongs at least partly to the generic observer-bound side.

What remains unexplained becomes the candidate quantum residue.

27.3 Generic observer-bound grammar

The generic grammar is:

Field → Declaration → Projection → Gate → Trace → Residual → Backreaction → Revision. (27.3)

A bounded observer does not access total reality.

It declares:

- boundary;
- feature map;
- horizon;
- admissible instruments;
- decision gate.

It projects a usable state.

It records an outcome.

The outcome changes future action and future observation.

This grammar appears in:

- finance;
- law;
- accounting;
- biology;
- AI agents;
- institutions;

- scientific measurement.

It should therefore be treated as a general observer grammar rather than a quantum-specific one.

27.4 Subtraction table

Candidate “quantum-like” feature	Reproduced by finance?	Removed by control case?
complex notation	yes	largely
phase	yes	largely
rotating frame	yes	largely
hidden complement	yes	partly
context dependence	yes	generic form
state-changing observation	yes	generic form
order sensitivity	yes	generic form
irreversible trace	yes	generic form
adaptive instrument choice	yes	generic form
redundant objectivity	yes	generic form
Born-rule probability	no	remains
entanglement	no	remains
Bell violation	no	remains
tensor-product nonseparability	no	remains
no-cloning	no	remains
quantum interference	no	remains

This table is methodological rather than final.

A generic feature may still interact with a specifically quantum law.

The subtraction method asks which component is genuinely irreducible.

27.5 Three levels of subtraction

Level 1 — Vocabulary subtraction

Remove words that appear mysterious only because they sound quantum.

Examples:

- phase;
- collapse;
- observer;
- hidden axis.

If a classical model reproduces them, the vocabulary carries little evidential weight.

Level 2 — Operational subtraction

Remove behaviours reproducible by classical contextual systems.

Examples:

- order effects;
- measurement backreaction;
- adaptive observation;
- irreversible trace.

Level 3 — Mathematical subtraction

Test whether the remaining correlations or probability structures admit a classical joint model.

This is where:

- Bell inequalities;
- contextuality inequalities;
- tensor-product structure;
- noncommutative probability;

become decisive.

The article's strongest physics contribution lies at Level 1 and Level 2.

It prepares the question for Level 3 but does not solve it.

27.6 Why subtraction is stronger than reductionism

The method does not assume in advance that quantum mechanics reduces to classical finance-like statistics.

It allows two outcomes.

Outcome A

A surprising amount of quantum strangeness is reproduced by generic observer-bound structure.

Then the conceptual mystery narrows.

Outcome B

Certain mathematical features remain irreducibly quantum.

Then their distinctiveness becomes clearer because generic observer effects have already been removed.

The method is therefore neutral between:

- deeper generative explanation;
- irreducible quantum ontology;
- operational agnosticism.

27.7 Control-model adequacy

The subtraction is meaningful only if the control model is sufficiently strong.

A weak control model might omit:

- memory;
- contextual protocols;
- adaptive instruments;
- endogenous disturbance;
- noncommuting operations.

It would then falsely classify generic phenomena as quantum residue.

The finance control case is useful because it includes many of these structures explicitly.

But it still has limits.

It does not reproduce:

- quantum probability amplitudes;
- entangled composition;
- nonlocal Bell correlations;
- no-cloning.

Therefore, the subtraction stops at the boundary of what the model actually generates.

27.8 A disciplined conclusion

The correct conclusion is:

Finance reproduces a substantial grammar of observer-bound world construction, but not the full mathematical structure of quantum theory. (27.4)

The incorrect conclusion is:

Finance has shown that quantum mechanics is merely ordinary statistics. (27.5)

The first is supported by the framework.

The second is not.

---

28. Candidate Irreducibly Quantum Residue

28.1 Born-rule probability geometry

In quantum mechanics, outcome probabilities are determined by amplitudes.

For a state $|\psi\rangle$ and projector Π_a:

$P(a) = \langle\psi|\Pi_a|\psi\rangle$. (28.1)

For a pure amplitude component:

$P(a) = |\langle a|\psi\rangle|^2$. (28.2)

The square-modulus rule is not derived by Finance Geometry.

Finance Geometry has:

$A^2 = R^2 + Q^2$. (28.3)

But this Pythagorean identity does not imply:

$P(R) = R^2/A^2$ (28.4)

or:

$P(Q) = Q^2/A^2$. (28.5)

R and Q are coordinates of admitted value and retained pressure, not quantum probability amplitudes.

Therefore:

Pythagorean Geometry ≠ Born Probability Geometry. (28.6)

The Born rule remains on the quantum-residue side.

28.2 Interference of amplitudes

Quantum probabilities depend on the sum of amplitudes before squaring.

For two alternatives:

$\psi = \psi_1 + \psi_2$. (28.7)

Then:

$|\psi|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\mathrm{Re}(\psi_1^*\psi_2)$. (28.8)

The cross term creates interference.

Finance may contain interacting expectations, narratives, strategies, or filters.

But ordinary combination of financial variables does not automatically reproduce quantum amplitude interference.

A genuine finance analogue would require:

1. a precisely defined amplitude space;
2. coherent superposition before observation;
3. experimentally testable interference terms;
4. a probability rule linked to amplitude magnitude.

The present R + iQ model does not supply those conditions.

28.3 Tensor-product nonseparability

Classical joint systems can often be represented by joint variables:

State_AB = (State_A,State_B,Correlation_AB). (28.9)

Quantum composite systems use tensor products:

$\mathscr{H}$_AB = $\mathscr{H}$_A ⊗ $\mathscr{H}$_B. (28.10)

Some joint states cannot be factorized:

|ψ_AB⟩ ≠ |ψ_A⟩ ⊗ |ψ_B⟩. (28.11)

This nonseparability is stronger than ordinary statistical correlation.

Finance contains highly coupled systems:

- banks and collateral networks;
- counterparties;
- portfolios;
- funding chains.

But strong coupling does not by itself produce Hilbert-space entanglement.

Therefore:

Network Dependence ≠ Quantum Entanglement. (28.12)

28.4 Bell inequality violation

Local hidden-variable models impose constraints on observable correlations.

A common CHSH expression is:

S = E(a,b) + E(a,b′) + E(a′,b) − E(a′,b′). (28.13)

Local hidden-variable theories satisfy:

|S| ≤ 2. (28.14)

Quantum theory allows:

$|S| \leq 2\sqrt{2}$. (28.15)

Experimental Bell violations show that broad classes of local, measurement-independent hidden-variable models cannot reproduce quantum correlations.

The financial control case does not derive such violations.

Financial systems can exhibit:

- common causes;
- strategic dependence;
- information transmission;
- institutional coordination;
- contextual selection.

These mechanisms can produce strong correlations but do not automatically reproduce loophole-controlled Bell-inequality violations.

Therefore:

Bell Violation Remains a Major Quantum Residue. (28.16)

28.5 Irreducible contextuality

Classical context dependence may arise because:

- the measurement changes the object;
- the protocol selects different features;
- the system retains memory;
- hidden variables are omitted.

Quantum contextuality is stronger.

It concerns the impossibility of assigning consistent pre-existing values to all observables while preserving the functional relations required by the theory.

Finance Geometry reproduces:

Protocol Dependence. (28.17)

It does not yet reproduce:

No Global Noncontextual Value Assignment. (28.18)

That stronger structure remains to be tested rather than assumed.

28.6 No-cloning

Quantum mechanics forbids a universal operation that copies an arbitrary unknown quantum state.

There is no universal unitary U satisfying:

$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ (28.19)

for every $|\psi\rangle$.

Financial records can generally be copied.

A price, rating, valuation model, or ledger entry can be duplicated, subject to ordinary practical limits.

There may be legal restrictions or strategic difficulty, but these are not the no-cloning theorem.

Therefore:

Institutional Noncopyability ≠ Quantum No-Cloning. (28.20)

No-cloning remains specifically quantum.

28.7 Quantum disturbance

In finance, measurement may disturb the system because agents react.

This disturbance may be reduced by:

- secrecy;
- delayed publication;
- smaller trade size;
- better sampling;
- alternative market design.

Quantum measurement disturbance is related to the mathematical incompatibility of observables and cannot generally be eliminated by improving institutional design.

For canonical variables:

$\Delta x \Delta p \geq \hbar/2$. (28.21)

Finance Geometry does not derive such a universal lower bound.

Therefore:

Practical Measurement Cost ≠ Quantum Uncertainty Relation. (28.22)

28.8 Identical-particle statistics

Quantum particles obey Bose–Einstein or Fermi–Dirac statistics.

Fermionic states exhibit exclusion.

Bosonic states permit multiple occupancy.

Financial agents may exhibit crowding, exclusion, capacity limits, or herding.

But these behavioural similarities do not derive quantum exchange symmetry.

Therefore:

Crowding or Exclusion Analogy ≠ Quantum Particle Statistics. (28.23)

28.9 Quantum tunnelling

A financial state may cross a barrier through:

- external funding;
- legal exception;
- strategic intervention;
- rare event;
- hidden pathway.

This may be metaphorically called tunnelling.

But physical quantum tunnelling follows from wavefunction dynamics through a classically forbidden region.

The R + iQ framework does not derive tunnelling amplitudes.

Therefore:

Unexpected Barrier Crossing ≠ Quantum Tunnelling. (28.24)

28.10 The residue map

The candidate quantum residue includes at least:

QR = {Born Geometry, Amplitude Interference, Tensor Nonseparability, Bell Violation, Irreducible Contextuality, No-Cloning, Quantum Disturbance}. (28.25)

This set may not be complete.

Its purpose is to mark the point where macro analogy stops.

28.11 The strongest negative conclusion

The article's strongest negative conclusion is:

Complex phase, observer dependence, order sensitivity, commitment, trace, and backreaction are insufficient to establish quantum ontology. (28.26)

This conclusion strengthens rather than weakens the quantum comparison.

It prevents generic observer grammar from being mistaken for specifically quantum structure.

---

29. The Constructor Problem

29.1 The object-oriented analogy

In object-oriented programming, a constructor establishes an object's initial state and usable interfaces.

A simplified constructor may be written:

Object = Constructor(Inputs,Rules,Dependencies). (29.1)

After construction, the object may operate through methods:

Output = Object.Method(Input). (29.2)

But the object does not ordinarily stand outside its own construction process.

Its access is limited to:

- stored state;
- exposed interfaces;
- permitted methods;
- available records.

This provides an analogy for observer-bound worlds.

29.2 Generative constructor

Let Σ denote a deeper generative process or relational field.

Let $\mathcal{C}$ denote a constructor-like map:

$\mathcal{C}$: Σ → (O,$\mathscr{H}$,$\mathscr{J}$,L). (29.3)

Where:

- O = observer;
- $\mathscr{H}$ = usable state space;
- $\mathscr{J}$ = admissible instrument set;
- L = record architecture.

The observer investigates Σ using instruments in $\mathscr{J}$.

But $\mathscr{J}$ was itself produced by $\mathcal{C}$.

Thus:

Observer Access to Constructor = Constructor-Provided Access. (29.4)

This creates a self-referential epistemic condition.

29.3 Financial version of the constructor

A financial institution is constructed through:

- legal charter;
- accounting rules;
- risk models;
- information systems;
- market access;
- reporting obligations;
- authority structure.

These establish:

- what the institution can observe;
- what it can recognize;
- what actions it can take;
- what records it preserves.

The institution then investigates the financial world using those constructed interfaces.

For example, a bank measures risk through categories that its regulatory and internal architecture makes available.

It may have difficulty observing risks that do not fit its own categories.

Thus:

Declared Observer → Declared Blind Spots. (29.5)

29.4 Physical observer version

A physical observer may similarly access reality only through:

- available interactions;
- measurable observables;
- finite energy;
- finite memory;
- finite spatial and temporal resolution;

- physically realizable instruments.

If the observer's measurement interface is part of the same physical world being investigated, it cannot simply step outside the world and inspect the full generative process from an external standpoint.

The constructor analogy therefore suggests:

Some mystery may arise because the observer attempts to reconstruct its own world-forming process using interfaces produced by that process. (29.6)

29.5 Interface limitation

One possibility is interface limitation.

A more complete underlying state exists:

$\lambda \in \Lambda$. (29.7)

But the observer accesses only:

$o = M_c(\lambda)$, (29.8)

where c is the measurement context.

Different contexts disclose different projections.

The observer's inability to recover λ may arise from:

- insufficient instruments;
- finite precision;
- information loss;
- inaccessible variables;
- computational limits.

Finance supplies many examples of this type.

The full balance-sheet or network state may exist, but one observer cannot access it.

29.6 Structural nonexistence

A stronger possibility is that no context-independent classical state λ exists with all the assumed simultaneous properties.

Then the problem is not merely:

The observer cannot access the full assignment. (29.9)

It is:

The requested global assignment is incompatible with the structure of the theory. (29.10)

Quantum contextuality and Bell results place strong constraints on ordinary hidden-state pictures.

Therefore, the constructor analogy must not be used to assume the existence of a classical hidden object merely because access is limited.

29.7 Three epistemic possibilities

The article should distinguish three cases.

Case A — Hidden but classical

A complete classical state exists but is inaccessible.

Case B — Contextually generative

Outcomes arise from a deeper process whose state cannot be represented as one fixed context-independent assignment.

Case C — Operationally irreducible

The quantum formalism may be the most complete available description, without a deeper representable state accessible to the theory.

Finance clearly demonstrates Case A.

It may provide conceptual intuition for Case B.

It does not decide between Case B and Case C in physics.

29.8 Constructor opacity

Define constructor opacity as:

Ω_C = Information Needed to Reconstruct $\mathcal{C}$ − Information Accessible through $\mathcal{J}$. (29.11)

If:

$\Omega_C > 0$, (29.12)

the observer cannot fully reconstruct its constructor.

This is a conceptual measure, not yet a numerical physical quantity.

Opacity may arise from:

- inaccessible initial conditions;
- lossy projection;
- finite memory;
- branch erasure;

- self-reference;
- computational incompleteness.

29.9 Self-measurement limits

An observer that measures itself must allocate part of itself to the measuring process.

Let:

O = O_target + O_instrument. (29.13)

The instrument is not external.

Measurement may change:

- memory;
- energy;
- state;
- future policy.

Therefore:

Self-Observation is an Intervention in the Observer. (29.14)

Finance again supplies an analogy.

A bank performing a stress test may change its own behaviour because the test alters capital planning.

An institution auditing itself may change workflows during the audit.

This does not prove a quantum limit, but it illustrates the generic self-reference problem.

29.10 Constructor and trace

The ledger gives the observer partial access to its own construction history.

Let:

L = {Record$_0$,Record$_1$,...}. (29.15)

The observer may infer:

$\mathcal{C}$ = InferConstructor(L,CurrentInterfaces). (29.16)

But inference is constrained because the ledger contains only admitted traces.

Rejected branches, erased states, and inaccessible alternatives may be absent.

Therefore:

Observed History = Constructor-Filtered History. (29.17)

The observer reconstructs its origin from traces already shaped by its own world-forming rules.

29.11 The philosophical consequence

The constructor problem does not show that reality is unknowable.

It gives a more precise claim:

Knowledge is produced through interfaces that are themselves part of the object of inquiry. (29.18)

Scientific objectivity therefore cannot mean complete escape from all interfaces.

It must mean:

- declared interfaces;
- reproducible transformations;
- cross-observer invariance;
- residual honesty;
- revision under failed predictions.

This is consistent with the effective-world approach developed throughout the article.

---

30. What Could Exist Beneath the Observable Quantum World?

30.1 The hypothesis space

Once generic observer-bound structure has been separated from specifically quantum residue, several possibilities remain.

A deeper layer might be:

1. ordinary hidden classical statistics;
2. contextual generative statistics;
3. nonlocal relational dynamics;
4. globally constrained histories;
5. process-first rather than state-first structure;
6. no deeper representable layer at all.

The finance analogy supports the general possibility of effective-world formation.

It does not select one physical ontology.

30.2 Ordinary hidden statistics

The simplest hypothesis is:

Observable quantum outcomes arise from hidden classical variables. (30.1)

Let:

$\lambda \in \Lambda$. (30.2)

Measurement outcome is:

$a = A(x,\lambda)$, (30.3)

where x is the measurement setting.

A probability distribution $\rho(\lambda)$ generates observed statistics.

This class of model is constrained by Bell-type results if it also assumes:

- locality;
- measurement independence;
- ordinary outcome factorization.

Therefore, an unrestricted statement that quantum mechanics is “just hidden statistics” is too weak and potentially misleading.

One must declare which assumptions the hidden model preserves or abandons.

30.3 Contextual generative statistics

A broader hypothesis allows the generative process to depend structurally on the measurement context.

The outcome is:

$a = A(x,\lambda,\Gamma_x)$, (30.4)

where Γ_x represents the full context.

The underlying process may not assign values to all incompatible measurements simultaneously.

This moves beyond ordinary noncontextual hidden variables.

A finance analogue is a market state whose effective value is constituted partly by the protocol applied to it.

But in physics, contextual generation must still reproduce exact quantum probability structures.

The analogy alone is insufficient.

30.4 Nonlocal relational dynamics

A deeper process may be nonlocal in the relevant sense.

The joint outcome may depend on a global relational state:

(a,b) = F(x,y,Λ_global). (30.5)

This can evade simple local hidden-variable models.

But it raises further questions:

- How is relativistic causal structure preserved?
- Why can the correlations not be used for ordinary superluminal signalling?
- What mathematical law governs the global relation?

Finance contains global constraints and network dependence, but these are not equivalent to physical quantum nonlocality.

30.5 Global consistency rather than forward causation

Another possibility is that the deeper process is constrained across an entire history.

Instead of:

Initial State → Local Forward Evolution → Outcome, (30.6)

one may have:

Global Boundary Conditions → Consistent History. (30.7)

The observable event is selected as part of a globally admissible solution.

Such approaches may involve:

- retrocausal models;
- two-time boundary conditions;
- path-integral constraints;
- all-at-once formulations.

The effective observer experiences local sequential time even if the generative description is globally constrained.

This resembles the article's distinction between internal phase time and world-forming structure, but it remains speculative.

30.6 Process-first ontology

A deeper layer may not consist of persistent hidden objects.

It may consist of:

- relations;

- transformations;
- interaction events;
- compositional processes;
- constraint networks.

Then observable “states” are effective summaries of process.

Schematically:

Process Network → Observer-Compatible State Space. (30.8)

The constructor does not reveal a hidden object.

It compiles a usable state representation from relational dynamics.

This is compatible with the article’s effective-world perspective:

A state may be secondary to the process through which it becomes observable. (30.9)

30.7 Measurement-dependent possibility fields

A stronger contextual model may hold that the possibility field itself changes with the admissible measurement family.

Let:

Σ_x = possibility field under context x. (30.10)

Then:

$\Sigma_x \neq \Sigma_y$ (30.11)

for incompatible contexts x and y.

The observer is not selecting among properties already jointly present in one universal state space.

The declaration partly defines which possibilities become meaningful.

Finance provides a macro analogy:

collateral value and accounting value are not always two simultaneously operative properties inside one universal financial frame.

But the physical version requires a rigorous probability and composition law.

30.8 Emergent Hilbert space

A deeper generative process might produce an effective Hilbert space rather than operating inside one fundamentally.

The sequence would be:

Relational Process → Stable Composition Grammar → Complex State Space → Born Geometry. (30.12)

To become a serious physical theory, such a proposal would need to derive:

- linearity;
- inner products;
- unitary evolution;
- tensor products;
- Born probabilities;
- observed symmetries;
- relativistic quantum field behaviour.

Finance Geometry derives none of these.

It only demonstrates the more modest possibility that low-dimensional complex state spaces can emerge from declared projection.

30.9 No deeper layer

It is also possible that the search for an underlying generative substrate is misguided.

Quantum theory may not be a projection of hidden classical structure.

The formalism may be irreducible, or deeper description may require concepts so different that "underlying state" is the wrong question.

The subtraction method remains useful even in this case.

It clarifies which aspects of quantum mystery arise from generic observer structure and which belong to the irreducible formalism.

30.10 What finance establishes

Finance establishes only the following structural possibility:

A high-dimensional, irregular, context-sensitive primary process can be compiled into a lower-dimensional effective world with complex coordinates, phase order, gates, trace, and backreaction. (30.13)

Finance does not establish:

- that physical quantum theory is derived this way;
- that the deeper physical layer is classical;
- that locality is restored;
- that Bell correlations are explained;

- that the Born rule emerges.

The article must maintain this boundary.

30.11 A hierarchy of claims

Safe claim

Some structures often described as quantum-like are generic features of bounded observers and effective-world construction. (30.14)

Research claim

Quantum foundations may benefit from explicitly subtracting those generic structures before identifying irreducible residue. (30.15)

Speculative claim

The observable quantum world may itself be an effective state space compiled from a deeper contextual or relational generative process. (30.16)

Unsupported claim

Quantum mechanics has been shown to reduce to ordinary hidden classical statistics. (30.17)

Only the first two claims are directly supported by the present framework.

The third is a research direction.

The fourth should be rejected.

30.12 The revised location of mystery

The framework relocates mystery.

The mystery is not primarily that complex numbers appear.

Classical systems use complex numbers.

The mystery is not primarily that measurement changes the system.

Macroscopic systems exhibit measurement backreaction.

The mystery is not primarily that context matters.

Classical institutional systems are context-dependent.

The deeper mystery lies in the exact mathematical residue:

- why amplitudes determine probabilities through squared norms;
- why composite systems use tensor products;
- why entanglement violates Bell inequalities;

- why no global noncontextual assignment exists;
- why no-cloning follows;
- why relativistic causality and quantum nonlocality coexist.

The subtraction method therefore changes the question from:

Why is quantum mechanics strange? (30.18)

to:

Which part of quantum mechanics remains strange after generic observer-bound world construction has been removed? (30.19)

That is a smaller question.

But it is a sharper one.

Part VIII — Empirical and Falsification Programme

31. A Minimal Dynamic Finance Experiment

31.1 Why the framework must face data

The mathematical identities of Finance Geometry are exact once A, R, and θ have been declared consistently.

But exact algebra does not establish practical value.

The research question is not merely whether:

$A^2 = R^2 + Q^2$. (31.1)

The more important question is:

Does the dynamic state $(A,R,Q,\theta,\omega_F,\Lambda_F,\varepsilon_{dyn},H_{loop})$ explain, predict, or govern anything better than existing financial variables? (31.2)

The original Finance Geometry framework explicitly places diagnostic gain above geometric elegance. A complex coordinate that does not improve explanation, attribution, stress analysis, model comparison, or intervention remains a re-expression of information already available elsewhere.

The dynamic extension must therefore be tested against mature alternatives.

31.2 The minimum experimental object

A first experiment should be deliberately narrow.

It should not attempt to validate the entire effective-world theory at once.

A suitable initial object is one of the following:

- a liquid corporate-bond sample;

- a panel of listed equities;
- a credit portfolio with observed spreads;
- a set of project valuations with periodically revised cash-flow forecasts;
- a collateralized asset class with observable haircut changes.

The best starting case is one where:

1. a declared pre-filter amplitude can be estimated repeatedly;
2. an admitted market or model value is observable;
3. a mature filter already exists;
4. event records are available;
5. standard comparison variables are available.

Corporate bonds are especially useful because they allow several channels to be compared:

- base-discounted amplitude;
- market value;
- credit spread;
- liquidity;
- rating events;
- default or downgrade gates;
- recovery outcomes.

Equities are also useful, but the meaning of the reference amplitude may be more model-sensitive.

31.3 Declared experimental protocol

The experimental protocol should be published before results are examined.

Let:

$$P_{exp} = (B,q,\varphi,h,F,G,T,U). \quad (31.3)$$

A concrete declaration might be:

$$B = \text{one asset class and one trading venue.} \quad (31.4)$$

$$q = \text{base discount-rate or benchmark model.} \quad (31.5)$$

$$\varphi = \text{cash flow, rate, spread, beta, liquidity, and event features.} \quad (31.6)$$

$$h = \text{fixed valuation horizon } T. \quad (31.7)$$

$$F = \text{CAPM, credit-spread, or certainty-equivalent filter.} \quad (31.8)$$

G = declared event gates such as downgrade, margin call, or default. (31.9)

T = sampling interval and ledger-record rule. (31.10)

U = admissible actions or stress interventions used in simulation. (31.11)

The protocol must also define:

- data frequency;
- missing-data treatment;
- derivative estimator;
- smoothing rule;
- outlier policy;
- transaction-cost assumptions;
- train–test split;
- failure threshold.

Without a fixed protocol, the researcher can unintentionally tune θ, Q, or residual until the desired story appears.

31.4 Required primary series

For each asset or episode, estimate:

A_t = declared pre-filter amplitude. (31.12)

R_t = admitted value. (31.13)

$\theta_t = \arccos(R_t/A_t)$. (31.14)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (31.15)

The basic domain condition is:

$0 \le R_t/A_t \le 1$. (31.16)

If the ratio lies outside this interval, the researcher must not silently force it back into range.

Possible explanations include:

- an invalid baseline;
- a negative-value state;
- a liability-like object;
- inconsistent units;
- speculative market value exceeding the declared amplitude;
- a geometry requiring signed or hyperbolic coordinates.

Such observations should be reported as domain failures rather than hidden through clipping.

31.5 Dynamic variables

Estimate radial growth:

g_A,t = Δ ln A_t/Δt. (31.17)

Estimate angular velocity:

ω_F,t = Δθ_t/Δt. (31.18)

Estimate angular repricing load:

Λ_F,t = Q_tω_F,t. (31.19)

The predicted change in admitted value is:

ΔR^_t = g_A,tR_tΔt − Λ_F,tΔt. (31.20)

The observed dynamic residual is:

ε_R,t = ΔR_t/Δt − g_A,tR_t + Λ_F,t. (31.21)

The pressure-coordinate residual is:

ε_Q,t = ΔQ_t/Δt − g_A,tQ_t − ω_F,tR_t. (31.22)

The complex residual is:

ε_dyn,t = ε_R,t + iε_Q,t. (31.23)

A normalized closure error is:

r_dyn,t = |ε_dyn,t|/[|ΔZ_t/Δt| + δ]. (31.24)

31.6 Derivative estimation

Finite differences amplify noise.

A naïve estimator is:

ω_F,t ≈ (θ_t − θ_{t−1})/Δt. (31.25)

But a practical experiment should compare several estimators:

- simple finite difference;
- central difference;
- rolling local-linear regression;
- robust spline derivative;
- state-space filter;

- Kalman-smoothed phase;
- total-variation regularized derivative.

The estimator must be chosen without using future test outcomes.

A derivative method that produces impressive crisis signals only after retrospective tuning is not valid evidence.

The article should therefore recommend:

Estimate Level Variables First, Derivatives Second, Claims Last. (31.26)

31.7 Event ledger

The experiment should not use only price or valuation time series.

It should construct a ledger of relevant events:

$L = \{Record_1, Record_2, \ldots, Record_n\}$. (31.27)

Each record should contain:

$Record_k = (EventType_k, Time_k, Evidence_k, Threshold_k, Authority_k, Residual_k)$. (31.28)

Possible event types include:

- earnings revision;
- spread jump;
- rating action;
- covenant breach;
- liquidity suspension;
- margin call;
- impairment;
- restructuring;
- default;
- regulatory intervention.

The ledger enables testing of whether:

- phase movement precedes gates;
- residual rises before regime change;
- similar phase paths produce different outcomes because of prior trace;
- apparent phase closure fails to restore the earlier world.

31.8 Fundamental versus filter-driven repricing

The primary attribution test is:

Observed Repricing = Amplitude Contribution + Angular Contribution + Residual. (31.29)

That is:

$\Delta R_t \approx g_{A,t} R_t \Delta t - Q_t \omega_{F,t} \Delta t + \varepsilon_{R,t} \Delta t$. (31.30)

Define the amplitude share:

$s_{A,t} = |g_{A,t} R_t| / [|g_{A,t} R_t| + |Q_t \omega_{F,t}| + |\varepsilon_{R,t}| + \delta]$. (31.31)

Define the angular share:

$s_{\theta,t} = |Q_t \omega_{F,t}| / [|g_{A,t} R_t| + |Q_t \omega_{F,t}| + |\varepsilon_{R,t}| + \delta]$. (31.32)

Define the residual share:

$s_{\varepsilon,t} = |\varepsilon_{R,t}| / [|g_{A,t} R_t| + |Q_t \omega_{F,t}| + |\varepsilon_{R,t}| + \delta]$. (31.33)

These shares classify an episode as:

- amplitude-dominant;
- frame-dominant;
- residual-dominant.

The classification should be compared with independent event narratives and analyst attribution.

31.9 Prediction targets

The framework should be tested against several target classes.

Target A — Future admitted-value change

Predict:

$\Delta R_{t \to t+h}$. (31.34)

Target B — Stress event

Predict whether:

$G_{t+h} = 1$. (31.35)

Examples include downgrade, margin event, or liquidity freeze.

Target C — Regime transition

Predict whether:

$P_{t+h} \neq P_t$. (31.36)

Target D — Residual escalation

Predict whether:

$r_{dyn,t+h} > \kappa_{dyn}$. (31.37)

Target E — Post-cycle memory

Predict:

$H_{loop} \neq 0$. (31.38)

Different variables may help different targets.

Q may be useful for pressure state.

Λ_F may be useful for active repricing.

ε_{dyn} may be useful for model breakdown.

H_{loop} may be useful for path-dependent recovery.

31.10 Out-of-sample design

A credible experiment should separate:

- estimation period;
- validation period;
- final test period.

The final test should be untouched until:

- baseline;
- smoothing;
- thresholds;
- feature definitions;
- event windows;
- loss functions;

have been fixed.

A rolling-origin design may be used:

Train on $[t_0,t_1]$. (31.39)

Validate on $[t_1,t_2]$. (31.40)

Test on $[t_2,t_3]$. (31.41)

Then move the origin forward without rewriting earlier results.

The framework should also be tested across:

- calm periods;
- tightening periods;
- crisis periods;
- recovery periods.

A model that works only during one famous crisis may be historically descriptive but not operationally general.

31.11 Cross-sectional design

The same time series may be tested across many assets.

For asset i:

$Z_{i,t} = A_{i,t}\exp(i\theta_{i,t})$. (31.42)

The cross-sectional model may ask whether:

$Q_{i,t}/R_{i,t}, \omega_{F,i,t}, \Lambda_{F,i,t}, r_{dyn,i,t}$ (31.43)

explain future:

- drawdown;
- spread widening;
- downgrade;
- liquidity deterioration;
- recovery speed.

Asset fixed effects, sector effects, duration, size, leverage, and rating must be controlled.

Otherwise, Q may merely proxy for already-known risk classifications.

31.12 Episode matching

One powerful test is matched-episode comparison.

Select pairs with similar observed price declines:

$\Delta R_a \approx \Delta R_b$. (31.44)

But require different decomposition:

$s_{A,a} \gg s_{\theta,a}$. (31.45)

$s_{\theta,b} \gg s_{A,b}$. (31.46)

Then compare future outcomes.

The hypothesis is:

Amplitude-Dominant Declines and Frame-Dominant Declines Have Different Recovery, Default, and Intervention Profiles. (31.47)

If the outcomes are indistinguishable after controlling for standard variables, the decomposition may add little value.

31.13 Crisis and recovery cycles

To test loop residual, identify episodes where a major scalar variable returns near its initial level.

For example:

$|R_T - R_0|/R_0 \le \kappa_{return}$. (31.48)

or:

$|\theta_T - \theta_0| \le \kappa_\theta$. (31.49)

Then test whether:

$L_T \neq L_0$ (31.50)

and whether the retained ledger difference predicts:

- higher future spread;
- lower liquidity;
- altered risk limits;
- greater sensitivity to later shocks;
- slower recovery.

Define a measurable memory vector:

$H_{loop} = (\Delta Leverage, \Delta Ownership, \Delta LiquidityDepth, \Delta CapitalRule, \Delta RatingHistory, \ldots)$. (31.51)

The hypothesis is:

Visible State Recovery Does Not Eliminate Ledgered Fragility. (31.52)

---

32. Candidate Research Questions

32.1 Does Λ_F distinguish fundamental deterioration from filter-driven repricing?

The first hypothesis is:

H_1: Λ_F improves attribution of visible-value change beyond ΔA and standard factor returns.

(32.1)

A test model may be:

$\Delta R_t = \alpha + \beta_1 \Delta A_t + \beta_2 \Lambda_{F,t} + \beta_3 X_{control,t} + u_t$. (32.2)

The null hypothesis is:

$H_0: \beta_2 = 0$ after standard controls. (32.3)

A stronger test asks whether adding Λ_F improves:

- out-of-sample R^2;
- mean absolute error;
- log score;
- event classification;
- economic decision quality.

Statistical significance alone is insufficient.

32.2 Does $Q\omega_F$ predict stress better than Q or ω_F alone?

The second hypothesis is interaction-based:

H_2: Pressure stock and frame velocity become most informative through their product. (32.4)

Compare four models:

M_0 = controls only. (32.5)

M_1 = controls + Q. (32.6)

M_2 = controls + ω_F. (32.7)

M_3 = controls + Q + ω_F + $Q\omega_F$. (32.8)

The interaction is useful if M_3 improves out-of-sample performance consistently.

A failure of $Q\omega_F$ does not invalidate the geometry.

It would invalidate the stronger claim that angular repricing load is a useful empirical stress variable.

32.3 Does dynamic residual rise before regime change?

The third hypothesis is:

H_3: Persistent ε_{dyn} growth precedes failure of the current valuation protocol. (32.9)

Let regime change occur at t^*.

Define the pre-transition window:

$$W_{pre} = [t^* - h, t^*). \quad (32.10)$$

Compare residual behaviour with matched non-transition windows.

Possible measures include:

$$\text{MeanResidual}_{pre} = \text{mean}(|\varepsilon_{dyn,t}|, t \in W_{pre}). \quad (32.11)$$

$$\text{ResidualSlope}_{pre} = \text{slope}(|\varepsilon_{dyn,t}|, t \in W_{pre}). \quad (32.12)$$

$$\text{ResidualPersistence}_{pre} = \text{mean}(\bar{E}_{dyn,t}, t \in W_{pre}). \quad (32.13)$$

The hypothesis is supported only if residual signals survive:

- alternate smoothing;
- alternate baselines;
- alternate horizons;
- out-of-sample testing.

32.4 Does loop residual capture post-crisis memory?

The fourth hypothesis is:

$$H_4: H_{loop} \text{ predicts future fragility after scalar recovery.} \quad (32.14)$$

Suppose:

$$R_T \approx R_0. \quad (32.15)$$

Then test whether H_{loop} predicts:

- future drawdown;
- funding spread;
- market depth;
- rating sensitivity;
- intervention probability.

A conventional mean-reversion model may treat the system as restored.

The loop-residual model predicts that two visually similar states can differ because their ledgers differ.

32.5 Does phase indexing improve event alignment?

The fifth hypothesis is:

$$H_5: \text{Events cluster more consistently by phase distance than by calendar duration.} \quad (32.16)$$

For event j, record phase distance from episode start:

$\Delta\theta_j = \theta(t_j) - \theta(t_{start})$. (32.17)

Compare dispersion of event timing in:

- calendar units;
- phase units.

If:

$\mathrm{Var}(\Delta\theta_{event}) < \mathrm{Var}(\Delta t_{event\ scaled})$, (32.18)

phase may provide a more stable internal clock.

The comparison must account for circularity because θ may already use variables associated with the event.

32.6 Does the framework improve cross-functional explanation?

Not all value is predictive.

The framework may improve communication between:

- valuation;
- treasury;
- credit;
- liquidity;
- risk;
- accounting;
- management.

A controlled study may present identical cases using:

1. scalar valuation only;
2. conventional risk dashboard;
3. radial–angular decomposition with residual.

Participants may be tested on:

- causal attribution;
- intervention choice;
- uncertainty recognition;
- confidence calibration;

- agreement across teams.

The hypothesis is:

H_6: The geometric state reduces attribution error without creating false precision. (32.19)

32.7 Does Q provide first-order warning near low-pressure states?

The local approximation gives:

$Q \approx A\theta$. (32.20)

$A - R \approx A\theta^2/2$. (32.21)

This motivates:

H_7: Q responds detectably before the scalar haircut becomes economically large. (32.22)

A valid test must compare detection delay while controlling false alarms.

Earlier movement alone is not useful if Q is excessively noisy.

32.8 Are apparent world transitions merely existing regime-switching signals?

The framework claims that a declaration change is more than a parameter shift.

The empirical question is:

H_8: World-transition indicators provide information beyond conventional structural-break and regime-switching models. (32.23)

A candidate comparison includes:

- hidden Markov regimes;
- Markov-switching volatility;
- Bayesian change-point detection;
- threshold autoregression;
- dynamic factor regimes;
- residual-based declaration failure.

If existing methods capture the same transitions more simply, the world-language may add interpretation but not statistical power.

32.9 Do noncommuting filters have measurable economic consequences?

The order-sensitivity hypothesis is:

H9: Sequential application of filters produces outcome differences not explained by the final levels of the underlying variables. (32.24)

For filters F_a and F_b:

κ_ab(X) = ‖F_aF_b(X) − F_bF_a(X)‖/[‖F_aF_b(X)‖ + ‖F_bF_a(X)‖ + δ]. (32.25)

The test should examine whether κ_ab predicts:

- forced-sale magnitude;
- recovery path;
- collateral loss;
- capital use;
- legal outcome.

If order effects vanish after ledger and state variables are fully included, the apparent noncommutation was merely omitted-state dependence.

32.10 Does observer disagreement reveal projection residual?

Let several departments or models produce aligned states:

Z_a,Z_b,...,Z_n. (32.26)

After translation into a common frame, define:

ε_tri = Dispersion(M_a→*Z_a,...,M_n→*Z_n). (32.27)

The hypothesis is:

H10: Rising cross-observer dispersion precedes material omitted-structure events. (32.28)

But disagreement may also arise from simple inconsistency, poor governance, or different goals.

The protocol must distinguish legitimate multi-frame diversity from analytical error.

---

33. Null Models and Existing Finance Benchmarks

33.1 Why null models matter

A new variable can appear informative because it is a nonlinear transformation of an established variable.

For example, θ may be a transformation of discount rate.

Q may be a transformation of R/A.

Λ_F may be related to duration-like sensitivity.

Therefore, the relevant question is not:

Does the new variable correlate with stress? (33.1)

It is:

Does the new variable improve performance beyond the mature quantity from which it was constructed? (33.2)

33.2 Duration benchmark

For fixed cash flow and yield y:

$D_{mod} = -(1/R)(\partial R/\partial y)$. (33.3)

Duration already measures price sensitivity to yield.

The phase-sensitivity relation is:

$-(1/R)(dR/d\theta) = Q/R$. (33.4)

A CAPM-based experiment must therefore compare:

$Q\omega_F$ (33.5)

with:

Duration × YieldChange. (33.6)

If the angular repricing load is merely a nonlinear restatement of duration exposure, its contribution may be interpretive rather than predictive.

That may still be useful, but it must be stated honestly.

33.3 Convexity benchmark

Bond repricing includes second-order yield effects:

$\Delta R/R \approx -D_{mod}\Delta y + \frac{1}{2}C(\Delta y)^2$. (33.7)

Because Finance Geometry uses trigonometric curvature, it may capture nonlinear change already represented by convexity.

Tests should compare:

- linear duration;
- duration plus convexity;
- phase decomposition;
- combined model.

The complex geometry earns value only if it improves either performance or causal

transparency.

33.4 Beta and factor benchmarks

For equities, compare with:

- market beta;
- multifactor exposures;
- momentum;
- size;
- value;
- quality;
- volatility;
- sector returns.

A CAPM-derived θ may largely reflect βERP.

The test must therefore determine whether θ or Q adds information beyond the underlying beta and risk premium.

33.5 Credit-spread benchmark

For bonds, compare with:

- option-adjusted spread;
- hazard rate;
- expected loss;
- recovery expectation;
- distance to default;
- rating;
- spread duration.

A credit-derived Q is constructed from risky versus default-free value.

Its incremental contribution may come from:

- nonlinear normalization;
- interaction with phase velocity;
- cross-filter comparison;
- residual and ledger integration.

If those additions do not improve results, the credit spread remains the simpler tool.

33.6 Volatility benchmark

Compare with:

- realized volatility;
- implied volatility;
- downside volatility;
- volatility-of-volatility;
- jump measures.

Q is not volatility.

But a rising Q may correlate with volatility because both respond to pressure.

The experiment must test whether Q retains value after volatility controls.

33.7 VaR and Expected Shortfall

Value at Risk estimates a loss quantile.

Expected Shortfall estimates expected loss beyond that quantile.

Finance Geometry does not replace either.

A useful comparison asks whether:

- Q identifies pressure state before VaR rises;
- Λ_F identifies active repricing;
- ε_{dyn} identifies model instability;
- H_{loop} identifies post-event fragility.

These are different tasks from tail-loss quantification.

33.8 Liquidity benchmarks

Compare with:

- bid–ask spread;
- Amihud illiquidity;
- price impact;
- market depth;

- turnover;
- order-book imbalance;
- redemption pressure.

A liquidity-derived Q may simply transform one of these variables.

The dynamic framework is useful only if it organizes their relation to admitted value and gate behaviour more effectively.

33.9 Regime-switching models

Compare declaration transitions with:

- hidden Markov models;
- Markov-switching autoregression;
- Bayesian change points;
- threshold models;
- structural-break tests.

A regime-switching model may detect state changes statistically.

The effective-world framework adds:

- explicit protocol declaration;
- gate and authority;
- ledger consequence;
- residual attachment;
- cross-world translation.

Its contribution may therefore be governance and interpretation even if change-point detection is statistically similar.

33.10 State-space and Kalman-filter benchmarks

The dynamic equations can themselves be represented as a state-space model.

For example:

$Y_{t+1} = A_tY_t + \eta_t.$ (33.8)

$Observation_t = H_tY_t + \nu_t.$ (33.9)

The framework should compare its hand-constructed state variables with data-driven latent-state estimates.

If an unconstrained state-space model performs much better, the Finance Geometry coordinates

may be too restrictive.

If the geometric model performs similarly with greater interpretability, that is a practical advantage.

33.11 Information criteria and complexity penalty

A richer model should be penalized for complexity.

Compare:

AIC = 2k – 2ln L. (33.10)

BIC = k ln n – 2ln L. (33.11)

Where:

- k = number of estimated parameters;
- n = sample size;
- L = likelihood.

Out-of-sample performance remains more important, but complexity penalties help prevent decorative state expansion.

33.12 Economic-value benchmark

A statistically better model may still be economically useless.

Evaluate:

EconomicGain = Utility(NewDecision) – Utility(BaselineDecision). (33.12)

Possible measures include:

- lower hedging cost;
- fewer false liquidations;
- earlier capital action;
- lower drawdown;
- improved recovery;
- better allocation;
- reduced disagreement.

The adoption test is:

Incremental Economic Gain > Implementation and Governance Cost. (33.13)

---

34. Falsification Ladder

34.1 Why several levels of failure are needed

A theory can fail algebraically, empirically, operationally, or conceptually.

The falsification ladder distinguishes these failure types.

This prevents a practical failure from being disguised as a philosophical success.

34.2 Level 0 — Algebraic consistency

Check:

$A^2 = R^2 + Q^2$. (34.1)

$R = A \cos \theta$. (34.2)

$Q = A \sin \theta$. (34.3)

$\theta = \arccos(R/A)$. (34.4)

The implementation tolerance is:

$|A^2 - R^2 - Q^2| \le \kappa_{alg}$. (34.5)

Failure at Level 0 means:

- coding error;
- unit mismatch;
- invalid domain;
- inconsistent inputs.

No higher claim should be considered until Level 0 passes.

34.3 Level 1 — Protocol validity

Check whether:

- A has a declared meaning;
- R comes from a mature filter;
- baseline q is explicit;
- horizon h is fixed;
- units are consistent;
- market and model values are not mixed;

- filter identity is attached to Q;
- domain restrictions are respected.

Define:

ProtocolValid(P) ∈ {0,1}. (34.6)

If:

ProtocolValid(P) = 0, (34.7)

the geometry is not interpretable even if the algebra is correct.

34.4 Level 2 — Dynamic closure

Test whether:

r_dyn,t ≤ κ_dyn (34.8)

for a sufficiently large fraction of the declared episode.

Define closure rate:

C_close = Number of valid periods/Total periods. (34.9)

If closure is poor, the single-Q dynamic world fails.

The appropriate response may be:

- revise A;
- revise θ;
- add channels;
- shorten the episode;
- reject the model.

34.5 Level 3 — Incremental diagnostic value

Compare baseline B with extension E.

Let:

Loss_B = out-of-sample loss of baseline. (34.10)

Loss_E = out-of-sample loss with Finance Geometry variables. (34.11)

Define improvement:

ΔLoss = Loss_B − Loss_E. (34.12)

The practical criterion is:

$\Delta \text{Loss} > \kappa_{\text{gain}}$. (34.13)

The framework fails at this level if it does not improve the declared task.

34.6 Level 4 — Null-model superiority

The framework must outperform or complement mature benchmarks.

Compare with:

- duration;
- convexity;
- factor models;
- spreads;
- volatility;
- liquidity measures;
- regime-switching models.

If a simpler benchmark performs equally well and explains the same mechanism, the new geometry should not be preferred merely because it is more inspirational.

34.7 Level 5 — Protocol robustness

Vary admissible choices:

$q \to q'$. (34.14)

$h \to h'$. (34.15)

$\text{smoothing} \to \text{smoothing}'$. (34.16)

$\text{asset sample} \to \text{sample}'$. (34.17)

A useful claim should remain qualitatively stable.

Define robustness:

$\text{Robust} = P(\text{Sign or Ranking Preserved under Admissible Variations})$. (34.18)

If small reasonable changes reverse the conclusion, the framework is frame-fragile.

34.8 Level 6 — Residual honesty

The model must allow:

$\text{Large Residual} \to \text{Model Rejection}$. (34.19)

It fails intellectually if every large residual is relabelled as:

- hidden pressure;
- deeper Q;
- unobserved phase;
- new world evidence.

Residual must remain capable of saying:

The proposed world does not exist under this protocol. (34.20)

34.9 Level 7 — Intervention test

A useful diagnosis should improve action.

Let u_B be the baseline intervention.

Let u_E be the intervention selected using the extended framework.

Compare outcomes:

Outcome(u_E) – Outcome(u_B). (34.21)

The framework fails operationally if it produces more complexity without better action.

34.10 Level 8 — Cross-observer reproducibility

Independent teams should reproduce:

- coordinates;
- events;
- residuals;
- conclusions;

within declared tolerances.

Define agreement:

A_agree = Similarity(Results_a,Results_b). (34.22)

Low agreement may indicate:

- ambiguous protocol;
- unstable data;
- excessive judgement;
- non-reproducible transformations.

34.11 Level 9 — Quantum boundary discipline

The framework fails conceptually if it claims to derive quantum phenomena it has not produced.

The following do not establish quantum theory:

- complex notation;
- phase;
- contextual projection;
- order dependence;
- commitment;
- trace;
- backreaction.

A quantum claim requires specifically quantum mathematical evidence.

The boundary rule is:

No Born Geometry, Entanglement, Bell Structure, or Equivalent Result → No Claim of Quantum Realization. (34.23)

34.12 Rejection criteria

The framework should be rejected for a declared application if any of the following persists:

1. A cannot be defined without circularity.
2. R/A frequently leaves the valid domain.
3. Q merely duplicates an existing variable.
4. derivative estimates are unstable.
5. residual remains large.
6. out-of-sample gain is negligible.
7. protocol variations reverse conclusions.
8. interventions do not improve.
9. independent teams cannot reproduce results.
10. quantum language adds rhetoric without structure.

A theory becomes stronger when it states how it can lose.

---

35. Governance and Interpretive Discipline

35.1 Every Q must name its originating filter

Never report:

Q = 40. (35.1)

Report:

Q_credit = 40 under protocol P_credit. (35.2)

or:

Q_liquidity = 40 under protocol P_liquidity. (35.3)

Without the filter label, the coordinate has no stable meaning.

35.2 Every A must declare its baseline

A report must state whether A means:

- base-discounted cash flow;
- default-free value;
- expected cash flow;
- theoretical executable value;
- unconstrained economic value;
- latent option value.

The rule is:

No Baseline Metadata → No Valid Amplitude. (35.4)

35.3 Market-implied and model-implied states must remain distinct

Let:

Z_market = R_market + iQ_market. (35.5)

Z_model = R_model + iQ_model. (35.6)

They may be compared only after:

- horizon alignment;
- baseline alignment;
- unit alignment;
- liquidity adjustment;

- option adjustment.

Otherwise:

Z_market – Z_model (35.7)

has no stable interpretation.

35.4 Residual must remain attached to trace

A committed record should include:

$Record_k = (Outcome_k, Evidence_k, Protocol_k, Residual_k)$. (35.8)

A record without residual metadata may become misleading when reused later.

35.5 Protocol revision must be visible

When:

$P_{k+1} \neq P_k$, (35.9)

the ledger must record:

$RevisionRecord_k = (OldP_k, NewP_k, Reason_k, Mapping_k, UntranslatedResidual_k)$. (35.10)

Silent protocol change destroys comparability.

35.6 Authority must be declared

Financial worlds are partly institutional.

A gate may depend on authority.

The same analytical result can have different consequences depending on whether it is issued by:

- trader;
- risk committee;
- auditor;
- regulator;
- court;
- rating agency;
- board.

Therefore:

GateOutcome = Function(State,Rule,Authority). (35.11)

Authority should be recorded, not treated as background.

35.7 Phase should not be forced into a clock

The internal-time interpretation should be suspended when:

- θ is non-monotonic;
- reversals are excessive;
- phase is not identifiable;
- residual is large;
- multiple angles conflict.

The rule is:

No Stable Ordering → No Internal Clock Claim. (35.12)

35.8 Analogy must stop where diagnostic gain stops

The framework may use analogies with:

- phasors;
- oscillators;
- fields;
- quantum measurement;
- constructors;
- gauge frames.

Each analogy must answer:

What operational distinction does this analogy add? (35.13)

If the answer is none, remove it.

The framework should become simpler when analogy ceases to improve reasoning.

35.9 Versioning

Every published result should include:

- protocol version;
- model version;
- data vintage;

- code hash;
- parameter set;
- event definitions;
- residual thresholds.

A minimal verification footer is:

VerificationFooter = (P_version,DataHash,CodeHash,Thresholds,Date,Reviewer). (35.14)

This converts the framework from philosophical narrative into an auditable research object.

35.10 Not investment advice

The coordinates introduced here are theoretical and experimental.

They do not, by themselves, justify:

- buying;
- selling;
- borrowing;
- lending;
- changing capital;
- changing accounting treatment;
- changing legal strategy.

Any real-world financial use requires:

- validated data;
- domain experts;
- regulatory compliance;
- independent risk review;
- comparison with established methods.

The framework is a proposed research language, not a guarantee of profitable or safe decisions.

Part IX — Synthesis

36. When Does Valuation Become a World?

36.1 From an output to an operational environment

A valuation begins as an answer to a question.

What is this asset worth under the declared assumptions? (36.1)

The answer may be:

R = 80. (36.2)

At this stage, the valuation is a scalar output.

It does not yet constitute a world.

A world begins to emerge when the output becomes part of a stable system of distinctions, transitions, events, records, and consequences.

The progression is:

Number → Coordinate → State → Dynamics → Event → Ledger → Backreaction → Revision. (36.3)

Each step adds a new form of operational reality.

36.2 Stage 1 — Number

The simplest financial object is one scalar:

R. (36.4)

It may be:

- price;
- present value;
- collateral value;
- recognized value;
- capital-admissible value.

The scalar is useful because it supports comparison and commitment.

But it hides how the result was produced.

A number alone does not reveal:

- its baseline;
- its filter;
- its horizon;
- its pressure complement;
- its residual;
- its event history.

Therefore:

Scalar Value = Admitted Result without Full State Geometry. (36.5)

36.3 Stage 2 — Coordinate

Finance Geometry completes the scalar into:

$Z = R + iQ$. (36.6)

The state is constrained by:

$A^2 = R^2 + Q^2$. (36.7)

The scalar output now belongs to a coordinate system.

It becomes possible to distinguish:

- admitted value R;
- retained pressure Q;
- pre-filter amplitude A;
- filter angle θ.

The valuation is no longer merely a point on one axis.

It is a location in a declared geometry.

36.4 Stage 3 — Dynamic state

When A and θ vary:

$Z(t) = A(t)\exp[i\theta(t)]$. (36.8)

The state acquires internal dynamics:

$dZ/dt = (g_A + i\omega_F)Z + \varepsilon_{dyn}$. (36.9)

The visible coordinate evolves as:

$dR/dt = g_A R - Q\omega_F + \mathrm{Re}(\varepsilon_{dyn})$. (36.10)

The financial object now possesses a causal decomposition.

A change in R can be attributed to:

- amplitude change;
- frame rotation;
- model residual.

The coordinate system has become a dynamic state space.

36.5 Stage 4 — Event grammar

Continuous state change is still insufficient for worldhood.

The system must distinguish between:

- movement;
- threshold;
- commitment.

A gate performs this distinction:

$G_k(\hat{Z}_k) \rightarrow \text{Commit}_k$, Reject_k, or Defer_k. (36.11)

An event is born when a possibility passes through a declared gate.

Examples include:

- a quote becoming a trade;
- deterioration becoming default;
- a potential loss becoming impairment;
- latent optionality becoming exercised investment.

The world now has an event grammar.

36.6 Stage 5 — Ledger

A committed event becomes history when it is recorded:

$L_{k+1} = L_k \sqcup (\text{Record}_k, \text{Evidence}_k, \text{GateMetadata}_k, \text{Residual}_k)$. (36.12)

The ledger gives the world memory.

Without the ledger, the same phase position may recur without distinction.

With the ledger, repeated positions belong to different histories.

Therefore:

State Recurrence + Different Ledger = Different Effective World. (36.13)

36.7 Stage 6 — Intervention

A world must support action.

Let:

$u_k \in U(P_k)$. (36.14)

An intervention may:

- hedge;
- refinance;
- sell;
- buy;
- restructure;
- post collateral;
- tighten a gate;
- provide liquidity;
- revise recognition.

The state is no longer merely observed.

It organizes admissible decisions.

36.8 Stage 7 — Backreaction

A financially effective state acts upon its primary field:

$X_{k+1} = \mathcal{B}(X_k, Z_k, L_{k+1}, u_k)$. (36.15)

This is the decisive move from representation to operational reality.

A price changes collateral.

A rating changes funding.

An impairment changes capital.

A stress result changes portfolio action.

The financial object participates in producing the next state it will later measure.

Therefore:

Causal Efficacy is a Core Criterion of Effective Worldhood. (36.16)

36.9 Stage 8 — Revision

A mature world must also be able to recognize its own limits.

When residual becomes persistent:

$P_{k+1} = \mathcal{U}(P_k, L_{k+1}, E_k)$. (36.17)

The protocol may change:

- baseline;

- horizon;
- filter;
- gate;
- feature map;
- intervention set.

The world is not defined by never changing.

It is defined by changing without erasing the trace that justified revision.

36.10 The complete emergence ladder

The full emergence ladder is:

$$
\begin{aligned}
&R \\
&\rightarrow (A,R,Q,\theta) \\
&\rightarrow Z(t) \\
&\rightarrow G_k \\
&\rightarrow L_k \\
&\rightarrow U(P) \\
&\rightarrow \boldsymbol{\mathcal{B}} \\
&\rightarrow \boldsymbol{\mathcal{U}}. \quad (36.18)
\end{aligned}
$$

Or in words:

Scalar Admission
→ Pressure-Preserving State
→ Internal Dynamics
→ Event Commitment
→ Historical Memory
→ Admissible Action
→ Causal Backreaction
→ Trace-Preserving Revision. (36.19)

A representation becomes increasingly world-like as it moves upward through this ladder.

36.11 Worldhood is not binary

A system need not be either “a world” or “not a world” in an absolute sense.

It may be world-like relative to:

- one task;
- one observer;
- one horizon;
- one intervention scale.

A CAPM world may be adequate for market-risk attribution but inadequate for liquidity collapse.

A credit world may be adequate before default but inadequate after legal restructuring.

Therefore:

Worldhood is Protocol-Relative and Task-Relative. (36.20)

This does not weaken the concept.

It prevents a local effective world from being mistaken for total reality.

36.12 An operational worldhood vector

Define an operational worldhood vector:

$\Omega_W = (D,C,E,G,L,U,B,R,V)$. (36.21)

Where:

- D = stable distinctions;
- C = coordinate adequacy;
- E = transition-law closure;
- G = gate validity;
- L = ledger integrity;
- U = intervention capacity;
- B = backreaction relevance;
- R = residual honesty;
- V = revision validity.

A simple diagnostic score may be:

$Score_W = \Sigma_j w_j\Omega_{W,j}$. (36.22)

The weights depend on the task.

This score should not be interpreted as a universal measure of reality.

It is a governance device for asking whether a representation is sufficiently world-like for operational use.

36.13 Approximate closure

No financial world is perfectly closed.

The correct condition is:

Residual ≤ Declared Tolerance. (36.23)

The world remains usable when:

- its transition laws remain sufficiently accurate;
- its gates remain meaningful;
- its ledger retains relevant consequences;
- its interventions produce bounded effects;
- its residual remains visible.

Thus:

Effective Closure ≠ Absolute Isolation. (36.24)

It means that omitted structure does not invalidate the declared task within the declared horizon.

36.14 The world as a compiler

The effective-world process can be interpreted as compilation.

The primary field contains more structure than the financial observer can use directly.

The protocol compiles that field into:

- readable states;
- executable decisions;
- recordable events.

The compilation map is:

Compiler_P: (X,L) → (Z,G,U,Record). (36.25)

This analogy emphasizes that an effective world is not a simple photograph.

It is a runtime environment.

It selects which operations are possible.

It assigns types to events.

It rejects inadmissible states.

It preserves trace.

It changes the conditions of the next execution cycle.

36.15 Why the effective world is real

The world is real in at least four operational senses.

Predictive reality

Its states help forecast what may happen next.

Institutional reality

Its categories are recognized by authorities and contracts.

Causal reality

Its outputs alter action and future states.

Historical reality

Its records persist and constrain later possibilities.

A financial world need not be fundamental to be real.

Its reality is effective, relational, and consequential.

36.16 The most compact criterion

The article's most compact worldhood criterion is:

A Representation Becomes an Effective World When It Can Remember and Act Back. (36.26)

Coordinates provide structure.

Dynamics provide continuity.

Gates provide events.

Ledgers provide memory.

Backreaction provides causal efficacy.

Revision provides survival.

---

37. Constructed Reality Without Relativism

37.1 The apparent dilemma

Financial reality is clearly constructed.

Prices depend on:

- institutions;
- market design;
- contracts;
- models;
- regulations;

- reporting rules;
- observer horizons.

But financial reality is not merely subjective.

A completed trade cannot be made nonexistent merely because one observer dislikes it.

A default may be disputed, but it has legal and financial consequences.

An accounting entry may depend on rules, yet the rules can be audited.

The apparent dilemma is:

If reality is protocol-dependent, is it still objective? (37.1)

The answer developed here is yes.

But objectivity must be defined through invariance, declaration, and trace rather than through observer-free access to total reality.

37.2 Protocol-relative truth

Let proposition C be evaluated under protocol P.

Write:

Truth_P(C) ∈ {Supported,Rejected,Residual}. (37.2)

For example:

C = “This bond is admissible as collateral.” (37.3)

The proposition is incomplete without specifying:

- which institution;
- which jurisdiction;
- which rule;
- which date;
- which haircut framework.

Once P is declared, the proposition can be objectively tested.

Therefore:

Protocol Dependence ≠ Arbitrary Truth. (37.4)

It means that the domain of the claim has been specified.

37.3 Objectivity as reproducible constraint

A result is objective under P when independent admissible observers reproduce it within

tolerance.

Let observers a and b implement equivalent protocols:

$P_a \simeq P_b$. (37.5)

Then objective agreement requires:

$\text{Distance}[Z_a, Z_b] \leq \kappa_{\text{frame}}$. (37.6)

For a declared invariant $\mathcal{I}$:

$\mathcal{I}(Z_a, L_a) \approx \mathcal{I}(Z_b, L_b)$. (37.7)

Objectivity therefore arises from:

- shared declaration;
- reproducible operations;
- stable invariants;
- auditable records;
- disclosed residual.

It does not require access to a view from nowhere.

37.4 Equivalent frames

Two protocols need not be identical to be equivalent.

They may use:

- different units;
- different coordinate bases;
- different but translatable feature maps;
- different implementations of the same filter.

Let:

$M_{ab}: \mathcal{Z}_{P_a} \to \mathcal{Z}_{P_b}$ (37.8)

be a translation map.

The frames are operationally equivalent when:

$M_{ab}[\Pi_{P_a}(X)] \approx \Pi_{P_b}(X)$ (37.9)

for the relevant class of X.

This allows objectivity across plural representations.

37.5 Invariance

An invariant is a relation that survives admissible frame change.

In the minimal geometry:

$A^2 = R^2 + Q^2$ (37.10)

is invariant under ordinary rotation of the R–Q coordinates.

In finance, useful invariants may include:

- legal obligation;
- cash-flow identity;
- conservation of ownership quantity;
- accounting equality;
- verified transaction record;
- ordering of stress severity;
- sign of residual escalation.

A mature theory should state:

What changes with the frame? (37.11)

What remains invariant? (37.12)

Without this distinction, protocol dependence becomes confusion.

37.6 Four types of disagreement

When observers disagree, at least four explanations are possible.

Type 1 — Legitimate frame difference

The observers ask different valid questions.

Example:

market value versus liquidation value.

Type 2 — Data disagreement

The protocols are aligned, but inputs differ.

Type 3 — Implementation failure

The same protocol is applied incorrectly.

Type 4 — Projection residual

The observers encounter omitted structure that neither frame adequately captures.

The diagnostic sequence is:

Align Declaration → Align Data → Audit Implementation → Inspect Residual. (37.13)

This is more useful than immediately declaring one observer correct and the other subjective.

37.7 Constructed categories can have hard consequences

A credit rating is constructed.

Yet a downgrade may trigger contractual consequences.

A collateral category is constructed.

Yet exclusion may remove borrowing capacity.

A legal entity is constructed.

Yet it can own property and incur liabilities.

A currency is constructed.

Yet it organizes payment, debt, and taxation.

The lesson is:

Constructed Objects Can Become Causally Hard. (37.14)

Their hardness arises from coordinated recognition, gate authority, ledger continuity, and enforcement.

37.8 The ledger stabilizes objectivity

A public ledger allows observers to agree on committed events even when they disagree about interpretation.

Observers may disagree about whether a trade was wise.

They can still agree that the trade occurred.

They may disagree about the economic meaning of a default.

They can still agree that a declaration was issued at a recorded time.

Therefore:

Ledger Agreement Can Precede Interpretive Agreement. (37.15)

This gives the effective world a stable historical core.

37.9 Residual prevents false objectivity

A system can appear objective by suppressing uncertainty.

For example:

R = 80 (37.16)

may appear precise.

But the stronger record is:

R = 80 under P, with residual E. (37.17)

Objectivity does not mean pretending residual is absent.

It means making the residual reproducibly visible.

Therefore:

Residual Disclosure is Part of Objectivity, Not Its Opposite. (37.18)

37.10 Authority and objectivity

Financial facts often depend on authorized gates.

An analyst may estimate that a covenant has been breached.

A court, trustee, or contractual authority may determine whether the breach is legally operative.

This creates two distinct states:

Analytical State. (37.19)

Institutionally Committed State. (37.20)

Both can be objective within their protocols.

But they must not be conflated.

Objectivity requires not only correct calculation, but correct typing of authority.

37.11 Reflexivity does not eliminate reality

A price may affect the asset it prices.

This reflexivity does not make price unreal.

It means the observable is part of the causal loop.

Similarly, a measurement that changes its object is not necessarily invalid.

Its validity depends on whether the intervention and backreaction are modelled.

The mature equation is not:

Observation = Passive Copy. (37.21)

It is:

Observation + Commitment + Backreaction = Effective State Transition. (37.22)

37.12 Objectivity as governed recursion

In a reflexive world, objectivity is not a once-and-for-all correspondence between model and passive object.

It is a governed recursive process:

Observe → Declare → Project → Record → Compare → Revise. (37.23)

A system remains objective when revisions:

- preserve trace;
- disclose error;
- improve prediction;
- maintain cross-frame invariance;
- avoid retrospective manipulation.

This produces a dynamic definition:

Objectivity = Reproducible Self-Correction under Trace-Preserving Constraints. (37.24)

37.13 Against two extremes

The framework rejects two extremes.

Extreme 1 — Naïve realism

Financial values exist fully formed and protocols merely reveal them.

This ignores the constitutive role of discounting, legal status, liquidity, recognition, and action.

Extreme 2 — Unrestricted relativism

Every valuation is equally valid because all values are constructed.

This ignores protocol quality, evidence, invariance, predictive performance, and consequence.

The middle position is:

Financial Reality is Protocol-Constructed, Empirically Constrained, and Causally Operative. (37.25)

37.14 Constructive objectivity

The proposed concept is **constructive objectivity**.

A financial object is constructively objective when:

1. its protocol is declared;
2. its operations are reproducible;
3. its invariants survive admissible translation;
4. its records are auditable;
5. its residual is preserved;
6. its revisions are traceable;
7. its consequences are empirically observable.

In compact form:

Constructive Objectivity = Declaration + Invariance + Trace + Falsifiability. (37.26)

---

38. The Deeper Lesson for Physics

38.1 The first subtraction: complex numbers

The appearance of i is not the deepest mystery.

Finance Geometry uses:

$Z = R + iQ$. (38.1)

Classical circuits use complex impedance.

Classical waves use complex phase.

Signal processing uses analytic signals.

Therefore:

Complex Representation is a General Tool for Coupled Orthogonal Structure. (38.2)

Quantum theory uses complex numbers in a much more specific way, especially through amplitude interference and Born probability.

The symbol i alone carries no proof of quantumness.

38.2 The second subtraction: phase

Phase is not uniquely quantum.

Financial filters can generate:

$\theta = \arccos(R/A)$. (38.3)

A monotonic θ can order an effective history.

Classical systems can therefore possess:

- phase;
- phase velocity;
- phase reversal;
- phase-indexed internal laws.

The quantum residue must lie beyond the mere existence of phase.

38.3 The third subtraction: observer dependence

Finance demonstrates that valid values may depend on the measurement protocol.

The same primary field can yield:

$Z_a = \Pi_{P_a}(X)$. (38.4)

$Z_b = \Pi_{P_b}(X)$. (38.5)

This is observer-relative projection.

But it remains classically representable.

Therefore:

Observer Dependence is Not Sufficient for Quantum Contextuality. (38.6)

The quantum question concerns whether all contextual outcomes can be embedded in one consistent underlying assignment.

38.4 The fourth subtraction: collapse-like commitment

A financial gate converts possibility into recorded actuality:

Potential → Gate → Commit → Ledger. (38.7)

This resembles the language of measurement collapse.

But institutional selection, biological decision, legal judgment, and AI verification can all exhibit the same grammar.

Therefore:

Outcome Commitment is a General World-Formation Operation. (38.8)

Quantum measurement may contain this generic grammar plus additional irreducible mathematical structure.

38.5 The fifth subtraction: measurement backreaction

Financial observation can alter its target:

Price → Collateral → Leverage → Price. (38.9)

Rating → Funding → Liquidity → Rating. (38.10)

This shows that state-changing observation is not uniquely quantum.

The quantum residue lies in the exact structure of disturbance, incompatibility, and correlation —not in the general fact that observation has consequences.

38.6 The sixth subtraction: order sensitivity

Finance supports:

$F_aF_b(X) \neq F_bF_a(X)$. (38.11)

The reason may be:

- memory;
- gate activation;
- changed state space;
- changed action set;
- path dependence.

Therefore:

Noncommutation of Effective Procedures is Not Sufficient for Quantum Operator Structure. (38.12)

The specifically quantum issue is whether order effects can be reproduced by a classical contextual process with adequate hidden state.

38.7 The seventh subtraction: irreversibility

A financial ledger creates:

$L_{k+1} \neq L_k$. (38.13)

Even when phase returns:

$\theta_T = \theta_0$, (38.14)

the history may not:

$L_T \neq L_0$. (38.15)

Irreversible trace is therefore not uniquely quantum.

It is a general consequence of event-bearing ledgers.

The measurement problem cannot be solved merely by saying that records are irreversible.

One must explain how quantum probabilities, entanglement, and definite records relate.

38.8 The central reversal

The effective financial world possesses a simple internal law:

$dZ/d\theta = iZ.$ (38.16)

The more complicated structures appear at:

- declaration;
- projection;
- gate;
- trace;
- residual;
- backreaction.

This suggests a general possibility:

Observed Internal Law May Be Simpler than the World-Forming Process that Makes the Law Usable. (38.17)

The handoff document identifies precisely this reversal: the classical nature of the internal R–Q dynamics relocates observer-related strangeness from the internal field law to the boundary that declares, projects, records, and revises the effective world.

38.9 Measurement constraint as a source of strangeness

Part of apparent mystery may arise because a bounded observer cannot access the full generative structure.

The observer has:

- limited instruments;
- limited memory;
- limited resolution;
- context-dependent interaction;
- constructor-supplied interfaces.

The accessible state is:

$Z = \Pi_P(X).$ (38.18)

The inaccessible structure contributes to:

$\varepsilon_{\Pi} = X - \tilde{X}(Z \mid P)$. (38.19)

This may explain some forms of:

- incomplete reconstruction;
- frame dependence;
- apparent discontinuity;
- hidden complement;
- observer disagreement.

38.10 Why measurement constraint is not the whole answer

Measurement limitation alone does not derive:

- Born probabilities;
- Bell violations;
- entanglement;
- no-cloning;
- quantum interference.

Therefore:

Measurement Constraint May Explain Generic Strangeness but Not Automatically Quantum Residue. (38.20)

A deeper model must reproduce the exact mathematical signatures.

The handoff explicitly cautions that any deeper layer need not be ordinary local classical statistics; it could be relational, contextual, nonlocal, globally constrained, or otherwise unlike familiar hidden-variable models.

38.11 The constructor question

A deeper question remains:

Can an observer fully reconstruct the process that constructed its own usable state space? (38.21)

Let:

$\mathcal{C}: \Sigma \rightarrow (O, \mathscr{H}, \mathscr{J}, L)$. (38.22)

The observer O investigates Σ through instruments $\mathscr{J}$.

But $\mathscr{J}$ was produced by $\mathcal{C}$.

This self-reference may create unavoidable epistemic limits.

However, the constructor idea must remain a research question.

It cannot substitute for quantitative physical derivation.

38.12 The quantum residue

After generic observer-bound structures are removed, the candidate residue includes:

QR = {Born Rule, Interference, Tensor Nonseparability, Bell Violation, Irreducible Contextuality, No-Cloning, Quantum Disturbance}. (38.23)

The purpose of the finance control model is not to dissolve QR through analogy.

It is to stop generic structures from being mistaken for QR.

38.13 A refined research equation

The comparative research programme can be expressed as:

Observed Quantum Strangeness = G_world + Q_residue. (38.24)

Where:

G_world = generic bounded-observer world-formation grammar. (38.25)

Q_residue = specifically quantum mathematical structure. (38.26)

The task is:

1. maximize the explanatory strength of G_world;
2. subtract only what the control model actually reproduces;
3. preserve the remainder honestly;
4. seek a theory that derives Q_residue without destroying established quantum predictions.

38.14 What would count as progress?

The programme would advance if it produced any of the following:

Progress A

A precise classification of quantum-measurement features into generic and irreducible components.

Progress B

A formal bounded-observer model reproducing contextual projection, trace, and backreaction.

Progress C

A proof that some alleged quantum mystery is classically reproducible under explicit memory and gate conditions.

Progress D

A no-go result showing that the macro grammar cannot reproduce a specific quantum signature.

Progress E

A derivation connecting a generative constructor to Hilbert-space structure and the Born rule.

The present article primarily contributes to Progress A and provides a conceptual platform for Progress B.

38.15 What would count as failure?

The physics programme fails if:

- analogy replaces derivation;
- finance terminology is presented as physical evidence;
- Bell constraints are ignored;
- hidden classical variables are assumed without a model;
- the Born rule is treated as a Pythagorean identity;
- contextuality is reduced to ordinary frame dependence;
- every quantum feature is declared generic without reproduction.

The methodological discipline is:

Subtract Only What the Control Model Can Actually Generate. (38.27)

38.16 The deeper lesson

The deeper lesson for physics is not:

The universe is a financial market. (38.28)

It is:

Effective worlds may be constructed through bounded projection, event gates, trace, and backreaction, while their internal laws remain simpler than the generative boundary that sustains them. (38.29)

Finance makes this possibility visible because its world-forming mechanisms are unusually explicit:

- contracts;

- prices;
- ledgers;
- authorities;
- filters;
- gates;
- revisions.

The same architecture may inspire more disciplined questions about physical observation without claiming material identity between the domains.

---

39. Conclusion: A World Is What Can Remember and Act Back

39.1 The starting problem

Modern finance repeatedly converts possibility into scalar value.

A future economic field enters a mature filter and exits as:

R. (39.1)

This scalar is operationally necessary.

But it can conceal the pressure implied by the filter that produced it.

Finance Geometry restores the complement:

$Z = R + iQ$. (39.2)

Where:

$A^2 = R^2 + Q^2$. (39.3)

$R = A \cos \theta$. (39.4)

$Q = A \sin \theta$. (39.5)

The original contribution was pressure-preserving valuation geometry.

The present article asked what happens when that geometry begins to move.

39.2 The dynamic contribution

Allow:

$A = A(t)$. (39.6)

$\theta = \theta(t)$. (39.7)

Then:

$Z(t) = A(t)\exp[i\theta(t)]$. (39.8)

The dynamic state satisfies:

$dZ/dt = (g_A + i\omega_F)Z + \varepsilon_{dyn}$. (39.9)

Where:

$g_A = (1/A)(dA/dt)$. (39.10)

$\omega_F = d\theta/dt$. (39.11)

The visible value evolves as:

$dR/dt = g_A R - Q\omega_F + Re(\varepsilon_{dyn})$. (39.12)

This produces the central attribution:

Visible Repricing = Radial Economic Change − Angular Filter Load + Residual. (39.13)

The angular repricing load is:

$\Lambda_F = Q\omega_F$. (39.14)

This distinction makes it possible, at least in principle, to separate:

- deterioration in economic amplitude;
- tightening of financial admissibility;
- failure of the declared model.

39.3 The temporal contribution

The article distinguishes three clocks:

Calendar Time = t. (39.15)

Phase Time = θ. (39.16)

Ledger Time = k. (39.17)

Calendar time measures duration.

Phase measures internal financial progression.

Ledger time advances when consequential events are committed.

The relation is:

$t \to \theta(t) \to G_k \to L_{k+1}$. (39.18)

Phase becomes a meaningful internal clock only when it provides stable local order.

Phase becomes historical only when gates and ledgers preserve consequence.

Therefore:

Phase Order + Gate + Trace = Effective Financial Time. (39.19)

39.4 The effective-world contribution

The complex state becomes world-like only after it is embedded in a larger runtime:

$W_P = (\mathcal{Z}_P, \mathcal{D}_P, \mathcal{G}_P, \boldsymbol{\mathcal{L}}_P, \mathcal{U}_P, \boldsymbol{\mathcal{B}}_P)$. (39.20)

The world contains:

- states;
- dynamics;
- gates;
- ledgers;
- interventions;
- backreaction.

Its full cycle is:

$X_k \to \Pi_{\{P_k, L_k\}} \to Z_k \to G_k \to L_{k+1} \to \boldsymbol{\mathcal{B}} \to X_{k+1} \to \boldsymbol{\mathcal{U}} \to P_{k+1}$. (39.21)

The financial world does not merely represent the primary field.

It acts upon it.

A price alters collateral.

A rating alters funding.

A ledger alters eligibility.

A model alters behaviour.

The world becomes causally operative.

39.5 The classical surprise

Under constant amplitude and stable declaration:

$dZ/d\theta = iZ$. (39.22)

Therefore:

$dR/d\theta = -Q$. (39.23)

$dQ/d\theta = R$. (39.24)

$d^2R/d\theta^2 = -R$. (39.25)

$d^2Q/d\theta^2 = -Q$. (39.26)

The internal effective law is classical.

This is not a failed result.

It identifies the division of labour within the theory.

Classical rotational dynamics describe movement inside the effective world.

Declaration, projection, gate, trace, residual, and backreaction describe how that world is formed.

The strongest thesis of the handoff is precisely that a mature finance filter can generate a protocol-bound complex state whose internal world is classical, while observer-related strangeness remains concentrated at its world-forming boundary.

39.6 The residual contribution

The article distinguishes:

Q = retained pressure inside the declared world. (39.27)

ε_{dyn} = failure of internal dynamic closure. (39.28)

ε_{Π} = omitted primary-field structure. (39.29)

E_L = unresolved consequence attached to a ledger record. (39.30)

H_{loop} = memory retained after apparent phase closure. (39.31)

This taxonomy prevents hidden pressure from becoming an unfalsifiable explanation for every discrepancy.

The framework remains honest only when residual can say:

The declared world no longer closes. (39.32)

39.7 The memory contribution

A financial cycle may return to its starting visible coordinate:

$R_T = R_0$. (39.33)

It may return to its starting phase:

$\theta_T = \theta_0$. (39.34)

But its ledger may remain changed:

$L_T \neq L_0$. (39.35)

Therefore:

Same Price ≠ Same History. (39.36)

Same Phase ≠ Same World. (39.37)

The loop residual captures:

- leverage scarring;
- ownership change;
- liquidity memory;
- legal history;
- regulatory change;
- altered expectations.

A world persists through what it remembers.

39.8 The declaration contribution

Not every crisis is movement inside an unchanged world.

Sometimes:

$P_{k+1} \neq P_k$. (39.38)

Going-concern valuation becomes liquidation valuation.

Ordinary liquidity becomes emergency liquidity.

Market pricing becomes model pricing.

A tradable asset becomes frozen collateral.

The world's coordinate grammar changes.

A legitimate revision must:

- preserve trace;
- disclose residual;
- translate old states where possible;
- identify what cannot be translated;
- resist retrospective manipulation.

Thus:

World Revision is Governed Reconstruction, Not Historical Erasure. (39.39)

39.9 The objectivity contribution

The effective financial world is constructed.

But it is not arbitrary.

Its objectivity arises from:

- declared protocols;
- reproducible operations;
- cross-frame invariants;
- authoritative gates;
- auditable ledgers;
- falsifiable predictions;
- trace-preserving revision.

The resulting position is:

Constructed Reality + Empirical Constraint + Causal Consequence = Effective Objectivity. (39.40)

This avoids both naïve realism and unrestricted relativism.

39.10 The physics contribution

Finance can reproduce, without quantum ontology:

- complex coordinates;
- phase;
- rotating frames;
- hidden complements;
- contextual outputs;
- state-changing observation;
- order effects;
- collapse-like commitment;
- irreversible trace;
- backreaction;
- phase-indexed internal time.

Therefore, none of these features is sufficient to identify uniquely quantum behaviour.

The subtraction method is:

Quantum Phenomenon = Generic Observer-Bound World Formation + Irreducibly Quantum Residue. (39.41)

The candidate residue includes:

- Born-rule probability;
- amplitude interference;
- tensor-product nonseparability;
- Bell violations;
- irreducible contextuality;
- no-cloning;
- quantum disturbance.

Finance does not derive these.

It makes their distinctiveness easier to locate.

39.11 The empirical boundary

The framework's future depends on evidence.

Its variables must be tested against:

- duration;
- convexity;
- beta;
- credit spread;
- volatility;
- liquidity measures;
- VaR;
- Expected Shortfall;
- regime-switching models;
- conventional factor decompositions.

The adoption rules remain:

No Mature Filter → No Valid Q. (39.42)

No Declared Protocol → No Interpretable θ. (39.43)

No Residualization → No Valid Multi-Q. (39.44)

No Diagnostic Gain → No Adoption. (39.45)

These cautions preserve the discipline of the original Finance Geometry framework, which treats Q as useful only where it helps identify, compare, communicate, or govern pressure hidden by

scalar valuation.

39.12 The complete runtime

The entire article can be compressed into one sequence:

Primary Field
→ Declared Protocol
→ Mature Filter
→ Admitted Value R
→ Retained Pressure Q
→ Phase θ
→ Internal Dynamics
→ Commitment Gate
→ Ledger Trace
→ Intervention
→ Backreaction
→ Residual Test
→ Protocol Revision. (39.46)

In symbolic form:

X_k
$\to P_k$
$\to \Pi_{P_k,L_k}$
$\to Z_k = R_k + iQ_k$
$\to \mathcal{D}_{P_k}$
$\to G_k$
$\to L_{k+1}$
$\to u_k$
$\to \boldsymbol{\mathcal{B}}$
$\to E_k$
$\to \boldsymbol{\mathcal{U}}$
$\to P_{k+1}$. (39.47)

39.13 Final propositions

The article's main propositions are:

Proposition 1

A mature scalar finance filter can be completed into a pressure-preserving complex state. (39.48)

Proposition 2

Dynamic repricing can be decomposed into radial amplitude change, angular filter movement, and residual. (39.49)

Proposition 3

A monotonic filter angle can serve as a local internal ordering coordinate. (39.50)

Proposition 4

Phase becomes historical time only through gate-passed, ledgered consequence. (39.51)

Proposition 5

The internal R–Q phase law is classical. (39.52)

Proposition 6

Observer-related complexity lies mainly in declaration, projection, gate, trace, residual, and backreaction. (39.53)

Proposition 7

A reduced financial representation becomes world-like when it can remember and act back. (39.54)

Proposition 8

Finance provides a non-quantum control case for separating generic observer-bound strangeness from irreducibly quantum structure. (39.55)

Proposition 9

Every practical and physics-facing claim remains conditional upon falsification, protocol robustness, and residual honesty. (39.56)

39.14 Final reflection

A price is not the whole asset.

It is what a declared financial world admits.

A hidden pressure coordinate is not another price.

It is the geometric complement required to preserve what the filter did not place on the visible axis.

A phase is not automatically time.

It becomes time-like when it orders transitions, passes through gates, and leaves records.

A ledger is not merely storage.

It bends the next projection by carrying forward what the world has committed.

A valuation is not merely descriptive when it changes collateral, funding, capital, behaviour, or law.

At that point, valuation participates in the reality it measures.

The deepest result of the article is therefore not that finance resembles physics.

It is that a mature system of filters, gates, records, and feedback can construct a secondary world

whose internal laws are simpler than the boundary that creates it.

That world is not fundamental.

It is not complete.

It is not arbitrary.

It is effective because agents can live, decide, remember, and suffer consequences inside it.

The final principle is:

A World Is Not Merely What Can Be Observed. (39.57)

A World Is What Can Preserve a Difference, Commit an Event, Remember the Consequence, and Act Back upon What Comes Next. (39.58)

Appendix A — Complete Notation Dictionary

A.1 Purpose of the notation system

The notation distinguishes five layers that must not be collapsed into one another:

1. the primary economic field;
2. the declared observer protocol;
3. the projected financial state;
4. the ledgered historical state;
5. the residual left outside successful closure.

The original Finance Geometry variables A, R, Q, θ, and Z remain the geometric core. The expanded notation adds the declaration, projection, gate, trace, backreaction, and revision structures required for a dynamic effective world.

A.2 Primary-field notation

X

X denotes the larger primary economic or financial state before compression into the selected effective world.

$X \in \mathcal{X}$. (A.1)

Depending on the application, X may contain:

- expected cash flows;
- balance-sheet variables;
- contractual rights;
- market conditions;

- counterparties;
- liquidity networks;
- legal constraints;
- investor behaviour;
- regulatory conditions;
- latent scenarios.

X should not be interpreted as a claim that the researcher possesses the complete ontological state of the financial system.

It denotes the larger state relative to the selected projection.

$\mathcal{X}$

$\mathcal{X}$ is the declared primary-state space.

It may be:

- a vector space;
- a probability space;
- a database schema;
- a state-space model;
- a scenario family;
- a graph or network;
- a hybrid continuous–discrete state space.

The declaration of $\mathcal{X}$ determines which kinds of primary structure the model considers available.

$\tilde{X}$

$\tilde{X}$ is a protocol-relative reconstruction of the primary field from the effective state and its ledger.

$\tilde{X} = \mathcal{R}_P(Z,L)$. (A.2)

In most applications:

$\tilde{X} \neq X$. (A.3)

The mismatch defines projection residual.

A.3 Protocol notation

P

P is the declared world-forming protocol:

$$P = (B,q,\varphi,h,F,G,T,U). \quad (A.4)$$

A protocol determines what the observer treats as:

- inside the system;
- relevant evidence;
- the baseline;
- the valuation horizon;
- the mature filter;
- a committed event;
- a valid record;
- an admissible intervention.

B

B is the system boundary.

It specifies:

- which assets are included;
- which legal entities are included;
- which markets or jurisdictions matter;
- which counterparties belong to the analysed system;
- which external variables are treated as environment.

Changing B can change the meaning of every downstream variable.

---

q

q is the baseline or reference state.

Examples include:

- risk-free rate;
- default-free value;
- expected cash flow;
- frictionless executable value;
- unconstrained economic value;
- pre-crisis reference state.

The baseline is not necessarily neutral.

It is a declared comparison point.

φ

φ is the feature map.

φ: X → Feature Space. (A.5)

It determines which primary-field structures are made available to the filter.

For example:

φ(X) = (CashFlow,Rate,Beta,Spread,Liquidity,Rating). (A.6)

A feature omitted by φ may later appear as projection residual.

---

h

h is the observation, prediction, or valuation horizon.

Examples include:

- one trading day;
- one quarter;
- bond maturity;
- project life;
- stress-testing horizon;
- immediate liquidation horizon.

Changing h may alter A, R, θ, Q, and the meaning of the gate.

F

F is the mature finance filter.

Examples include:

- CAPM;
- certainty-equivalent adjustment;
- credit-spread valuation;
- liquidity haircut;
- accounting recognition;
- collateral eligibility;

- capital-admissibility rule;
- option-exercise framework.

The filter must exist before θ and Q are defined.

G

G is the admission or commitment gate.

G: Candidate State → {Commit,Reject,Defer}. (A.7)

Examples include:

- trade-execution rule;
- default threshold;
- covenant test;
- impairment test;
- rating decision;
- collateral-eligibility decision;
- exercise decision.

T

T is the trace rule.

T determines:

- what enters the ledger;
- which evidence is attached;
- how residual is preserved;
- how revisions are recorded;
- whether records may be corrected or reversed.

U

U is the admissible intervention set.

u ∈ U. (A.8)

Possible interventions include:

- trade;
- hedge;

- refinance;
- liquidate;
- provide liquidity;
- post collateral;
- exercise;
- restructure;
- change a limit;
- revise the protocol.

A.4 Projection notation

Π

Π is the projection from the primary field into the effective financial state.

$Z_k = \Pi_\{P_k,L_k\}(X_k)$. (A.9)

The subscripts emphasize that projection depends on:

- the current declaration;
- the accumulated ledger.

A history-bearing observer does not necessarily project the same X identically after a major event.

$\mathscr{R}$

$\mathscr{R}$ is the reconstruction map.

$\tilde{X} = \mathscr{R}_P(Z,L)$. (A.10)

$\mathscr{R}$ need not be a true inverse of Π.

Projection can be many-to-one:

$\Pi_P(X_a) = \Pi_P(X_b)$ even when $X_a \neq X_b$. (A.11)

A.5 Core Finance Geometry variables

A

A is the declared pre-filter value amplitude.

$A \geq 0$. (A.12)

A may represent:

- base-discounted value;
- expected-value amplitude;
- default-free value;
- ideal executable value;
- unconstrained economic value;
- latent option value.

A does not automatically mean metaphysically true or intrinsic value.

R

R is admitted value.

R is the component that passes through the declared mature filter and becomes visible on the real axis.

In the minimal non-negative domain:

$0 \le R \le A$. (A.13)

Q

Q is retained pressure under the declared filter.

$Q = \sqrt{(A^2 - R^2)}$. (A.14)

In the minimal model:

$Q \ge 0$. (A.15)

Q must be labelled by its source when ambiguity is possible:

Q_{CAPM}. (A.16)

Q_{credit}. (A.17)

$Q_{liquidity}$. (A.18)

$Q_{capital}$. (A.19)

θ

θ is the finance-filter angle.

$\theta = \arccos(R/A)$. (A.20)

In the minimal domain:

$0 \le \theta \le \pi/2$. (A.21)

θ is not freely chosen.

It is induced by the mature filter ratio R/A.

Z

Z is the complex effective financial state:

$Z = R + iQ$. (A.22)

Equivalently:

$Z = A \exp(i\theta)$. (A.23)

Its magnitude is:

$|Z| = A$. (A.24)

H

H is the scalar haircut:

$H = A - R$. (A.25)

It must not be confused with Q.

Near $\theta = 0$:

$H \approx Q^2/(2A)$. (A.26)

A.6 Dynamic notation

t

t is external calendar time.

It may be continuous or sampled discretely.

---

k

k indexes ledger events.

$k \to k + 1$ (A.27)

occurs when a new committed record enters the ledger.

---

g_A

g_A is the proportional radial growth rate:

$g_A = (1/A)(dA/dt)$. (A.28)

It captures change in the declared economic amplitude.

ω_F

ω_F is the angular velocity of the financial filter:

$\omega_F = d\theta/dt$. (A.29)

Positive ω_F represents tightening under the chosen sign convention.

Negative ω_F represents relaxation.

α_F

α_F is angular acceleration:

$\alpha_F = d^2\theta/dt^2$. (A.30)

It measures whether frame movement itself is accelerating.

Π_Q

Π_Q is the pressure ratio:

$\Pi_Q = Q/R = \tan\theta$. (A.31)

It is the retained-pressure intensity per unit of admitted value.

Λ_F

Λ_F is angular repricing load:

$\Lambda_F = Q\omega_F$. (A.32)

The angular contribution to admitted-value change is:

$dR/dt|_{angular} = -\Lambda_F$. (A.33)

$\mathcal{D}_P$

$\mathcal{D}_P$ is the internal dynamic law under protocol P.

A general form is:

$dZ/dt = \mathcal{D}_P(Z,u,t) + \varepsilon_{dyn}$. (A.34)

In the minimal model:

$\mathcal{D}_P(Z) = (g_A + i\omega_F)Z$. (A.35)

A.7 Temporal notation

θ-time

When θ is locally monotonic, it may serve as an internal order coordinate.

$dY/d\theta = (dY/dt)/(d\theta/dt)$. (A.36)

This remains local because θ may reverse or become ambiguous.

τ_F

τ_F denotes an effective financial-time coordinate.

One possible construction is:

$d\tau_F = w(\theta,Q,G,L)|d\theta|$. (A.37)

The weight w determines which phase increments become historically significant.

τ_L

τ_L denotes ledger time.

It is the ordered sequence of committed records.

$\tau_L = order(L_0,L_1,L_2,...)$. (A.38)

A.8 Ledger notation

L

L is the accumulated ledger.

$L_{k+1} = L_k \sqcup Record_k$. (A.39)

A stronger record is:

$L_{k+1} = L_k \sqcup (Record_k,Evidence_k,GateMetadata_k,Residual_k)$. (A.40)

⊔

⊔ is the trace-preserving ledger-join operation.

It does not imply ordinary numerical addition.

It means that a new record is attached without erasing prior history.

$Record_k$

$Record_k$ is a committed event.

A general record is:

$Record_k = (Outcome_k, Evidence_k, Authority_k, ProtocolVersion_k, Residual_k)$. (A.41)

M_G,k

M_G,k is gate metadata:

$M_{G,k} = (Authority_k, Evidence_k, Threshold_k, Time_k, ProtocolVersion_k)$. (A.42)

A.9 Residual notation

ε_dyn

ε_dyn is calendar-time dynamic residual:

$\varepsilon_{dyn} = dZ/dt - (g_A + i\omega_F)Z$. (A.43)

Write:

$\varepsilon_{dyn} = \varepsilon_R + i\varepsilon_Q$. (A.44)

ε_θ

ε_θ is phase-time residual:

$\varepsilon_\theta = dZ/d\theta - [(d \ln A/d\theta) + i]Z$. (A.45)

For constant A:

$\varepsilon_\theta = dZ/d\theta - iZ$. (A.46)

ε_Π

ε_Π is projection residual:

$\varepsilon_\Pi = X - \tilde{X}$. (A.47)

It contains primary-field structure omitted by the effective projection.

E_L

E_L is ledger residual.

It records unresolved consequence after a gate has committed an event.

H_loop

H_loop is loop residual or financial memory:

H_loop = $\mathcal{S}$(X_T,L_T,P_T) − $\mathcal{S}$(X0,L0,P0). (A.48)

It may remain nonzero even when:

R_T = R0 (A.49)

or:

θ_T = θ0. (A.50)

ε_transition

ε_transition is cross-world transition residual:

ε_transition = Z_{P′}(t*) − M_{P→P′}[Z_P(t*−)]. (A.51)

It measures structure in the new world that cannot be reconstructed from the old one.

A.10 Backreaction and revision notation

$\mathcal{B}$

$\mathcal{B}$ is the backreaction map:

$X_{k+1} = \mathcal{B}(X_k,Z_k,L_{k+1},P_k,u_k)$. (A.52)

It models how projected and ledgered financial states alter the next primary state.

$\mathcal{U}$

$\mathcal{U}$ is the protocol-revision operator:

$P_{k+1} = \mathcal{U}(P_k,L_{k+1},E_k)$. (A.53)

A valid revision should preserve trace and disclose what changed.

M_{P→P′}

M_{P→P′} is a partial translation between effective worlds:

M_{P→P′}: $\mathcal{Z}$_P → $\mathcal{Z}$_{P′}. (A.54)

It should specify:

- preserved variables;
- transformed variables;
- discarded variables;
- untranslated residual.

A.11 Multi-pressure notation

$\vec{Q}$

$\vec{Q}$ is the pressure vector:

$\vec{Q}$ = (Q_market,Q_credit,Q_liquidity,Q_capital,Q_tail,Q_model,...). (A.55)

G

When used in multi-pressure geometry, G is the pressure metric.

This use must be distinguished from the gate symbol G by context or by writing G_metric.

The generalized magnitude is:

$A^2 = R^2 + \vec{Q}^{\top} G_metric \vec{Q}$. (A.56)

$\vec{\omega}$

$\vec{\omega}$ is the vector of pressure-channel angular velocities:

$\vec{\omega}$ = (ω_market,ω_credit,ω_liquidity,...). (A.57)

The total angular repricing load is:

Λ_total = $\vec{\omega}^{\top}\vec{Q}$. (A.58)

C_Q

C_Q is the pressure-coupling matrix.

The multi-pressure dynamics may include:

$d\vec{Q}/dt = g_A\vec{Q} + \vec{\omega}R + C_Q\vec{Q} + \varepsilon_Q$. (A.59)

A.12 Quantum-comparison notation

G_world

G_world denotes generic observer-bound world-forming structure.

It includes:

- declaration;
- contextual projection;
- gate;

- trace;
- residual;
- backreaction;
- revision.

Q_residue

Q_residue denotes candidate irreducibly quantum structure.

It must not be confused with financial pressure Q.

A safer full notation is:

$\mathcal{Q}$_residue. (A.60)

The comparative decomposition is:

Observed Quantum Strangeness = G_world + $\mathcal{Q}$_residue. (A.61)

$\mathcal{C}$

$\mathcal{C}$ is the constructor-like generative map:

$\mathcal{C}$: Σ → (O,$\mathscr{H}$,$\mathscr{J}$,L). (A.62)

Where:

- Σ = deeper generative process;
- O = observer;
- $\mathscr{H}$ = usable state space;
- $\mathscr{J}$ = admissible instrument set;
- L = record architecture.

This is a conceptual research model, not a derived physical law.

Appendix B — Derivation of the Radial–Angular Equations

B.1 Starting representation

Begin with the dynamic complex state:

Z(t) = A(t)exp[iθ(t)]. (B.1)

Where:

$A(t) \geq 0$. (B.2)

The corresponding Cartesian coordinates are:

$R(t) = A(t)\cos\theta(t)$. (B.3)

$Q(t) = A(t)\sin\theta(t)$. (B.4)

B.2 Direct complex differentiation

Differentiate equation (B.1):

$dZ/dt = (dA/dt)\exp(i\theta) + A\exp(i\theta)i(d\theta/dt)$. (B.5)

Factor out $A\exp(i\theta)$:

$dZ/dt = [(1/A)(dA/dt) + i(d\theta/dt)]A\exp(i\theta)$. (B.6)

Since:

$A\exp(i\theta) = Z$, (B.7)

we obtain:

$dZ/dt = [(1/A)(dA/dt) + i(d\theta/dt)]Z$. (B.8)

Define:

$g_A = (1/A)(dA/dt)$. (B.9)

$\omega_F = d\theta/dt$. (B.10)

Then:

$dZ/dt = (g_A + i\omega_F)Z$. (B.11)

With dynamic residual:

$dZ/dt = (g_A + i\omega_F)Z + \varepsilon_{dyn}$. (B.12)

B.3 Cartesian expansion

Write:

$Z = R + iQ$. (B.13)

Then:

$(g_A + i\omega_F)Z = (g_A + i\omega_F)(R + iQ)$. (B.14)

Expanding:

$(g_A + i\omega_F)(R + iQ) = g_AR + ig_AQ + i\omega_FR - \omega_FQ$. (B.15)

Collect real and imaginary terms:

$(g_A + i\omega_F)Z = (g_{AR} - \omega_{FQ}) + i(g_{AQ} + \omega_{FR})$. (B.16)

Therefore:

$dR/dt = g_{AR} - \omega_{FQ} + \mathrm{Re}(\varepsilon_{dyn})$. (B.17)

$dQ/dt = g_{AQ} + \omega_{FR} + \mathrm{Im}(\varepsilon_{dyn})$. (B.18)

B.4 Direct Cartesian verification

Starting from:

$R = A\cos\theta$, (B.19)

differentiate:

$dR/dt = (dA/dt)\cos\theta - A\sin\theta(d\theta/dt)$. (B.20)

Use:

$R = A\cos\theta$, (B.21)

$Q = A\sin\theta$, (B.22)

$g_A = (1/A)(dA/dt)$, (B.23)

$\omega_F = d\theta/dt$. (B.24)

Then:

$(dA/dt)\cos\theta = g_{AR}$. (B.25)

$A\sin\theta(d\theta/dt) = Q\omega_F$. (B.26)

Therefore:

$dR/dt = g_{AR} - Q\omega_F$. (B.27)

Similarly, starting from:

$Q = A\sin\theta$, (B.28)

differentiate:

$dQ/dt = (dA/dt)\sin\theta + A\cos\theta(d\theta/dt)$. (B.29)

Therefore:

$dQ/dt = g_{AQ} + R\omega_F$. (B.30)

This verifies the complex derivation.

B.5 Magnitude evolution

The squared magnitude is:

$A^2 = R^2 + Q^2$. (B.31)

Differentiate:

$2A(dA/dt) = 2R(dR/dt) + 2Q(dQ/dt)$. (B.32)

Divide by 2:

$A(dA/dt) = R(dR/dt) + Q(dQ/dt)$. (B.33)

Insert the ideal dynamic equations:

$R(dR/dt) + Q(dQ/dt) = R(g_AR - \omega_FQ) + Q(g_AQ + \omega_FR)$. (B.34)

The angular cross terms cancel:

$-R\omega_FQ + Q\omega_FR = 0$. (B.35)

Therefore:

$R(dR/dt) + Q(dQ/dt) = g_A(R^2 + Q^2)$. (B.36)

Since:

$R^2 + Q^2 = A^2$, (B.37)

we obtain:

$A(dA/dt) = g_AA^2$. (B.38)

Using:

$g_A = (1/A)(dA/dt)$, (B.39)

the identity is satisfied.

This shows:

- radial change alters magnitude;
- angular change redistributes R and Q while preserving magnitude.

B.6 Constant-amplitude case

If:

$dA/dt = 0$, (B.40)

then:

$g_A = 0$. (B.41)

The dynamics become:

$dZ/dt = i\omega_FZ$. (B.42)

$dR/dt = -\omega_FQ$. (B.43)

$$dQ/dt = \omega_{FR}. \quad (B.44)$$

The magnitude is conserved:

$$d(R^2 + Q^2)/dt = 0. \quad (B.45)$$

B.7 Phase-coordinate law

If:

$$\omega_F = d\theta/dt \neq 0, \quad (B.46)$$

then:

$$dZ/d\theta = (dZ/dt)/(d\theta/dt). \quad (B.47)$$

Using:

$$dZ/dt = (g_A + i\omega_F)Z, \quad (B.48)$$

we obtain:

$$dZ/d\theta = [(g_A/\omega_F) + i]Z. \quad (B.49)$$

But:

$$g_A/\omega_F = [(1/A)(dA/dt)]/(d\theta/dt). \quad (B.50)$$

By the chain rule:

$$g_A/\omega_F = (1/A)(dA/d\theta) = d \ln A/d\theta. \quad (B.51)$$

Therefore:

$$dZ/d\theta = [(d \ln A/d\theta) + i]Z. \quad (B.52)$$

For constant amplitude:

$$d \ln A/d\theta = 0. \quad (B.53)$$

Hence:

$$dZ/d\theta = iZ. \quad (B.54)$$

B.8 Phase-coordinate Cartesian equations

Write:

$$dZ/d\theta = dR/d\theta + i(dQ/d\theta). \quad (B.55)$$

The right-hand side is:

$$[(d \ln A/d\theta) + i](R + iQ). \quad (B.56)$$

Expanding:

$$[(d \ln A/d\theta) + i](R + iQ) = [(d \ln A/d\theta)R - Q] + i[(d \ln A/d\theta)Q + R]. \quad (B.57)$$

Therefore:

$$dR/d\theta = (d \ln A/d\theta)R - Q. \quad (B.58)$$

$$dQ/d\theta = (d \ln A/d\theta)Q + R. \quad (B.59)$$

For constant A:

$$dR/d\theta = -Q. \quad (B.60)$$

$$dQ/d\theta = R. \quad (B.61)$$

B.9 Second-order constant-amplitude equations

Differentiate:

$$dR/d\theta = -Q. \quad (B.62)$$

Then:

$$d^2R/d\theta^2 = -dQ/d\theta. \quad (B.63)$$

Using:

$$dQ/d\theta = R, \quad (B.64)$$

we obtain:

$$d^2R/d\theta^2 = -R. \quad (B.65)$$

Thus:

$$d^2R/d\theta^2 + R = 0. \quad (B.66)$$

Similarly:

$$d^2Q/d\theta^2 + Q = 0. \quad (B.67)$$

These are classical harmonic-oscillator equations in phase coordinates.

B.10 Logarithmic admitted-value equation

From:

$$dR/dt = g_AR - \omega_FQ + \varepsilon_R, \quad (B.68)$$

divide by R:

$$(1/R)(dR/dt) = g_A - \omega_F(Q/R) + \varepsilon_R/R. \quad (B.69)$$

Since:

$$Q/R = \tan \theta, \quad (B.70)$$

we obtain:

$d \ln R/dt = g_A - \omega_F \tan \theta + \varepsilon_R/R$. (B.71)

This gives:

Visible-Value Growth = Radial Growth − Angular Pressure Rate + Normalized Residual. (B.72)

B.11 Angular repricing load

Define:

$\Lambda_F = Q\omega_F$. (B.73)

Then:

$dR/dt = g_AR - \Lambda_F + \varepsilon_R$. (B.74)

The integrated form over $[t_0,t_1]$ is:

$R(t_1) - R(t_0) = \int_{t_0}^{t_1} g_A(t)R(t)dt - \int_{t_0}^{t_1} Q(t)\omega_F(t)dt + \int_{t_0}^{t_1} \varepsilon_R(t)dt$. (B.75)

Because:

$\omega_F dt = d\theta$, (B.76)

the angular contribution is:

$-\int_{t_0}^{t_1} Q(t)\omega_F(t)dt = -\int_{\theta_0}^{\theta_1} Q(\theta)d\theta$. (B.77)

Under constant A:

$Q(\theta) = A \sin \theta$. (B.78)

Therefore:

$-\int_{\theta_0}^{\theta_1} A \sin \theta \, d\theta = A[\cos \theta_1 - \cos \theta_0]$. (B.79)

Since:

$R_j = A \cos \theta_j$, (B.80)

we recover:

$-\int_{\theta_0}^{\theta_1} Q d\theta = R_1 - R_0$. (B.81)

Thus, under constant amplitude and zero residual, the entire visible-value change is exactly the integrated angular repricing load.

B.12 Discrete-time approximation

For sampled observations separated by Δt:

$g_{A,t} \approx [\ln A_t - \ln A_{t-1}]/\Delta t$. (B.82)

$\omega_{F,t} \approx (\theta_t - \theta_{t-1})/\Delta t$. (B.83)

A midpoint approximation is:

$\bar{R}_t = (R_t + R_{t-1})/2$. (B.84)

$\bar{Q}_t = (Q_t + Q_{t-1})/2$. (B.85)

Then:

$\Delta R_t \approx g_{A,t}\bar{R}_t\Delta t - \bar{Q}_t\omega_{F,t}\Delta t + \varepsilon_{R,t}\Delta t$. (B.86)

Equivalently:

$\Delta R_t \approx g_{A,t}\bar{R}_t\Delta t - \bar{Q}_t\Delta\theta_t + \varepsilon_{R,t}\Delta t$. (B.87)

For large phase moves, numerical integration of $Q(\theta)$ is preferable to a single midpoint approximation.

B.13 Angular-load acceleration

Starting from:

$\Lambda_F = Q\omega_F$, (B.88)

differentiate:

$d\Lambda_F/dt = (dQ/dt)\omega_F + Q(d\omega_F/dt)$. (B.89)

Define:

$\alpha_F = d\omega_F/dt$. (B.90)

Using:

$dQ/dt = g_A Q + \omega_F R + \varepsilon_Q$, (B.91)

we obtain:

$d\Lambda_F/dt = g_A Q\omega_F + \omega_F^2 R + \varepsilon_Q\omega_F + Q\alpha_F$. (B.92)

Since:

$\Lambda_F = Q\omega_F$, (B.93)

then:

$d\Lambda_F/dt = g_A\Lambda_F + \omega_F^2 R + Q\alpha_F + \varepsilon_Q\omega_F$. (B.94)

This equation separates growth in angular repricing load into:

- radial scaling;
- sustained rotation;
- angular acceleration;
- pressure-coordinate residual.

Appendix C — Worked CAPM Numerical Example

C.1 Purpose of the example

This example illustrates three distinct paths leading to similar admitted value:

1. pure frame tightening;
2. pure amplitude deterioration;
3. mixed amplitude and frame change.

The example is not intended as an investment model.

It shows how the radial–angular decomposition changes the interpretation of the same visible price.

C.2 Initial declaration

Assume a future cash flow:

$CF_T = 121.$ (C.1)

Assume a two-year horizon:

$T = 2.$ (C.2)

Assume a declared base rate:

$r_* = 0.05.$ (C.3)

The base-discounted amplitude is:

$A_0 = 121/(1.05)^2.$ (C.4)

Therefore:

$A_0 \approx 109.7506.$ (C.5)

Assume:

$r_f = 0.03.$ (C.6)

$\beta = 1.00.$ (C.7)

$ERP_0 = 0.04.$ (C.8)

The initial CAPM rate is:

$r_{CAPM,0} = r_f + \beta ERP_0.$ (C.9)

Therefore:

$r_{CAPM,0} = 0.07.$ (C.10)

C.3 Initial admitted value

The initial CAPM-admitted value is:

$R_0 = 121/(1.07)^2$. (C.11)

Therefore:

$R_0 \approx 105.6861$. (C.12)

The filter ratio is:

$R_0/A_0 \approx 0.9630$. (C.13)

The initial angle is:

$\theta_0 = \arccos(R_0/A_0)$. (C.14)

Therefore:

$\theta_0 \approx 0.2730$ radians. (C.15)

The initial pressure coordinate is:

$Q_0 = \sqrt{(A_0^2 - R_0^2)}$. (C.16)

Therefore:

$Q_0 \approx 29.5912$. (C.17)

The initial complex state is:

$Z_0 \approx 105.6861 + i29.5912$. (C.18)

The initial pressure ratio is:

$Q_0/R_0 \approx 0.2800$. (C.19)

C.4 Scenario A — Pure frame tightening

Suppose expected cash flow and the base rate remain unchanged.

Therefore:

$A_1 = A_0 \approx 109.7506$. (C.20)

Suppose the equity-risk premium rises:

$ERP_1 = 0.07$. (C.21)

The new CAPM rate is:

$r_{CAPM,1} = 0.03 + 1.00(0.07)$. (C.22)

Therefore:

$r_{CAPM,1} = 0.10$. (C.23)

The new admitted value is:

$R_1 = 121/(1.10)^2$. (C.24)

Therefore:

$R_1 = 100.0000$. (C.25)

The new angle is:

$\theta_1 = \arccos(R_1/A_1)$. (C.26)

Therefore:

$\theta_1 \approx 0.4247$ radians. (C.27)

The new pressure coordinate is:

$Q_1 = \sqrt{(A_1^2 - R_1^2)}$. (C.28)

Therefore:

$Q_1 \approx 45.2237$. (C.29)

The new complex state is:

$Z_1 \approx 100.0000 + i45.2237$. (C.30)

C.5 Endpoint changes in Scenario A

The amplitude change is:

$\Delta A = A_1 - A_0 = 0$. (C.31)

The admitted-value change is:

$\Delta R = R_1 - R_0$. (C.32)

Therefore:

$\Delta R \approx -5.6861$. (C.33)

The pressure change is:

$\Delta Q = Q_1 - Q_0$. (C.34)

Therefore:

$\Delta Q \approx 15.6326$. (C.35)

The phase change is:

$\Delta\theta = \theta_1 - \theta_0$. (C.36)

Therefore:

$\Delta\theta \approx 0.1517$ radians. (C.37)

This is a pure frame event under the declared protocol:

$\Delta A = 0$. (C.38)

$\Delta\theta > 0$. (C.39)

R falls while Q rises.

C.6 Exact angular attribution in Scenario A

Because A is constant and residual is assumed zero:

$\Delta R = -\int_{\theta_0}^{\theta_1} Q(\theta)\,d\theta$. (C.40)

Using:

$Q(\theta) = A \sin\theta$, (C.41)

we obtain:

$\Delta R = -\int_{\theta_0}^{\theta_1} A \sin\theta\, d\theta$. (C.42)

Therefore:

$\Delta R = A(\cos\theta_1 - \cos\theta_0)$. (C.43)

Since:

$A\cos\theta_1 = R_1$, (C.44)

$A\cos\theta_0 = R_0$, (C.45)

then:

$\Delta R = R_1 - R_0 \approx -5.6861$. (C.46)

The entire visible decline is angular under this declaration.

C.7 Midpoint approximation

The midpoint pressure is:

$\bar{Q} = (Q_0 + Q_1)/2$. (C.47)

Therefore:

$\bar{Q} \approx 37.4075$. (C.48)

The midpoint estimate of the angular change is:

$\Delta R_{angular} \approx -\bar{Q}\Delta\theta$. (C.49)

Therefore:

$\Delta R_{angular} \approx -37.4075(0.1517)$. (C.50)

So:

$\Delta R_{angular} \approx -5.675$. (C.51)

The approximation is close to the exact decline of approximately −5.6861.

The small difference arises because Q changes along the phase path.

C.8 Scenario A interpretation

A scalar account says:

The value fell from approximately 105.69 to 100. (C.52)

The geometric account says:

- the declared economic amplitude remained constant;
- the market-risk frame tightened;
- the finance angle increased;
- retained market-risk pressure rose;
- the visible decline was angular rather than radial.

This suggests an intervention focus on:

- risk-premium exposure;
- discount-rate sensitivity;
- hedging;
- funding and timing;
- market-frame normalization.

It does not, by itself, imply deterioration in the expected cash flow.

C.9 Scenario B — Pure amplitude deterioration

Return to the original CAPM rate:

$r_{CAPM} = 0.07$. (C.53)

Keep:

$r_* = 0.05$. (C.54)

$T = 2$. (C.55)

Now reduce the expected future cash flow to:

$CF_{T,1} = 114.49$. (C.56)

The new amplitude is:

$A_1 = 114.49/(1.05)^2$. (C.57)

Therefore:

$A_1 \approx 103.8458$. (C.58)

The new admitted value is:

$R_1 = 114.49/(1.07)^2$. (C.59)

Therefore:

$R_1 \approx 100.0000$. (C.60)

Thus, the final visible value is approximately the same as in Scenario A.

C.10 Scenario B angle and pressure

Because both values use the original rate ratio:

$R_1/A_1 = [(1.05)/(1.07)]^2$. (C.61)

Therefore:

$\theta_1 = \theta_0 \approx 0.2730$ radians. (C.62)

The new pressure coordinate is:

$Q_1 = \sqrt{(A_1^2 - R_1^2)}$. (C.63)

Therefore:

$Q_1 \approx 28.013$. (C.64)

The new complex state is approximately:

$Z_1 \approx 100.0000 + i28.013$. (C.65)

C.11 Endpoint changes in Scenario B

The amplitude declines:

$\Delta A \approx 103.8458 - 109.7506$. (C.66)

Therefore:

$\Delta A \approx -5.9048$. (C.67)

The angle remains approximately unchanged:

$\Delta\theta \approx 0$. (C.68)

The admitted value declines:

$\Delta R \approx -5.6861$. (C.69)

The pressure coordinate also declines:

$\Delta Q \approx 28.013 - 29.5912$. (C.70)

Therefore:

$\Delta Q \approx -1.578$. (C.71)

This is radial contraction.

Both R and Q shrink while their ratio remains approximately stable.

C.12 Scenario B interpretation

The scalar endpoint is again approximately:

$R_1 = 100$. (C.72)

But the geometry differs from Scenario A.

Scenario A

$A \approx 109.7506$. (C.73)

$Q \approx 45.2237$. (C.74)

$\theta \approx 0.4247$. (C.75)

Scenario B

$A \approx 103.8458$. (C.76)

$Q \approx 28.013$. (C.77)

$\theta \approx 0.2730$. (C.78)

The same admitted value belongs to different effective financial states.

Scenario A is pressure-heavy.

Scenario B is amplitude-reduced.

C.13 Comparative table

Variable	Initial	Scenario A: Frame Tightening	Scenario B: Amplitude Decline
CF_T	121.0000	121.0000	114.4900
A	109.7506	109.7506	103.8458
r_CAPM	7.00%	10.00%	7.00%
R	105.6861	100.0000	100.0000
θ	0.2730	0.4247	0.2730

Variable	Initial	Scenario A: Frame Tightening	Scenario B: Amplitude Decline
Q	29.5912	45.2237	≈28.013
Main movement	initial	angular	radial

The table demonstrates:

Same R Does Not Imply Same A, Q, θ, or Causal History. (C.79)

C.14 Scenario C — Mixed improvement and tightening

Suppose expected future cash flow rises to:

$CF_{T,1} = 127.05$. (C.80)

This is a 5% increase from 121.

The new amplitude is:

$A_1 = 127.05/(1.05)^2$. (C.81)

Therefore:

$A_1 \approx 115.2381$. (C.82)

Suppose the CAPM rate simultaneously rises to:

$r_{CAPM,1} = 0.12$. (C.83)

The admitted value becomes:

$R_1 = 127.05/(1.12)^2$. (C.84)

Therefore:

$R_1 \approx 101.283$. (C.85)

The asset's economic amplitude has grown.

Yet admitted value remains below the initial value of approximately 105.6861.

The tightening filter more than offsets part of the growth.

C.15 Scenario C geometry

The angle is:

$\theta_1 = \arccos(R_1/A_1)$. (C.86)

The pressure coordinate is:

$Q_1 = \sqrt{(A_1^2 - R_1^2)}$. (C.87)

The qualitative attribution is:

$g_A > 0$. (C.88)

$\omega_F > 0$. (C.89)

dR/dt may be negative if:

$Q\omega_F > g_{AR}$. (C.90)

Thus:

Growing Economic Capacity + Faster Filter Tightening → Falling Admitted Value. (C.91)

This is a common but often misunderstood market condition.

C.16 Scenario D — Deterioration masked by frame relaxation

Suppose future cash flow declines while the CAPM rate falls sharply.

Then:

$g_A < 0$. (C.92)

$\omega_F < 0$. (C.93)

The admitted value can rise when:

$-Q\omega_F > |g_{AR}|$. (C.94)

In words:

The positive effect of frame relaxation exceeds the negative effect of amplitude deterioration. (C.95)

A rising market value would then not necessarily indicate stronger economic capacity.

This is the inverse attribution error.

C.17 Why endpoint geometry is not enough

Two endpoints provide:

A_0, R_0, Q_0, θ_0 (C.96)

and:

A_1, R_1, Q_1, θ_1. (C.97)

But they do not reveal the intermediate path.

The path may contain:

- phase reversal;
- gate events;

- temporary residual spikes;
- liquidity interruption;
- protocol revision.

Therefore, a full empirical analysis requires:

- intermediate observations;
- event ledger;
- protocol version history;
- residual estimation.

Endpoint geometry describes state difference.

Runtime geometry describes how the world moved.

C.18 Minimum audit footer for the example

A reproducible implementation should publish:

Protocol = CAPM filter relative to fixed 5% base rate. (C.98)

Horizon = two years. (C.99)

Cash-flow convention = one terminal cash flow. (C.100)

Beta convention = constant $\beta = 1$. (C.101)

ERP convention = independently declared premium. (C.102)

Compounding = annual discrete compounding. (C.103)

Q convention = non-negative retained market-risk pressure. (C.104)

Residual assumption = zero for the illustrative derivation. (C.105)

This metadata is as important as the numerical result.

Without it, θ and Q cannot be interpreted consistently.

Appendix D — Certainty-Equivalent and Credit Examples

D.1 Why two further examples are necessary

CAPM provides a clean discount-rate laboratory, but it does not cover every way mature finance filters value.

Two additional examples clarify the distinction between:

- filtering the expected cash-flow numerator;

- filtering the discount or survival structure;
- continuous pressure movement;
- discrete gate commitment.

The first example uses certainty-equivalent valuation.

The second uses risky debt relative to a default-free reference claim.

Both preserve the rule:

Mature Filter → Admitted Ratio → θ → Q. (D.1)

The angle is derived from an established financial relation. It is not selected independently. This filter-first discipline is central to the original Finance Geometry framework.

D.2 Certainty-Equivalent Valuation

D.2.1 Declared structure

Let:

$\mu_{CF} = E(CF_T)$. (D.2)

Let:

CE_T = certainty-equivalent cash flow at horizon T. (D.3)

Assume:

$0 \le CE_T \le \mu_{CF}$. (D.4)

Let r_* be the declared base discount rate.

The expected-value amplitude is:

$A_{CE} = \mu_{CF}/(1 + r_*)^T$. (D.5)

The certainty-admitted value is:

$R_{CE} = CE_T/(1 + r_*)^T$. (D.6)

Therefore:

$R_{CE}/A_{CE} = CE_T/\mu_{CF}$. (D.7)

Define:

$c_{CE} = CE_T/\mu_{CF}$. (D.8)

Then:

$\cos \theta_{CE} = c_{CE}$. (D.9)

$\theta_{CE} = \arccos(c_{CE})$. (D.10)

The retained certainty pressure is:

$$Q_{CE} = A_{CE}\sqrt{(1 - c_{CE}^2)}. \quad (D.11)$$

The complex state is:

$$Z_{CE} = R_{CE} + iQ_{CE}. \quad (D.12)$$

Equivalently:

$$Z_{CE} = A_{CE} \exp(i\theta_{CE}). \quad (D.13)$$

D.2.2 Interpretation

The variables mean:

- A_{CE} = expected-value amplitude;
- R_{CE} = value admitted after certainty adjustment;
- Q_{CE} = pressure associated with the gap between expectation and certainty admission;
- θ_{CE} = orientation of the certainty filter.

This pressure may contain:

- uncertainty about realization;
- risk aversion;
- downside asymmetry;
- confidence deterioration;
- state dependence;
- incomplete hedgability.

It should not be labelled simply “risk” without specifying how CE_T was obtained.

D.2.3 Numerical example

Assume:

$$\mu_{CF} = 120. \quad (D.14)$$

$$CE_T = 96. \quad (D.15)$$

$$T = 2. \quad (D.16)$$

$$r_* = 0.05. \quad (D.17)$$

Then:

$$A_{CE} = 120/(1.05)^2. \quad (D.18)$$

Therefore:

$A_{CE} \approx 108.8435$. (D.19)

The admitted value is:

$R_{CE} = 96/(1.05)^2$. (D.20)

Therefore:

$R_{CE} \approx 87.0748$. (D.21)

The certainty ratio is:

$c_{CE} = 96/120 = 0.8$. (D.22)

Therefore:

$\theta_{CE} = \arccos(0.8)$. (D.23)

So:

$\theta_{CE} \approx 0.6435$ radians. (D.24)

The pressure coordinate is:

$Q_{CE} = A_{CE}\sqrt{(1 - 0.8^2)}$. (D.25)

Therefore:

$Q_{CE} \approx 65.3061$. (D.26)

The complex certainty-equivalent state is:

$Z_{CE} \approx 87.0748 + i65.3061$. (D.27)

D.2.4 Certainty deterioration as angular movement

Suppose μ_{CF} and r_* remain fixed, but CE_T declines from 96 to 84.

Then:

$c_{CE,new} = 84/120 = 0.7$. (D.28)

The new angle is:

$\theta_{CE,new} = \arccos(0.7)$. (D.29)

Therefore:

$\theta_{CE,new} \approx 0.7954$ radians. (D.30)

The new admitted value is:

$R_{CE,new} = 84/(1.05)^2$. (D.31)

Therefore:

$R_{CE,new} \approx 76.1905$. (D.32)

The new pressure coordinate is:

$Q_{CE,new} = A_{CE}\sqrt{(1 - 0.7^2)}$. (D.33)

Therefore:

$Q_{CE,new} \approx 77.7421$. (D.34)

The amplitude remains fixed:

$\Delta A_{CE} = 0$. (D.35)

The certainty filter tightens:

$\Delta\theta_{CE} > 0$. (D.36)

The movement is primarily angular.

D.2.5 Expected-value deterioration as radial movement

Return to the original certainty ratio:

$CE_T/\mu_{CF} = 0.8$. (D.37)

Now reduce expected cash flow proportionally:

$\mu_{CF,new} = 108$. (D.38)

$CE_{T,new} = 86.4$. (D.39)

Then:

$A_{CE,new} = 108/(1.05)^2$. (D.40)

$R_{CE,new} = 86.4/(1.05)^2$. (D.41)

The ratio remains:

$R_{CE,new}/A_{CE,new} = 0.8$. (D.42)

Therefore:

$\theta_{CE,new} = \theta_{CE}$. (D.43)

Both R_{CE} and Q_{CE} contract proportionally.

This is radial deterioration.

D.2.6 Mixed certainty dynamics

Let:

$A_{CE}(t) = \mu_{CF}(t)/(1 + r_*)^T$. (D.44)

Let:

$\theta_{CE}(t) = \arccos[CE_T(t)/\mu_{CF}(t)]$. (D.45)

Then:

$Z_{CE}(t) = A_{CE}(t)\exp[i\theta_{CE}(t)]$. (D.46)

Its dynamics are:

$dZ_{CE}/dt = (g_{CE} + i\omega_{CE})Z_{CE} + \varepsilon_{CE}$. (D.47)

Where:

$g_{CE} = d \ln A_{CE}/dt$. (D.48)

$\omega_{CE} = d\theta_{CE}/dt$. (D.49)

The admitted-value equation is:

$dR_{CE}/dt = g_{CE} R_{CE} - Q_{CE}\omega_{CE} + Re(\varepsilon_{CE})$. (D.50)

The certainty-equivalent framework therefore separates:

- changing expected economic amount;
- changing confidence or certainty admission;
- residual beyond the declared model.

D.2.7 Possible empirical targets

Candidate tests include whether Q_{CE} or $Q_{CE}\omega_{CE}$ helps predict:

- forecast revision;
- project cancellation;
- covenant pressure;
- capital withdrawal;
- downside realization;
- exercise delay;
- management abandonment.

The null comparison should include:

- variance;
- downside deviation;
- skewness;
- expected shortfall;

- conventional certainty-equivalent spread;
- subjective probability revision.

If the complex variables add nothing, the ordinary certainty-equivalent ratio remains sufficient.

D.3 Credit Geometry

D.3.1 Declared reference claim

Let V_0 denote the value of a default-free or reference claim.

Let V_{risk} denote the value of the risky claim.

Assume:

$$0 \le V_{risk} \le V_0. \quad (D.51)$$

Define:

$$A_{credit} = V_0. \quad (D.52)$$

$$R_{credit} = V_{risk}. \quad (D.53)$$

Then:

$$\cos \theta_{credit} = V_{risk}/V_0. \quad (D.54)$$

The credit angle is:

$$\theta_{credit} = \arccos(V_{risk}/V_0). \quad (D.55)$$

The retained credit-pressure coordinate is:

$$Q_{credit} = V_0\sqrt{[1 - (V_{risk}/V_0)^2]}. \quad (D.56)$$

The complex credit state is:

$$Z_{credit} = V_{risk} + iQ_{credit}. \quad (D.57)$$

D.3.2 Yield-spread construction

Let one terminal contractual payment CF_T be discounted at a reference yield y_0.

Then:

$$V_0 = CF_T/(1 + y_0)^T. \quad (D.58)$$

Let s denote the risky spread.

Then:

$$V_{risk} = CF_T/(1 + y_0 + s)^T. \quad (D.59)$$

Therefore:

$$\cos \theta_{credit} = [(1 + y_0)/(1 + y_0 + s)]^T. \quad (D.60)$$

The credit state is:

$$Z_{credit} = V_0 \exp(i\theta_{credit}). \quad (D.61)$$

As s increases:

$$s \uparrow \rightarrow \theta_{credit} \uparrow \rightarrow R_{credit} \downarrow \rightarrow Q_{credit} \uparrow. \quad (D.62)$$

D.3.3 Numerical example

Assume:

$$CF_T = 110. \quad (D.63)$$

$$T = 2. \quad (D.64)$$

$$y_0 = 0.04. \quad (D.65)$$

Then:

$$V_0 = 110/(1.04)^2. \quad (D.66)$$

Therefore:

$$V_0 \approx 101.7012. \quad (D.67)$$

Suppose:

$$s = 0.03. \quad (D.68)$$

Then:

$$V_{risk} = 110/(1.07)^2. \quad (D.69)$$

Therefore:

$$V_{risk} \approx 96.0783. \quad (D.70)$$

The ratio is:

$$V_{risk}/V_0 \approx 0.9447. \quad (D.71)$$

The credit angle is:

$$\theta_{credit} \approx \arccos(0.9447). \quad (D.72)$$

Therefore:

$$\theta_{credit} \approx 0.3340 \text{ radians}. \quad (D.73)$$

The credit-pressure coordinate is:

$$Q_{credit} = \sqrt{(V_0^2 - V_{risk}^2)}. \quad (D.74)$$

Therefore:

$Q_{credit} \approx 33.346$. (D.75)

The credit state is:

$Z_{credit} \approx 96.0783 + i33.346$. (D.76)

D.3.4 Spread widening

Suppose the spread widens from 3% to 7%.

Then:

$s_{new} = 0.07$. (D.77)

The new risky value is:

$V_{risk,new} = 110/(1.11)^2$. (D.78)

Therefore:

$V_{risk,new} \approx 89.2785$. (D.79)

The new ratio is:

$V_{risk,new}/V_0 \approx 0.8779$. (D.80)

The new credit angle is:

$\theta_{credit,new} \approx \arccos(0.8779)$. (D.81)

Therefore:

$\theta_{credit,new} \approx 0.4990$ radians. (D.82)

The new pressure coordinate is:

$Q_{credit,new} = \sqrt{(V_0^2 - V_{risk,new}^2)}$. (D.83)

Therefore:

$Q_{credit,new} \approx 48.683$. (D.84)

The reference amplitude is unchanged, but the credit filter tightens.

The movement is angular.

D.3.5 Expected-loss construction

A simplified risky-debt value may instead be written:

$V_{risk} = V_0 - EL - LP - RP$. (D.85)

Where:

- EL = expected loss;

- LP = liquidity premium effect;
- RP = other required-risk-premium effect.

This additive decomposition is not automatically compatible with treating all three as orthogonal Q channels.

The ratio:

V_{risk}/V_0 (D.86)

can generate one total credit-related angle.

But separating:

$Q_{default}, Q_{liquidity}, Q_{risk\text{-}premium}$ (D.87)

requires a residualized multi-pressure construction.

Otherwise, the same economic burden may be counted more than once.

D.3.6 Credit pressure versus expected loss

Q_{credit} is not expected loss.

Expected loss is commonly:

$EL = PD \times LGD \times EAD.$ (D.88)

Where:

- PD = probability of default;
- LGD = loss given default;
- EAD = exposure at default.

Q_{credit} is:

$Q_{credit} = \sqrt{(V_0^2 - V_{risk}^2)}.$ (D.89)

The two quantities have different units and meanings.

Expected loss is a probabilistic loss estimate.

Q_{credit} is an orthogonal coordinate implied by the risky-to-reference value ratio.

A valid empirical programme must compare them rather than treat them as synonyms.

D.3.7 Pre-default phase and default gate

Credit deterioration may evolve continuously:

$\theta_{credit}(t) \uparrow.$ (D.90)

$Q_{credit}(t) \uparrow.$ (D.91)

R_credit(t) ↓. (D.92)

But default is committed through a gate:

G_default(X,Z,L,P) → Commit(Default). (D.93)

Possible gate criteria include:

- payment failure;
- insolvency filing;
- distressed exchange;
- contractual event of default;
- regulatory determination;
- credit-event committee decision.

The gate record should include:

DefaultRecord_k = (EventType,Time,Authority,Evidence,ContractReference,Residual). (D.94)

D.3.8 Post-default world transition

Before default, the relevant filter may be:

F_pre = Risky Going-Concern Credit Valuation. (D.95)

After default, it may become:

F_post = Recovery and Legal-Claim Valuation. (D.96)

Therefore:

P_pre ≠ P_post. (D.97)

The transition is not merely:

θ_credit ↑ further. (D.98)

It is:

Credit World → Recovery World. (D.99)

A cross-world translation may preserve:

- contractual amount;
- priority;
- collateral;
- expected recovery.

But it may discard or replace:

- ordinary yield;
- ordinary spread;
- going-concern beta;
- pre-default liquidity assumptions.

D.3.9 Credit loop residual

Suppose the bond spread later returns to its pre-crisis level:

$s_T \approx s_0$. (D.100)

The credit angle may also return:

$\theta_{credit,T} \approx \theta_{credit,0}$. (D.101)

But the ledger may contain:

- prior downgrade;
- covenant amendment;
- refinancing;
- collateral release;
- rating-watch history;
- investor-base change.

Therefore:

$H_{loop,credit} \neq 0$. (D.102)

The bond may occupy the same present spread but not the same financial world.

D.3.10 Credit experiment

A practical credit experiment can test whether:

$Q_{credit}/R_{credit}, \omega_{credit}, Q_{credit}\omega_{credit}, \varepsilon_{credit}$ (D.103)

improve prediction of:

- spread widening;
- downgrade;
- default;
- recovery;
- liquidity loss;

- collateral ineligibility.

Comparison models should include:

- spread level;
- spread change;
- spread duration;
- rating;
- hazard rate;
- expected loss;
- distance to default;
- liquidity;
- volatility.

Corporate credit is likely to be one of the strongest initial empirical domains because both continuous pressure and discrete gate events are observable. The Finance Geometry source similarly identifies credit and related mature filters as suitable domains for testing whether Q adds diagnostic value.

Appendix E — Three-Clock Event Log Template

E.1 Purpose

A conventional time series records values against calendar time.

The effective-world framework requires a richer event log containing:

- calendar time;
- phase position;
- ledger sequence;
- gate status;
- residual;
- protocol version;
- backreaction.

The three-clock log distinguishes:

t = external duration. (E.1)

θ = internal phase order. (E.2)

k = committed ledger order. (E.3)

The governing chain is:

$t \to \theta(t) \to G_k \to L_{k+1}$. (E.4)

E.2 Minimum event-log schema

Each observation row should include:

Field	Meaning
observation_id	unique observation identifier
asset_id	asset, firm, project, or portfolio
calendar_time	external timestamp t
protocol_version	active declaration P
amplitude_A	declared pre-filter amplitude
admitted_R	admitted value
pressure_Q	retained pressure
phase_theta	finance angle
radial_rate_gA	amplitude growth rate
angular_velocity_omega	dθ/dt
angular_load_Lambda	Qω_F
dynamic_residual	ε_dyn
gate_status	Commit, Reject, or Defer
event_type	trade, downgrade, default, etc.
ledger_index	event counter k
ledger_residual	unresolved consequence
intervention	action taken
backreaction_target	variable expected to change
evidence_reference	source supporting the observation

E.3 Observation row

A non-event observation may be represented as:

Observation_t = (t,P,A,R,Q,θ,g_A,ω_F,Λ_F,ε_dyn). (E.5)

The ledger index does not advance:

k_t = k_{t−1}. (E.6)

The gate status is:

G_t = Reject or Defer. (E.7)

E.4 Event row

A gate-passed event is:

$\text{Event}_k = (t_k,\theta_k,k,\text{EventType}_k,\text{Authority}_k,\text{Evidence}_k,\text{Residual}_k)$. (E.8)

The ledger update is:

$L_{k+1} = L_k \sqcup \text{Event}_k$. (E.9)

The event row should preserve both:

- the state immediately before commitment;
- the state immediately after commitment.

Thus:

$\text{State_pre} = (A^-, R^-, Q^-, \theta^-, P^-)$. (E.10)

$\text{State_post} = (A^+, R^+, Q^+, \theta^+, P^+)$. (E.11)

E.5 Protocol-transition row

When:

$P^+ \neq P^-$, (E.12)

the log should record:

$\text{ProtocolTransition} = (P^-, P^+, \text{Reason}, \text{Mapping}, \text{UntranslatedResidual})$. (E.13)

The transition residual is:

$\varepsilon_{\text{transition}} = Z_{P^+}(t^*) - M_{P^- \to P^+}[Z_{P^-}(t^{*-})]$. (E.14)

The row must state whether the event was:

- parameter recalibration;
- structural filter replacement;
- gate-rule revision;
- boundary change;
- horizon change;
- intervention-set change.

E.6 Illustrative credit episode

t	θ_credit	k	Event	Gate	Ledger effect
Day 1	0.25	10	ordinary observation	Defer	no new record
Day 5	0.31	10	spread widening	Defer	monitoring note
Day 8	0.39	10	rating watch	Commit	k → 11
Day 12	0.47	11	collateral haircut raised	Commit	k → 12
Day 14	0.61	12	covenant breach	Commit	k → 13
Day 15	discontinuity	13	default declaration	Commit	k → 14

t	θ_credit	k	Event	Gate	Ledger effect
Day 16	new protocol	14	recovery valuation begins	Commit	P_pre → P_post

The table shows that:

- calendar time advances continuously;
- phase accelerates;
- ledger time advances only at committed events;
- the final event changes the world declaration.

E.7 Phase-distance field

For each observation:

$\Delta\theta_t = \theta_t - \theta_{episode\text{-}start}$. (E.15)

For each event:

$\Delta\theta_{event,k} = \theta(t_k) - \theta_{episode\text{-}start}$. (E.16)

This supports comparisons such as:

- average phase distance to downgrade;
- average phase distance from warning to default;
- dispersion of event position in phase versus calendar time.

E.8 Phase-direction field

Because the same θ may occur during tightening and relaxation, record:

$d_k = \mathrm{sign}(\omega_{F,k})$. (E.17)

Where:

$d_k = +1$ for tightening. (E.18)

$d_k = -1$ for relaxation. (E.19)

$d_k = 0$ for approximately stationary phase. (E.20)

A phase-addressed state becomes:

$\mathrm{PhaseAddress}_k = (\theta_k, d_k, n_{cycle}, L_k)$. (E.21)

This prevents repeated phase values from being mistaken for identical histories.

E.9 Event-causality field

Each committed record should distinguish:

- triggering variables;
- declared gate condition;
- authority;
- downstream consequences.

A causal event entry is:

$\text{CausalRecord}_k = (\text{Trigger}_k, \text{Gate}_k, \text{Commitment}_k, \text{Consequences}_k)$. (E.22)

For example:

Trigger = rating falls below BBB−. (E.23)

Gate = collateral policy threshold. (E.24)

Commitment = asset becomes ineligible. (E.25)

Consequence = borrowing capacity declines. (E.26)

This supports backreaction analysis.

E.10 Backreaction trace

A backreaction record may be written:

$\text{BRecord}_k = (\text{SourceRecord}_k, \text{TargetVariable}, \text{ExpectedSign}, \text{Lag}, \text{ObservedEffect})$. (E.27)

Example:

SourceRecord = collateral haircut increase. (E.28)

TargetVariable = forced sales. (E.29)

ExpectedSign = positive. (E.30)

Lag = one trading day. (E.31)

ObservedEffect = measured sale volume. (E.32)

The ledger can then test whether the declared causal loop actually occurred.

E.11 Residual fields

The event log should contain separate fields for:

Dynamic residual:

$\varepsilon_{\text{dyn},k}$. (E.33)

Projection residual:

ε_Π,k. (E.34)

Ledger residual:

E_L,k. (E.35)

Transition residual:

ε_transition,k. (E.36)

Loop residual:

H_loop,episode. (E.37)

These should not be collapsed into one general “uncertainty” field.

E.12 Evidence status

Each record should classify evidence:

EvidenceStatus ∈ {Verified,Provisional,Disputed,Missing}. (E.38)

A confidence score may be attached:

c_evidence ∈ [0,1]. (E.39)

But the original evidence status must remain visible.

A later increase in confidence should create a new record rather than silently rewrite the old one.

E.13 Reversal record

If an event is reversed:

ReversalRecord_k = (OriginalRecordID,ReversalReason,Authority,NewResidual). (E.40)

The ledger becomes:

$L_{new} = L_{old} \sqcup \text{ReversalRecord}_k$. (E.41)

It does not erase the original record.

This preserves historical irreversibility even when practical consequences are partly undone.

E.14 Episode-closure record

At the end of a declared episode, record:

EpisodeSummary = (t_start,t_end,θ_start,θ_end,k_start,k_end,H_loop,FinalResidual). (E.42)

The episode may be classified as:

- phase-closed and ledger-closed;

- phase-closed but ledger-open;
- phase-open and ledger-open;
- protocol-terminated.

E.15 Three-clock consistency checks

The log should test:

Calendar consistency:

$t_{j+1} \geq t_j$. (E.43)

Ledger consistency:

$k_{j+1} \geq k_j$. (E.44)

Gate consistency:

$k_{j+1} = k_j + 1$ only if a commit event occurred. (E.45)

Protocol consistency:

P changes only through a recorded revision event. (E.46)

Phase consistency:

θ must be derived from the declared A and R. (E.47)

Residual consistency:

Unresolved residual cannot disappear without a resolution record. (E.48)

Appendix F — Empirical Data Schema and Verification Footer

F.1 Research-object design

The empirical implementation should be treated as an auditable research object rather than a collection of charts.

The minimum architecture contains:

1. protocol table;
2. asset table;
3. observation table;
4. event ledger;
5. intervention table;

6. model-output table;
7. residual table;
8. verification footer.

The tables may be stored in:

- relational database;
- columnar research file;
- versioned spreadsheet;
- reproducible data package.

The logical relations should remain the same.

F.2 Protocol table

A protocol table should contain:

Field	Description
protocol_id	unique identifier
protocol_version	version number
effective_start	start timestamp
effective_end	end timestamp
boundary_B	declared system boundary
baseline_q	reference state
feature_map_phi	included features
horizon_h	valuation horizon
filter_F	mature finance filter
gate_G	commitment rule
trace_T	ledger rule
intervention_U	admissible actions
residual_threshold	world-validity tolerance
revision_reason	reason for new version
parent_protocol	preceding version

F.3 Asset table

The asset table should contain stable identifiers and metadata:

Field	Description
asset_id	permanent identifier
asset_type	equity, bond, project, loan, etc.
issuer_id	issuer or counterparty
currency	valuation currency
jurisdiction	governing jurisdiction

Field	Description
sector	economic sector
maturity	contractual maturity
seniority	legal ranking
collateral_type	collateral classification
accounting_class	recognition category
rating	contemporaneous rating
data_source	origin of static metadata

F.4 Observation table

The observation table should include:

Field	Description
observation_id	unique row
asset_id	link to asset
protocol_id	active protocol
timestamp	calendar time
A	pre-filter amplitude
R	admitted value
Q	retained pressure
theta	finance angle
g_A	radial growth
omega_F	phase velocity
Lambda_F	angular repricing load
epsilon_R	real dynamic residual
epsilon_Q	pressure dynamic residual
closure_score	normalized residual
data_quality	quality flag
source_timestamp	data vintage

The algebraic audit is:

$A^2 - R^2 - Q^2 \approx 0$. (F.1)

The domain audit is:

$0 \le R/A \le 1$. (F.2)

F.5 Event table

The event table should contain:

Field	Description
event_id	unique event
asset_id	affected object
timestamp	event time

Field	Description
ledger_index	k
event_type	downgrade, default, etc.
gate_result	Commit, Reject, Defer
authority	decision authority
threshold	applied threshold
evidence_id	supporting evidence
pre_state_id	observation before event
post_state_id	observation after event
protocol_before	P^-
protocol_after	P^+
ledger_residual	unresolved consequence
reversal_of	original event if reversed

F.6 Intervention table

The intervention table should contain:

Field	Description
intervention_id	unique action
event_id	triggering event
action_type	hedge, sale, refinance, etc.
decision_time	action time
actor	decision-maker
target	variable intended to change
expected_sign	stabilizing or amplifying
expected_lag	anticipated delay
cost	implementation cost
observed_outcome	measured result
outcome_window	evaluation period

This table supports the intervention-level falsification test.

F.7 Residual table

The residual table should distinguish:

Residual type	Symbol
dynamic residual	ε_{dyn}
phase residual	ε_{θ}
projection residual	ε_{Π}
ledger residual	E_L
loop residual	H_{loop}
transition residual	$\varepsilon_{transition}$
cross-observer residual	ε_{tri}

Each residual entry should contain:

- measurement method;
- unit;
- threshold;
- confidence;
- resolution status;
- linked record;
- ageing.

F.8 Model-comparison table

Each model run should store:

Field	Description
run_id	unique run
training_window	estimation range
validation_window	tuning range
test_window	untouched test range
baseline_model	mature comparator
extended_model	model with Finance Geometry
features	included variables
hyperparameters	fixed settings
loss_metric	evaluation loss
calibration_metric	calibration score
economic_metric	decision value
complexity_metric	AIC, BIC, or parameter count
code_hash	exact implementation
data_hash	exact input data
result_timestamp	execution time

F.9 Minimum verification footer

Every published table, chart, or conclusion should carry a footer containing:

Protocol ID: P_x. (F.3)

Protocol Version: v_n. (F.4)

Data Vintage: YYYY-MM-DD. (F.5)

Data Hash: H_data. (F.6)

Code Hash: H_code. (F.7)

Baseline Definition: q. (F.8)

Filter Definition: F. (F.9)

Horizon: h. (F.10)

Derivative Method: D_method. (F.11)

Residual Threshold: κ_ε. (F.12)

Train Window: W_train. (F.13)

Test Window: W_test. (F.14)

Reviewer: Reviewer_ID. (F.15)

F.10 Compact audit footer

A compact publication footer may be:

VerificationFooter = (P_version,DataHash,CodeHash,Horizon,DerivativeMethod,Thresholds,TestWindow,Reviewer). (F.16)

This footer should accompany claims such as:

- Q predicted stress;
- residual preceded a regime transition;
- phase alignment improved event timing;
- loop memory predicted fragility.

Without the footer, the result cannot be reliably rerun.

F.11 Data-quality gates

Before computing θ or Q, observations should pass:

Unit consistency. (F.17)

Baseline availability. (F.18)

Horizon consistency. (F.19)

Non-missing A and R. (F.20)

Domain validity. (F.21)

Market-executability check where relevant. (F.22)

Protocol-version match. (F.23)

If any gate fails:

ObservationStatus = Invalid for Geometry. (F.24)

The system should not silently impute a valid θ.

F.12 Reproducibility rule

A result is reproducible when an independent implementation using the declared data and protocol obtains:

Distance(Result_independent,Result_original) ≤ κ_rep. (F.25)

Reproducibility should be tested for:

- A;
- R;
- θ;
- Q;
- ω_F;
- Λ_F;
- residual;
- event classification.

F.13 Revision history

Every revision should append:

Revision_n = (PreviousVersion,ChangedFields,Reason,Evidence,ImpactAssessment). (F.26)

The revision history must distinguish:

- data correction;
- code correction;
- parameter recalibration;
- structural model revision;
- protocol change.

A corrected result should not overwrite the existence of the earlier error.

The ledger principle applies to research itself:

Correction Extends the Record; It Does Not Erase the History of the Claim. (F.27)

Appendix G — Finance–Classical Field–Quantum

Comparison Matrix

G.1 Purpose of the comparison

The purpose of this appendix is not to prove material identity among finance, classical field systems, and quantum mechanics.

It is to separate three different levels of correspondence:

1. shared mathematical form;
2. shared operational role;
3. distinct physical realization.

The central discipline is:

Functional Homology ≠ Material Identity. (G.1)

Two systems may use similar coordinates or transition grammars while differing completely in:

- substrate;
- causal mechanism;
- probability law;
- composition rule;
- experimental signature.

Finance is therefore used as a **macroscopic control case**. It helps identify which apparently strange features can arise from ordinary contextual projection, gating, trace, memory, and backreaction. Only the remainder should be treated as candidate quantum residue. This subtraction strategy follows the comparative programme developed in the handoff document.

G.2 Three-column distinction

The three domains are:

Finance Geometry

A declared mature financial filter maps a larger economic state into admitted value and retained pressure.

$Z_F = R + iQ$. (G.2)

Classical Field or Control System

A classical system may use complex coordinates to represent quadrature, phase, oscillation, impedance, or coupled state variables.

$Z_C = X + iY$. (G.3)

Quantum Mechanics

A quantum state is represented in a complex Hilbert space and generates probabilities through the Born rule.

$|\psi\rangle \in \mathscr{H}$. (G.4)

$P(a) = \langle\psi|\Pi_a|\psi\rangle$. (G.5)

The use of complex numbers appears in all three domains.

Its meaning is not the same.

G.3 Comparison matrix

Concept	Finance Geometry	Classical Field or Control	Quantum Mechanics
primary object	economic or institutional state X	classical physical state x	quantum state
declared boundary	protocol P	system boundary and boundary conditions	preparation and measurement context
visible coordinate	admitted value R	measured state component X	measurement outcome or expectation
orthogonal complement	retained pressure Q	quadrature Y	phase-dependent amplitude component
complex state	$Z = R + iQ$	$Z = X + iY$	
magnitude	declared amplitude A	classical amplitude or norm	state norm
phase	finance-filter angle θ	oscillator or phasor phase	quantum relative phase
internal law	$dZ/d\theta = iZ$	rotation or harmonic motion	Schrödinger-type unitary evolution
measurement frame	mature finance filter	sensor or coordinate basis	observable or measurement basis
event gate	trade, default, impairment, etc.	threshold detector or switch	measurement outcome registration
trace	ledger record	data log or physical memory	macroscopic measurement record
backreaction	valuation alters funding or action	controller alters plant	measurement interaction disturbs state
order sensitivity	sequential filters with memory	noncommuting procedures or hysteresis	noncommuting observables
probability rule	external statistical model	classical probability	Born rule
composition	portfolios, networks, contracts	Cartesian product or coupled system	tensor product
irreducible nonseparability	not established	not generally	entanglement
Bell violation	not established	not under ordinary local classical models	experimentally observed
no-cloning	absent as physical theorem	absent	fundamental theorem
ontology claim	none required	classical	quantum

G.4 Complex coordinates

Finance

$Z_F = R + iQ$. (G.6)

The imaginary coordinate preserves pressure implied by the mature filter.

Classical field

$Z_C = X + iY$. (G.7)

The two coordinates may represent:

- in-phase and quadrature components;
- real and reactive power;
- position and momentum-like oscillator coordinates;
- amplitude and lag.

Quantum mechanics

A quantum amplitude may be written:

$\psi = a + ib$. (G.8)

But the probability is not a or b separately.

It is:

$|\psi|^2 = a^2 + b^2$. (G.9)

The shared use of i establishes mathematical similarity only.

It does not transfer the Born rule into finance.

G.5 Magnitude preservation

In Finance Geometry:

$A^2 = R^2 + Q^2$. (G.10)

In an ideal classical oscillator:

$A_C^2 = X^2 + Y^2$. (G.11)

In normalized quantum mechanics:

$\langle\psi|\psi\rangle = 1$. (G.12)

These equations all preserve a norm-like quantity.

But their interpretations differ.

- A is a declared pre-filter financial amplitude.
- A_C is a classical state magnitude.
- ⟨ψ|ψ⟩ is quantum-state normalization.

Therefore:

Norm Preservation ≠ Common Ontology. (G.13)

G.6 Phase

Finance phase

θ_F = arccos(R/A). (G.14)

It is derived from a mature financial ratio.

Classical phase

For:

Z_C = A_C exp(iθ_C), (G.15)

θ_C locates the oscillator or phasor in its cycle.

Quantum phase

For:

|ψ⟩ = Σ_j c_j exp(iφ_j)|j⟩, (G.16)

relative phases affect interference probabilities.

The finance angle influences visible projection and pressure sensitivity.

It does not automatically create interference among probability amplitudes.

G.7 Internal evolution

Finance

For constant A:

dZ_F/dθ_F = iZ_F. (G.17)

Classical rotation

dZ_C/dθ_C = iZ_C. (G.18)

Quantum unitary evolution

iℏ(d|ψ⟩/dt) = H|ψ⟩. (G.19)

The finance and classical equations directly express geometric rotation.

The quantum equation includes:

- Planck's constant;
- Hamiltonian dynamics;
- Hilbert-space structure;
- experimentally constrained unitary evolution.

The resemblance:

$dZ/d\theta = iZ$ (G.20)

does not derive:

$i\hbar(d|\psi\rangle/dt) = H|\psi\rangle$. (G.21)

The generator, time variable, state space, and empirical meaning are different.

G.8 Measurement setting

Finance

P_a and P_b may produce:

Z_a = Π_{P_a}(X). (G.22)

Z_b = Π_{P_b}(X). (G.23)

Classical control

Different sensors or coordinate bases produce different observed components:

y_a = H_ax. (G.24)

y_b = H_bx. (G.25)

Quantum mechanics

Different observables correspond to different measurement operators.

P(a|A) = Tr(ρΠ_a^A). (G.26)

All three systems are setting-sensitive.

Only the quantum case necessarily involves the quantum probability and incompatibility structure.

G.9 Hidden complements

Finance may suppress Q when only R is reported.

A classical sensor may observe one component of a vector while omitting another.

Quantum mechanics may prohibit a single noncontextual assignment of definite values to all incompatible observables.

Therefore:

Unobserved Classical Component ≠ Irreducibly Indefinite Quantum Observable. (G.27)

The distinction is between:

- hidden because unreported or inaccessible;
- undefined as a joint classical assignment under the quantum structure.

G.10 Gate and collapse

Finance gate

$G_F(\hat{Z}) \rightarrow$ Commit, Reject, or Defer. (G.28)

Classical detector

$G_C(x) \rightarrow$ Trigger or No Trigger. (G.29)

Quantum measurement

A measurement yields an outcome a with probability:

$P(a) = Tr(\rho \Pi_a)$. (G.30)

and conditionally updates the state according to the measurement model.

All three contain outcome commitment.

But only the quantum case is tied to quantum probability amplitudes and instrument maps.

Therefore:

Gate-Passed Commitment is Generic. (G.31)

Quantum State Update is More Specific. (G.32)

G.11 Trace and objectivity

Finance

$L_{k+1} = L_k \sqcup Record_k$. (G.33)

Classical measurement

$Memory_{k+1} = Memory_k \sqcup Data_k$. (G.34)

Quantum measurement

A microscopic interaction becomes a macroscopic record through apparatus and environmental coupling.

In all three cases, redundant records can stabilize public agreement.

But quantum decoherence and quantum Darwinism concern specific dynamics of quantum-environment interaction.

A financial ledger demonstrates the generic role of redundant trace, not the physical mechanism of decoherence.

G.12 Backreaction

Finance

$X_{k+1} = \mathcal{B}(X_k, Z_k, L_{k+1}, u_k)$. (G.35)

Classical control

$\dot{x} = f(x,u)$. (G.36)

$u = K(y)$. (G.37)

The observation y determines a control input u that changes the plant.

Quantum measurement

The interaction between system and apparatus changes the quantum state.

The presence of backreaction is common.

The specifically quantum content lies in:

- incompatible observables;
- measurement operators;
- quantum disturbance limits;
- entangled system–apparatus evolution.

G.13 Order dependence

Finance

$[F_a, F_b]X = F_a[F_b(X)] - F_b[F_a(X)]$. (G.38)

Classical procedures

Two state-changing operations can satisfy:

$T_aT_bx \neq T_bT_ax$. (G.39)

Quantum mechanics

$[A,B] = AB - BA$. (G.40)

When:

$[A,B] \neq 0$, (G.41)

the observables are incompatible in the quantum formalism.

The same commutator notation can describe different mechanisms.

In finance, noncommutation may arise because the first operation changes:

- ledger;
- collateral;
- population;
- legal status;
- gate conditions.

Quantum noncommutation is built into operator algebra.

G.14 Probability structures

Classical finance probability

A financial outcome may be modelled by:

$P_F(E|X,P)$. (G.42)

This is usually an ordinary Kolmogorov probability distribution.

Classical control probability

Noise and uncertainty may be represented by:

$P_C(x_t|x_{t-1},u_t)$. (G.43)

Quantum probability

Quantum probability is generated through amplitudes and operators:

$P_Q(a) = Tr(\rho\Pi_a)$. (G.44)

The decisive difference is not that all three systems are uncertain.

It is the geometry and composition law of their probabilities.

G.15 Composition

Financial composition

A portfolio may be represented by:

X_portfolio = (X1,X2,...,X_n,Dependencies). (G.45)

Classical composition

Two systems may use:

$\mathcal{X}$_AB = $\mathcal{X}$_A × $\mathcal{X}$_B. (G.46)

Quantum composition

Quantum systems use:

$\mathscr{H}$_AB = $\mathscr{H}$_A ⊗ $\mathscr{H}$_B. (G.47)

The tensor product permits entangled states that cannot be represented as ordinary mixtures of independent subsystem states.

A financial network can be highly nonseparable operationally.

But that does not establish Hilbert-space entanglement.

G.16 Entanglement comparison

A financial system may contain:

- shared collateral;
- mutual obligations;
- nonlinear contagion;
- common information;
- reflexive expectations.

These can create strong joint dependence.

Yet an ordinary joint classical distribution may still exist:

P(X_A,X_B). (G.48)

Quantum entanglement concerns states for which no separable decomposition exists:

ρ_AB ≠ Σ_j p_jρ_A,j ⊗ ρ_B,j. (G.49)

Therefore:

Systemic Financial Coupling ≠ Quantum Entanglement. (G.50)

G.17 Observer agreement

Finance may achieve agreement through:

- common data;
- shared standards;
- audited ledgers;
- institutional authority.

Classical science may achieve agreement through:

- reproducible measurement;
- calibration;
- common reference frames.

Quantum experiments achieve agreement through:

- shared preparation procedures;
- measurement statistics;
- stable classical records.

The generic structure is:

Shared Protocol + Reproducible Trace → Cross-Observer Agreement. (G.51)

The quantum-specific question is how definite public records emerge from quantum dynamics.

G.18 Time

Finance

Three clocks are distinguished:

t = calendar time. (G.52)

θ = phase time. (G.53)

k = ledger time. (G.54)

Classical systems

Time may be external, or an internal phase may parameterize dynamics.

Quantum mechanics

Time is generally a parameter in non-relativistic quantum mechanics, while time observables and quantum gravity raise deeper issues.

The finance framework demonstrates only that internal ordering can emerge from phase and

trace.

It does not resolve the quantum problem of time.

G.19 Residual

Finance

ε_dyn records failure of internal closure.

ε_Π records omitted primary structure.

Classical modelling

Residual measures discrepancy between model and observed system.

Quantum mechanics

Experimental residual measures mismatch between theoretical prediction and data.

But quantum uncertainty is not simply model residual.

Therefore:

Quantum Indeterminacy ≠ Ordinary Estimation Error. (G.55)

The distinction must remain explicit.

G.20 World revision

Finance allows:

$P_k \rightarrow P_{k+1}$. (G.56)

A classical model may similarly undergo regime or model revision.

Quantum theory normally treats the formal state and measurement structure as fixed by the physical model, although experimental context changes.

A financial protocol revision is therefore not analogous to physical-law revision inside an individual experiment.

It is closer to changing the effective model through which the observer organizes the system.

G.21 Comparison by claim strength

Level A — Mathematical resemblance

Examples:

- complex state;

- norm;
- phase;
- rotation;
- commutator notation.

Level B — Operational homology

Examples:

- contextual projection;
- gate-passed outcome;
- trace;
- backreaction;
- order dependence.

Level C — Physical equivalence

Examples would require demonstrating identical:

- state-space structure;
- probability law;
- composition;
- invariants;
- experimental signatures.

The present article establishes selected Level A and Level B correspondences.

It does not establish Level C.

G.22 The subtraction result

The comparison allows the following structures to be provisionally assigned to generic world formation:

G_world = {Complex Coordinates, Frame Dependence, Phase Order, Gate, Trace, Backreaction, Classical Order Effects}. (G.57)

The candidate quantum residue remains:

Q_residue = {Born Rule, Interference, Tensor Nonseparability, Bell Violation, Irreducible Contextuality, No-Cloning, Quantum Disturbance}. (G.58)

The subtraction is:

Q_residue = Observed Quantum Structure – G_world. (G.59)

This is a research classification, not a completed physical reduction.

G.23 Comparison checklist

Before describing any financial or macroscopic phenomenon as quantum-like, ask:

1. Is the similarity merely notational?
2. Is there a mature operational correspondence?
3. Does the system possess a Hilbert-space state?
4. Is a Born-type probability rule present?
5. Is tensor-product composition present?
6. Are Bell or contextuality inequalities relevant?
7. Could memory and ordinary path dependence explain the order effect?
8. Could a classical hidden state jointly specify the observables?
9. What empirical result would distinguish the quantum claim?
10. Does the analogy add prediction or only vocabulary?

If questions 3–9 cannot be answered, the claim should remain analogical.

Appendix H — Residual Taxonomy and Diagnostic Decision Tree

H.1 Why residual requires a governance architecture

Residual is often treated as one undifferentiated error term.

That is inadequate for an effective-world model.

A deviation can arise because:

- the data are noisy;
- the financial state is mismeasured;
- the dynamic law is incomplete;
- the projection omitted important structure;
- an event has committed unresolved consequences;
- the system completed a path without historical closure;
- the protocol changed;
- two observers used incompatible frames.

These failures require different responses.

The residual taxonomy is therefore:

$$E_{total} = (\varepsilon_{measure}, \varepsilon_{dyn}, \varepsilon_{\Pi}, E_L, H_{loop}, \varepsilon_{transition}, \varepsilon_{tri}). \quad (H.1)$$

The handoff framework treats residual not as disposable noise but as a boundary signal that shows where the effective world no longer closes.

H.2 Measurement residual

Definition

Measurement residual captures error introduced while estimating observable variables.

Examples include:

- stale prices;
- bid–ask noise;
- missing data;
- rounding;
- asynchronous timestamps;
- unstable derivative estimation.

Write:

$$\varepsilon_{measure} = \text{Observed Input} - \text{Best Estimated Input}. \quad (H.2)$$

Diagnostic signs

- residual falls when data quality improves;
- disagreement is concentrated around illiquid observations;
- alternative estimators converge after smoothing;
- protocol and model remain stable.

Response

- improve sampling;
- synchronize timestamps;
- use robust estimation;
- report uncertainty intervals;
- avoid structural revision unless the error persists.

H.3 Algebraic residual

Definition

The geometric identity should satisfy:

$\varepsilon_{alg} = A^2 - R^2 - Q^2$. (H.3)

A normalized form is:

$r_{alg} = |A^2 - R^2 - Q^2|/(A^2 + \delta)$. (H.4)

Diagnostic signs

- coding or unit error;
- incorrect sign convention;
- inconsistent values;
- invalid transformation.

Response

Reject the observation until corrected.

No economic interpretation should be attempted.

H.4 Dynamic residual

Definition

$\varepsilon_{dyn} = dZ/dt - (g_A + i\omega_F)Z$. (H.5)

Or in phase coordinates:

$\varepsilon_\theta = dZ/d\theta - [(d \ln A/d\theta) + i]Z$. (H.6)

Diagnostic signs

- observed state does not follow declared radial–angular dynamics;
- residual rises before a regime transition;
- large unexplained movement remains after amplitude and frame attribution.

Possible causes

- incorrect A;
- incorrect θ;
- omitted pressure channel;

- nonlinear coupling;
- jump event;
- backreaction;
- protocol drift.

Response

1. inspect data quality;
2. test alternative derivative estimators;
3. inspect gate events;
4. test additional pressure channels;
5. evaluate whether P remains valid.

H.5 Projection residual

Definition

$\varepsilon_\Pi = X - \mathscr{R}_P(Z,L)$. (H.7)

It measures primary-field structure not recoverable from the effective state.

Examples

- off-balance-sheet exposure;
- hidden collateral chains;
- legal restrictions;
- unobserved strategic behaviour;
- concentration;
- political intervention.

Diagnostic signs

- different primary states produce the same Z but different future outcomes;
- cross-observer disagreement persists after frame alignment;
- interventions fail because omitted variables dominate.

Response

- expand feature map φ;
- revise boundary B;

- add a specialized observer;
- reduce the claim scope;
- retain the residual explicitly.

H.6 Ledger residual

Definition

E_L,k = unresolved consequence remaining after event k is committed. (H.8)

Examples

- uncertain recovery;
- disputed valuation;
- pending litigation;
- incomplete settlement;
- provisional authority;
- unresolved model disagreement.

Diagnostic signs

- official event exists but consequence remains uncertain;
- later decisions depend on unresolved assumptions;
- residual ages without resolution.

Response

- attach E_L,k to Record_k;
- assign owner and review date;
- record evidence updates;
- create a resolution record rather than editing history silently.

H.7 Loop residual

Definition

H_loop = $\mathcal{S}$(X_T,L_T,P_T) − $\mathcal{S}$(X0,L0,P0). (H.9)

It measures historical non-closure after an apparently closed state path.

Diagnostic signs

- R or θ returns but leverage, ownership, trust, or regulation does not;
- recovery remains fragile;
- the same visible state produces a different future response.

Response

- model hysteresis;
- include ledger variables in future projection;
- avoid treating price recovery as system restoration;
- revise stress baselines.

H.8 Transition residual

Definition

$\varepsilon_{transition} = Z_{P'}(t^*) - M_{P \to P'}[Z_P(t^{*-})]$. (H.10)

It measures what the old world cannot translate into the new one.

Diagnostic signs

- going-concern model suddenly replaced by liquidation;
- market price disappears after suspension;
- old coordinates lose meaning;
- new variables become decisive.

Response

- document protocol change;
- identify preserved and discarded variables;
- start a new world episode;
- preserve the old ledger;
- do not compare old and new Z without translation metadata.

H.9 Cross-observer residual

Definition

Let aligned observer states be:

$Z_{a^*} = M_{a \to *}(Z_a)$. (H.11)

$Z_b^* = M_{b\to *}(Z_b)$. (H.12)

Then:

$\varepsilon_{tri} = \text{Dispersion}(Z_a^*, Z_b^*, \ldots, Z_n^*)$. (H.13)

Possible causes

- legitimate frame difference;
- inconsistent inputs;
- implementation error;
- omitted primary structure;
- conflicting authorities.

Response

1. align protocols;
2. align data;
3. audit calculations;
4. compare authority;
5. classify remaining disagreement as material residual.

H.10 Residual severity vector

For each residual type j, define:

$\text{Severity}_j = \text{Magnitude}_j \times \text{Persistence}_j \times \text{Consequence}_j$. (H.14)

Where each component may be normalized to [0,1].

A total governed residual score is:

$S_E = \Sigma_j w_j \text{Severity}_j$. (H.15)

The weights w_j depend on:

- decision stakes;
- reversibility;
- horizon;
- capital buffer;
- legal consequence.

This score is not a universal truth measure.

It is a prioritization tool.

H.11 Residual state classification

Residual state	Condition	Interpretation
Green	low magnitude, low persistence	normal model noise
Amber	moderate or rising	diagnostic review required
Red	high and persistent	declared world may be failing
Black	protocol meaning lost	suspend model and revise world

The thresholds must be declared before evaluation.

H.12 First diagnostic split

When unexplained behaviour appears, ask:

Is the geometric identity valid? (H.16)

No

Investigate:

- data;
- coding;
- unit;
- domain.

Do not proceed.

Yes

Ask:

Does the dynamic law close within tolerance? (H.17)

H.13 Second diagnostic split

If dynamic closure fails, ask:

Did a gate event or discontinuity occur? (H.18)

Yes

Classify the deviation as possible:

ε_gate or ε_transition. (H.19)

Inspect:

- event authority;

- pre-state;
- post-state;
- protocol version.

No

Ask:

Can measurement or estimation error explain the failure? (H.20)

H.14 Third diagnostic split

If measurement error is insufficient, ask:

Is one pressure coordinate inadequate? (H.21)

Indicators include:

- credit and liquidity variables diverge;
- Q changes without coherent θ;
- residual correlates with an omitted mature risk measure.

Yes

Move to:

- multi-pressure model;
- residualization;
- coupling estimation.

No

Ask:

Has the baseline, horizon, or filter changed implicitly? (H.22)

H.15 Fourth diagnostic split

If declaration drift is present:

$P_{actual} \neq P_{recorded}$. (H.23)

Response:

- suspend historical comparability;
- create revision record;

- translate old states where possible;
- compute ε_transition.

If declaration remains stable, inspect projection residual:

Does the effective state omit a causally important variable? (H.24)

H.16 Fifth diagnostic split

If projection residual is material, choose among:

Expand the world

Add features or pressure channels.

Narrow the claim

Reduce the domain in which the model is treated as valid.

Add an observer

Use legal, liquidity, accounting, or operational expertise.

Reject the model

Conclude that no useful closure exists under P.

The last option must remain available.

H.17 Event-residual decision tree

For every committed event, ask:

1. Was the gate authority valid?
2. Was the evidence sufficient?
3. Was the applied protocol version correct?
4. Did the event change the action space?
5. What consequence remains unresolved?
6. Who owns the residual?
7. When will it be reviewed?
8. What evidence would resolve it?
9. Can the event be reversed?
10. Would reversal erase or merely extend the ledger?

The expected answer to question 10 is normally:

Reversal Extends the Ledger. (H.25)

H.18 Loop-closure decision tree

When a variable returns near its initial level, ask:

Is $R_T \approx R_0$? (H.26)

Is $\theta_T \approx \theta_0$? (H.27)

Is $A_T \approx A_0$? (H.28)

Is $L_T \simeq L_0$? (H.29)

Is $P_T = P_0$? (H.30)

If the first three are true but the last two are false, the system has achieved geometric return without world return.

Record:

$H_{loop} \neq 0$. (H.31)

H.19 Residual-age management

For ledger residual item j:

$Age_j(t) = t - t_j$. (H.32)

Define ageing burden:

$B_{age} = \Sigma_j w_j Age_j |E_{L,j}|$. (H.33)

Residuals should be escalated when:

- age exceeds threshold;
- evidence quality declines;
- responsible owner disappears;
- later decisions depend on the unresolved item.

H.20 Residual interaction

Residual channels may interact.

For example:

Projection residual → dynamic residual. (H.34)

Dynamic residual → protocol revision. (H.35)

Protocol revision → transition residual. (H.36)

Transition residual → ledger residual. (H.37)

A residual propagation matrix may be introduced:

E_{t+1} = M_EE_t + ξ_t. (H.38)

Where E contains the residual vector.

Large off-diagonal terms indicate that one unresolved failure generates others.

H.21 Residual containment

A residual is contained when:

1. it is identified;
2. it is attached to a state or record;
3. its consequence is bounded;
4. an owner is assigned;
5. a review condition exists;
6. it does not silently contaminate later claims.

Containment does not mean elimination.

It means governed non-closure.

H.22 Residual escalation rule

A residual should trigger protocol review when:

S_E > κ_review. (H.39)

It should trigger protocol suspension when:

S_E > κ_suspend (H.40)

or when one critical residual exceeds a hard gate:

Severity_critical > κ_hard. (H.41)

The thresholds should vary by decision stakes.

H.23 Residual resolution rule

Residual j is resolved only when:

ResolutionEvidence_j satisfies declared criterion C_j. (H.42)

The resolution update is:

$L_{new} = L_{old} \sqcup ResolutionRecord_j$. (H.43)

The original residual remains historically visible.

H.24 Residual and falsifiability

The theory remains falsifiable only if the following conclusion is admissible:

No stable effective financial world exists under the tested declaration. (H.44)

This conclusion may follow when:

- residual remains high;
- protocol variations change everything;
- no phase closure exists;
- no diagnostic gain occurs;
- interventions fail.

Residual is therefore not a weakness added to the model.

It is the mechanism that prevents the model from becoming self-sealing.

H.25 Compact diagnostic algorithm

The operational sequence is:

Validate Algebra
→ Validate Protocol
→ Validate Data
→ Estimate Dynamics
→ Inspect Gate Events
→ Classify Residual
→ Test Projection Adequacy
→ Test Ledger Memory
→ Continue, Expand, Narrow, Revise, or Reject. (H.45)

In pseudoequation form:

Decision(E,P) =
Continue, if $E \le \kappa_{low}$;
Review, if $\kappa_{low} < E \le \kappa_{high}$;
Revise, if $E > \kappa_{high}$ and translation exists;
Reject, if no admissible closure exists. (H.46)

Appendix I — Multi-Q Residualization and Metric Construction

I.1 The multi-pressure problem

The single-pressure geometry is:

$A^2 = R^2 + Q^2$. (I.1)

A realistic financial system may contain:

$\vec{Q}_{raw} = (Q_{market}, Q_{credit}, Q_{liquidity}, Q_{capital}, Q_{tail}, Q_{model}, \ldots)$. (I.2)

The naïve extension is:

$A^2 = R^2 + \Sigma_j Q_j^2$. (I.3)

This is usually invalid because the channels may overlap.

For example:

- credit spread may contain liquidity premium;
- volatility may contain credit information;
- capital pressure may respond to rating;
- tail risk may already enter market price.

Adding raw Q channels can therefore count the same pressure repeatedly.

I.2 Common-factor model

Let the raw pressure vector satisfy:

$\vec{Q}_{raw} = BF + u$. (I.4)

Where:

- F = common pressure factors;
- B = loading matrix;
- u = channel-specific residual pressure.

Estimate:

$\hat{F} = \text{argmin}_F \|\vec{Q}_{raw} - BF\|^2$. (I.5)

The residualized pressure is:

$\vec{Q}_{res} = \vec{Q}_{raw} - B\hat{F}$. (I.6)

Only $\vec{Q}_{res}$ should be treated as channel-specific orthogonal pressure.

I.3 Sequential residualization

Suppose liquidity pressure is partly explained by credit pressure.

Estimate:

$$Q_{liquidity} = \alpha + \beta Q_{credit} + u_{liquidity}. \quad (I.7)$$

Define residual liquidity pressure:

$$Q_{liquidity \perp credit} = u_{liquidity}. \quad (I.8)$$

Similarly:

$$Q_{capital \perp credit,liquidity} = \text{residual from regression on credit and liquidity pressure}. \quad (I.9)$$

The sequence must be economically justified because sequential residualization can depend on order.

A symmetric method such as principal components or joint covariance decomposition may be preferable when no natural hierarchy exists.

I.4 Covariance metric

Let Σ_Q be the covariance matrix of residualized pressure channels.

A whitening metric is:

$$G_{metric} = \Sigma_Q^{-1}. \quad (I.10)$$

Then the generalized pressure magnitude is:

$$M_Q^2 = \vec{Q}_{res}^{T} \Sigma_Q^{-1} \vec{Q}_{res}. \quad (I.11)$$

The full geometry becomes:

$$A^2 = R^2 + \kappa_Q M_Q^2. \quad (I.12)$$

Where κ_Q calibrates units between admitted value and normalized pressure distance.

This calibration cannot be chosen merely for visual convenience.

It requires economic interpretation or empirical fitting.

I.5 Positive semidefinite requirement

The metric must satisfy:

$$v^T G_{metric} v \geq 0 \quad (I.13)$$

for every pressure vector v.

Therefore:

$$G_{metric} \succeq 0. \quad (I.14)$$

A metric violating this condition can produce negative squared pressure magnitude and invalid

geometry.

I.6 Economic interaction metric

A covariance metric measures statistical overlap.

An economic interaction metric may instead be derived from intervention consequences.

Let loss response to pressure be:

$$\Delta \text{Loss} \approx J_Q \Delta \vec{Q}. \quad \text{(I.15)}$$

A consequence metric may be:

$$G_{\text{consequence}} = J_Q^T W_{\text{loss}} J_Q. \quad \text{(I.16)}$$

Where W_{loss} weights different consequence dimensions.

This metric treats two pressure channels as close when they produce similar economic effects.

I.7 Hybrid metric

A hybrid construction may combine statistical and consequence information:

$$G_{\text{hybrid}} = \lambda G_{\text{cov}} + (1 - \lambda) G_{\text{consequence}}. \quad \text{(I.17)}$$

Where:

$$0 \le \lambda \le 1. \quad \text{(I.18)}$$

The parameter λ must be validated out of sample.

I.8 Multi-pressure state

Define:

$$Y = [R, \vec{Q}^T]^T. \quad \text{(I.19)}$$

The dynamic law is:

$$dY/dt = \mathfrak{A} Y + \varepsilon_Y. \quad \text{(I.20)}$$

A structured generator is:

$$\mathfrak{A} = \begin{bmatrix} g_A & -\vec{\omega}^T \\ \vec{\omega} & g_{AI} + C_Q \end{bmatrix}. \quad \text{(I.21)}$$

Then:

$$dR/dt = g_{AR} - \vec{\omega}^T \vec{Q} + \varepsilon_R. \quad \text{(I.22)}$$

$$d\vec{Q}/dt = g_A \vec{Q} + \vec{\omega} R + C_Q \vec{Q} + \varepsilon_Q. \quad \text{(I.23)}$$

I.9 Total angular repricing load

Define:

$\Lambda_{total} = \omega^{\top} Q$. (I.24)

Then:

$dR/dt = g_{AR} - \Lambda_{total} + \varepsilon_R$. (I.25)

The channel contribution is:

$\Lambda_j = \omega_j Q_j$. (I.26)

But these contributions are interpretable only after overlap is controlled.

I.10 Interaction contribution

If pressure channels interact nonlinearly, introduce:

$\Lambda_{interaction} = Q^{\top} K_{\omega} Q$. (I.27)

Then:

$dR/dt = g_{AR} - \omega^{\top} Q - Q^{\top} K_{\omega} Q + \varepsilon_R$. (I.28)

This extension should be used only when interaction terms improve out-of-sample performance and remain interpretable.

I.11 Channel-dominance index

Define absolute load share:

$s_j = |\Lambda_j| / [\Sigma_l |\Lambda_l| + \delta]$. (I.29)

The dominant channel is:

$j^* = \operatorname{argmax}_j s_j$. (I.30)

Dominance is declared when:

$s_{j^*} \geq \kappa_{dom}$. (I.31)

If no channel dominates, the system is in a mixed-pressure regime.

I.12 Pressure concentration

Define a Herfindahl-type pressure concentration index:

$HHI_Q = \Sigma_j s_j^2$. (I.32)

High HHI_Q indicates one dominant pressure channel.

Low HHI_Q indicates distributed pressure.

A concentrated regime may admit targeted intervention.

A distributed regime may require system-wide action.

I.13 Metric instability

The pressure metric may change across regimes:

G_metric,t ≠ G_metric,t+1. (I.33)

For example, credit and liquidity pressure may become much more correlated during crisis.

A fixed metric may then underestimate systemic coupling.

Define metric drift:

D_G(t) = ‖G_metric,t − G_metric,t−1‖. (I.34)

Large D_G may be a declaration-revision signal.

I.14 Residualization failure

Residualization fails when:

- channels remain highly correlated;
- signs change unpredictably;
- results depend heavily on ordering;
- metric is unstable;
- pressure contributions are not reproducible.

The correct conclusion may be:

No Valid Multi-Q Orthogonalization Exists for This Episode. (I.35)

The researcher should then report the channels separately rather than force them into one geometric magnitude.

I.15 Multi-Q verification checklist

A valid multi-pressure model should declare:

1. raw pressure definitions;
2. common-factor method;
3. residualization order;
4. metric construction;

5. unit calibration;
6. covariance stability;
7. interaction terms;
8. out-of-sample validation;
9. residual after orthogonalization;
10. conditions under which the model is rejected.

The original Finance Geometry programme similarly requires residualization and careful interpretation before multiple hidden-pressure channels can be combined.

Appendix J — Noncommuting Filter Examples

J.1 Purpose

Financial filters are often applied sequentially.

A credit model may be applied before a liquidity haircut.

A rating action may occur before a collateral decision.

An impairment test may occur before a covenant test.

A stress model may trigger sales that alter the inputs of the next stress model.

If the first operation changes the state upon which the second operation acts, reversing their order may produce a different result.

The general condition is:

$$F_a[F_b(X)] \neq F_b[F_a(X)]. \quad (J.1)$$

This is **financial order sensitivity**.

It can arise from:

- threshold gates;
- ledger updates;
- changed legal status;
- changed collateral;
- changed asset population;
- changed intervention capacity;
- path-dependent behaviour.

It does not, by itself, establish quantum noncommutation. The handoff framework uses such macroscopic examples precisely to separate generic order dependence from the stronger mathematical structures specific to quantum observables.

J.2 Effective filter commutator

Define the financial filter commutator:

$$[F_a,F_b]_P(X,L) = F_a[F_b(X,L),L_b] - F_b[F_a(X,L),L_a]. \quad (J.2)$$

Here:

- L_a is the ledger after operation a;
- L_b is the ledger after operation b;
- P is the declared protocol governing both paths.

If:

$$[F_a,F_b]_P(X,L) = 0, \quad (J.3)$$

the filters commute for the declared state.

If:

$$[F_a,F_b]_P(X,L) \neq 0, \quad (J.4)$$

the final effective state depends on sequence.

A normalized order-sensitivity index is:

$$\kappa_{ab} = \|[F_a,F_b]_P(X,L)\|/[\|F_aF_b(X,L)\| + \|F_bF_a(X,L)\| + \delta]. \quad (J.5)$$

Where:

$$\delta > 0 \quad (J.6)$$

prevents instability near zero-valued outputs.

J.3 Three sources of noncommutation

Financial operations can fail to commute for at least three distinct reasons.

J.3.1 State transformation

The first filter alters an input used by the second.

J.3.2 Gate transformation

The first filter activates or disables a threshold governing the second.

J.3.3 Ledger transformation

The first event creates a record that changes how the second filter interprets an otherwise similar state.

These mechanisms should be separately identified.

The generic decomposition is:

[F_a,F_b]_P = C_state + C_gate + C_ledger + C_residual. (J.7)

Where:

- C_state = order effect caused by changed continuous state;
- C_gate = effect caused by threshold activation;
- C_ledger = effect caused by historical trace;
- C_residual = unexplained order effect.

J.4 Example 1 — Credit Filter Followed by Liquidity Filter

J.4.1 Initial state

Assume a default-free reference amplitude:

A = 100. (J.8)

The initial credit filter admits:

R_credit = 90. (J.9)

Suppose a liquidity filter applies a 10% executable-value reduction:

F_liq(R) = 0.90R. (J.10)

If credit is applied first:

R_credit→liq = F_liq(F_credit(A)). (J.11)

Therefore:

R_credit→liq = 0.90 × 90 = 81. (J.12)

J.4.2 Reverse sequence without state interaction

Suppose the liquidity filter first reduces A:

F_liq(A) = 90. (J.13)

If the credit filter simply applies the same proportional 10% reduction:

F_credit(R) = 0.90R. (J.14)

Then:

R_liq→credit = 0.90 × 90 = 81. (J.15)

Under purely proportional filters:

$F_{credit,F_{liq}} = 0$. (J.16)

The operations commute.

J.4.3 Gate-dependent credit–liquidity interaction

Now suppose the credit filter includes a collateral gate.

If value after liquidity adjustment falls below 95:

$G_{collateral} = 1 \text{ if } R < 95$. (J.17)

Once the gate activates, the credit admission ratio falls from 0.90 to 0.75.

Credit first

Credit admission occurs while $A = 100$:

$R_{credit} = 0.90 \times 100 = 90$. (J.18)

Liquidity then applies:

$R_{credit \to liq} = 0.90 \times 90 = 81$. (J.19)

Liquidity first

Liquidity reduces the state to:

$R_{liq} = 0.90 \times 100 = 90$. (J.20)

This crosses the collateral threshold.

The harsher credit filter applies:

$R_{liq \to credit} = 0.75 \times 90 = 67.5$. (J.21)

Therefore:

$F_{credit,F_{liq}} = 81 - 67.5 = 13.5$. (J.22)

The filters no longer commute.

The difference arises because the first operation changes which credit gate governs the second.

J.4.4 Financial interpretation

The first path is:

Credit Adjustment → Ordinary Liquidity Haircut. (J.23)

The second path is:

Liquidity Shock → Collateral Ineligibility → Harsher Credit Adjustment. (J.24)

The final values differ even though the same named filters appear in both paths.

The sequence changes the applicable world.

J.5 Example 2 — Rating Action and Collateral Eligibility

J.5.1 Declared rules

Suppose a bond is eligible collateral only while its rating is investment grade.

Let:

$G_{collateral}(\rho) = 1$ if $\rho \geq BBB-$. (J.25)

Assume the bond begins at:

$\rho_0 = BBB$. (J.26)

Its market value is:

$R_0 = 100$. (J.27)

The ordinary collateral haircut is 5%.

Therefore:

$CollateralValue_0 = 0.95R_0 = 95$. (J.28)

J.5.2 Path A — Downgrade first

Suppose the rating is downgraded to BB+.

Then:

$G_{collateral}(BB+) = 0$. (J.29)

The asset becomes ineligible.

Its collateral value is:

$CollateralValue_A = 0$. (J.30)

A forced-sale process begins.

Assume forced selling lowers market value to:

$R_A = 85$. (J.31)

The path is:

Downgrade → Ineligibility → Forced Sale → Price Decline. (J.32)

J.5.3 Path B — Collateral support first

Suppose an emergency liquidity facility first changes the collateral rule.

The temporary gate becomes:

$G_{collateral}^{emergency}(\rho) = 1$ if $\rho \geq BB$. (J.33)

The bond remains eligible after the downgrade.

Assume the emergency haircut is 20%.

Then:

$CollateralValue_B = 0.80 \times 100 = 80$. (J.34)

Because forced selling is avoided, market value remains near:

$R_B = 100$. (J.35)

The path is:

Emergency Gate Revision → Downgrade → Continued Eligibility → No Forced Sale. (J.36)

J.5.4 Order effect

The two paths differ because the intervention changes the gate before the rating event.

The effective commutator is:

$F_{rating},F_{collateral} \neq 0$. (J.37)

The difference is not created by mysterious quantum incompatibility.

It is created by:

- institutional authority;
- gate timing;
- changed action space;
- backreaction.

J.6 Example 3 — Accounting Impairment and Covenant Test

J.6.1 Initial balance-sheet state

Assume:

Reported Assets = 150. (J.38)

Reported Liabilities = 100. (J.39)

Reported Equity = 50. (J.40)

A covenant requires:

Equity/Liabilities ≥ 0.40. (J.41)

Initially:

50/100 = 0.50. (J.42)

The covenant passes.

J.6.2 Path A — Impairment first

Suppose an impairment of 15 is recognized.

Then:

Assets_A = 135. (J.43)

Equity_A = 35. (J.44)

The covenant ratio becomes:

35/100 = 0.35. (J.45)

The covenant is breached.

The breach may trigger:

- higher interest;
- restricted dividends;
- accelerated repayment;
- renegotiation;
- default classification.

Thus:

Impairment → Covenant Breach → Financing Consequence. (J.46)

J.6.3 Path B — Covenant renegotiation first

Suppose the borrower renegotiates the covenant threshold from 0.40 to 0.30 before impairment is recognized.

After impairment:

Equity_B = 35. (J.47)

The ratio remains:

35/100 = 0.35. (J.48)

The revised covenant passes.

The path is:

Covenant Revision → Impairment → No Breach. (J.49)

J.6.4 Ledger dependence

The two paths may end with identical accounting numbers:

$$\text{Assets}_A = \text{Assets}_B = 135. \quad (J.50)$$

$$\text{Equity}_A = \text{Equity}_B = 35. \quad (J.51)$$

But the ledgers differ.

Path A contains:

- breach record;
- possible default record;
- renegotiation under distress.

Path B contains:

- prior consensual amendment;
- no breach.

Therefore:

$$\text{Same Balance Sheet} \neq \text{Same Financial World}. \quad (J.52)$$

The order effect is preserved in the ledger even when final scalar values match.

J.7 Example 4 — VaR Shock and Forced Sale

J.7.1 Risk-model rule

Suppose a portfolio has value:

$$R_0 = 100. \quad (J.53)$$

A VaR limit requires:

$$\text{VaR}/R \leq 0.10. \quad (J.54)$$

Initially:

$$\text{VaR}_0 = 8. \quad (J.55)$$

The ratio is:

$$0.08. \quad (J.56)$$

No sale is required.

J.7.2 Path A — Volatility shock first

Suppose volatility rises and VaR increases to:

$VaR_1 = 14$. (J.57)

The ratio becomes:

0.14. (J.58)

The limit is breached.

The institution sells assets.

Assume the forced sale reduces market value to:

$R_A = 92$. (J.59)

The next VaR calculation may increase further because volatility and liquidity worsened.

The path is:

Volatility Shock → VaR Breach → Forced Sale → Price Impact → Higher Next VaR. (J.60)

J.7.3 Path B — Position reduction first

Suppose the institution voluntarily reduces exposure before volatility rises.

The portfolio value becomes:

$R_{pre} = 80$. (J.61)

Assume the reduced position produces:

$VaR_{pre} = 6$. (J.62)

After the same proportional volatility shock:

$VaR_B = 10.5$. (J.63)

The ratio becomes:

$10.5/80 \approx 0.1313$. (J.64)

A breach may still occur, but the absolute forced-sale burden is smaller.

If liquidity support is also available, the gate may not trigger at all.

The final loss differs because position reduction altered the state before the risk-model event.

J.7.4 Reflexive model operation

VaR is not only measuring risk.

Through the limit gate, it changes position size.

The changed position changes market impact.

Market impact changes the next VaR.

Therefore:

Risk Model → Gate → Action → Market → Next Risk Model. (J.65)

The operation is self-referential and order-sensitive.

J.8 Example 5 — Option Exercise and Financing Approval

J.8.1 Latent option state

Suppose a project has latent economic value:

$A_{option} = 120$. (J.66)

Its current exercise-admitted value is:

$R_{option} = 90$. (J.67)

The retained option pressure is:

$Q_{option} = \sqrt{(120^2 - 90^2)}$. (J.68)

Therefore:

$Q_{option} \approx 79.3725$. (J.69)

The project requires financing approval before exercise.

J.8.2 Path A — Financing approval first

A financing commitment is obtained while market conditions remain favourable.

The project is then exercised.

The exercise cost is locked.

The post-exercise world contains:

- contractual debt;
- committed capital expenditure;
- operational obligations.

The path is:

Financing Approval → Exercise → Project World. (J.70)

J.8.3 Path B — Exercise decision first

Suppose management announces exercise before financing is secured.

The announcement alters:

- market expectations;
- negotiating leverage;
- credit exposure;
- perceived commitment.

Lenders may demand a higher spread.

The financing filter tightens.

The project may become uneconomic or require equity issuance.

The path is:

Exercise Commitment → Financing Repricing → Altered Project Economics. (J.71)

J.8.4 Noncommutation

Let:

F_finance = financing-admission filter. (J.72)

Let:

G_exercise = exercise gate. (J.73)

Then:

G_exerciseF_finance(X) ≠ F_financeG_exercise(X). (J.74)

The exercise gate changes the state that the financing filter evaluates.

The financing decision changes the world in which exercise occurs.

J.9 Example 6 — Regulatory Stress Test and Capital Raising

J.9.1 Path A — Stress result first

A stress test identifies a capital shortfall.

The result becomes public.

The institution attempts to raise capital after the disclosure.

The disclosure may depress the share price and increase the cost of new equity.

The path is:

Stress Disclosure → Price Decline → More Expensive Capital Raise. (J.75)

J.9.2 Path B — Capital raise first

The institution raises capital before the stress result is published.

The stronger balance sheet changes the stress result.

The eventual disclosure may show no shortfall.

The path is:

Capital Raise → Improved Stress State → No Adverse Disclosure. (J.76)

J.9.3 Observer intervention

The stress test is both:

- measurement;
- potential causal intervention.

Its timing relative to capital action changes the outcome.

This is a classical example of observer backreaction combined with order-sensitive gates.

J.10 Continuous noncommutation

J.10.1 Vector-field interpretation

Suppose two financial transformations generate continuous flows.

Let:

V_a(X) (J.77)

and:

V_b(X) (J.78)

be vector fields representing two pressure processes.

Their Lie bracket is:

V_a,V_b = J_{V_b}(X)V_a(X) − J_{V_a}(X)V_b(X). (J.79)

Where J denotes the Jacobian.

If:

V_a,V_b = 0, (J.80)

small sequential 0, (J.80)

small sequential movements commute to first order.

If:

V_a,V_b ≠ 0, (J.81)

the order of infinitesimal transformations matters.

This can occur in ordinary nonlinear classical systems.

J.10.2 Financial interpretation

A nonzero Lie bracket may indicate that:

- credit pressure changes liquidity sensitivity;
- liquidity pressure changes credit sensitivity;
- capital pressure changes market-risk exposure;
- market movement changes regulatory response.

The bracket measures local path dependence.

It does not imply a quantum operator algebra.

J.11 Ledger-conditioned filters

J.11.1 State-plus-ledger filter

A mature filter should often be written:

F_a(X,L). (J.82)

not merely:

F_a(X). (J.83)

After event a:

L_a = L ⊔ Record_a. (J.84)

The second filter becomes:

F_b(X_a,L_a). (J.85)

In the reverse order:

F_a(X_b,L_b). (J.86)

Because:

L_a ≠ L_b, (J.87)

the operations may differ even when:

$X_a \approx X_b$. (J.88)

J.11.2 History-generated order effects

A system may therefore exhibit order sensitivity without any hidden instantaneous physical incompatibility.

The difference can be entirely carried by trace:

Order Effect = Ledger Difference. (J.89)

This is one reason financial order effects should not be called quantum merely because they resemble noncommuting operators.

J.12 Gate-induced branching

J.12.1 Branch structure

Suppose operation a may trigger gate G_a.

The state branches:

$X \rightarrow X_a^{commit}$ if $G_a = 1$. (J.90)

$X \rightarrow X_a^{defer}$ if $G_a = 0$. (J.91)

The second filter acts on different branches.

Therefore:

$F_b(X_a^{commit}) \neq F_b(X_a^{defer})$. (J.92)

The reverse path may never enter the same branches.

J.12.2 Branch erasure

Some branches become unavailable after commitment.

For example:

- a completed sale cannot be treated as an unsold position;
- default cannot be analysed as though no default declaration occurred;
- exercised optionality is no longer unexercised optionality.

Thus:

Gate Commitment Removes Some Future Alternatives. (J.93)

This creates irreversible path dependence.

J.13 Experimental test for financial noncommutation

J.13.1 Declared question

The experiment should ask:

Does order produce outcome differences after final observable states and ordinary control variables are matched? (J.94)

A weak test merely observes different outcomes after different sequences.

A stronger test controls for:

- initial state;
- final scalar values;
- external shocks;
- decision authority;
- time interval;
- liquidity;
- market regime.

J.13.2 Paired-path design

Identify episodes:

Path A: a → b. (J.95)

Path B: b → a. (J.96)

Estimate:

Δ_ab = Outcome(a → b) − Outcome(b → a). (J.97)

Then control for:

C = (InitialState,FinalLevels,Environment,Protocol). (J.98)

The order effect is supported when:

E(Δ_ab | C) ≠ 0. (J.99)

J.13.3 Ledger mediation test

Test whether the order effect disappears after ledger variables are included.

Model 1:

Outcome = α + βOrder + γControls + u. (J.100)

Model 2:

Outcome = α + β′Order + γControls + ηLedger + u. (J.101)

If:

β′ ≈ 0, (J.102)

the order effect is largely mediated by recorded history.

If:

β′ remains material, (J.103)

additional state or gate mechanisms may exist.

J.13.4 Gate mediation test

Include gate indicators:

Outcome = α + βOrder + γControls + ηLedger + ζGate + u. (J.104)

If the order coefficient disappears after Gate is included, the noncommutation is gate-induced.

J.13.5 Residual order effect

Define:

ε_order = Δ_ab − Δ̂_state − Δ̂_gate − Δ̂_ledger. (J.105)

A persistent ε_order may indicate:

- omitted nonlinear interaction;
- incorrect matching;
- strategic anticipation;
- protocol inconsistency;
- genuine unresolved path dependence.

It should not automatically be called quantum residue.

J.14 Conditions under which filters commute

Financial filters may commute when:

1. both are linear;
2. both act on independent coordinates;

3. neither changes the gate or ledger;
4. no backreaction occurs between applications;
5. the same baseline and horizon are preserved.

For linear operators:

$F_aF_b = F_bF_a$. (J.106)

This may hold when one filter adjusts an independent cash-flow component while another adjusts a separate, non-interacting component.

Commutation should be tested rather than assumed.

J.15 Conditions under which order effects are misleading

An apparent order effect may be spurious when:

- the initial states were different;
- the time intervals differed;
- the second event was anticipated;
- the protocols were not aligned;
- the data omitted an external shock;
- one path occurred only during crisis;
- selection bias determined which path was observed.

The causal claim requires more than observing two historical sequences.

J.16 Quantum boundary

Financial noncommutation demonstrates:

Classical Context + Memory + Gates + Backreaction → Order-Sensitive Operations. (J.107)

It does not demonstrate:

- incompatible quantum observables;
- Hilbert-space operator structure;
- uncertainty relations;
- contextuality inequalities;
- Bell violations.

The correct comparative statement is:

Appendix K — Scope, Non-Claims, and Financial Disclaimer

K.1 Purpose of this appendix

The framework combines:

- exact geometric identities;
- finance interpretations;
- empirically testable hypotheses;
- broader observer-theory conjectures.

These levels must remain visibly separated.

The original Finance Geometry article presents the complex coordinate as a disciplined extension of mature financial filters, not a replacement for established finance and not a claim that markets arestems. filecite turn4file1

The dynamic handoff extends that framework toward effective-world formation while repeatedly preserving the distinction between generic observer-bound structures and isults. filecite turn4file0

This appendix defines the interpretive boundaries of the completed article.

K.2 Claim classification

K.2.1 Exact representational claims

The following are exact once A, R, and the domain are declared:

$A^2 = R^2 + Q^2$. (K.1)

$R = A \cos \theta$. (K.2)

$Q = A \sin \theta$. (K.3)

$Z = R + iQ = A \exp(i\theta)$. (K.4)

$\theta = \arccos(R/A)$. (K.5)

For differentiable $A(t)$ and $\theta(t)$:

$dZ/dt = [(1/A)(dA/dt) + i(d\theta/dt)]Z$. (K.6)

These are mathematical identities within the chosen representation.

They do not establish that the representation is economically useful.

K.2.2 Interpretive finance claims

The following are protocol-dependent interpretations:

- A represents declared pre-filter financial amplitude;
- R represents admitted value;
- Q represents retained pressure;
- θ represents orientation of the mature financial filter;
- g_A represents radial economic change;
- ω_F represents financial-frame velocity;
- Qω_F represents angular repricing load.

These interpretations are valid only when:

- the baseline is declared;
- the filter is mature;
- the horizon is fixed;
- the units are aligned;
- the ratio R/A is meaningful.

K.2.3 Empirical hypotheses

The following are hypotheses rather than identities:

- Q may identify pressure before a large scalar haircut appears;
- Qω_F may improve stress attribution;
- dynamic residual may rise before model or regime failure;
- phase indexing may align events better than calendar time;
- loop residual may predict fragility after visible recovery;
- noncommuting filter measures may improve path-dependence diagnosis.

Each claim requires out-of-sample testing.

K.2.4 Observer-theory propositions

The following are theoretical interpretations:

- a protocol-bounded representation can become an effective world;

- ledgered trace contributes to internal historical time;
- backreaction makes financial observables causally operative;
- world-forming boundaries may be more complex than internal laws;
- finance can serve as a non-quantum control case for observer-bound structure.

These claims are intended as coherent conceptual proposals.

They are not established physical laws.

K.2.5 Speculative physics claims

The following remain speculative:

- the observable quantum world may be generated by a deeper contextual process;
- measurement constraints may explain part of quantum strangeness;
- a constructor-like process may generate usable observer state spaces;
- Hilbert-space structure may emerge from a deeper relational grammar.

The article does not derive these possibilities.

It uses finance to sharpen the questions.

K.3 What the framework does not prove

K.3.1 It does not prove that markets are quantum

The use of:

$Z = R + iQ$ (K.7)

does not imply that financial assets occupy quantum states.

Complex coordinates occur throughout classical mathematics and engineering.

K.3.2 It does not derive the Born rule

The identity:

$A^2 = R^2 + Q^2$ (K.8)

does not imply:

$P(R) = R^2/A^2$. (K.9)

R and Q are financial coordinates, not probability amplitudes.

K.3.3 It does not establish entanglement

Financial dependence, contagion, and shared collateral do not establish tensor-product nonseparability.

Systemic coupling is not automatically quantum entanglement.

K.3.4 It does not explain Bell violations

The framework does not reproduce loophole-controlled Bell correlations.

Any deeper physical proposal must confront Bell's theorem explicitly.

K.3.5 It does not establish quantum contextuality

Protocol dependence is weaker than the impossibility of a global noncontextual value assignment.

Ordinary financial context dependence can be represented through classical hidden state, memory, and institutions.

K.3.6 It does not prove hidden classical variables

The constructor analogy does not imply that a complete local classical state exists beneath quantum observations.

Any such claim must address:

- Bell constraints;
- contextuality;
- measurement independence;
- locality;
- relativistic causality.

K.3.7 It does not solve the measurement problem

Financial gates show that possibility-to-record commitment is a general operational grammar.

They do not explain how quantum amplitudes produce definite outcomes.

K.3.8 It does not solve the quantum problem of time

A financial phase can function as a local clock.

This does not resolve time in quantum gravity or the status of time in quantum mechanics.

K.4 What the framework does not replace

The framework does not replace:

- discounted cash-flow analysis;
- CAPM;
- multifactor models;
- stochastic discount factors;
- certainty equivalents;
- option pricing;
- duration;
- convexity;
- structural credit models;
- reduced-form credit models;
- liquidity measures;
- VaR;
- Expected Shortfall;
- stress testing;
- accounting standards;
- legal analysis;
- regulatory capital frameworks.

Its role is secondary:

Mature Finance Output → Pressure-Preserving and Runtime Interpretation. (K.10)

The underlying finance method remains responsible for generating R and its governing ratio.

K.5 Domain restrictions

K.5.1 Minimal geometric domain

The basic construction assumes:

$A \geq 0$. (K.11)

$0 \leq R \leq A$. (K.12)

$0 \leq \theta \leq \pi/2$. (K.13)

$Q \geq 0$. (K.14)

Cases involving:

- negative value;
- liabilities;
- short positions;
- market values above the declared amplitude;
- signed directional pressure;

may require a different geometry.

K.5.2 Baseline sensitivity

Changing A changes:

- θ;
- Q;
- pressure ratio;
- dynamic decomposition.

Therefore:

A New Baseline Creates a New Financial Geometry. (K.15)

Results from different baselines should not be compared without an explicit translation.

K.5.3 Horizon sensitivity

Changing T may change:

- discount ratios;
- angle;
- pressure;
- gate relevance.

Therefore:

θ_T_1 and θ_T_2 are different objects unless a horizon transformation is declared. (K.16)

K.5.4 Filter specificity

A CAPM-derived Q is not automatically comparable with a liquidity-derived Q.

The filter label must remain attached.

K.5.5 Derivative instability

Variables such as:

$\omega_F = d\theta/dt$ (K.17)

and:

$\alpha_F = d^2\theta/dt^2$ (K.18)

may be highly sensitive to noise.

Derivative-based conclusions require:

- smoothing;
- uncertainty intervals;
- robustness checks;
- untouched test data.

K.6 Q is not a universal risk score

Q depends on:

- A;
- R;
- filter;
- horizon;
- baseline.

Therefore, a larger Q does not universally imply:

- greater expected loss;
- higher volatility;
- higher default probability;
- worse investment quality.

A large Q means:

A larger retained pressure coordinate under the declared filter. (K.19)

Its economic meaning must be interpreted through that filter.

K.7 Residual is not automatically signal

A large residual may represent:

- noise;
- data error;
- omitted variable;
- model failure;
- protocol change;
- gate discontinuity.

It should not automatically be treated as:

- hidden alpha;
- new risk factor;
- evidence of a deeper world;
- evidence of quantum behaviour.

The correct rule is:

Residual First Questions the Model; Only Later May It Support a New Model. (K.20)

K.8 Phase is not globally time

A filter angle may function as a local internal clock only when:

- θ is identifiable;
- θ is locally monotonic;
- the state closes adequately in phase coordinates;
- relevant events can be ordered;
- gates and trace preserve consequence.

When these conditions fail:

Phase ≠ Usable Internal Time. (K.21)

K.9 World language is operational, not absolute ontology

The phrase “effective financial world” means a protocol-bounded operational state space.

It does not mean:

- a separate physical universe;
- a metaphysically independent reality;
- an observer-created world without external constraint.

The financial world remains embedded in a larger primary field.

Its worldhood is:

- task-relative;
- horizon-relative;
- protocol-relative;
- causally consequential.

K.10 Causality limitations

Backreaction examples show plausible causal loops.

But empirical correlation does not establish causation.

For example:

Rating Change ↔ Funding Cost (K.22)

may involve:

- common economic causes;
- anticipation;
- selection bias;
- policy response.

Causal claims require:

- timing;
- identification strategy;
- natural experiment;
- instrumental variable;
- structural model;
- intervention test.

K.11 Multi-Q limitations

The multi-pressure extension requires:

- non-overlapping channel definitions;
- residualization;
- metric construction;
- stability testing;
- interaction validation.

The expression:

$A^2 = R^2 + Q^{\uparrow}GQ^{\downarrow}$ (K.23)

is not valid merely because several risk measures are available.

If the channels cannot be separated cleanly:

Report Them Separately. (K.24)

Do not force geometric orthogonality.

K.12 Noncommutation limitations

Financial order effects may be caused by:

- omitted state;
- different initial conditions;
- different timing;
- different external shocks;
- gate activation;
- ledger differences.

The existence of:

$F_aF_b \neq F_bF_a$ (K.25)

does not establish quantum operator incompatibility.

The correct interpretation is classical path dependence unless specifically quantum structure is demonstrated.

K.13 Objectivity limitations

Protocol-relative objectivity depends on:

- equivalent frames;
- shared data;
- stable implementation;

- auditable authority;
- declared residual.

Agreement created by one dominant institution is not necessarily epistemic objectivity.

It may reflect:

- legal authority;
- market power;
- enforced standardization.

The article distinguishes:

Institutional Commitment (K.26)

from:

Empirical Truth. (K.27)

They can overlap, but they are not identical.

K.14 Model-risk statement

Any implementation of Finance Geometry creates model risk.

Possible failure sources include:

- incorrect amplitude;
- unstable baseline;
- misidentified filter;
- over-smoothed phase;
- noisy derivatives;
- omitted pressure channels;
- false gate thresholds;
- uncontrolled backreaction;
- retrospective protocol revision.

A model-risk register should accompany any practical use.

K.15 Minimum conditions for applied use

Before operational adoption, the implementation should demonstrate:

1. a mature originating filter;

2. clear meaning of A and R;
3. stable domain validity;
4. reproducible θ and Q;
5. controlled derivative estimation;
6. incremental diagnostic value;
7. comparison with null models;
8. residual disclosure;
9. intervention relevance;
10. independent review.

Failure of these conditions should block operational reliance.

K.16 Minimum conditions for publication

A published empirical claim should disclose:

- protocol;
- baseline;
- horizon;
- filter;
- data source;
- code version;
- derivative method;
- residual threshold;
- train–test split;
- null models;
- failed tests;
- sensitivity analysis.

Selective reporting of successful angles or assets would invalidate the evidence.

K.17 Financial disclaimer

This article is provided for theoretical, academic, and research purposes.

It is not:

- investment advice;
- trading advice;
- tax advice;
- legal advice;
- accounting advice;
- credit advice;
- regulatory advice;
- a recommendation to buy, sell, hold, borrow, lend, hedge, or restructure.

The equations and examples are illustrative.

They may rely on simplifying assumptions including:

- fixed cash flows;
- stable rates;
- continuous differentiability;
- zero transaction costs;
- zero residual;
- idealized gate rules.

Real financial decisions require qualified professional analysis, current data, institutional authorization, and compliance with applicable law and regulation.

K.18 Physics disclaimer

The article does not claim experimental evidence for a new physical theory.

It does not claim to supersede:

- quantum mechanics;
- quantum field theory;
- general relativity;
- statistical mechanics;
- established quantum-foundations results.

Its physics contribution is methodological:

Finance provides a macro control case for distinguishing generic observer-bound world-forming structures from candidate irreducibly quantum structure. (K.28)

K.19 Interpretation firewall

The following firewall should be applied throughout the article:

Mathematical Identity ≠ Empirical Success. (K.29)

Empirical Success ≠ Universal Validity. (K.30)

Functional Analogy ≠ Material Identity. (K.31)

Observer Dependence ≠ Quantum Contextuality. (K.32)

Order Dependence ≠ Quantum Noncommutation. (K.33)

Complex Coordinates ≠ Quantum State. (K.34)

Gate Commitment ≠ Physical Wavefunction Collapse. (K.35)

Hidden Residual ≠ Hidden Classical Reality. (K.36)

Constructed World ≠ Arbitrary Fiction. (K.37)

Effective Reality ≠ Fundamental Ontology. (K.38)

K.20 Claim-status table

Claim	Status
$A^2 = R^2 + Q^2$ under the declared geometry	exact identity
$dZ/dt = (g_A + i\omega_F)Z$ for differentiable A and θ	exact identity
Q is retained pressure	protocol-dependent interpretation
$Q\omega_F$ is angular repricing load	derived interpretation
Q may give early warning	empirical hypothesis
phase may serve as internal time	conditional hypothesis
ledger creates historical irreversibility	operational proposition
valuation can backreact on finance	observable macro mechanism
finance is a non-quantum control case	methodological proposition
quantum reality is generated by a deeper layer	speculative conjecture
quantum mechanics is ordinary hidden statistics	not established

---

K.21 Final adoption rule

The entire framework can be governed by one sequence:

Declare
→ Derive
→ Measure
→ Compare
→ Intervene
→ Audit
→ Revise or Reject. (K.39)

The framework should be retained only when:

Interpretive Gain + Predictive Gain + Governance Gain > Complexity Cost + Model Risk. (K.40)

If this inequality is not supported:

Use the Mature Finance Model Without the Complex Extension. (K.41)

K.22 Final boundary statement

The completed theory makes a disciplined proposal.

A mature financial filter may be represented as a world-forming operation that:

- admits value;
- retains pressure;
- generates phase;
- supports local internal dynamics;
- commits events;
- records history;
- acts back upon its parent field;
- revises itself under residual pressure.

This proposal remains valuable only while its boundaries remain visible.

The article therefore ends with four safeguards:

No Declared Filter → No Valid θ. (K.42)

No Declared Amplitude → No Valid Q. (K.43)

No Empirical Gain → No Financial Adoption. (K.44)

No Irreducibly Quantum Signature → No Quantum Ontology Claim. (K.45)

These safeguards do not weaken the framework.

They are what allow its inspirational reach to remain compatible with scientific and financial discipline.

Appendix L — Protocol Declaration Worksheet and Minimal Reproducible Experiment

L.1 Purpose

The framework developed in this article is useful only when its declarations are made before interpretation.

A researcher should not begin with an attractive chart and then invent:

- the amplitude A;
- the finance angle θ;
- the pressure coordinate Q;
- the event gate G;
- the residual threshold;
- the relevant benchmark.

The correct order is:

Declare → Compute → Test → Compare → Accept, Revise, or Reject. (L.1)

This appendix converts the theory into a reusable research protocol.

Its purposes are:

1. to prevent retrospective interpretation;
2. to make different implementations comparable;
3. to distinguish mathematical identity from empirical success;
4. to ensure that residual can invalidate the model;
5. to produce a minimal experiment that another researcher can rerun.

The central protocol remains:

P = (B,q,φ,h,F,G,T,U). (L.2)

Where:

- B = system boundary;
- q = baseline;
- φ = feature map;
- h = horizon;
- F = mature finance filter;
- G = commitment gate;
- T = trace rule;
- U = admissible intervention set.

The worksheet should be completed before the final test data are examined.

L.2 Research Registration Header

Every experiment should begin with a registration header.

L.2.1 Project identity

Project title:
[Enter title]

Protocol ID:
[Enter unique identifier]

Protocol version:
[Enter version]

Research team:
[Enter names or identifiers]

Registration date:
[YYYY-MM-DD]

Planned completion date:
[YYYY-MM-DD]

Asset class or system:
[Enter asset, firm, portfolio, project, or market]

Research question:
[Enter one precise question]

Primary hypothesis:
[Enter one falsifiable statement]

Primary outcome:
[Enter one measurable outcome]

Primary null model:
[Enter mature benchmark]

Final untouched test period:
[Enter date range]

L.2.2 Claim classification

The researcher should classify the intended claim.

Mark one or more:

- exact mathematical identity;
- descriptive interpretation;
- diagnostic claim;
- predictive claim;
- causal claim;
- intervention claim;

- observer-theory claim;
- quantum-comparison claim.

The strongest permitted conclusion should be declared before testing.

For example:

MaximumClaim = “Λ_F improves out-of-sample downgrade prediction beyond spread level and spread change.” (L.3)

The experiment should not later be presented as proving:

- universal financial geometry;
- quantum market behaviour;
- hidden physical ontology;
- general causal law;

unless those claims were explicitly designed and tested.

L.3 One-Page Protocol Declaration

The minimum protocol worksheet is:

Component	Declaration
B — Boundary	What system is included and excluded?
q — Baseline	Relative to what reference is amplitude defined?
φ — Feature map	Which variables enter the filter?
h — Horizon	Over what period is value defined?
F — Filter	Which mature financial model generates R?
G — Gate	What counts as a committed event?
T — Trace	What evidence and residual enter the ledger?
U — Intervention	Which actions are admissible?

The protocol should be expressed as:

P_exp = (B_exp,q_exp,φ_exp,h_exp,F_exp,G_exp,T_exp,U_exp). (L.4)

A protocol is incomplete if any component remains undefined.

L.4 System Boundary Worksheet

L.4.1 Boundary declaration

The system boundary B should specify:

Primary object:
[Asset, issuer, project, portfolio, institution, or market]

Included legal entities:
[Enter list]

Excluded legal entities:
[Enter list]

Included markets or venues:
[Enter list]

Included jurisdictions:
[Enter list]

Included counterparties:
[Enter class or list]

Included liabilities:
[Enter list]

Included off-balance-sheet exposures:
[Enter list]

Excluded exposures:
[Enter list]

Environmental variables treated as external:
[Enter variables]

L.4.2 Boundary test

The boundary should pass three tests.

Relevance test

Does every included variable affect the declared outcome?

Sufficiency test

Are any excluded variables likely to dominate the result?

Stability test

Can B remain approximately stable during the declared episode?

Define:

BoundaryValid = Relevance ∧ Sufficiency ∧ Stability. (L.5)

If:

BoundaryValid = False, (L.6)

the protocol should be revised before geometry is computed.

L.4.3 Boundary-change rule

A boundary change occurs when:

$B_{k+1} \neq B_k$. (L.7)

Examples include:

- a subsidiary enters consolidation;
- a guarantee becomes enforceable;
- a market closes;
- a jurisdiction imposes restrictions;
- a previously external counterparty becomes systemically relevant.

The researcher must declare:

BoundaryChangeTrigger = [Enter condition]. (L.8)

A boundary change creates a protocol transition.

It should not be treated as an ordinary movement in θ.

L.5 Baseline Worksheet

L.5.1 Baseline identity

The baseline q defines the reference from which A is constructed.

Possible baselines include:

- risk-free discounting;
- default-free value;
- expected cash flow;
- frictionless executable value;
- pre-crisis state;
- unconstrained economic value;
- certainty-neutral value;
- strategic value before exercise restrictions.

The worksheet should state:

Baseline name:
[Enter]

Economic meaning:
[Enter]

Formula:

[Enter]

Data source:
[Enter]

Update frequency:
[Enter]

Units:
[Enter]

Reason for selection:
[Enter]

L.5.2 Baseline equation

The amplitude should be defined explicitly.

For example:

$A_t = CF_{T,t}/(1 + r_{*,t})^T$. (L.9)

Or:

$A_t = V_{default\text{-}free,t}$. (L.10)

Or:

$A_t = E_t(CF_T)/(1 + r_*)^T$. (L.11)

The researcher must not define A merely as:

A = “true value.” (L.12)

That phrase has no operational content.

L.5.3 Baseline sensitivity plan

Before testing, declare alternative admissible baselines:

$q_1, q_2, \ldots, q_m$. (L.13)

The robustness test is:

Result(q_j) retains sign, ranking, or conclusion across admissible q_j. (L.14)

Record:

Primary baseline:
[Enter]

Alternative baseline 1:
[Enter]

Alternative baseline 2:

[Enter]

Baseline rejection condition:
[Enter]

L.5.4 Baseline failure

The baseline fails if:

- it is not observable or estimable;
- it changes meaning during the episode;
- it depends circularly on R;
- it places many observations outside $0 \le R/A \le 1$;
- small reasonable changes reverse all conclusions.

The researcher should predeclare:

BaselineFailureThreshold = [Enter]. (L.15)

L.6 Feature-Map Worksheet

L.6.1 Feature declaration

The feature map is:

$\varphi: X \rightarrow x_feature$. (L.16)

List all features used to generate A, R, gates, or residual.

Feature	Meaning	Source	Frequency	Transformation
[Feature 1]	[Meaning]	[Source]	[Frequency]	[Transformation]
[Feature 2]	[Meaning]	[Source]	[Frequency]	[Transformation]
[Feature 3]	[Meaning]	[Source]	[Frequency]	[Transformation]

L.6.2 Feature classes

Features should be classified as:

Amplitude features

Variables used to estimate A.

Filter features

Variables used by F to generate R.

Gate features

Variables used to determine G.

Residual-monitoring features

Variables not included in the core state but monitored for omitted structure.

Control variables

Variables included in benchmark or regression models.

The same feature may appear in more than one class, but the overlap should be declared.

L.6.3 Leakage test

A feature leaks future information when it uses data unavailable at the prediction time.

Define:

AvailableTime(feature_j) ≤ DecisionTime. (L.17)

Every feature must satisfy this condition.

Examples of leakage include:

- revised financial statements used as though known earlier;
- final recovery values used in pre-default prediction;
- future rating actions embedded in current labels;
- smoothed derivatives using future test observations.

The leakage rule is:

No Future Availability → No Valid Predictive Feature. (L.18)

L.6.4 Feature omission register

The researcher should list important excluded variables.

Omitted variable	Reason omitted	Expected consequence	Monitoring proxy
[Variable]	[Reason]	[Consequence]	[Proxy]

This register provides a starting point for projection residual.

L.7 Horizon Worksheet

L.7.1 Horizon declaration

The horizon h should specify:

Valuation horizon:
[Enter]

Prediction horizon:
[Enter]

Event-detection horizon:
[Enter]

Intervention-evaluation horizon:
[Enter]

Sampling interval:
[Enter]

These horizons may differ, but each must be named.

L.7.2 Horizon consistency

A and R must refer to compatible horizons.

The consistency condition is:

h_A = h_R. (L.19)

If:

h_A ≠ h_R, (L.20)

the ratio R/A may be meaningless.

The researcher should state:

HorizonAlignmentMethod = [Enter]. (L.21)

L.7.3 Multi-horizon analysis

If several horizons are tested:

$h \in \{h_1, h_2, \ldots, h_m\}$. (L.22)

the result should be reported separately for each horizon.

Do not combine:

θ_{h_1} and θ_{h_2} (L.23)

as though they were one coordinate.

A horizon-indexed state is:

$Z_h = A_h \exp(i\theta_h)$. (L.24)

L.8 Mature Filter Worksheet

L.8.1 Filter identity

The filter F must come from an established finance rule or model.

Filter name:
[CAPM, credit-spread valuation, certainty equivalent, liquidity haircut, etc.]

Formal equation:
[Enter]

Parameters:
[Enter]

Parameter sources:
[Enter]

Calibration procedure:
[Enter]

Economic purpose:
[Enter]

Known limitations:
[Enter]

L.8.2 Filter admissibility test

The filter is admissible when:

1. it has an established financial interpretation;
2. it produces a reproducible R;
3. the baseline comparison is meaningful;
4. its parameters are observable or estimable;
5. it is appropriate for the asset and horizon.

Define:

FilterValid(F) = 1 (L.25)

only when all five conditions pass.

L.8.3 Filter ratio

The filter must generate:

$c_t = R_t/A_t$. (L.26)

The basic domain is:

$0 \le c_t \le 1$. (L.27)

The angle is then:

$\theta_t = \arccos(c_t)$. (L.28)

The pressure coordinate is:

$Q_t = A_t\sqrt{(1 - c_t^2)}$. (L.29)

No separate subjective choice of θ is permitted.

L.8.4 Domain-failure policy

Declare what will happen when:

$c_t < 0$ (L.30)

or:

$c_t > 1$. (L.31)

Permitted responses include:

- classify as invalid observation;
- revise baseline;
- use a signed extension;
- use liability geometry;
- use hyperbolic geometry;
- narrow the application domain.

Prohibited response:

Clip c_t silently into $[0,1]$. (L.32)

L.9 Gate Worksheet

L.9.1 Gate identity

The event gate is:

$G(\text{state},\text{evidence},\text{authority},\text{protocol}) \rightarrow \{\text{Commit},\text{Reject},\text{Defer}\}$. (L.33)

Complete:

Event type:
[Enter]

Commit threshold:
[Enter]

Reject condition:

[Enter]

Defer condition:
[Enter]

Required evidence:
[Enter]

Decision authority:
[Enter]

Review or appeal process:
[Enter]

Expected consequence of commitment:
[Enter]

L.9.2 Gate examples

Default gate

Commit if contractual default criteria are met.

Downgrade gate

Commit if the authorized rating process issues a formal action.

Impairment gate

Commit if recognition criteria are satisfied.

Trading gate

Commit if an executable match produces a completed trade.

Margin gate

Commit if collateral deficiency exceeds the contractual threshold.

L.9.3 Gate-prediction target

For a prediction experiment, define:

$Y_{t+h} = 1$ if G commits within horizon h. (L.34)

Otherwise:

$Y_{t+h} = 0$. (L.35)

The event label must not depend on retrospective narrative alone.

It should depend on the declared gate.

L.9.4 Gate error costs

Declare:

Cost_false_commit = [Enter]. (L.36)

Cost_false_noncommit = [Enter]. (L.37)

Cost_delay = [Enter]. (L.38)

Cost_irreversibility = [Enter]. (L.39)

These costs determine whether precision, recall, calibration, or expected utility should be the primary metric.

L.10 Trace Worksheet

L.10.1 Trace rule

The trace rule T determines what is preserved.

A minimum record is:

$Record_k = (Outcome_k, Time_k, Protocol_k)$. (L.40)

A stronger record is:

$Record_k = (Outcome_k, Evidence_k, Authority_k, GateMetadata_k, Residual_k, Protocol_k)$. (L.41)

Complete:

Ledger location:
[Enter]

Record schema:
[Enter]

Evidence retention rule:
[Enter]

Correction rule:
[Enter]

Reversal rule:
[Enter]

Residual attachment rule:
[Enter]

Version-control rule:
[Enter]

L.10.2 Non-erasure rule

A correction should produce:

L_new = L_old ⊔ CorrectionRecord. (L.42)

It should not produce:

L_new = Rewrite(L_old as though error never occurred). (L.43)

This applies both to financial events and to research results.

L.10.3 Residual ageing rule

For unresolved residual j:

Age_j(t) = t − t_j. (L.44)

Declare:

Review interval:
[Enter]

Escalation age:
[Enter]

Suspension age:
[Enter]

Responsible owner:
[Enter]

L.11 Intervention Worksheet

L.11.1 Intervention set

List the admissible interventions:

U = {u1,u2,…,u_m}. (L.45)

Examples include:

- hedge;
- sell;
- refinance;
- provide liquidity;
- post collateral;
- revise limit;
- restructure;
- wait;

- collect more evidence.

Complete:

Intervention	Trigger	Expected effect	Cost	Reversibility
[u_1]	[Trigger]	[Effect]	[Cost]	[High/Medium/Low]
[u_2]	[Trigger]	[Effect]	[Cost]	[High/Medium/Low]

L.11.2 Intervention policy

The policy is:

$u_t = \pi(Z_t, L_t, E_t, P_t)$. (L.46)

Declare:

Decision rule:
[Enter]

Human override:
[Enter]

Approval authority:
[Enter]

Maximum exposure:
[Enter]

Stop-loss or suspension condition:
[Enter]

L.11.3 Intervention evaluation

The intervention should be evaluated against a baseline action.

Let:

u_B = baseline action. (L.47)

u_E = action selected using Finance Geometry. (L.48)

The economic gain is:

$\Delta\text{Utility} = \text{Utility}(u_E) - \text{Utility}(u_B)$. (L.49)

The framework adds operational value only if:

$E(\Delta\text{Utility}) > \text{ImplementationCost} + \text{ModelRiskCost}$. (L.50)

L.12 Core Geometry Worksheet

L.12.1 Required variables

For every valid observation t, compute:

A_t = [Declared amplitude]. (L.51)

R_t = [Admitted value]. (L.52)

$c_t = R_t/A_t$. (L.53)

$\theta_t = \arccos(c_t)$. (L.54)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (L.55)

$Z_t = R_t + iQ_t$. (L.56)

$\Pi_{Q,t} = Q_t/R_t$. (L.57)

L.12.2 Algebraic audit

The geometric identity is:

$A_t^2 = R_t^2 + Q_t^2$. (L.58)

Define:

$\varepsilon_{alg,t} = A_t^2 - R_t^2 - Q_t^2$. (L.59)

The observation passes when:

$|\varepsilon_{alg,t}|/(A_t^2 + \delta) \leq \kappa_{alg}$. (L.60)

Declare:

κ_{alg} = [Enter]. (L.61)

L.12.3 Interpretation record

Each Q should be labelled:

$Q_{FilterName,t}$. (L.62)

Examples:

$Q_{CAPM,t}$. (L.63)

$Q_{credit,t}$. (L.64)

$Q_{liquidity,t}$. (L.65)

The worksheet should state:

Q meaning:
[Enter]

What Q does not include:
[Enter]

Units:
[Enter]

Valid domain:
[Enter]

L.13 Dynamic Estimation Worksheet

L.13.1 Radial growth

Estimate:

$g_{A,t} = d \ln A_t/dt$. (L.66)

Discrete approximation:

$g_{A,t} \approx [\ln A_t - \ln A_{t-1}]/\Delta t$. (L.67)

Declare:

Estimator:
[Enter]

Smoothing method:
[Enter]

Minimum data requirement:
[Enter]

L.13.2 Angular velocity

Estimate:

$\omega_{F,t} = d\theta_t/dt$. (L.68)

Discrete approximation:

$\omega_{F,t} \approx (\theta_t - \theta_{t-1})/\Delta t$. (L.69)

Declare:

Derivative method:
[Enter]

Window length:
[Enter]

Boundary handling:
[Enter]

Uncertainty estimate:
[Enter]

L.13.3 Angular repricing load

Compute:

Λ_F,t = Q_tω_F,t. (L.70)

The predicted visible-value change is:

dR^_t/dt = g_A,tR_t − Λ_F,t. (L.71)

The real residual is:

ε_R,t = dR_t/dt − g_A,tR_t + Λ_F,t. (L.72)

The pressure residual is:

ε_Q,t = dQ_t/dt − g_A,tQ_t − ω_F,tR_t. (L.73)

The complex residual is:

ε_dyn,t = ε_R,t + iε_Q,t. (L.74)

L.13.4 Closure score

Define:

r_dyn,t = |ε_dyn,t|/[|dZ_t/dt| + δ]. (L.75)

Declare:

κ_close = [Enter]. (L.76)

A period is dynamically closed when:

r_dyn,t ≤ κ_close. (L.77)

Define closure rate:

C_close = Number of closed periods/Number of valid periods. (L.78)

Declare the minimum acceptable closure rate:

C_close ≥ [Enter]. (L.79)

L.13.5 Derivative robustness

The primary result should be recomputed using at least two derivative estimators.

For example:

ω_F,t^{(1)} = finite difference. (L.80)

ω_F,t^{(2)} = local-linear derivative. (L.81)

The result is robust when:

Sign[Λ_F^{(1)}] = Sign[Λ_F^{(2)}] (L.82)

for a declared fraction of observations.

Declare:

DerivativeAgreementThreshold = [Enter]. (L.83)

L.14 Residual Worksheet

L.14.1 Residual categories

Record separately:

ε_measure. (L.84)

ε_alg. (L.85)

ε_dyn. (L.86)

ε_Π. (L.87)

E_L. (L.88)

H_loop. (L.89)

ε_transition. (L.90)

ε_tri. (L.91)

No general “error” field should replace this classification.

L.14.2 Residual thresholds

Complete:

Residual	Green	Amber	Red	Suspension
ε_measure	[ ]	[ ]	[ ]	[ ]
ε_dyn	[ ]	[ ]	[ ]	[ ]
ε_Π	[ ]	[ ]	[ ]	[ ]
E_L	[ ]	[ ]	[ ]	[ ]
H_loop	[ ]	[ ]	[ ]	[ ]
ε_transition	[ ]	[ ]	[ ]	[ ]

Thresholds should be fixed before final evaluation.

L.14.3 Residual response rule

The response function is:

Response(E) =

Continue, if $E \le \kappa_{low}$;
Review, if $\kappa_{low} < E \le \kappa_{high}$;
Revise, if $E > \kappa_{high}$ and admissible revision exists;
Reject, if no stable closure exists. (L.92)

The researcher should define:

Review authority:
[Enter]

Revision authority:
[Enter]

Rejection authority:
[Enter]

L.15 Event-Ledger Worksheet

L.15.1 Event record

For each event, record:

$\text{Event}_k = (t_k, \theta_k, k, \text{Type}_k, \text{Authority}_k, \text{Evidence}_k, \text{Residual}_k)$. (L.93)

The ledger update is:

$L_{k+1} = L_k \sqcup \text{Event}_k$. (L.94)

L.15.2 Three-clock record

Each event should contain:

$\text{CalendarTime} = t_k$. (L.95)

$\text{PhaseTime} = \theta_k$. (L.96)

$\text{LedgerTime} = k$. (L.97)

Also record phase direction:

$d_k = \text{sign}(\omega_F, k)$. (L.98)

The complete phase address is:

$\text{PhaseAddress}_k = (\theta_k, d_k, n_{cycle}, L_k)$. (L.99)

L.15.3 Event-log template

Field	Entry
event_id	[Enter]
calendar_time	[Enter]

Field	Entry
phase_theta	[Enter]
ledger_index	[Enter]
event_type	[Enter]
gate_result	[Enter]
authority	[Enter]
evidence	[Enter]
pre_state	[A,R,Q,θ,P]
post_state	[A,R,Q,θ,P]
residual	[Enter]
intervention	[Enter]
downstream effect	[Enter]

L.16 Null-Model Worksheet

L.16.1 Primary benchmark

The primary benchmark should be the mature method most likely to explain the same outcome.

Examples include:

- duration;
- convexity;
- spread level;
- spread change;
- beta;
- factor model;
- volatility;
- liquidity measure;
- VaR;
- Expected Shortfall;
- hidden Markov model;
- Bayesian change-point model.

Complete:

Primary null model:
[Enter]

Why it is relevant:
[Enter]

Primary benchmark variables:

[Enter]

L.16.2 Required model comparison

At minimum compare:

M_0 = intercept or naïve baseline. (L.100)

M_1 = mature finance benchmark. (L.101)

M_2 = Finance Geometry variables only. (L.102)

M_3 = mature benchmark + Finance Geometry. (L.103)

The incremental contribution is:

$\Delta\text{Performance} = \text{Performance}(M_3) - \text{Performance}(M_1)$. (L.104)

The framework adds value only if ΔPerformance is positive under the declared metric and test set.

L.16.3 Redundancy test

Test whether θ, Q, or Λ_F are merely transformations of existing variables.

For example:

$Q_t = f(\text{Spread}_t, A_t)$. (L.105)

If a simpler variable explains the same target with equal performance, the geometry may provide interpretation without incremental prediction.

The result should be classified as:

- predictive gain;
- diagnostic gain only;
- communication gain only;
- no meaningful gain.

L.17 Hypothesis Worksheet

L.17.1 Primary hypothesis

Use one sentence:

H_1: [Finance Geometry variable] improves [declared outcome] beyond [benchmark] over [horizon] in [population]. (L.106)

Example:

H_1: $Q_{credit}\omega_{credit}$ improves six-month downgrade prediction beyond spread level, spread change, rating, and duration in investment-grade corporate bonds. (L.107)

L.17.2 Null hypothesis

H_0: The incremental contribution of the Finance Geometry variables is zero or economically immaterial. (L.108)

Define the materiality threshold:

κ_{gain} = [Enter]. (L.109)

L.17.3 Secondary hypotheses

Limit secondary hypotheses before testing.

Examples:

H_2: Persistent r_{dyn} rises before protocol transition. (L.110)

H_3: Phase distance aligns gate events more tightly than calendar duration. (L.111)

H_4: H_{loop} predicts fragility after price recovery. (L.112)

Each hypothesis should have:

- one outcome;
- one horizon;
- one primary metric;
- one rejection criterion.

L.18 Train–Validation–Test Worksheet

L.18.1 Data split

Declare:

Training window:

$W_{train} = [t_0,t_1]$. (L.113)

Validation window:

$W_{validation} = [t_1,t_2]$. (L.114)

Final test window:

$W_{test} = [t_2,t_3]$. (L.115)

The final test data must not be used for:

- feature selection;
- smoothing selection;
- threshold setting;
- model choice;
- chart selection.

L.18.2 Rolling-origin design

A rolling design may use:

$\text{Train}_1 \rightarrow \text{Test}_1$. (L.116)

$\text{Train}_2 \rightarrow \text{Test}_2$. (L.117)

$\text{Train}_3 \rightarrow \text{Test}_3$. (L.118)

All historical predictions should remain preserved.

Do not replace earlier failed predictions with recalibrated hindsight estimates.

L.18.3 Crisis balance

Declare whether the data contain:

- calm regime;
- tightening regime;
- crisis regime;
- recovery regime.

A model tested only in one type of regime should not be claimed as generally valid.

L.19 Evaluation-Metric Worksheet

L.19.1 Prediction metrics

For continuous outcomes:

- mean absolute error;
- root mean squared error;
- out-of-sample R^2;
- log predictive score.

For event outcomes:

- area under ROC curve;
- precision;
- recall;
- Brier score;
- log loss;
- calibration slope.

For ranking:

- Spearman correlation;
- top-decile event rate;
- concordance index.

L.19.2 Economic metrics

Declare one economic metric:

EconomicMetric = [Enter]. (L.119)

Examples include:

- avoided loss;
- lower hedging cost;
- reduced forced sale;
- earlier intervention;
- improved capital allocation;
- reduced false alarm cost.

L.19.3 Governance metrics

Possible governance outcomes include:

- analyst agreement;
- confidence calibration;
- residual recognition;
- intervention consistency;
- audit completeness;
- reduction in unsupported causal narratives.

A framework may provide governance value even if predictive improvement is modest.

That value should be measured rather than asserted.

L.20 Falsification Worksheet

L.20.1 Predeclared failure conditions

The protocol should be rejected for the declared application if any of the following occurs:

- A cannot be defined without circularity.
- R/A frequently lies outside the valid domain.
- θ is unstable under reasonable baselines.
- Q duplicates a mature variable without additional value.
- derivative estimates are estimator-dependent.
- closure residual remains above threshold.
- no out-of-sample gain is observed.
- conclusions reverse across reasonable horizons.
- event gates are not reproducible.
- intervention quality does not improve.
- independent researchers cannot replicate results.

L.20.2 Quantitative rejection rules

Complete:

Reject if:

C_{close} < [Enter]. (L.120)

Reject if:

ΔPerformance ≤ [Enter]. (L.121)

Reject if:

RobustnessRate < [Enter]. (L.122)

Reject if:

ReplicationAgreement < [Enter]. (L.123)

Reject if:

EconomicGain ≤ ImplementationCost. (L.124)

L.20.3 Claim downgrade rules

If predictive gain fails but diagnostic clarity improves, the claim should be downgraded from:

Predictive Technology (L.125)

to:

Interpretive Framework. (L.126)

If interpretive clarity also fails:

Reject the complex extension for that application. (L.127)

L.21 Minimal Reproducible Experiment

L.21.1 Objective

The minimum experiment should test one narrow claim.

Recommended first question:

Does angular repricing load improve attribution or prediction beyond the mature finance variable from which it is derived? (L.128)

L.21.2 Minimum dataset

The minimum dataset should contain:

- one clearly defined asset class;
- at least one stable baseline;
- one mature filter;
- repeated A and R observations;
- at least one event type;
- standard benchmark variables;
- enough history for untouched testing.

A minimal observation row is:

Data_t = (AssetID,t,A_t,R_t,Benchmark_t,Event_{t+h}). (L.129)

L.21.3 Minimum computation

For each observation:

$c_t = R_t/A_t$. (L.130)

$\theta_t = \arccos(c_t)$. (L.131)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (L.132)

$g_{A,t} = \Delta \ln A_t/\Delta t$. (L.133)

$\omega_{F,t} = \Delta\theta_t/\Delta t$. (L.134)

$\Lambda_{F,t} = Q_t\omega_{F,t}$. (L.135)

$\varepsilon_{dyn,t} = \Delta Z_t/\Delta t - (g_{A,t} + i\omega_{F,t})Z_t$. (L.136)

L.21.4 Minimum comparison

Estimate:

BaselineModel:

$Y_{t+h} = f(Benchmark_t)$. (L.137)

ExtendedModel:

$Y_{t+h} = f(Benchmark_t, Q_t, \omega_{F,t}, \Lambda_{F,t}, r_{dyn,t})$. (L.138)

Compare on untouched test data.

L.21.5 Minimum success condition

The experiment succeeds only if:

1. algebraic validity passes;
2. protocol validity passes;
3. closure is adequate;
4. the extended model improves the declared metric;
5. the gain survives alternative baselines;
6. results are reproducible.

Formally:

$\text{Success} = \text{Algebra} \wedge \text{Protocol} \wedge \text{Closure} \wedge \text{Gain} \wedge \text{Robustness} \wedge \text{Replication}$. (L.139)

L.21.6 Minimum failure conclusion

A valid failure conclusion is:

Under protocol P_{exp}, the Finance Geometry variables did not provide sufficient incremental value beyond the mature benchmark. (L.140)

This is a scientifically useful result.

It narrows the domain of the theory.

L.22 Suggested First Credit Experiment

L.22.1 Research question

Does credit angular repricing load improve prediction of rating downgrade beyond conventional spread variables?

L.22.2 Protocol

Boundary:

B = publicly traded corporate bonds in one currency and rating range. (L.141)

Baseline:

A_t = matched default-free bond value. (L.142)

Filter:

R_t = observed risky bond value. (L.143)

Angle:

$\theta_{credit,t} = \arccos(R_t/A_t)$. (L.144)

Pressure:

$Q_{credit,t} = \sqrt{(A_t^2 - R_t^2)}$. (L.145)

Gate:

G = formal rating downgrade within six months. (L.146)

Trace:

T = rating record, evidence date, protocol version, and residual. (L.147)

Intervention:

U = monitoring escalation or credit-limit review. (L.148)

L.22.3 Benchmarks

Baseline variables should include:

- spread level;
- spread change;

- spread duration;
- rating;
- leverage;
- volatility;
- liquidity;
- sector;
- maturity.

The extended variables are:

Q_credit/R_credit. (L.149)

ω_credit. (L.150)

Q_creditω_credit. (L.151)

r_dyn,credit. (L.152)

L.22.4 Success test

The extended model should improve:

- out-of-sample Brier score;
- calibration;
- economically weighted early-warning value.

A positive AUC change alone should not be treated as sufficient.

L.23 Suggested First CAPM Experiment

L.23.1 Research question

Can radial–angular decomposition distinguish cash-flow-driven equity declines from risk-premium-driven declines?

L.23.2 State construction

Amplitude:

A_t = forecast cash flow discounted at declared base rate. (L.153)

Admitted value:

R_t = CAPM-filtered value or aligned market-implied value. (L.154)

Angle:

$\theta_{CAPM,t} = \arccos(R_t/A_t)$. (L.155)

Angular load:

$\Lambda_{CAPM,t} = Q_{CAPM,t}\omega_{CAPM,t}$. (L.156)

L.23.3 Episode classification

Classify:

Amplitude-dominant if:

$|g_{AR}| > \kappa_{dom}|Q\omega_F|$. (L.157)

Frame-dominant if:

$|Q\omega_F| > \kappa_{dom}|g_{AR}|$. (L.158)

Residual-dominant if:

$|\varepsilon_R| > \kappa_{dom} \max(|g_{AR}|,|Q\omega_F|)$. (L.159)

Declare κ_{dom} before testing.

L.23.4 Outcome comparison

Compare future:

- earnings revision;
- recovery speed;
- volatility;
- drawdown;
- analyst estimate revision.

The hypothesis is:

Amplitude-Dominant and Frame-Dominant Declines Have Different Subsequent Paths. (L.160)

L.24 Reproducibility Package

L.24.1 Required files

A reproducible release should contain:

1. protocol declaration;
2. data dictionary;

3. raw-data references;
4. cleaned data;
5. transformation code;
6. model code;
7. event definitions;
8. residual thresholds;
9. train–validation–test split;
10. failed-model log;
11. output tables;
12. verification footer.

L.24.2 Hashes

Record:

DataHash = H_data. (L.161)

CodeHash = H_code. (L.162)

ProtocolHash = H_protocol. (L.163)

ResultHash = H_result. (L.164)

These hashes make silent revision easier to detect.

L.24.3 Environment record

Record:

- programming language;
- version;
- package versions;
- operating system;
- random seed;
- numerical precision;
- hardware where relevant.

A result that depends on hidden implementation details is not fully reproducible.

L.25 Independent Replication Worksheet

L.25.1 Replication target

An independent team should reproduce:

A_t. (L.165)

R_t. (L.166)

θ_t. (L.167)

Q_t. (L.168)

$\omega_{F,t}$. (L.169)

$\Lambda_{F,t}$. (L.170)

$\varepsilon_{dyn,t}$. (L.171)

Primary outcome metric. (L.172)

L.25.2 Agreement metric

Define:

$Agreement_j = 1 - Distance(Result_j^A, Result_j^B)/Scale_j$. (L.173)

Declare:

MinimumAgreement = [Enter]. (L.174)

Disagreement should be classified as:

- data mismatch;
- protocol mismatch;
- implementation mismatch;
- irreducible judgement difference;
- projection residual.

L.26 Protocol Revision Worksheet

L.26.1 Revision trigger

A revision may be triggered by:

- persistent residual;
- domain failure;

- structural break;
- new evidence;
- invalid gate;
- changed baseline;
- changed horizon;
- failed intervention;
- replication disagreement.

The trigger is:

RevisionTrigger = [Enter]. (L.175)

L.26.2 Revision record

A revision should contain:

RevisionRecord_n = (P_old,P_new,Reason,Evidence,Mapping,UntranslatedResidual). (L.176)

Complete:

Old protocol:
[Enter]

New protocol:
[Enter]

Changed fields:
[Enter]

Reason:
[Enter]

Evidence:
[Enter]

Effect on prior conclusions:
[Enter]

Variables no longer comparable:
[Enter]

L.26.3 Revision discipline

A revision may improve future performance.

It must not be used to rewrite the original preregistered result.

The correct publication record is:

Original Result + Revision Record + New Result. (L.177)

Not:

New Result Presented as Though It Had Always Been the Original Protocol. (L.178)

L.27 Publication Claim Template

L.27.1 Supported claim

A properly bounded claim is:

Under protocol [P], over horizon [h], in population [B], variable [V] improved [metric] relative to [benchmark] by [amount] on untouched test data, with residual [level] and robustness across [declared alternatives]. (L.179)

L.27.2 Unsupported claim

Avoid statements such as:

- Q reveals the true hidden value;
- θ is the market's real internal time;
- the model proves finance is quantum;
- residual proves an unseen field exists;
- the geometry universally predicts crises.

These claims exceed the permitted evidence.

L.27.3 Negative-result template

A valid negative result is:

Under the declared protocol, the Finance Geometry extension was algebraically valid but did not improve diagnostic, predictive, intervention, or governance performance sufficiently to justify its added complexity. (L.180)

L.28 Compact Pre-Registration Form

The following form may be copied directly into a research registration.

Project

Title:
[ ]

Protocol ID and version:
[ ]

Research question:
[ ]

Primary hypothesis:
[ ]

Primary outcome:
[ ]

Protocol

B — Boundary:
[ ]

q — Baseline:
[ ]

φ — Feature map:
[ ]

h — Horizon:
[ ]

F — Mature filter:
[ ]

G — Event gate:
[ ]

T — Trace rule:
[ ]

U — Intervention set:
[ ]

Geometry

A definition:
[ ]

R definition:
[ ]

θ formula:
[ ]

Q formula:
[ ]

Valid domain:
[ ]

Dynamics

g_A estimator:
[ ]

ω_F estimator:
[ ]

Λ_F definition:
[ ]

Residual definition:
[ ]

Closure threshold:
[ ]

Data

Data source:
[ ]

Sampling frequency:
[ ]

Training window:
[ ]

Validation window:
[ ]

Final test window:
[ ]

Benchmarks

Primary null model:
[ ]

Secondary benchmark:
[ ]

Primary metric:
[ ]

Economic-value metric:
[ ]

Falsification

Algebraic failure condition:
[ ]

Protocol failure condition:
[ ]

Dynamic failure condition:
[ ]

Predictive failure condition:
[ ]

Robustness failure condition:
[ ]

Replication failure condition:
[ ]

Publication

Maximum permitted claim:
[ ]

Planned negative-result statement:
[ ]

Code and data release plan:
[ ]

L.29 Minimal Runtime Algorithm

The complete experimental algorithm is:

Step 1 — Declare the world

Fix:

$P = (B,q,\varphi,h,F,G,T,U)$. (L.181)

Step 2 — Construct the state

Compute:

A_t,R_t,θ_t,Q_t,Z_t. (L.182)

Step 3 — Validate algebra and domain

Require:

$A_t^2 \approx R_t^2 + Q_t^2$. (L.183)

$0 \le R_t/A_t \le 1$. (L.184)

Step 4 — Estimate dynamics

Compute:

g_A,t,ω_F,t,Λ_F,t,ε_dyn,t. (L.185)

Step 5 — Record events

Update:

$L_{k+1} = L_k \sqcup \text{Record}_k$. (L.186)

Step 6 — Compare with benchmarks

Estimate:

M_1 = mature benchmark. (L.187)

M_3 = mature benchmark + Finance Geometry. (L.188)

Step 7 — Evaluate untouched data

Compute:

ΔPerformance_test. (L.189)

Step 8 — Test robustness

Vary admissible:

q,h,estimator,sample. (L.190)

Step 9 — Test intervention value

Compare:

Utility(u_E) − Utility(u_B). (L.191)

Step 10 — Decide

Continue, revise, narrow, or reject. (L.192)

L.30 Final Protocol Object

The full reproducible experiment may be represented as:

Experiment = (P,D,M,H,K,R,V). (L.193)

Where:

- P = declared protocol;
- D = versioned data;
- M = models and null models;
- H = hypotheses;
- K = gates and event ledger;

- R = residual architecture;
- V = verification package.

A complete experiment is not merely a fitted model.

It is:

Declaration + Data + Comparison + Trace + Falsification. (L.194)

L.31 Final Checklist

Before publication, confirm:

- The system boundary is explicit.
- The baseline is economically interpretable.
- The feature map contains no future leakage.
- The horizon is consistent across A and R.
- The filter is mature and reproducible.
- θ is derived rather than invented.
- Q is labelled by its source.
- Gate authority is recorded.
- Residual categories remain separate.
- Test data were untouched.
- Mature benchmarks were included.
- Failed specifications were preserved.
- Intervention costs were considered.
- Robustness tests were completed.
- Independent replication was attempted.
- Quantum claims remain within evidence.
- Negative results are publishable.
- Protocol revisions preserve history.

L.32 Closing Principle

Finance Geometry becomes scientifically valuable only when its declarations are made visible enough to fail.

A complex coordinate is easy to construct.

A trustworthy effective world is harder.

It requires:

- explicit boundaries;
- stable baselines;
- mature filters;
- reproducible gates;
- honest residual;
- untouched tests;
- benchmark comparison;
- trace-preserving revision.

The protocol worksheet therefore protects the framework from becoming decorative mathematics.

Its final rule is:

No Predeclared Protocol → No Interpretable Experiment. (L.195)

No Untouched Test → No Predictive Claim. (L.196)

No Benchmark Gain → No Applied Adoption. (L.197)

No Residual-Based Rejection Path → No Scientific World Model. (L.198)

The minimal reproducible experiment is complete only when another observer can reconstruct not merely the final result, but the declared world in which that result was allowed to become meaningful.

Appendix M — CAPM Grand-Unification Mapping Tables: Quantum, Special-Relativistic, and General-Relativistic Layers

M.1 Purpose

The preceding article began with a narrow finance construction:

A mature financial filter admits a scalar value R while implying a retained pressure coordinate Q.

The core state is:

$Z_F = R + iQ = A \exp(i\theta)$. (M.1)

Where:

$A^2 = R^2 + Q^2$. (M.2)

$R = A \cos \theta$. (M.3)

$Q = A \sin \theta$. (M.4)

For CAPM:

$r_i = r_f + \beta_i ERP$. (M.5)

$A_i = CF_T/(1 + r_*)^T$. (M.6)

$R_i = CF_T/(1 + r_i)^T$. (M.7)

Therefore:

$R_i/A_i = [(1 + r_*)/(1 + r_i)]^T$. (M.8)

The original Finance Geometry construction is exact as a coordinate completion once the baseline, horizon, and mature filter have been declared. Its use of a complex coordinate is intended to preserve pressure hidden by scalar valuation, not to claim that a financial asset is physically quantum.

Subsequent discussion uncovered three larger mathematical correspondences:

1. a **quantum-like observer and complex-state layer**;
2. a **special-relativistic layer of relative valuation frames**;
3. a **general-relativistic layer of state-dependent financial geometry**.

This appendix records those mappings so that a later article can continue the derivation without confusing:

- exact finance mathematics;
- constructed geometric embeddings;
- functional analogies;
- speculative physical implications.

Its central question is:

Can one CAPM-derived model contain QM-like, SR-like, and GR-like structures without internal mathematical contradiction? (M.9)

The qualified answer is:

> Yes, as a layered effective theory—provided each structure occupies a distinct mathematical role and incompatible coordinate interpretations are not silently identified.

This appendix does not claim that CAPM derives physical quantum mechanics, special relativity, general relativity, or quantum gravity.

M.2 Status Legend

The mapping tables use four status levels.

Status	Meaning

Status	Meaning
Exact	Follows algebraically from the declared CAPM geometry
Constructed	A mathematically consistent extension introduced by definition
Functional	Systems perform corresponding operational roles
Unresolved	Requires new derivation or empirical verification

The distinction is essential:

Exact Identity ≠ Physical Identity. (M.10)

Constructed Compatibility ≠ Empirical Confirmation. (M.11)

Functional Homology ≠ Material Equivalence. (M.12)

M.3 The CAPM Effective-World Kernel

The complete CAPM-derived runtime developed in the article is:

X_k
→ Declare P_k
→ Project Π_{P_k,L_k}
→ $Z_k = R_k + iQ_k$
→ Internal Dynamics
→ Gate G_k
→ Ledger L_{k+1}
→ Backreaction $\boldsymbol{\mathcal{B}}$
→ Revised Primary State X_{k+1}
→ Protocol Revision $\boldsymbol{\mathcal{U}}$. (M.13)

In compact form:

$X_k \to \Pi_{P_k,L_k} \to Z_k \to G_k \to L_{k+1} \to \boldsymbol{\mathcal{B}} \to X_{k+1} \to \boldsymbol{\mathcal{U}} \to P_{k+1}$. (M.14)

Within a stable local episode:

$Z = A \exp(i\theta)$. (M.15)

The general dynamic law is:

$dZ/dt = (g_A + i\omega_F)Z + \varepsilon_{dyn}$. (M.16)

Where:

$g_A = d \ln A/dt$. (M.17)

$\omega_F = d\theta/dt$. (M.18)

Under constant amplitude:

$dZ/d\theta = iZ$. (M.19)

The handoff analysis emphasizes that this internal phase law is classical; the observer-sensitive structures enter mainly through declaration, projection, gate, trace, residual, and backreaction.

M.4 Master Mapping Table

CAPM effective-world layer	Quantum-mechanical comparison	Special-relativistic comparison	General-relativistic comparison
primary state X	premeasurement state or preparation field	event or object before frame transformation	matter or state distribution on a manifold
protocol P	preparation and measurement context	declared reference frame	coordinate chart, local observer, and boundary conditions
filter F	measurement instrument or observable context	frame transformation	local metric-dependent projection
admitted value R	registered or expectation-level output	light-cone projection or contracted coordinate	locally measured scalar under g^F
retained pressure Q	orthogonal unresolved component	boost-related transverse coordinate	pressure or stress component affecting geometry
complex state Z = R + iQ	complex state representation	not required by SR itself	complex field living on curved state space
phase θ	quantum-like relative phase	circular precursor, not boost rapidity	local internal coordinate affected by curvature
discount rapidity ρ	no direct QM equivalent	additive boost parameter	local rapidity varying across the manifold
gate G	outcome registration or effective collapse boundary	event commitment in a frame	horizon or threshold crossing
ledger L	classical measurement record	frame-indexed event history	history affecting the local metric
residual ε	model-state mismatch, not quantum uncertainty	failure of flat-frame approximation	curvature or model inadequacy signal
backreaction $\mathcal{B}$	measurement or observer disturbance	frame-sensitive response	stress changes geometry, geometry redirects motion
protocol revision $\mathcal{U}$	instrument adaptation	change of reference frame or model	chart, metric, or regime transition
local CAPM law	effective complex-state law	local flat-frame law	tangent-space approximation
multi-pressure metric	covariance or state-space metric	Minkowski-like quadratic form	state-dependent curved metric

This table should be read vertically as a layered construction, not horizontally as an assertion that the entries are materially identical.

M.5 CAPM–Quantum Mechanics Mapping Table

M.5.1 Primary correspondence

CAPM or effective-world	Quantum-mechanical object	Correspondence type	Limitation

object			
primary field X	prepared physical state	Functional	X is usually classically representable
protocol P	preparation plus measurement setting	Functional	no Hilbert-space requirement
Π_P(X)	state projection or measurement map	Functional	finance projection can be ordinary classical filtering
Z = R + iQ	complex state amplitude	Mathematical resemblance	no Born probability follows
A =	Z		state norm or amplitude
θ	phase	Mathematical resemblance	finance θ is induced by a filter ratio
dZ/dθ = iZ	unitary-like phase evolution	Mathematical resemblance	this is classical rotation
G	measurement outcome gate	Functional	financial gate is institutional or contractual
L	macroscopic record	Functional	not a derivation of decoherence
$\mathcal{B}$	measurement disturbance	Functional	financial disturbance is behavioural and institutional
ε_Π	inaccessible or omitted structure	Functional	not equivalent to quantum indeterminacy
adaptive P_{k+1}	later instrument selected from previous trace	Functional	does not imply quantum self-reference

The strongest defensible relationship is:

CAPM Effective World = Classical Complex State + Contextual Projection + Gate + Trace + Backreaction. (M.20)

It does not imply:

CAPM Effective World = Quantum System. (M.21)

M.5.2 The complex-state correspondence

Finance uses:

Z_F = R + iQ. (M.22)

Quantum mechanics uses complex amplitudes such as:

ψ = a + ib. (M.23)

Both contain:

- magnitude;
- phase;
- orthogonal components;

- complex evolution.

But the probability structures differ.

Finance Geometry gives:

$A^2 = R^2 + Q^2$. (M.24)

Quantum mechanics gives, for an amplitude ψ:

$P = |\psi|^2$. (M.25)

The first equation is a valuation geometry.

The second is a probability rule.

Therefore:

Pythagorean Completion ≠ Born Rule. (M.26)

M.5.3 The phase correspondence

The financial phase is:

$\theta_F = \arccos(R/A)$. (M.27)

It represents the orientation of a mature financial filter.

A quantum relative phase affects interference among amplitudes.

The financial phase affects:

- admitted value;
- retained pressure;
- phase sensitivity;
- angular repricing.

Thus:

Finance Phase → Valuation Orientation. (M.28)

Quantum Phase → Amplitude-Interference Structure. (M.29)

The same word “phase” should not conceal the difference.

M.5.4 The gate correspondence

The CAPM effective-world gate is:

$G_P(\hat{Z}, \text{Evidence}, L) \rightarrow$ Commit, Reject, or Defer. (M.30)

A quantum measurement instrument may be represented by a family of outcome maps.

The common operational grammar is:

Possibility → Contextual Interaction → Registered Outcome → Record. (M.31)

The finance model demonstrates that this grammar is not uniquely quantum.

The quantum residue still includes:

- Born probabilities;
- irreducible contextuality;
- entanglement;
- Bell violations;
- no-cloning;
- quantum interference.

The source handoff explicitly recommends subtracting macro-reproducible observer grammar from the quantum phenomenon before identifying candidate quantum residue.

M.5.5 Quantum-subtraction table

Structure	CAPM effective world reproduces it?	Uniquely quantum?
complex state notation	yes	no
phase evolution	yes	no
contextual projection	yes	no
changing measurement frame	yes	no
gate-passed outcome	yes	no
irreversible ledger	yes	no
observer backreaction	yes	no
order-sensitive operations	yes	no
adaptive instrument choice	yes	no
Born probability geometry	no	candidate residue
coherent interference	no	candidate residue
tensor-product nonseparability	no	candidate residue
Bell violation	no	candidate residue
no-cloning	no	candidate residue

The comparative rule is:

Quantum Phenomenon − Macro-Reproducible World Grammar = Candidate Quantum Residue. (M.32)

M.6 CAPM–Special Relativity Mapping Table

M.6.1 Original Euclidean pressure geometry

The CAPM geometry begins with:

$A^2 = R^2 + Q_E^2$. (M.33)

The subscript E identifies the original Euclidean pressure coordinate.

Rearrange:

$R^2 = A^2 - Q_E^2$. (M.34)

Define:

$b_Q = Q_E/A$. (M.35)

Define:

$\gamma_Q = A/R$. (M.36)

Then:

$\gamma_Q = 1/\sqrt{(1 - b_Q^2)}$. (M.37)

This has the exact algebraic form of a Lorentz factor.

Define a Q-preserving rapidity:

$\eta_Q = \operatorname{artanh}(b_Q)$. (M.38)

Then:

$A = R \cosh \eta_Q$. (M.39)

$Q_E = R \sinh \eta_Q$. (M.40)

The invariant is:

$A^2 - Q_E^2 = R^2$. (M.41)

This is a valid hyperbolic representation of the existing A–R–Q_E relation.

M.6.2 Q-preserving Lorentz mapping

Special-relativistic quantity	Q-preserving financial quantity	Status
normalized velocity v/c	$b_Q = Q_E/A$	Exact algebraic mapping
Lorentz factor γ	$\gamma_Q = A/R$	Exact algebraic mapping
rapidity η	$\eta_Q = \operatorname{artanh}(Q_E/A)$	Exact construction
temporal coordinate	A	Constructed interpretation
spatial coordinate	Q_E	Constructed interpretation
proper interval	R	Constructed interpretation
Minkowski invariant	$A^2 - Q_E^2 = R^2$	Exact identity
collinear boost composition	$\eta_{total} = \eta_1 + \eta_2$	Not derived from CAPM filtering

The final row is crucial.

The Q-preserving rapidity exists mathematically, but CAPM does not automatically prove that successive filters compose by Lorentz boosts.

M.6.3 Discount-ratio rapidity

CAPM also supplies a second natural quantity:

$\rho_D = \ln(A/R)$. (M.42)

Using the CAPM discount ratio:

$\rho_D = T \ln[(1 + r_i)/(1 + r_*)]$. (M.43)

For valuation frames a, b, and c:

$\rho_{ab} = T \ln[(1 + r_b)/(1 + r_a)]$. (M.44)

$\rho_{bc} = T \ln[(1 + r_c)/(1 + r_b)]$. (M.45)

Therefore:

$\rho_{ac} = \rho_{ab} + \rho_{bc}$. (M.46)

This additive property resembles rapidity composition.

But ρ_D is not identical to η_Q.

Their relationship is:

$\rho_D = \ln \cosh \eta_Q$. (M.47)

Because:

$A/R = \cosh \eta_Q$. (M.48)

Thus:

$\eta_Q = \operatorname{arcosh}[\exp(\rho_D)]$. (M.49)

The two variables solve different problems:

- η_Q preserves the original Q_E as a hyperbolic coordinate;
- ρ_D preserves multiplicative discount-frame composition.

M.6.4 Light-cone construction

Define:

$U = A \cosh \rho_D$. (M.50)

$V = A \sinh \rho_D$. (M.51)

Then:

$U^2 - V^2 = A^2$. (M.52)

Define light-cone coordinates:

$R_- = U - V.$ (M.53)

$R_+ = U + V.$ (M.54)

Since:

$U - V = A \exp(-\rho_D),$ (M.55)

and:

$\rho_D = \ln(A/R),$ (M.56)

we obtain:

$R_- = R.$ (M.57)

The conjugate coordinate is:

$R_+ = A \exp(\rho_D) = A^2/R.$ (M.58)

Under a boost-like shift $\Delta\rho$:

$R_-' = \exp(-\Delta\rho)R_-.$ (M.59)

$R_+' = \exp(+\Delta\rho)R_+.$ (M.60)

The invariant is:

$R_-R_+ = A^2.$ (M.61)

This gives a cleaner Lorentz-group construction because ρ_D is additive.

M.6.5 Hyperbolic pressure coordinate

In the light-cone construction, define:

$Q_H = V = A \sinh \rho_D.$ (M.62)

This is not the original pressure Q_E.

Using:

$R = A \exp(-\rho_D),$ (M.63)

we obtain:

$Q_H = \tfrac{1}{2}(A^2/R - R).$ (M.64)

Since:

$Q_E^2 = A^2 - R^2,$ (M.65)

then:

$Q_H = Q_E^2/(2R).$ (M.66)

This equation is important.

It proves that the Euclidean pressure and the hyperbolic pressure can coexist without contradiction because they are distinct functions of the same A and R:

$Q_E = \sqrt{(A^2 - R^2)}$. (M.67)

$Q_H = (A^2 - R^2)/(2R)$. (M.68)

They encode different geometries.

M.6.6 Two Lorentz embeddings

Property	Q-preserving embedding	Discount-composition embedding
rapidity	$\eta_Q = \operatorname{artanh}(Q_E/A)$	$\rho_D = \ln(A/R)$
original Q preserved	yes	no
additive under CAPM discount ratios	not automatically	yes
invariant	$A^2 - Q_E^2 = R^2$	$U^2 - V^2 = A^2$
financial admitted value	proper interval-like R	light-cone coordinate R_-
hyperbolic pressure	Q_E	$Q_H = Q_E^2/(2R)$
best use	preserving Finance Geometry	constructing relative valuation boosts

The next article should choose one embedding for each purpose rather than declaring them identical.

M.6.7 CAPM beta is not relativistic beta

CAPM uses:

$\beta_i = \mathrm{Cov}(r_i,r_m)/\mathrm{Var}(r_m)$. (M.69)

Relativity commonly uses $\beta_{SR} = v/c$.

These are unrelated quantities.

Therefore:

$\beta_{CAPM} \neq \beta_{SR}$. (M.70)

A bounded financial frame velocity can instead be defined from rapidity:

$u_D = \tanh \rho_D$. (M.71)

Then:

$-1 < u_D < 1$. (M.72)

For relative frames:

$u_{ac} = (u_{ab} + u_{bc})/(1 + u_{ab} u_{bc})$. (M.73)

This composition follows mathematically from additive ρ_D.

It remains a constructed financial-frame law, not an observed universal market law.

M.6.8 SR mapping summary

SR structure	CAPM-derived construction	Status
inertial frame	baseline or benchmark valuation frame	Functional
relative frame	alternative discount or risk frame	Functional
boost parameter	ρ_{ab}	Constructed and additive
bounded relative velocity	$\tanh \rho_{ab}$	Constructed
Lorentz factor	$\cosh \rho_{ab}$ in light-cone embedding	Constructed
light-cone scaling	$R \to \exp(-\rho)R$	Exact from discount ratio
invariant interval	$U^2 - V^2 = A^2$	Constructed exact identity
universal speed c	not derived	Unresolved
causal cone	not derived from CAPM alone	Unresolved
local Lorentz invariance	possible in curved extension	Constructed

---

M.7 CAPM–General Relativity Mapping Table

M.7.1 Transition from flat to curved finance geometry

Special-relativistic structure uses a fixed metric:

$ds^2 = \eta_{\mu\nu}dx^\mu dx^\nu$. (M.74)

A GR-like financial theory requires:

$ds_F^2 = g^F_{\mu\nu}(x,L,P)dx^\mu dx^\nu$. (M.75)

Where:

- x is the financial-state location;
- L is the accumulated ledger;
- P is the active protocol;
- $g^F_{\mu\nu}$ is the financial metric.

The defining transition is:

Fixed Valuation Geometry → State-Dependent Valuation Geometry. (M.76)

M.7.2 GR mapping table

General-relativistic object	CAPM grand-unification object	Status
spacetime manifold $\mathcal{M}$	financial state manifold $\mathcal{M}_F$	Constructed

General-relativistic object	CAPM grand-unification object	Status
event coordinates x^μ	ledger time, rapidity, pressure, liquidity, leverage, etc.	Constructed
metric $g_{\mu\nu}$	state-dependent financial-distance metric $g^F_{\mu\nu}$	Constructed
local inertial frame	local CAPM-valid valuation frame	Functional
Minkowski tangent metric	local Lorentz-like discount geometry	Constructed
Christoffel connection Γ	change of valuation coordinates across state space	Constructed
geodesic	locally natural unforced financial trajectory	Functional
curvature tensor	path-dependent distortion of nearby financial trajectories	Constructed
geodesic deviation	differential divergence of initially similar assets	Functional and testable
stress-energy tensor $T_{\mu\nu}$	leverage, liquidity, obligations, and flow tensor $T^F_{\mu\nu}$	Speculative construction
Einstein tensor $G_{\mu\nu}$	curvature summary $G^F_{\mu\nu}$	Speculative construction
equivalence principle	common market shock locally removable by co-moving frame	Functional hypothesis
tidal force	differential beta, liquidity, collateral, or gate exposure	Functional
proper time	local phase or event-bearing financial time	Constructed
time dilation	state-dependent phase or ledger cadence	Functional hypothesis
event horizon	irreversible gate changing the future action set	Functional analogy
matter curves spacetime	financial stress changes market metric	Core GR-like hypothesis
spacetime guides matter	changed metric redirects later valuation and capital flow	Core GR-like hypothesis

---

M.7.3 Suggested financial manifold

A candidate financial coordinate vector is:

$$x^\mu = (\tau_L, \rho_D, q^1, q^2, \ldots, q^n). \quad (M.77)$$

Where:

- τ_L = ledgered event time;
- ρ_D = discount-frame rapidity;
- q^a = pressure coordinates.

Possible pressure coordinates include:

$$q^1 = \text{credit pressure}. \quad (M.78)$$

$$q^2 = \text{liquidity pressure}. \quad (M.79)$$

q^3 = leverage pressure. (M.80)

q^4 = collateral pressure. (M.81)

q^5 = capital pressure. (M.82)

A possible line element is:

$ds_F^2 = c_F^2 d\tau_L^2 - G_{ab}(x,L) dq^a dq^b$. (M.83)

The matrix G_{ab} is allowed to vary across financial states.

This variability is the source of curvature.

M.7.4 Local CAPM as a tangent-space law

At a financial state x_0, choose local coordinates such that:

$g^F_{\mu\nu}(x_0) \approx \eta_{\mu\nu}$. (M.84)

And:

$\Gamma^\mu_{\alpha\beta}(x_0) \approx 0$. (M.85)

Within a sufficiently small neighbourhood:

$r_i \approx r_f + \beta_i ERP$. (M.86)

$\rho_i \approx T \ln[(1 + r_i)/(1 + r_*)]$. (M.87)

Thus:

CAPM = Local Flat Approximation of Curved Financial Geometry. (M.88)

This would explain why CAPM may work reasonably in stable regimes but fail during:

- liquidity collapse;
- leverage spirals;
- collateral regime change;
- market closure;
- legal restructuring;
- major protocol transition.

M.7.5 Financial equivalence principle

A candidate financial equivalence principle is:

> Within a sufficiently small financial neighbourhood, the common first-order effect of a shared market field can be removed by transforming to an appropriate co-moving valuation frame.

Let nearby asset returns be:

dr_i = CommonMarketComponent + DifferentialComponent_i. (M.89)

A local frame transformation removes the common component:

dr_i′ = dr_i − dr_common. (M.90)

What remains is:

dr_i′ ≈ DifferentialCredit_i + DifferentialLiquidity_i + DifferentialGate_i + Residual_i. (M.91)

The residual differences play the role of financial tidal effects.

M.7.6 Geodesic equation

A locally natural financial trajectory may satisfy:

$d^2x^\mu/d\lambda^2 + \Gamma^\mu_{\alpha\beta}(dx^\alpha/d\lambda)(dx^\beta/d\lambda) = 0$. (M.92)

With intervention:

$d^2x^\mu/d\lambda^2 + \Gamma^\mu_{\alpha\beta}u^\alpha u^\beta = f^\mu_U$. (M.93)

Where f^μ_U may represent:

- liquidity injection;
- forced sale;
- refinancing;
- capital raising;
- regulatory action;
- hedging;
- covenant enforcement.

The distinction is:

Geometry-Induced Motion ≠ External Intervention. (M.94)

M.7.7 Geodesic deviation

Let ξ^μ measure the separation between two initially similar financial trajectories.

Then a GR-like deviation equation is:

$D^2\xi^\mu/D\lambda^2 = -R^\mu_{\ \nu\alpha\beta}u^\nu\xi^\alpha u^\beta$. (M.95)

Possible interpretation:

Two firms with similar initial cash flows diverge because their locations differ in:

- collateral dependence;
- funding maturity;
- investor concentration;
- liquidity depth;
- proximity to a legal or rating gate.

This differential divergence is more GR-like than ordinary common-market movement.

M.7.8 Financial source tensor

A speculative financial stress tensor is:

T^F_{μν} = Stress and Flow Distribution over $\mathscr{M}$_F. (M.96)

Possible components are:

T^F_{00} = obligation, leverage, or stored-pressure density. (M.97)

T^F_{0a} = funding, liquidity, information, or collateral flow. (M.98)

T^F_{ab} = directional market stress and constraint coupling. (M.99)

A candidate field equation is:

G^F_{μν} + Λ_F^G g^F_{μν} = κ_FT^F_{μν} + Ξ^F_{μν}. (M.100)

Where:

- G^F_{μν} is derived from financial curvature;
- Λ_F^G is a background market-geometry term;
- κ_F is a coupling scale;
- Ξ^F_{μν} is unexplained geometric residual.

This equation is only a research template.

It has not been derived from CAPM.

M.7.9 Backreaction criterion

A genuinely GR-like finance theory requires both directions:

T^F → g^F. (M.101)

g^F → financial trajectories. (M.102)

In words:

Financial stress changes the metric. (M.103)

The changed metric redirects future financial motion. (M.104)

If only the second direction exists, the metric is an externally imposed statistical geometry.

If only the first direction exists, stress changes descriptors but not future motion.

GR-like backreaction requires the loop:

Stress → Geometry → Motion → New Stress. (M.105)

M.8 Electromagnetic and Gauge Bridge

Although this appendix focuses on QM, SR, and GR, the electromagnetic-like layer supplies an important bridge.

Let $\mathcal{A}_\mu$ be a financial connection.

Define its field strength:

$\mathscr{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu$. (M.106)

A financial electric-like component may represent:

- direct valuation gradient;
- immediate risk-premium force;
- directional price pressure.

A financial magnetic-like component may represent:

- circulation;
- path-dependent capital flow;
- loop effects;
- rotation among pressure channels.

The gauge layer describes how local frames connect.

The metric layer describes financial distance and curvature.

These are distinct roles:

Gauge Connection $\mathcal{A}_\mu$ → orientation and circulation. (M.107)

Metric g^F_{μν} → distance, causal structure, and curvature. (M.108)

Conflating the connection with the metric would create unnecessary contradiction.

M.9 Layered Mathematical Architecture

M.9.1 Base manifold

Let:

$\mathcal{M}_F$ = financial state manifold. (M.109)

Each point contains a local financial state:

$x \in \mathcal{M}_F$. (M.110)

M.9.2 GR-like metric layer

Assign:

$g^F_{\mu\nu}(x,L,P)$. (M.111)

This defines:

- local interval;
- local inertial frames;
- geodesics;
- curvature.

M.9.3 SR-like tangent-space layer

At each point x, the tangent space $T_x\mathcal{M}_F$ has a local flat metric:

$g^F_{\mu\nu}(x) \to \eta_{ab}$ under a local frame $e^a_\mu(x)$. (M.112)

Thus:

SR-Like Valuation Kinematics = Local Tangent Limit of GR-Like Financial Geometry. (M.113)

This prevents SR and GR from competing.

SR becomes the local approximation of GR, as it should.

M.9.4 Gauge layer

Define a connection:

$\mathcal{A}_\mu(x,L,P)$. (M.114)

It controls:

- local financial phase comparison;
- path-dependent transport;
- pressure circulation;
- frame alignment.

The covariant derivative is:

$D_\mu = \nabla_\mu + i\mathcal{A}_\mu$. (M.115)

Where ∇_μ is metric-compatible differentiation.

M.9.5 QM-like complex-state layer

Let:

$\Psi_F(x,\tau)$ (M.116)

be a complex effective state or field over $\mathscr{M}_F$.

A general evolution template is:

$i\hbar_F D_\tau \Psi_F = \hat{H}_F[g^F,\mathcal{A},P,L]\Psi_F + \mathcal{N}_F(\Psi_F) + \varepsilon_\Psi$. (M.117)

Where:

- $\hbar_F$ is a scaling constant, not physical Planck's constant;
- $\hat{H}_F$ is an effective financial evolution operator;
- $\mathcal{N}_F$ represents nonlinear feedback;
- ε_Ψ is model residual.

The CAPM complex state Z may be a low-dimensional local reduction of Ψ_F.

M.9.6 Gate and ledger layer

The complex field does not itself define a committed financial event.

The observer boundary performs:

$\text{Outcome}_k = G_{P_k}[\Psi_F,X_k,L_k]$. (M.118)

$L_{k+1} = L_k \sqcup (\text{Outcome}_k,\text{Evidence}_k,\text{Residual}_k)$. (M.119)

The gate–ledger layer must remain separate from the smooth field equation unless a future theory derives it from the same action.

M.10 Unified Action Template

A mathematically organized grand-unification model may begin from:

$S_{total} = S_{geom} + S_{gauge} + S_{state} + S_{gate} + S_{residual}$. (M.120)

Where:

$S_{geom}[g^F]$ = geometry action. (M.121)

$S_{gauge}[g^F,\mathcal{A}]$ = connection and circulation action. (M.122)

$S_{state}[g^F, \mathcal{A}, \Psi_F]$ = complex-state dynamics. (M.123)

$S_{gate}[P, L, \Psi_F]$ = commitment and ledger rule. (M.124)

$S_{residual}$ = penalty or disclosure term for non-closure. (M.125)

Variation with respect to the metric gives a geometry equation:

$\delta S_{total}/\delta g^F_{\mu\nu} = 0$. (M.126)

Variation with respect to the connection gives a gauge equation:

$\delta S_{total}/\delta \mathcal{A}_\mu = 0$. (M.127)

Variation with respect to the complex state gives a state equation:

$\delta S_{total}/\delta \Psi_F^* = 0$. (M.128)

This architecture allows the layers to interact without requiring them to be the same mathematical object.

M.11 Compatibility Limits

The unified model should recover simpler layers under declared limits.

M.11.1 Scalar CAPM limit

When:

$Q \to$ unreported, (M.129)

$g^F \to$ fixed, (M.130)

$\mathcal{A} \to 0$, (M.131)

$\varepsilon \to$ small, (M.132)

the model reduces to:

$r_i = r_f + \beta_{iERP}$. (M.133)

M.11.2 Complex Finance Geometry limit

When curvature and multi-state structure are negligible:

$Z_F = R + iQ_E$. (M.134)

$A^2 = R^2 + Q_{E^2}$. (M.135)

M.11.3 SR-like limit

When:

$\partial\lambda g^{F}\{\mu\nu\} \approx 0$, (M.136)

the geometry is locally flat and relative financial frames may be described through:

$\rho_{ab} = T \ln[(1 + r_b)/(1 + r_a)]$. (M.137)

M.11.4 GR-like limit

When g^{F} varies materially:

$\partial\lambda g^{F}\{\mu\nu\} \neq 0$, (M.138)

the connection and curvature become necessary.

M.11.5 QM-like local-field limit

When the complex state evolves smoothly on the declared geometry:

$i\hbar_{FD}\tau\Psi_F = \hat{H}_F\Psi_F + \varepsilon_\Psi$. (M.139)

This remains QM-like rather than physically quantum unless the model derives the relevant quantum probability and composition structures.

M.12 Can the CAPM Grand Unification Avoid Mathematical Contradiction?

M.12.1 Yes, at the level of layered formal compatibility

There is no immediate mathematical contradiction in combining:

- a curved base manifold;
- local Lorentzian tangent spaces;
- a gauge connection;
- a complex state field;
- observer-dependent projection;
- event gates;
- ledgered backreaction.

These objects already belong to mathematically compatible categories:

Metric Geometry → g^{F}. (M.140)

Local Lorentz Symmetry → tangent frames. (M.141)

Gauge Structure → $\mathcal{A}_\mu$. (M.142)

Complex State → Ψ_F. (M.143)

Projection and Commitment → Π_P and G_P. (M.144)

Trace and Backreaction → L and $\mathcal{B}$. (M.145)

The absence of contradiction comes from assigning different functions to different structures.

M.12.2 SR does not compete with GR

The proper relationship is:

SR-Like Layer = Local Flat Limit of GR-Like Layer. (M.146)

It should not be:

SR Metric + Independent GR Metric on the Same Coordinates without a transition rule. (M.147)

The local tangent metric η_ab and curved metric g^F_{μν} are connected by a local frame:

g^F_{μν} = e^a_μe^b_νη_ab. (M.148)

This keeps the two layers compatible.

M.12.3 QM-like states can live on curved geometry

A complex field can be defined on a curved manifold.

Its derivatives must become covariant:

∂_μ → D_μ. (M.149)

Thus the state equation depends on:

- the metric;
- the connection;
- the local observer frame.

At this formal level:

Complex State Dynamics + Curved Geometry (M.150)

is not contradictory.

The unresolved issue is not basic compatibility.

It is whether a complete theory can derive:

- the correct probability law;
- state composition;
- backreaction;

- measurement outcomes;
- a quantum-consistent geometry.

M.12.4 The CAPM model can demonstrate architectural compatibility

The CAPM construction may serve as a toy model showing that:

1. one scalar state can be completed into a complex state;
2. relative filter frames can be assigned boost-like parameters;
3. the frame law can become local when the metric varies;
4. complex states can evolve on the resulting curved state space;
5. gate and ledger processes can generate effective observations and historical backreaction.

This demonstrates:

Possible Coexistence of Roles without Algebraic Collision. (M.151)

It does not demonstrate:

Physical Unification of QM and GR. (M.152)

M.13 Main Sources of Potential Contradiction

M.13.1 Identifying Q_E and Q_H

The Euclidean pressure is:

$Q_E = \sqrt{(A^2 - R^2)}$. (M.153)

The light-cone hyperbolic pressure is:

$Q_H = Q_E^2/(2R)$. (M.154)

They must not be treated as the same coordinate.

M.13.2 Identifying η_Q and ρ_D

The Q-preserving rapidity is:

$\eta_Q = \operatorname{artanh}(Q_E/A)$. (M.155)

The discount rapidity is:

$\rho_D = \ln(A/R)$. (M.156)

Their relationship is:

$\rho_D = \ln \cosh \eta_Q$. (M.157)

They are not interchangeable.

M.13.3 Treating CAPM beta as velocity

CAPM β is a covariance sensitivity.

It is not a bounded speed.

The appropriate bounded frame variable is more plausibly:

$u_D = \tanh \rho_D$. (M.158)

M.13.4 Treating θ as every kind of phase

The circular finance angle θ should not simultaneously serve as:

- quantum phase;
- Lorentz rapidity;
- global time;
- spacetime coordinate;

without explicit maps.

The distinct variables should remain:

θ = circular finance-filter angle. (M.159)

η_Q = Q-preserving hyperbolic rapidity. (M.160)

ρ_D = additive discount rapidity. (M.161)

τ_L = ledger time. (M.162)

M.13.5 Treating c_F as a discovered constant

A normalized bound:

$|u_D| < 1$ (M.163)

does not prove a universal financial speed of light.

A genuine c_F would require:

- a financial distance;
- a propagation law;
- an invariant speed;

- a causal-cone interpretation;
- empirical universality.

M.13.6 Treating gate commitment as Born collapse

The financial gate can reproduce:

Contextual Commitment + Record. (M.164)

It does not derive:

$P(a) = \langle \psi | \Pi_a | \psi \rangle$. (M.165)

The probability law remains a missing layer.

M.13.7 Treating metric estimation as an Einstein equation

A state-dependent covariance metric may be useful.

But:

Estimated Curved Metric ≠ Derived Gravitational Field Equation. (M.166)

A true GR-like financial theory needs a source–geometry backreaction law.

M.13.8 Double-counting gauge and metric effects

A direct risk-premium gradient may belong to the gauge or potential layer.

A change in financial distance belongs to the metric layer.

The two should not both explain the same repricing term without residualization.

M.13.9 Confusing protocol transition with coordinate transformation

A coordinate transformation changes the description of one world.

A protocol transition may change the world itself:

$P_{k+1} \neq P_k$. (M.167)

The latter cannot always be represented as a smooth Lorentz or coordinate transformation.

M.14 Unified Mapping Table by Mathematical Role

Mathematical role	CAPM realization	QM-like role	SR-like role	GR-like role
scalar observable	R	registered output	frame-dependent coordinate	local measurement
norm or amplitude	A	state norm analogy	invariant scale	locally defined magnitude
orthogonal complement	Q_E	unresolved component	hyperbolic coordinate in one embedding	pressure component
circular phase	θ	relative-phase analogy	not true rapidity	local phase coordinate
additive rapidity	ρ_D	none required	boost parameter	local boost field
complex field	Z or Ψ_F	state-like field	scalar under boosts unless specified	section over curved manifold
metric	G or g^F	state-space geometry	η	g^F(x,L)
connection	filter transport	phase connection	inertial-frame relation	Levi–Civita plus gauge connection
curvature	loop and order residual	contextual path dependence	zero in flat limit	nonzero Riemann curvature
source	leverage and obligations	state intensity	not required	T^F_{μν}
observer boundary	P, Π, G	measurement context	frame declaration	local observer
record	L	outcome record	event history	geometry-affecting trace
backreaction	$\mathcal{B}$	disturbance	frame-sensitive action	stress–geometry loop
residual	ε	model inadequacy	non-flat correction	field-equation residual

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M.15 Proposed Grand-Unification Ladder

The next article may organize the derivation through the following ladder.

Layer 0 — Mature CAPM

$r_i = r_f + \beta_{iERP}$. (M.168)

Layer 1 — Scalar valuation ratio

$w_i = R_i/A_i$. (M.169)

Layer 2 — Complex finance geometry

$Z_i = R_i + iQ_{E,i}$. (M.170)

Layer 3 — Circular phase dynamics

$Z_i = A_i \exp(i\theta_i)$. (M.171)

Layer 4 — Lorentz-like Q embedding

A_i = R_i cosh η_Q,i. (M.172)

Q_E,i = R_i sinh η_Q,i. (M.173)

Layer 5 — Additive discount-frame boosts

ρ_ab = T ln[(1 + r_b)/(1 + r_a)]. (M.174)

Layer 6 — Light-cone valuation coordinates

R_- = A exp(−ρ_D). (M.175)

R_+ = A exp(+ρ_D). (M.176)

Layer 7 — Curved financial manifold

ds_F² = g^F_{μν}(x,L)dx^μdx^ν. (M.177)

Layer 8 — Gauge transport

D_μ = ∇_μ + i𝒜_μ. (M.178)

Layer 9 — Complex field on curved finance geometry

iℏ_FD_τΨ_F = Ĥ_F[g^F,𝒜]Ψ_F + ε_Ψ. (M.179)

Layer 10 — Gate, ledger, and recursive backreaction

Ψ_F → G_P → L_{k+1} → $\boldsymbol{\mathcal{B}}$ → g^F_{next},P_{next}. (M.180)

This ladder integrates the roles without claiming that every layer has already been derived.

M.16 What the Grand-Unification Model Could Demonstrate

A successful CAPM grand-unification model could demonstrate the following logical possibility:

> A bounded observer may inhabit a locally classical and Lorentz-compatible effective world, represent states through complex phase coordinates, experience a curved geometry generated by accumulated pressure and trace, and commit events through a contextual gate—without requiring these structures to contradict one another.

More specifically, it could show:

Demonstration 1 — Local and global laws can differ

Local CAPM may be simple even when the global financial manifold is curved.

Demonstration 2 — Complex state and Lorentz geometry can coexist

The complex state lives in a fibre or internal state space.

Lorentzian geometry governs the base manifold or frame transformations.

They need not compete for the same coordinate role.

Demonstration 3 — SR and GR can be nested

The SR-like layer is recovered in local flat regions of the GR-like layer.

Demonstration 4 — Observer projection can coexist with covariant dynamics

Smooth field evolution may occur inside the effective world while gates and ledgers operate at its observer boundary.

Demonstration 5 — Backreaction can close the architecture

State and ledger affect the metric, while the metric affects subsequent state motion.

M.17 What It Cannot Demonstrate by Analogy Alone

Even a mathematically consistent CAPM construction would not prove:

- that physical spacetime emerges from finance-like filters;
- that physical gravity is market-style backreaction;
- that quantum amplitudes are hidden valuation coordinates;
- that the Born rule follows from $A^2 = R^2 + Q^2$;
- that Bell violations follow from observer constraints;
- that full quantum gravity has been solved;
- that the physical universe is a CAPM effective world.

The framework can provide:

Architectural Demonstration. (M.181)

Conceptual Integration. (M.182)

Formal Toy Model. (M.183)

Research Heuristics. (M.184)

It cannot provide physical confirmation without physical derivation and experiment.

M.18 The Most Defensible Grand-Unification Thesis

The next article's strongest defensible thesis may be:

> CAPM can be extended into a protocol-bound toy geometry in which complex valuation states, relative Lorentz-like valuation frames, locally curved financial state

> space, gauge-like transport, observer projection, ledgered commitment, and recursive backreaction coexist within one layered mathematical architecture. This does not unify the physical forces, but it demonstrates how several structures associated with QM, SR, and GR can be assigned compatible roles without immediate algebraic contradiction.

A stronger speculative thesis is:

> The compatibility suggests that quantum state, local relativity, curved geometry, and observer-generated records may eventually be understood as layers of one world-forming process rather than mutually incompatible theories.

The stronger thesis remains conjectural.

M.19 Questions the Next LLM Session Must Resolve

The next derivation should not begin by inventing additional physics vocabulary.

It should answer the following questions in order.

M.19.1 Which Lorentz embedding is primary?

Choose among:

- Q-preserving rapidity η_Q;
- discount-composition rapidity ρ_D;
- a dual-coordinate theory retaining both.

M.19.2 What is the financial manifold?

Declare:

$\mathscr{M}_F$ = [state space]. (M.185)

Specify its coordinates and dimensions.

M.19.3 What is the invariant?

Possible candidates include:

- $R^2 = A^2 - Q_E^2$;
- $A^2 = R_- R_+$;
- proper financial interval ds_F^2;
- invariant ledgered obligation;
- action S_{total}.

Without an invariant, the relativity layer remains incomplete.

M.19.4 What generates the metric?

Specify:

$g^F_{\mu\nu} = g^F_{\mu\nu}(T^F,L,P)$. (M.186)

The source must be measurable or operationally defined.

M.19.5 What is the financial equivalence principle?

Define which common effects may be removed locally and which differential effects count as curvature.

M.19.6 What is the complex state space?

Decide whether Ψ_F is:

- one complex scalar;
- a vector of filter channels;
- a probability-amplitude-like object;
- a section of a complex bundle;
- an effective stochastic state.

M.19.7 What probability law governs outcomes?

The present theory has gates but no Born rule.

A grand-unification article must either:

- remain explicitly classical and stochastic;
- derive a new probability rule;
- treat the QM layer as structural analogy only.

M.19.8 How does backreaction work?

Derive both:

$T^F \rightarrow g^F$. (M.187)

$g^F \rightarrow \Psi_F$ or financial trajectories. (M.188)

M.19.9 How does the gate relate to smooth evolution?

Determine whether:

- the gate is an external effective boundary;
- the gate emerges from nonlinear dynamics;
- the gate is a stochastic instrument;
- the gate remains institutionally declared.

M.19.10 What would falsify the construction?

Define:

- flat-limit failures;
- invariant failures;
- metric instability;
- failure of geodesic prediction;
- absence of backreaction;
- no gain over conventional models.

M.20 Handoff Summary for the Next Article

The next session should inherit the following logic.

1. CAPM supplies a mature local valuation filter.
2. The filter ratio generates:

   R/A, θ, Q_E, and $Z = R + iQ_E$.
3. The internal complex dynamics are classical:

   $dZ/d\theta = iZ$. (M.189)
4. The original Euclidean geometry admits a Q-preserving hyperbolic embedding:

   $A = R \cosh \eta_Q$. (M.190)

   $Q_E = R \sinh \eta_Q$. (M.191)
5. CAPM discount ratios independently generate an additive frame parameter:

   $\rho_{ab} = T \ln[(1 + r_b)/(1 + r_a)]$. (M.192)
6. The additive parameter naturally acts on light-cone valuation coordinates:

   $R_{-}' = \exp(-\Delta\rho) R_{-}$. (M.193)

$R_+' = \exp(+\Delta\rho) R_+$. (M.194)

7. The two hyperbolic constructions are compatible but not identical:

   $\rho_D = \ln \cosh \eta_Q$. (M.195)

   $Q_H = Q_E^2/(2R)$. (M.196)

8. A GR-like extension promotes the fixed valuation geometry into:

   $g^F_{\mu\nu}(x,L,P)$. (M.197)

9. CAPM then becomes a local tangent-space approximation.

10. Leverage, liquidity, obligations, and ledgered events become candidate sources of curvature.

11. Complex financial states may evolve covariantly on this curved manifold through:

$D_\mu = \nabla_\mu + i\mathcal{A}_\mu$. (M.198)

12. Gate and ledger processes remain the observer-facing commitment layer.

13. The full architecture can be mathematically layered without immediate contradiction.

14. Its physical interpretation remains unproven.

M.21 Final Assessment

A CAPM grand-unification model can make a meaningful contribution even if it never becomes a literal physical theory.

Its value would be to provide a transparent constructed system in which:

- scalar valuation becomes complex state;
- complex state acquires phase;
- phase admits relative-frame transformations;
- relative frames become local tangent structures;
- local structures form a curved global geometry;
- accumulated stress changes that geometry;
- the changed geometry redirects future state evolution;
- observer gates convert possibilities into ledgered events;
- ledgered events feed back into the next geometry.

The complete conceptual loop is:

Filter
→ Complex State
→ Relative Frame
→ Local Lorentz Geometry

→ Curved Global Geometry
→ Covariant Complex Dynamics
→ Gate
→ Ledger
→ Backreaction
→ Revised Geometry. (M.199)

This architecture supports a limited but important conclusion:

> QM-like state representation, SR-like local frame symmetry, and GR-like state-dependent curvature do not have to be mutually contradictory when they are treated as distinct but coupled layers of one effective-world runtime.

The framework does not yet demonstrate that physical QM and GR arise from this mechanism.

It demonstrates something logically prior:

> There exists a coherent route by which complex state, relativity, curvature, observation, trace, and backreaction can be organized inside one declared model without forcing one concept to perform every mathematical role.

That is sufficient to justify a separate grand-unification article.

Its decisive discipline should remain:

Do Not Prove Unity by Reusing the Same Symbol. (M.200)

Prove Unity by Showing How Distinct Structures Transform, Couple, Reduce, and Remain Consistent under Shared Limits. (M.201)

Reference

- The Complex Residual Principle: How Phase, Projection, Residual, Trace, and Emergent Time Reappear across Quantum Physics, Financial Markets, and Large Language Models
https://osf.io/yucvm/files/osfstorage/6a53876497a8be0d215b9278

- Finance Geometry: Complex Valuation, Risk Pressure, and the Hidden Coordinate Behind Mature Finance Filters
https://osf.io/yucvm/files/osfstorage/6a4abb8fcaf0a0c36ddaa3e3

- Imaginary Time as Admissibility Depth: A Ledger Ontology of Wick Rotation, Macro Systems, and Physical Time
https://osf.io/mvq6e/files/osfstorage/6a405c693e12266e39804e08

- The True Nature of Technical Analysis - An Operator-First Interpretation of Market Charts, Volume, Waves, Gann Geometry, and Financial Self-Reference
https://osf.io/ne89a/files/osfstorage/6a3689cb33b86e3d1a86e142

- The Imaginary Axis of Technical Analysis: How Complex Numbers Turn Chart Folklore into Market Pressure Geometry
https://osf.io/yucvm/files/osfstorage/6a4b942006735c3ce6daa274

- A Rigorous Mathematical Grammar And Checklist That Ensure Nature-Inspired Systems Are Stable, Bounded, And Economically Viable
https://osf.io/hj8kd/files/osfstorage/6a500b7bbdb5870c2c7afb69

- From Fundamental Physics to Purpose-Matched AI Agents

4π Spinor Closure, Hidden Control Stacks, and Environment-Aware Runtime Design
https://osf.io/hj8kd/files/osfstorage/6a4f89f3eef0d1166c5b9338

- From Physics to AI Design: A Rosetta Stone for Runtime Architecture
https://osf.io/hj8kd/files/osfstorage/69d5023f5cdefa314c3eb654

- Proto-Eight Dynamics (P8D): a small, testable model of how growth actually works【先天八卦動力學】
https://osf.io/9rdsc/files/osfstorage/68b71c00b65e7b0e352c22f6

- From Interfaces to Isomorphisms: A Protocol-Bound Theory of World Formation
How Bounded Observers Turn Fields into Operational Worlds — and Why Physics, Life, Organizations, Finance, Law, and AI Reuse the Same Grammar
https://osf.io/ae8cy/files/osfstorage/69ffbfc888878a0f3e78fda2

- Philosophical Interface Engineering 1 - Turning Deep Ideas into Testable Worlds, Thought Experiments, and Civilizational Tools - A New Renaissance of Philosophy after AI
https://osf.io/ae8cy/files/osfstorage/69f777e12417f21f0f1e5206

- Philosophical Interface Engineering 2 - Turning Deep Ideas into Testable Worlds, Thought Experiments, and Civilizational Tools - A New Renaissance of Philosophy after AI
https://osf.io/ae8cy/files/osfstorage/69f777e12417f21f0f1e5206

- Philosophical Interface Engineering 3 - Turning Deep Ideas into Testable Worlds, Thought Experiments, and Civilizational Tools - A New Renaissance of Philosophy after AI
https://osf.io/ae8cy/files/osfstorage/69f777e12417f21f0f1e5206

- Life as a Dual Ledger: Signal – Entropy Conjugacy for the Body, the Soul, and Health
https://osf.io/s5kgp/files/osfstorage/690f973b046b063743fdcb12

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.